The Psychology of

JUDGMENT AND DECISION-MAKING

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PREFACE

As a master student at the VU University Amsterdam, I took a course called *Thinking and Deciding*. I eventually specialized in a different branch of psychology – cognitive neuroscience – but to this day this course remains one of my favorite educational experiences. In part, this was thanks to a young and passionate teacher, Mark Nieuwenstein. But in part, it was also because the course provided me with a new set of tools to understand the mysterious ways of people.

Our beliefs are shaped by those around us – our bubble. You don't need to be a psychologist to know this. But *Thinking and Deciding* helped me to think more systematically about how this comes to be. I learned about the availability heuristic – a powerful force that causes beliefs and behavior to be dominated by information that readily comes to mind, such as what we see on social media. I learned how we're far from the neutral observers we like to think we are. Instead, we're biased observers who seek out specific kinds of information: negative information, emotionally charged information, and information that confirms what we already believe. I learned how this biased information diet interacts with the availability heuristic to shape beliefs and behaviors. I learned how we identify causal relationships, and why and when we do, or do not, assign responsibility to people – praise and blame. And most important of all, I learned that none of these are failures. Instead, they are efficient and usually effective ways to reduce the complexity of the world to something we can manage. Something simple enough for our limited minds.

And I learned much, much more.

In 2024, we (the experimental psychology unit) redesigned our bachelor curriculum. I volunteered to set up an entirely new course. Or at least a course that was new to us: *Thinking and Deciding*. By then, Mark and I worked in the same department. I built the course based on what I learned from him as a colleague as well as a student.

But the world has changed a lot since I was a student. (I graduated in 2008.) Smartphones and social media dominate every aspect of our lives. If you are like most of your peers, you spend several hours each day looking at a luminous rectangle that projects information into your brain in a way that is only partly under your control. And even though it is too early to know for sure, it seems that AI will become an increasingly important part of our lives as well.

But one thing hasn't changed – the concepts and theories that I learned back in 2008 are just as important today to understand human cognition. The world may have changed, but the fundamental building blocks of human cognition have not.

With this in mind, I set out to redesign the course once again for 2026. One aspect of this redesign is what you're reading right now – a new, open textbook. This is a wild experiment. You may have doubts about it, and you'd be right to have them. Because I did not write all of the hundreds of pages myself. Instead, I composed a detailed outline, collected hundreds of primary sources (mainly research papers), and used AI to generate, review, and revise the text based on these ingredients. Finally, I edited the result myself.

In the process of creating this textbook, I learned more about human decision making than I have at any other point in my life, including when I was taking the course myself as a student. This may sound strange, because AI is generally seen as something that disrupts rather than enhances learning. And it usually is. If you let your eyes glaze over AI-generated text, you learn almost nothing. Unfortunately, that's how most students use AI, and there's a real risk of future generations (you!) having reduced intelligence as a result. (As we will see, average intelligence may already be declining, possibly as a result of social media and smartphone use, although there's a lot of uncertainty around this topic.)

But if you engage deeply with study material, you learn a lot. This is true for any material, regardless of whether it's human- or AI-generated. So: discuss it, question it, find weaknesses, complement it with your own thoughts. That's what I did. And in creating this textbook, I hope to set a positive example of how to use AI to enhance learning. With the course assignment – editing chapters in groups – I invite you to do the same.

Now let's get to work and learn about the human mind.

– Sebastiaan Mathôt

FOUNDATIONS

FOUNDATIONS

REASON AND INTUITION

Learning goals:

- Describe the two types of mental processing (System 1 and System 2) and their key features.
- Explain bounded rationality and why people rely so heavily on fast, automatic thinking.
- Distinguish between associative and rule-based reasoning.
- Describe the concept of ecological rationality and how it challenges the view that heuristics are errors.
- Explain how categorization connects to both types of processing.
- Describe what the Cognitive Reflection Test measures and why it matters for judgment and decision-making.

Two Ways of Thinking

Imagine you are cycling through Groningen on your way to a lecture. A car suddenly pulls out from a side street, and you swerve to avoid it. You did not calculate the angle of approach, estimate the car's speed, or weigh the costs and benefits of braking versus swerving. You just reacted. Now contrast this with another decision you might face today: choosing between two elective courses for next semester. You read the course descriptions, ask friends for their experiences, compare the schedules, and eventually settle on the one that fits best. These two situations feel very different. They rely on fundamentally different kinds of mental processing.

This distinction between fast, automatic thinking and slow, deliberate thinking is one of the most influential organizing frameworks in the study of judgment and decision-making. Researchers often refer to these as System 1 and System 2 (Evans & Stanovich, 2013). Because these are not literally separate brain systems, some researchers prefer the terms Type 1 and Type 2; in this book, however, we will usually use the more familiar labels System 1 and System 2. This is not just a descriptive distinction. Human cognition is fundamentally limited – in time, knowledge, and processing capacity – so fast, intuitive processing is often not a luxury but a necessity. Understanding how these two types of processing work, when each dominates, and how they interact is essential for understanding nearly every topic in this book.

System 1 and System 2

System 1 processing is fast, automatic, and effortless. It runs in the background without requiring your conscious attention. When you read the word "cat," you instantly understand its meaning. When you see a friend's facial expression, you immediately sense whether they are happy or upset. When you catch a ball, your hand moves to the right place without any deliberate calculation. All of these are examples of System 1 at work. So are the many habitual actions you perform every day without thinking: typing your password, cycling a familiar route, or braking automatically when a traffic light turns red. System 1 operates in parallel – it can handle multiple things at once – and it does not require working memory, the limited mental workspace you use when you hold a phone number in your head or solve a math problem step by step. System 1 is also closely linked to emotions and gut feelings. Affect is not just an accompaniment to intuition; it is often part of the information that intuitive processing uses. A sense of unease about a decision, a flash of excitement about an opportunity – these emotional signals feed directly into System 1 judgments, a point that will become important in later chapters on risk perception (see Risk Perception) and moral judgment (see Morality). System 1 is not always open to introspection: you may not be able to explain why something feels right or wrong, only that it does.

System 2 processing differs from System 1 in several characteristic ways. It is slow, effortful, and deliberate. It operates one step at a time, in a serial fashion, and it depends heavily on working memory. When you work through a statistics problem, draft an essay, or plan a holiday budget, you are engaging System 2. This type of processing is consciously accessible – you can reflect on what you are doing and why. One of its most distinctive features is what Evans & Stanovich (2013) call cognitive decoupling: the ability to think about things that are not currently happening. You can imagine hypothetical scenarios, reason about the future, or consider abstract possibilities. This capacity for detachment from the here and now is what allows you to plan, to think about "what if," and to evaluate arguments on their logical merits rather than their emotional appeal.

These two types of processing interact constantly. One influential view is the default-interventionist model, which holds that System 1 generates quick, default responses to the situations you encounter, and System 2 may then step in to check, correct, or override those responses (Evans, 2008; Evans & Stanovich, 2013). Sometimes System 2 endorses what System 1 suggests, and you go with your gut. Other times, System 2 recognizes that the intuitive answer is wrong and replaces it with a more carefully reasoned one. Yet other times, System 2 is not used at all, leaving the System 1 response unchecked.

The distinction between System 1 and System 2 is a useful simplification, not a precise description of the brain's architecture. There is no single "System 1 module" or "System 2 module" in the brain. Instead, Type 1 processing encompasses a diverse collection of automatic processes – learned habits, emotional reactions, perceptual routines – while Type 2 processing refers to the kind of reflective, working-memory-intensive thinking that these various automatic processes sometimes require as a check (Evans & Stanovich, 2013). Many real-world mental operations have aspects of both systems. An experienced chess player, for example, may "see" a strong move instantly (System 1), drawing on years of deliberate practice that originally required effortful analysis (System 2). The boundary between the two is not always sharp, but the distinction remains one of the most productive frameworks in cognitive science.

Bounded Rationality

Why do we rely so heavily on System 1? The concept of bounded rationality, introduced by Herbert Simon in the 1950s, provides the answer (Simon, 1956, 1972).

Classical models in economics assumed that people are fully rational decision-makers. In these models, a person facing a choice gathers all relevant information, considers every possible option, calculates the expected outcome of each, and selects the one that maximizes their benefit. Simon pointed out that this picture is unrealistic. In the real world, you almost never have access to all relevant information. Even when information is available, you lack the time and cognitive capacity to process it all. And the problems you face are often so complex that finding the truly optimal solution would require more computation than any human brain can perform.

Consider a concrete example from Simon's own work. He compared decision-making to playing chess (Simon, 1972). In chess, the rules are perfectly known and all information is visible on the board. Yet finding the optimal move at any point in a game would require evaluating something on the order of 10 to the power of 120 possible sequences of moves – a number so vast it exceeds the number of atoms in the observable universe. If perfect rationality is impossible even in chess, how could it be possible in the messier, more uncertain situations of everyday life?

Simon's answer was that people do not optimize; they satisfice. This term, a combination of "satisfy" and "suffice," describes the strategy of searching through options until you find one that is good enough, and then stopping. You do not need to examine every apartment in Groningen to find a place to live. You look at a few, and when you find one that meets your basic requirements, you take it. This is a sensible adaptation to the limits of your time, knowledge, and cognitive resources. Satisficing is a concept that will return in the chapter on how people choose between options (see Satisficing vs. Maximizing).

Bounded rationality explains the dominance of System 1 in daily life. System 2 is powerful but expensive – it requires attention, effort, and time, all of which are scarce. For the vast majority of decisions you face each day, from what to eat for breakfast to how to respond to a friend's text message, System 1 provides responses that are quick and usually good enough. People rely heavily on System 1 not out of laziness but out of necessity. System 2 is reserved for situations where the stakes are high, the problem is novel, or System 1's initial response is detected as potentially wrong.

Associative and Rule-Based Reasoning

The System 1/System 2 distinction characterizes how processing feels and operates – fast versus slow, automatic versus deliberate. But what kinds of reasoning do these systems actually carry out? Sloman (1996) offered a complementary characterization that helps answer this question, describing an associative system and a rule-based system that map closely onto the dual-process distinction.

Sloman's associative system corresponds roughly to System 1. It reasons through similarity and statistical regularities. It draws on patterns you have encountered before, making judgments based on how much a new situation resembles familiar ones. When you hear an unfamiliar melody and immediately sense that it sounds "jazzy," your associative system is matching the new input to stored patterns. When you meet someone at a party and quickly form an impression of their personality, you are drawing on associations between their appearance, behavior, and the kinds of people you have encountered in the past. Many of the heuristics and biases covered in this book reflect this kind of associative processing.

Sloman's rule-based system corresponds roughly to System 2. It reasons through explicit rules – logical, mathematical, social, or definitional. Instead of relying on resemblance, it checks whether specific criteria are met. A doctor diagnosing a patient might initially rely on pattern recognition (associative processing), but then verify the diagnosis by checking whether the patient's symptoms meet the formal criteria listed in a diagnostic manual (rule-based processing).

One of Sloman's key contributions was showing that these two systems can produce conflicting responses to the same problem, and that people can be aware of both responses at the same time. A simple illustration is the whale: most people will agree that a whale is more similar to a fish than to a human. That is the associative system at work – whales look like fish, live in the sea like fish, and swim like fish. But most people also know that a whale is in fact a mammal, more closely related to humans than to fish. That is the rule-based system applying biological classification criteria. You can hold both responses in mind simultaneously: the whale feels like a fish, but you know it is a mammal. This kind of conflict between the two systems is a recurring theme in the study of judgment and decision-making, and it underlies many of the biases we will encounter in later chapters (see Heuristics and Biases).

Ecological Rationality

Kahneman, Tversky, and Stanovich emphasize the systematic errors that intuitive processing can produce relative to normative standards, especially in laboratory tasks (Kahneman, 2011; Stanovich & West, 2000). In their view, System 1's reliance on shortcuts (called heuristics) often leads to systematic biases, and System 2 serves as the corrective mechanism that catches and fixes these mistakes.

Gerd Gigerenzer and colleagues offer a different perspective (Gigerenzer & Todd, 2000). They do not deny that people use simple heuristics, but they argue that many of these heuristics are not flawed shortcuts in need of correction. Instead, they are ecologically rational: well adapted to the structure of the environments in which people actually make decisions. A heuristic is ecologically rational when it exploits the statistical or informational structure of a particular environment to produce good outcomes, even if it ignores some information or violates the rules of logic or probability. Gigerenzer does not claim that heuristics never lead to poor outcomes; rather, he questions whether "bias" is the right label when a simple strategy performs well in the environment where it is actually used.

Consider the recognition heuristic: if you recognize one option but not the other, choose the recognized one. This sounds crude, but it can be remarkably effective. In a study described by Gigerenzer & Todd (2000), people were asked which of two cities has a larger population. German students, who recognized many American cities but not all, actually performed better on questions about American cities than American students did, because the German students could rely on recognition as a useful cue. If you have heard of one city but not the other, recognition is often a valid indicator of size, because larger cities are more likely to appear in the news, in films, and in conversation. The heuristic works because the environment is structured in a way that makes recognition informative.

Another example is the Take the Best heuristic: when comparing two options, search through cues in order of their validity, and decide based on the first cue that distinguishes between them. This approach ignores all remaining cues, which might seem reckless. But Gigerenzer & Todd (2000) showed that in many real-world prediction tasks – such as predicting which of two cities is larger based on features like whether the city has a university, a major sports team, or is a state capital – Take the Best performed as well as or better than complex strategies that weighed and integrated all available information. This is because real-world environments often have a structure where one or two cues carry most of the useful information, and adding more cues introduces noise rather than clarity.

This perspective sets up a central tension that runs throughout this book. When people make judgments that deviate from the standards of logic or probability, are they making genuine errors, or are they applying strategies that work well in the real world but look foolish when tested in a laboratory? The answer, as we will see, is often "both." Some biases clearly lead to poor outcomes, while others reflect sensible adaptations to the kinds of problems people typically face. Keeping this tension in mind will help you evaluate the research more critically.

Categorization as a Bridge

A useful place to see both the efficiency and the potential pitfalls of intuitive processing is categorization – one of the most fundamental things the mind does. You constantly sort objects, people, and situations into categories: this object is a chair, that person is a friend, this situation is dangerous. Categorization is what allows you to go beyond the specific instance in front of you and bring your past experience to bear on the present moment. It is also a place where the distinction between associative and rule-based processing becomes especially clear.

When System 1 categorizes, it does so by resemblance. You see a small, feathered animal perched on a branch, and you immediately recognize it as a bird. You did not check a list of defining features; instead, you matched what you saw to a stored prototype – a mental image of what a "typical" bird looks like. In some cases, categorization may rely less on an abstract prototype than on remembered specific examples: you might recognize an animal as a bird because it reminds you of particular birds you have seen before, such as sparrows or robins. These two approaches – prototype-based and exemplar-based categorization – are both forms of associative, System 1 processing. They are fast, efficient, and usually accurate. But they can go wrong. A penguin does not look much like a typical bird, and you might hesitate before categorizing it as one. An ostrich even less so. Categorization by resemblance is sensitive to how representative an instance is of its category, and this sensitivity has consequences throughout the study of judgment and decision-making.

When System 2 categorizes, it does so by checking criteria. A biologist classifies an animal as a bird not because it looks like one, but because it meets specific biological criteria: it has feathers, it lays eggs, it is warm-blooded, and so on. This approach is more reliable for edge cases (penguins and ostriches clearly qualify) but it is also slower and more effortful.

Why does this matter for a course on judgment and decision-making? Because many of the biases we will study can be understood as cases where categorization by resemblance leads people astray. The representativeness heuristic, which we will examine in detail later (see Representativeness), is essentially categorization by resemblance applied to probability judgments. When people judge that a quiet, bookish person is more likely to be a librarian than a farmer, they are matching the person's description to their prototype of a librarian, while ignoring the statistical fact that farmers vastly outnumber librarians. This is base-rate neglect, and it arises directly from the associative system's reliance on similarity rather than rules. Stereotyping, too, reflects the application of category-level expectations to individual people – a process that is fast and automatic but often inaccurate and unfair.

The Cognitive Reflection Test

If System 1 and System 2 are both always available, what determines which one wins? Part of the answer lies in the situation: time pressure, cognitive load, and emotional arousal all favor System 1. But part of the answer also lies in the person. Some people are more inclined than others to pause, reflect, and check their initial intuitive response.

Frederick (2005) developed a simple but powerful tool for measuring this tendency: the Cognitive Reflection Test (CRT). The original test consists of just three questions, each designed so that an appealing but wrong answer immediately springs to mind. The most famous item is the bat-and-ball problem: "A bat and a ball together cost €1.10. The bat costs €1.00 more than the ball. How much does the ball cost?" Most people's immediate, intuitive response is 10 cents. But if you pause and check, you will see that this cannot be right: if the ball costs 10 cents and the bat costs €1.00 more, the bat costs €1.10, and together they cost €1.20. The correct answer is 5 cents. Another item asks: "If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?" The intuitive answer is 100 minutes, but the correct answer is 5 minutes – each machine makes one widget in 5 minutes, regardless of how many machines are running in parallel. Since the original three-item test became widely known, researchers have developed expanded versions with additional items to reduce the problem of participants having already encountered the questions.

In a series of studies involving over 3,000 participants, Frederick (2005) found that higher CRT scores predicted more patient choices in decisions about money (for example, preferring to receive €3,800 next month rather than €3,400 today) and different patterns of risk-taking. People with higher CRT scores were more willing to accept gambles with higher expected value, even when those gambles involved risk, and they were less susceptible to the kinds of framing effects and biases that prospect theory describes (see Gains and Losses). Higher CRT scores also predict less acceptance of paranormal beliefs, suggesting that the tendency to question intuitive responses extends beyond financial decisions to how people evaluate extraordinary claims. The CRT correlates with other measures of cognitive ability, such as university entrance exams, and it overlaps with numeracy – basic skill with numbers and probabilities. It therefore captures something that is not purely about the disposition to reflect; it also taps into cognitive capacity. Still, the CRT measures something more specific than general intelligence alone: not just how much cognitive horsepower you have, but how willing you are to use it when your first instinct might be wrong.

This connects directly to a broader point about the relationship between intelligence and good thinking, a topic explored more fully in Intelligence vs. Rationality. Being smart, in the sense of having a high IQ, does not guarantee that you will think carefully about the problems you face. You also need the disposition to question your own intuitions, to recognize when a problem is harder than it seems, and to invest the effort required to reason it through.

Summary

Judgment and decision-making research is organized around the distinction between two types of mental processing. System 1 (Type 1) is fast, automatic, and effortless – it encompasses habits, emotional reactions, and perceptual routines, and operates through associations and gut feelings. System 2 (Type 2) is slow, deliberate, and effortful – it relies on working memory and explicit rules, and is capable of abstract, hypothetical reasoning. People depend heavily on System 1 because of bounded rationality: the limits of human knowledge, cognitive capacity, and time make full rational analysis impossible for most decisions. Sloman's characterization of associative and rule-based reasoning helps clarify what each system does – one reasons through similarity and pattern matching, the other through explicit criteria and logic – and these two systems can produce conflicting responses to the same problem. Gigerenzer's concept of ecological rationality challenges the assumption that heuristic thinking is always flawed, showing that simple strategies can be well adapted to the structure of real-world environments. This sets up a central tension in the field: are deviations from normative standards genuine errors, or adaptive strategies judged against an unrealistic benchmark?

Categorization illustrates how both systems operate on the same problem in different ways, and many biases covered in this book can be traced to categorization by resemblance overriding categorization by criteria. The Cognitive Reflection Test measures individual differences in the willingness to override intuitive but incorrect responses, though it also overlaps with numeracy and general cognitive ability. Together, these ideas form the foundation for understanding why people think the way they do – and when that thinking goes well or poorly. The heuristics and biases you will encounter in later chapters are largely products of System 1 processing. The normative standards used to evaluate judgment, such as Bayesian reasoning (see Bayesian Reasoning), are most naturally associated with deliberate, rule-based thought. And the prescriptive recommendations for improving decisions, from debiasing techniques (see Heuristics and Biases) to nudges (see Steering Other People's Choices), often work by either engaging System 2 more effectively or restructuring the environment so that System 1 leads to better outcomes. The goal of this course is not to convince you that System 1 is bad and System 2 is good. It is to help you understand when each type of processing is likely to serve you well, when it is likely to lead you astray, and what you can do about it.

BAYESIAN REASONING

Learning goals:

- Understand the Bayesian framework of updating prior beliefs with new evidence to form posterior beliefs.
- Know the components of Bayes' theorem: prior probability, likelihood, and posterior probability.
- Explain why people fail the mammogram problem and similar diagnostic reasoning tasks.
- Describe how natural frequencies improve Bayesian reasoning compared to conditional probabilities.
- Understand what ideal observer models are and how they connect perception to rational inference.
- Explain the key parameters of signal detection theory: sensitivity and criterion.
- Define calibration and explain how it relates to good judgment.
- Recognize how the major biases covered in this course can be understood as specific departures from Bayesian rationality.

Combining what you know with what you see

Imagine you are cycling home through Groningen late at night. In the distance, you see a figure that vaguely resembles your friend. Could it be them? Before you even process the visual details, you already have relevant background knowledge: the chances that your friend happens to be on this particular street at this particular time are very small. That background knowledge – your prior belief – keeps your confidence low, even though the silhouette bears some resemblance to your friend. As you get closer, a street light illuminates the figure's face. Now the visual evidence is strong and clear: it is someone you have never seen before. The strong evidence resolves the question decisively.

In that brief moment, you combined two things: a prior belief (your friend is very unlikely to be here) and incoming evidence (what the figure looks like). When the evidence was weak, the prior dominated and you remained uncertain. When the evidence became strong, it overwhelmed the prior and you reached a clear conclusion.

This act of combining what you already believe with new evidence is the core of Bayesian reasoning. Formalizing this process reveals something important: people are often surprisingly bad at it, especially when problems involve numbers. Understanding where and why people deviate from this standard is the central theme of this course. As we discussed in Reason and Intuition, much of our thinking relies on fast, automatic processes that do not always follow the rules of probability. Bayesian reasoning gives us a precise way to describe what those rules are and, by extension, a way to identify exactly how we break them. Throughout this book, Bayesian updating serves as the normative standard – the formal description of how a rational agent should revise beliefs in light of evidence. The many biases we will cover are specific, identifiable departures from this ideal.

The Bayesian framework

The Bayesian framework has three components. The first is the prior belief, or simply the prior. This is what you believe before you encounter new evidence, expressed as a probability. Your priors are shaped by base rates in the population, personal experience, cultural knowledge, and the mental models and schemas you carry around (see Mental Models). For example, if you know that about 30% of students in your program are Dutch, then before meeting a new classmate, your prior probability that this person is Dutch is 0.30.

The second component is the evidence, described formally by the likelihood. This is the new information you receive, and specifically how probable that information would be if your hypothesis were true versus if it were false. Suppose you hear your new classmate speaking Dutch on the phone. The likelihood asks: how probable is it that you would hear this person speaking Dutch if they were indeed Dutch? Very probable – say, 0.90. How probable is it that you would hear this person speaking Dutch if they were not Dutch? Much less probable – say, 0.05, since only a small fraction of non-Dutch students speak Dutch. The ratio between these two likelihoods (0.90 / 0.05 = 18) captures the strength of the evidence. A ratio of 18 means this evidence is eighteen times more likely if the person is Dutch than if they are not. That large asymmetry is what makes the evidence informative. If you instead observed something equally likely for Dutch and non-Dutch students – say, carrying a backpack – the ratio would be close to 1, and the observation would tell you almost nothing.

The third component is the posterior belief, or the posterior. This is your updated belief after you have combined the prior with the evidence. After hearing your classmate speak Dutch, your belief that they are Dutch should increase substantially. In the Bayesian framework, the posterior is a specific probability that can be calculated, not just a vague feeling of increased confidence. Importantly, this posterior then becomes the prior for the next piece of evidence you encounter. If you later learn that this classmate grew up in Germany, you update again. Bayesian reasoning is a continuous process of belief revision.

Bayes' theorem

The formal rule that describes how to combine priors and evidence is known as Bayes' theorem, named after Thomas Bayes, an 18th-century minister and mathematician who first described the logic of inverse probability (Bayes, 1763). The theorem states:

$$P(H|E) = \frac{P(E|H) \times P(H)}{P(E)}$$

In this equation, P(H|E) is the posterior probability of a hypothesis H given evidence E. P(E|H) is the likelihood of the evidence given that the hypothesis is true. P(H) is the prior probability of the hypothesis. And P(E) is the total probability of the evidence.

That last term, P(E), is the one students most often find confusing, but it plays a crucial role. It represents the overall probability of observing the evidence, considering all the ways it could have occurred – both when the hypothesis is true and when it is false. Formally, P(E) = P(E|H) × P(H) + P(E|¬H) × P(¬H). The denominator acts as a normalizing factor: it ensures that the posterior is a valid probability. When the evidence could easily occur even if the hypothesis is false (making P(E) large), the posterior will be pulled down. This is exactly what happens in the mammogram problem below.

The logic behind the equation is straightforward. Your updated belief depends on what you believed before (the prior) and how diagnostic the evidence is (the likelihood). Evidence is diagnostic to the extent that it is more probable under the hypothesis than under the alternative. If a piece of evidence is equally likely whether or not the hypothesis is true, it tells you nothing, and your posterior should equal your prior.

The mammogram problem

To see why this matters, consider a classic problem from medical decision-making. A woman receives a positive mammogram. The test has a sensitivity of 80%, meaning that if a woman has breast cancer, the test will correctly detect it 80% of the time: P(positive | cancer) = 0.80. The false-positive rate is 10%, meaning that if a woman does not have cancer, the test will incorrectly say she does 10% of the time: P(positive | no cancer) = 0.10. The base rate of breast cancer in the population of women being screened is 1%: P(cancer) = 0.01. Given all of this, what is the probability that this woman actually has cancer?

When this problem is posed to physicians, the vast majority dramatically overestimate the answer. In a well-known demonstration, 95 out of 100 physicians estimated that the probability was between 70% and 80% (Hoffrage & Gigerenzer, 1998). The correct answer, obtained by applying Bayes' theorem, is approximately 7.5%. The physicians were off by a factor of ten.

We can see where the correct answer comes from by plugging the numbers into Bayes' theorem:

$$P(\text{cancer}|\text{positive}) = \frac{P(\text{positive}|\text{cancer}) \times P(\text{cancer})}{P(\text{positive})}$$

The numerator is 0.80 × 0.01 = 0.008. The denominator is the total probability of testing positive, which includes both true positives and false positives: P(positive) = (0.80 × 0.01) + (0.10 × 0.99) = 0.008 + 0.099 = 0.107. So the posterior is 0.008 / 0.107 ≈ 0.075, or about 7.5%.

Why is the answer so low despite the test being fairly accurate? The denominator reveals the answer. Because the base rate of cancer is only 1%, the vast majority of women being tested are healthy. Even with a false-positive rate of just 10%, the sheer number of healthy women tested means that false positives (99 out of 990) vastly outnumber true positives (8 out of 10). The physicians made their error by confusing two different probabilities. They took the sensitivity of the test, P(positive | cancer) = 0.80, and treated it as though it were the answer to the question being asked, P(cancer | positive). But these are very different things. The probability that you test positive given that you have cancer is about the test's ability to detect disease. The probability that you have cancer given that you tested positive is about what the test result means for you. The second depends critically on the base rate – the prior.

This is not just an academic puzzle. Misunderstanding diagnostic test results has real consequences for patients who may undergo unnecessary and invasive follow-up procedures based on inflated risk estimates (see Medical Decision-Making). And the problem is not limited to medicine. Whenever people encounter probabilistic evidence and need to figure out what it means, they face the same challenge of integrating evidence with base rates.

Natural frequencies

If the mammogram problem is so difficult when presented as probabilities, is there a way to make it easier? Hoffrage & Gigerenzer (1998) showed that there is: present the information as natural frequencies rather than conditional probabilities. Natural frequencies are concrete counts of cases – the kind of information our ancestors would have encountered by observing events in the world.

Here is the mammogram problem restated in natural frequencies. Imagine 1,000 women who are screened. Of these 1,000, about 10 have breast cancer. Of those 10 women with cancer, 8 will test positive. Of the remaining 990 women without cancer, about 99 will also test positive. Now, of all the women who test positive (8 + 99 = 107), how many actually have cancer? The answer is 8 out of 107, which is about 7.5%. The natural frequency format preserves the base rate information in a way that is easy to grasp: you can picture the 1,000 women divided into groups and see directly that the 99 false positives vastly outnumber the 8 true positives. When the same information is expressed as conditional probabilities, each probability is stated within its own reference class, and the base rate disappears from view.

Hoffrage & Gigerenzer (1998) found that when physicians received problems in natural frequency format, 46% solved them correctly, compared to only 10% with the standard probability format. Sedlmeier (1997) went further and developed a computer-based training program that taught people to translate probability problems into natural frequency representations using visual tools: a frequency grid (a square grid where cases are marked as dots) and a frequency tree (a branching diagram that organizes the numbers hierarchically). In a study comparing this approach to traditional rule-based training, participants who learned the frequency method achieved correct solution rates of 70 to 80% immediately after training, compared to only 35% for the rule-based group. When tested again five weeks later, the frequency group's performance had improved to 90%, while the rule-based group had dropped to 20%. The frequency-based approach did not just help people plug numbers into a formula; it gave them a way of thinking about probability that stuck.

Despite this evidence, few medical schools train physicians to use natural frequencies. This gap between what research shows and what is practiced is a recurring theme in this course, and one that will return when we discuss Statistical vs. Intuitive Prediction.

Ideal observer models

So far, we have treated Bayesian reasoning as a normative tool for explicit judgment – a standard for how people should interpret test results or update beliefs. But the same logic also appears in models of how the mind processes information automatically, especially in perception. Griffiths et al. (2008) describe how Bayesian models of cognition provide a framework for understanding how humans make inferences under uncertainty, from recognizing objects to learning language to reasoning about causes. The basic idea is that the brain faces the same computational problem as a doctor interpreting a test result: it receives noisy, ambiguous evidence from the senses and must figure out what is most likely happening in the world.

An ideal observer model specifies how a perfectly rational agent would solve this problem. Such an agent would combine prior expectations about the world with incoming sensory evidence, weighting each according to its reliability. When the evidence is strong and clear, it should dominate. When the evidence is weak or ambiguous, prior expectations should have more influence.

Human perception often approximates this ideal. Consider a classic example from vision. When you look at a surface with circular patches of shading, you tend to see patches that are lighter on top as bumps and patches that are lighter on the bottom as dents. Why? Your visual system has a strong prior that light comes from above, built up from a lifetime of experience in a world where the sun and most artificial lights are overhead. This prior interacts with the ambiguous shading evidence to produce a definite perception. Rotate the image 180 degrees and bumps become dents, because the same shading pattern now conflicts with the light-from-above prior. The visual system is doing Bayesian inference: combining a prior expectation (light from above) with sensory evidence (the shading gradient) to arrive at the most probable interpretation (bump or dent).

This connection between perception and judgment matters for our purposes. Perceptual illusions arise when prior expectations dominate over ambiguous sensory evidence. Cognitive biases arise in the same way – when prior beliefs or heuristic shortcuts dominate over the actual evidence in a reasoning problem. The parallel suggests that biases are not random errors but systematic consequences of the same inference machinery that usually serves us well (see also Heuristics and Biases). Lieder & Griffiths (2020) develop this idea further, arguing that many apparent cognitive biases can be understood as resource-rational: they are the best solutions the brain can achieve given its limited computational resources. On this view, the brain is not failing at rationality; it is optimizing under constraints, much as an engineer builds the best bridge possible within a budget.

Signal detection theory

Another framework that complements Bayesian reasoning is signal detection theory (SDT). Where Bayes' theorem tells you how to update your beliefs, signal detection theory addresses the next step: how to make a decision based on those uncertain beliefs. The two frameworks are complementary – Bayes tells you what to believe, and SDT tells you what to do about it.

Imagine you are a radiologist examining an X-ray for signs of a tumor. The image is grainy, and what you see could be a real tumor or just normal tissue that happens to look unusual. You must make a decision: report a tumor or not. There are four possible outcomes:

	Tumor present	**Tumor absent**
Report "tumor"	Hit	False alarm
Report "no tumor"	Miss	Correct rejection

Signal detection theory describes this situation with two key parameters. The first is sensitivity (d', pronounced "d-prime"), which captures how well you can distinguish signal (tumor) from noise (normal tissue). This depends on the quality of the evidence. A high-resolution scan with a clear abnormality gives you high sensitivity; a blurry scan with a subtle shadow gives you low sensitivity. Sensitivity is a property of the evidence itself and is not under the decision-maker's control.

The second parameter is the criterion (also called bias). This is the threshold at which you decide to say "yes, there is a tumor." The criterion is under the decision-maker's control. If you set a low criterion, you will report tumors frequently, catching most real ones (many hits) but also producing many false alarms. If you set a high criterion, you will be more conservative, missing some real tumors (more misses) but producing fewer false alarms. A Bayesian decision-maker would set this criterion using both prior probabilities and the costs and benefits of each outcome. Missing a real tumor (a miss) is far worse than ordering an unnecessary follow-up scan (a false alarm), which pushes the optimal criterion lower. But the base rate of tumors is low, which pushes the criterion higher. The optimal criterion balances these competing pressures.

In practice, people do not always set optimal criteria. A radiologist who has just seen several cases of cancer may temporarily lower their criterion, reporting more tumors than usual, because the recent experience has made cancer seem more common (see Availability). A radiologist under time pressure may raise their criterion to avoid the effort of writing additional reports. These criterion shifts represent departures from the Bayesian ideal that have real consequences for patients.

Calibration

If Bayesian reasoning is the normative standard, how do we measure how close people actually come to it? One important measure is calibration. A person is well calibrated if their expressed confidence matches their actual accuracy. If you say you are 70% sure about something, then among all the things you say you are 70% sure about, roughly 70% should turn out to be true. If 90% of them turn out to be true, you are underconfident. If only 50% are true, you are overconfident.

Calibration can be assessed straightforwardly. You collect a large number of a person's probability judgments, group them by confidence level, and compare the stated confidence to the actual proportion correct. Weather forecasters are famously well calibrated: among all the days when they say there is a 30% chance of rain, it rains on roughly 30% of those days. This is no accident. Weather forecasters make many predictions, receive clear and immediate feedback, and work in a domain with well-defined probabilities – all conditions that support learning. Experts in domains with less clear feedback, such as political pundits or clinical psychologists making long-term prognoses, tend to be poorly calibrated. This connection between feedback quality and calibration will be relevant when we discuss why statistical prediction tends to outperform clinical judgment (see Statistical vs. Intuitive Prediction).

Good calibration is one important sign of sound probabilistic judgment, because it means a person's posteriors accurately reflect the evidence they have received. In practice, most people are overconfident: they express more certainty than their accuracy warrants. This pervasive finding will be examined in detail in Overconfidence.

Why this matters for the course

The Bayesian framework is not just one topic among many in this course. It is the thread that runs through nearly every chapter. Almost every bias and heuristic we will encounter can be understood as a specific, identifiable departure from the Bayesian ideal. The overview below may not mean much to you – yet! – because it refers to biases and heuristics that we haven't discussed yet. However, you can revisit the overview later. Doing so will help you identify common threads that run through the textbook. Herewith:

- Base-rate neglect, which we will discuss in Representativeness, occurs when people ignore the prior and judge probabilities based on how well the evidence matches a prototype. The mammogram problem is a classic example, because people tend to ignore how rare cancer is, and focus on the sensitivity of the screening.
- Anchoring, covered in Anchoring and Primacy, happens when people start with an initial value and fail to adjust sufficiently. In Bayesian terms, this is insufficient updating from the prior – the anchor acts as a sticky prior that resists revision.
- The availability heuristic, discussed in Availability, leads people to overestimate the probability of events that come easily to mind. In Bayesian terms, this is biased evidence sampling: the evidence that enters the calculation is not representative of the actual evidence in the world.
- Confirmation bias, the topic of Confirmation, is asymmetric updating: people revise their beliefs more in response to confirming evidence than disconfirming evidence, violating the Bayesian requirement to weight evidence according to its diagnosticity regardless of direction.
- Overconfidence, explored in Overconfidence, means that people's posteriors are too extreme or too precise relative to the evidence they actually have.
- Hindsight bias, covered in Hindsight, involves retroactively revising the prior after learning an outcome, making the outcome seem more predictable than it actually was.

- Belief perseverance, discussed in Belief Formation and Perseverance, is a failure to update from disconfirming evidence. People hold on to their prior even when the evidence clearly points the other way.
- The superiority of statistical over clinical prediction, the topic of Statistical vs. Intuitive Prediction, reflects the difference between informal evidence integration (which is prone to all the biases above) and formal evidence integration using explicit rules.

This is not to say that Bayesian reasoning is the only useful framework for understanding judgment and decision-making. As we saw in Reason and Intuition, the distinction between fast, intuitive processing and slow, deliberate processing captures something important that is not purely about probability. And as discussed in How It Is vs. How It Should Be, the relationship between normative standards (what we should do), descriptive findings (what we actually do), and prescriptive recommendations (what we practically can do to improve) is more nuanced than simply pointing out where people fail to be Bayesian (Swarna, 2017). But the Bayesian framework provides a powerful normative standard against which we can measure and understand the many ways in which human judgment falls short. In many of the following chapters, we will ask: how does this topic relate to the process of updating a prior belief with evidence to form a posterior? Sometimes the connection will be direct and obvious, as with base-rate neglect or confirmation bias. Sometimes it will require more interpretation, as when we consider how moral intuitions function as strong priors that resist updating (see Morality). And occasionally the Bayesian lens may not be the most natural way to think about a topic, in which case we will say so. The goal is not to force every phenomenon into a single framework, but to use the framework where it illuminates – which, as this chapter has shown, is in a surprisingly large number of places.

Summary

Bayesian reasoning is the process of combining prior beliefs with new evidence to form updated beliefs, or posteriors. Bayes' theorem formalizes this process and serves as the normative standard for rational belief updating throughout this course. People frequently violate this standard, as illustrated by the mammogram problem, where physicians dramatically overestimate the probability of disease after a positive test by ignoring the low base rate. Presenting information as natural frequencies rather than conditional probabilities makes Bayesian reasoning much more intuitive and produces lasting improvements in accuracy. The Bayesian framework extends beyond explicit reasoning to perception, where ideal observer models describe how the brain combines prior expectations with noisy sensory evidence, and to decision-making under uncertainty, where signal detection theory formalizes the tradeoff between hits, misses, false alarms, and correct rejections. Calibration – the degree to which a person's confidence matches their actual accuracy – provides a measurable standard for evaluating judgment quality, with most people being overconfident. Nearly every bias and heuristic covered in this course can be understood as a specific departure from Bayesian rationality, making this framework the central thread of the book.

BAYESIAN REASONING

HEURISTICS AND BIASES

Learning goals:

- Explain what a heuristic is and how attribute substitution works.
- Define cognitive bias and distinguish it from colloquial uses of the word "bias."
- Describe how cognitive biases parallel perceptual illusions.
- Explain the affect heuristic and how emotions serve as cues for judgment.
- Describe the bias blind spot and explain why it makes debiasing difficult.
- Evaluate the prospects and limitations of debiasing strategies.

What Is a Heuristic?

Imagine you are in a supermarket in Groningen, trying to decide which of two unfamiliar brands of olive oil to buy. You have no time to research reviews, compare nutritional labels, or calculate price per milliliter. Instead, you pick the one with the more elegant-looking bottle. You have just used a heuristic: a quick rule of thumb that gives you a workable answer without much effort.

The term "heuristic" was popularized by the mathematician George Pólya in his 1945 book How to Solve It, where he used it to describe useful strategies for solving mathematical problems. In the study of judgment and decision-making, the meaning has shifted. Kahneman & Frederick (2002) define a heuristic as a mental process in which a person answers a hard question by substituting an easier one. Instead of asking "Which oil has the best quality for its price?" you ask "Which oil looks nicer?" The answer to the easy question stands in for the answer to the hard one. This process is called attribute substitution: you replace a target attribute that is difficult to assess (quality) with a heuristic attribute that comes to mind more readily (appearance).

Attribute substitution is the common mechanism behind many different heuristics. When someone is asked "How happy are you with your life overall?" – a question that requires weighing many factors – they often answer based on their current mood instead. When someone is asked "How likely is a plane crash?" they answer based on how easily they can bring examples of plane crashes to mind. In each case, a hard question is swapped for an easier one, and the answer to the easier question is treated as the answer to the harder one.

Heuristics are a form of System 1 (Type 1) processing, as discussed in Reason and Intuition. They operate automatically and quickly, without deliberate effort. This is not a flaw. The world is far too complex for us to evaluate every decision through careful analysis. As the concept of bounded rationality suggests, our cognitive resources are limited, and heuristics allow us to make decisions that are good enough given those limits. You cannot spend twenty minutes on every item in the supermarket, and you do not need to. Most of the time, the olive oil in the elegant bottle is fine. Later in the chapter, we will interpret these shortcuts as rough, low-cost approximations to Bayesian inference.

The three classic heuristics identified by Tversky & Kahneman (1974) – representativeness, availability, and anchoring – are each instances of this general substitution process, and each is the focus of its own chapter later in this book (see Representativeness, Availability, and Anchoring and Primacy). For now, the important point is the general principle: when faced with a difficult judgment under uncertainty, people tend to simplify the problem by relying on a mental shortcut. These shortcuts are not random. They follow predictable patterns, and understanding these patterns is the central project of the heuristics and biases research program.

What Is a Bias?

A bias is a systematic error in thinking that pushes judgments or decisions in a particular direction. The word "systematic" is key: biases are not random mistakes. They are errors that occur reliably, across many people, in predictable situations.

Consider an example from Tversky & Kahneman (1974). Participants were given a description of a man named Steve: "very shy and withdrawn, invariably helpful, but with little interest in people, or in the world of reality. A meek and tidy soul, he has a need for order and structure, and a passion for detail." Then they were asked: is Steve more likely to be a librarian or a farmer? Most people said librarian, because the description resembles the stereotype of a librarian. But in many countries, farmers vastly outnumber librarians. The base rate – the relative frequency of farmers and librarians in the population – should heavily influence the answer, but people largely ignored it. This is base-rate neglect, a bias that arises from the representativeness heuristic.

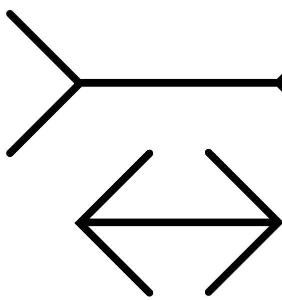
The error is systematic because it does not go in a random direction. People consistently overweight how well a description matches a stereotype and consistently underweight base rates. If each person made a random error, some would overestimate the probability that Steve is a librarian and others would underestimate it, and the errors would cancel out on average. But that is not what happens. Nearly everyone errs in the same direction. That is what makes it a bias.

Base-rate neglect is just one of many biases examined in this book. Others include overestimating the probability of dramatic events (see Availability), being too confident in your own predictions (see Overconfidence), and anchoring on irrelevant numbers (see Anchoring and Primacy). What they share is a systematic, directional deviation from a normative standard.

In everyday language, "bias" often refers to socially undesirable beliefs, such as stereotypes about gender or ethnicity. In the study of judgment and decision-making, a bias is simply a systematic deviation from a normative standard, regardless of whether it is socially desirable or not. Some biases do involve stereotypes, but many have nothing to do with social attitudes. Throughout this book, we use "bias" in this technical sense: a predictable, directional error in judgment or decision-making (see also How It Is vs. How It Should Be for more on normative standards).

Cognitive Illusions

One of the most useful ways to think about biases is by analogy with perceptual illusions. Consider the Müller-Lyer illusion: two horizontal lines of equal length, one with inward-pointing arrowheads (> – <) and the other with outward-pointing arrowheads (< – >). The line with outward-pointing arrowheads looks longer, even though you can measure and confirm that the two lines are the same length. The illusion does not disappear when you learn about it.



Biases in judgment work in much the same way. They are cognitive illusions: systematic errors in judgment that can persist even when people are aware of them. Just as the Müller-Lyer illusion reveals something about how the visual system normally processes depth and distance, cognitive biases reveal something about how the mind normally processes information. They are not signs of stupidity or carelessness. They are side effects of cognitive processes that usually serve us well.

This analogy carries an important implication. Visual illusions are not fixed by simply telling someone about them. You can stare at the Müller-Lyer figure for as long as you like, fully understanding the trick, and the illusion remains. Similarly, learning about a cognitive bias does not make you immune to it. You may still neglect base rates, anchor on irrelevant numbers, or feel overconfident in your predictions, even after reading this chapter. Some biases can be reduced through training or reframing, as we will discuss below, but many are stubbornly resistant. This is a humbling point, and it leads directly to the bias blind spot.

The Affect Heuristic

Not all heuristics involve cold, cognitive shortcuts. Some rely on feelings. The affect heuristic is the tendency to use your emotional reaction – your gut feeling of "good" or "bad" – as a cue for judgment (Slovic et al., 2007). When you need to evaluate something quickly, you often consult your feelings about it rather than carefully weighing evidence. In terms of attribute substitution: instead of asking "How risky is this technology?" you ask "How do I feel about it?"

Slovic et al. (2007) describe a striking pattern: people tend to see the risks and benefits of activities or technologies as inversely related. If they feel positively about something (say, solar energy), they judge it as high in benefit and low in risk. If they feel negatively about something (say, nuclear power), they judge it as low in benefit and high in risk. Objectively, risk and benefit are often positively correlated – activities that offer large benefits frequently carry large risks – but in people's minds, affect creates a negative correlation. If it feels good, it must be safe; if it feels bad, it must be dangerous.

In one experiment described by Slovic et al. (2007), researchers manipulated how participants felt about nuclear power by providing information that emphasized either its benefits or its risks. When people read about the benefits of nuclear power, they not only rated its benefits as higher but also judged its risks as lower – even though no new information about risk had been provided. The reverse happened when people read about the risks: benefit ratings dropped too. This shift was not warranted by the information provided: changing participants' evaluation of the benefits also changed their judgment of the risks, even though the evidence about risk itself had not changed. Because the information altered people's feelings about nuclear power, it shifted their judgments about both risk and benefit in an affectively consistent direction.

The affect heuristic is System 1 processing: fast, automatic, and largely outside conscious awareness. It can be highly useful – if something makes you feel uneasy, that feeling may reflect real dangers that you have not consciously identified. But it can also mislead, especially when feelings are manipulated through vivid images, emotional framing, or media coverage. The role of affect in judgment will come up repeatedly throughout this book, particularly in the chapters on Risk Perception and Morality.

The Bias Blind Spot

People typically believe that the biases described in this book apply more to other people than to themselves. This belief is itself a bias, known as the bias blind spot (Pronin et al., 2002).

Pronin et al. (2002) demonstrated this in a series of studies. In one study, Stanford University students were given descriptions of eight well-documented cognitive and motivational biases, such as the halo effect (letting a single positive trait influence your overall impression of someone) and self-serving attributions (taking credit for successes but blaming failures on circumstances). For each bias, participants rated how susceptible the "average American" was and how susceptible they themselves were. The result was clear: participants rated themselves as significantly less susceptible than the average American on every single bias.

You might wonder whether this is simply because the comparison group – the "average American" – feels abstract and distant. To test this, the researchers repeated the study but asked participants to compare themselves to their own classmates, people they knew personally. The bias blind spot remained. Participants still believed they were less biased than their peers.

In a second study, participants were first asked to rate themselves on positive and negative personality traits relative to their peers. As expected, most people rated themselves as better than average on desirable traits and less bad than average on undesirable ones – a well-documented phenomenon called the better-than-average effect. Participants were then told about this effect and asked whether it might have influenced their own ratings. The majority said no. Even when directly confronted with evidence of a specific bias, most people denied that it had affected them.

Why does the bias blind spot occur? Part of the answer lies in an asymmetry in how we evaluate ourselves versus others. When assessing whether we are biased, we tend to rely on introspection: we look inward, notice no conscious intention to be biased, and conclude that we are not. But when assessing whether other people are biased, we rely on their observable behavior and outcomes. Because most biases operate automatically through System 1 processing, they are not visible through introspection. You do not experience yourself neglecting base rates – it simply feels like you are making a reasonable judgment. This asymmetry between introspection and observation helps explain why people consistently see more bias in others than in themselves.

The bias blind spot is not just a curiosity. It has a serious practical consequence: it undermines the motivation to correct one's own biases. If you do not believe you are biased, why would you take steps to counteract your biases? This is one reason why the bias blind spot makes every bias discussed in this book personal. The biases described in the following chapters apply to you, the reader, not just to the anonymous "other people" you might be imagining.

Can Biases Be Corrected?

Because biases often persist even when we know about them – and because the bias blind spot reduces motivation to correct them – the obvious next question is whether debiasing is possible at all. The answer is: sometimes, but it is difficult.

The most promising approaches to debiasing share a common strategy: they shift thinking from System 1 processing to System 2 (Type 2) processing. Rather than relying on gut reactions and quick impressions, these approaches encourage deliberate, structured thinking. Some concrete techniques include:

- Consider the opposite. Before committing to a judgment, deliberately generate reasons why you might be wrong. This simple technique has been shown to reduce anchoring effects and overconfidence, because it forces you to engage with evidence that your initial impression might overlook.
- Use natural frequencies instead of probabilities. As discussed in Bayesian Reasoning, people find it much easier to reason correctly about base rates when information is presented as natural frequencies (e.g., "10 out of every 1,000 people") rather than as probabilities (e.g., "1%"). Reframing statistical information in this way can substantially reduce base-rate neglect.
- Delay decisions. Introducing a pause between receiving information and making a judgment gives System 2 processing time to engage. This can be as simple as sleeping on a decision before committing to it.
- Use structured procedures. Checklists, scoring rubrics, and decision aids force decision-makers to consider all relevant information systematically. In medical diagnosis, for example, structured protocols help clinicians avoid anchoring on an initial hypothesis (see Medical Decision-Making).

Some of these interventions can also be built into decision environments, a theme we revisit in Steering Other People's Choices.

However, debiasing faces several obstacles. The first is the bias blind spot itself: people who do not believe they are biased see no reason to use corrective strategies. The second is effort. System 2 processing is slow and demanding. In a busy day, you cannot subject every decision to careful analysis, and even when you try, fatigue and distraction can undermine your efforts. The third obstacle is that some biases are rooted in deeply held beliefs and emotional reactions, which are resistant to change even with deliberate effort (see Belief Formation and Perseverance).

This does not mean that studying biases is pointless. Awareness of biases is a necessary first step, even if it is rarely sufficient on its own. Understanding how biases work allows you to design better systems – better decision procedures, better information displays, better institutional checks – even if it does not make you personally immune to bias. The goal of this book is not to turn you into a perfectly rational thinker. It is to help you recognize the patterns in human judgment so that you can anticipate errors and create conditions that make good decisions more likely.

A Bayesian Lens

Throughout this book, we use Bayesian reasoning as a normative framework for how beliefs should be updated in light of evidence (see Bayesian Reasoning). From this perspective, heuristics can be understood as computationally cheap approximations to Bayesian inference. Bayesian updating requires combining a prior belief with the likelihood of the evidence to produce a posterior belief. This is often mathematically demanding. Heuristics skip the computation: instead of carefully weighing the prior (how common are farmers versus librarians?) against the likelihood (how well does this description match each group?), you simply go with the option that feels most representative.

When priors and evidence are typical – that is, when the shortcut happens to align with what a full Bayesian calculation would produce – heuristics lead to good decisions. This is why they persist: in the environments where human cognition evolved, and in many everyday situations, they are adequate. But when the shortcut diverges from proper integration of prior and likelihood, the result is a bias. Base-rate neglect, for instance, is a failure to adequately weight the prior. The conjunction fallacy is a failure to respect the basic rules of probability. These are predictable consequences of using a simplified process instead of the full Bayesian computation. In this sense, biases are the price we pay for the efficiency of heuristic thinking.

Summary

Heuristics are mental shortcuts that simplify difficult judgments by substituting an easier question for a harder one – a process called attribute substitution. They are a natural consequence of bounded rationality: the world is too complex for full rational analysis, and heuristics allow us to function efficiently. However, because heuristics skip important steps in reasoning, they produce systematic errors called biases. These biases are analogous to perceptual illusions: they reveal how the mind normally works, and they can persist even when we are aware of them. One important heuristic is the affect heuristic, in which people use their emotional reactions as cues for judgment, leading to predictable distortions in how they perceive risks and benefits. The bias blind spot – the tendency to see biases in others but not in oneself – makes debiasing particularly difficult, partly because people rely on introspection to assess their own objectivity and introspection cannot detect biases that operate automatically. While structured strategies such as considering the opposite, using natural frequencies, and applying checklists can help shift thinking from System 1 to System 2, awareness of biases alone is rarely enough to eliminate them. From a Bayesian perspective, heuristics are fast approximations to rational belief updating that work well in many situations but produce predictable errors when the shortcut diverges from proper integration of prior beliefs and evidence. Every bias discussed in this book applies to you, the reader – not just to other people.

HOW IT IS VS HOW IT SHOULD BE

Learning goals:

- Explain the difference between descriptive, normative, and prescriptive models of judgment and decision-making.
- Give examples of each type of model.
- Explain why prescriptive models are often a practical compromise between descriptive and normative models.
- Understand why people sometimes resist normative models, using effective altruism and QALYs as examples.
- Describe how the distinction between these three model types organizes the field of judgment and decision-making.

Three Ways to Think About Decisions

Imagine you are choosing what to eat for lunch. A food scientist could study what you actually pick and why. A nutritionist could tell you what you should eat to be as healthy as possible. And a public health official could redesign the cafeteria layout so that healthy options are easier to grab. These three perspectives – studying what people do, defining what is ideal, and finding practical ways to improve – correspond to the three types of models that organize the field of judgment and decision-making: descriptive, normative, and prescriptive models.

This three-part distinction recurs throughout the book. Descriptive chapters document patterns of judgment. Normative frameworks provide standards of correctness. And prescriptive discussions ask how to improve real decisions. Without being clear about all three, it is impossible to say whether something counts as a reasoning error, or to figure out what to do about it. When researchers show that people make systematic errors, they are comparing what people actually do (a descriptive observation) against some standard of what they should do (a normative benchmark). And once we know where people go wrong, we need prescriptive thinking to figure out how to help.

Descriptive Models

Descriptive models describe how people actually judge and decide, without evaluating whether those judgments are good or bad. The goal is accuracy: does the model capture what people really do?

The vast majority of models in the field of judgment and decision-making are descriptive. For example, dual-process theory, discussed in the chapter on Reason and Intuition, characterizes the interplay between fast, intuitive processes and slower, more deliberate ones. Dual-process models are primarily descriptive, even though some applications also evaluate which process is more reliable in a given context. Similarly, prospect theory, which is covered in Gains and Losses, describes how people evaluate outcomes relative to a reference point, treating losses as more painful than equivalent gains are pleasant. A classic demonstration involves what is known as the Asian disease problem: when a public health scenario is framed in terms of lives saved, people prefer a sure option, but when the same scenario is framed in terms of lives lost, they prefer a gamble – even though the expected outcomes are identical. Prospect theory does not claim that people should be loss averse or susceptible to framing. It documents that they are.

Descriptive models also include the many heuristics that people rely on, such as the availability heuristic, representativeness, and anchoring. Each of these describes a mental shortcut that people use to make judgments under uncertainty. Whether these shortcuts lead to good or bad outcomes depends on the situation. The heuristic itself is simply a description of how people think.

Normative Models

Normative models specify how people should judge and decide if they want to be logically consistent and achieve their goals as effectively as possible. Classical normative models often abstract away from limits in time, information, and cognitive capacity. They serve as benchmarks against which actual behavior can be evaluated.

Importantly, normative recommendations are always relative to some goal and set of assumptions. The best choice for maximizing expected value may not be the best choice for preserving fairness, loyalty, or peace of mind. When we call a model "normative," we mean it defines the correct answer given a particular objective – not that it captures everything a person might reasonably care about.

The most prominent normative model for decision-making under uncertainty is expected utility theory. When choosing between options, you should multiply the subjective utility of each possible outcome by its probability, and then choose the option with the highest expected utility. Note that utility here refers to the decision-maker's subjective valuation, not the objective payoff. If you are deciding whether to bike or take the bus to campus, and there is a 40% chance of rain, expected utility theory says you should weigh the personal costs and benefits of each option, taking into account the probability of getting wet. This framework is covered in more detail in Gains and Losses.

For belief updating, the normative standard is Bayes' theorem, discussed at length in Bayesian Reasoning. Bayes' theorem specifies exactly how you should revise your beliefs when you encounter new evidence. If you initially think there is a 10% chance that your headache is caused by dehydration, and you then learn that you drank very little water today, Bayes' theorem tells you how to revise that estimate, given how much more likely that evidence is if dehydration is the cause than if it is not.

These models are useful because they specify, given a set of assumptions, what the correct answer should be. But they are unrealistic as descriptions of human behavior. People rarely compute expected utilities or apply Bayes' theorem explicitly in everyday life. This is exactly why the distinction between normative and descriptive models matters: biases and errors are identified by comparing actual judgments with a normative benchmark.

Prescriptive Models

Prescriptive models bridge the gap between how people actually decide and how they ideally should. They offer practical recommendations that take human limitations into account. Where normative models specify what is optimal in theory, prescriptive models aim for what is achievable in practice.

Prescriptive tools come in many forms: external redesigns of the choice environment, but also decision aids, checklists, training programs, and rules of thumb that individuals can apply themselves.

Consider hand hygiene in hospitals. The relevant standard of correctness is clear: healthcare workers should wash their hands thoroughly with soap and water between every patient interaction. Descriptively, however, compliance is far from perfect, because handwashing takes time, sinks are not always nearby, and busy professionals cut corners. A prescriptive solution is to install hand sanitizer dispensers at every bedside. This does not achieve the ideal of full handwashing, but it dramatically improves hygiene in practice by making the desired behavior easier.

Another example comes from medical statistics. The normative way to reason about diagnostic test results involves applying Bayes' theorem to conditional probabilities. Descriptively, both patients and physicians struggle with this. They misinterpret what it means when a test comes back positive, often dramatically overestimating the probability of disease. A prescriptive solution, discussed in Bayesian Reasoning, is to present the same statistical information using natural frequencies instead of probabilities. Rather than saying "the sensitivity of the test is 90% and the base rate of the disease is 1%," you can say "out of 1,000 people, 10 have the disease, and of those 10, 9 will test positive; of the 990 who do not have the disease, about 99 will also test positive." Research has consistently shown that physicians and patients reason much more accurately when the same information is presented as counts out of 1,000 rather than as abstract conditional probabilities. The underlying math is the same, but the format changes how well people reason.

Prescriptive models also include nudges: small changes in how choices are presented that steer people toward better decisions without restricting their freedom. Default options, such as automatically enrolling employees in a pension plan unless they opt out, are a classic example. These interventions, discussed in Steering Other People's Choices, work because they use descriptive knowledge about human behavior (people tend to stick with defaults) to move people closer to better outcomes (saving for retirement).

But moving people toward a normative ideal is not only a technical problem. It is also a psychological and moral one, because people often resist being told that their intuitive judgments are suboptimal.

Resistance to Normative Models

If normative models define the best way to achieve one's goals, you might expect people to welcome them. In practice, the opposite often happens. People frequently resist normative approaches, finding them cold, unnatural, or even offensive. This resistance arises precisely because normative models can conflict with the heuristics and emotional responses that shape everyday judgment.

Effective altruism illustrates this tension well. Effective altruism is a movement that tries to answer a simple question: how can we do the most good with the resources we have? It is strongly informed by consequentialist and expected-value reasoning applied to charitable giving. Rather than donating to whichever cause tugs at your heartstrings, effective altruism recommends calculating where your money will have the largest impact. Organizations like GiveWell evaluate charities by estimating the cost per life saved or the cost per quality-adjusted life year (QALY) gained. A QALY is a metric used for comparing health interventions: one year of perfect health equals one QALY, and in QALY-based analyses, a year lived in less-than-full health may be assigned a weight below 1. By this logic, a medical intervention that gives 10 people an extra year at 0.8 QALYs each produces 8 QALYs, and you can compare this with other interventions to decide where resources should go.

This is, in principle, a straightforward application of normative reasoning to an important real-world problem. Yet many people find it deeply uncomfortable. One reason is that QALYs translate health outcomes into a common numerical metric, which many people experience as uncomfortably impersonal – as though a human life could be reduced to a number. Another reason is that effective altruism often recommends donating to causes that are geographically and psychologically distant, such as distributing malaria bed nets in sub-Saharan Africa, rather than helping a local homeless shelter. Imagine you are asked to choose between two options: donating to buy a hearing aid for a specific child at your local school, or donating the same amount to a program that will prevent blindness in dozens of children you will never meet, in a country you have never visited. Most people feel a stronger pull toward the local child, even if the second option does more total good. This clashes with the normative calculus, and it reflects our tendency to care more about identifiable, nearby individuals than about anonymous, distant ones – a pattern discussed in Risk Perception under the identifiable victim effect.

There is also a deeper issue. Normative models often omit emotional and social motives unless those motives are explicitly built into the objective being optimized. When you donate to a friend's fundraiser, you are not just trying to maximize global welfare; you are also maintaining a relationship, expressing solidarity, and responding to a specific person's story. These motivations are real and important to people, even if they are irrelevant from the perspective of a cost-effectiveness analysis. The result is that approaches like effective altruism are sometimes perceived as cold and calculating – not because they are wrong in any logical sense, but because they conflict with the intuitive, affect-driven processes that normally guide moral judgment. As discussed in the chapter on Morality, moral judgments often arise from fast intuitions rather than careful reasoning, and people experience discomfort when a normative framework asks them to override those intuitions.

This does not mean that normative models are useless or that resistance to them is always justified. It means that prescriptive models must take this resistance seriously. If you want people to actually make better decisions, it is not enough to tell them what the optimal choice is. You also need to understand why they are drawn to suboptimal choices, and design interventions that work with human psychology rather than against it.

How the Three Models Organize the Book

The distinction between descriptive, normative, and prescriptive models is not just an abstract classification. It provides a map for the rest of this book.

The early chapters in the Foundations section introduce both the descriptive architecture of human cognition (dual-process theory, heuristics, mental models) and the normative benchmark against which reasoning is evaluated (Bayesian updating, expected utility theory). These chapters establish the tools you will need to understand everything that follows.

The middle sections of the book – Uncertainty, Causality, Past and Future, Choice, and Belief – are primarily descriptive. They document recurring patterns in how people judge probability, infer causation, remember the past, predict the future, and form beliefs. Each chapter shows how a particular heuristic or bias produces a systematic departure from the normative standard. For instance, the availability heuristic leads people to overestimate the frequency of vivid events, and confirmation bias leads people to seek evidence that supports what they already believe.

The applied chapters – Real Life – emphasize prescriptive tools. The chapter on Medical Decision-Making discusses how decision support systems can help physicians avoid diagnostic errors. The chapter on Steering Other People's Choices explores how choice architecture can improve everyday decisions. And the chapter on Statistical vs. Intuitive Prediction makes the case for replacing clinical intuition with statistical models in high-stakes settings like judicial decisions and data-driven football recruitment.

Simon (1955) captured the essential insight that links these three perspectives. By showing that human rationality is bounded, he made clear why descriptive and normative models diverge. People do not fail to be rational because they are foolish. They fall short because the world is complex and the mind has limits. Once you accept this, the role of prescriptive models becomes obvious: they are tools for helping bounded minds make better decisions in a complex world.

Summary

The field of judgment and decision-making is organized around three types of models. Descriptive models capture how people actually think and decide. Normative models define how they should think and decide to be logically consistent and achieve their goals, given a set of assumptions. Prescriptive models offer practical ways to close the gap between the two. Most of the research in this field is descriptive, documenting the heuristics and biases that characterize human judgment. The primary normative standard is Bayesian reasoning, supplemented by expected utility theory. Prescriptive interventions – including natural frequencies, nudges, decision aids, and checklists – use descriptive knowledge about human psychology to help people make better decisions. People sometimes resist normative models, as illustrated by reactions to effective altruism and QALYs, because these models can conflict with the intuitive and emotional processes that normally guide judgment. Understanding this resistance is itself an important part of building effective prescriptive tools.

MENTAL MODELS

Learning goals:

- Describe what framework theories are and how they organize understanding in core domains such as physics, psychology, and biology.
- Explain the three stances (mechanical, design, and intentional) and how each leads to a different type of explanation.
- Define schemas and scripts, and explain how they guide comprehension, memory, and behavior.
- Distinguish between artifacts and natural kinds, and explain how each activates a different default stance.
- Relate the layered structure of stances, schemas, and scripts to a Bayesian framework of prior beliefs and prediction errors.

How Background Knowledge Shapes Everything

Imagine you are cycling through Groningen and you see a dog sprint across the road, narrowly avoiding a car. You immediately construct a rich interpretation of what just happened. The dog wanted to reach the other side. The car moved forward because its engine was running and its driver pressed the accelerator. The near-miss was dangerous because collisions between solid objects cause damage. Each of these interpretations draws on a different kind of background knowledge: knowledge about animal minds, about how machines work, and about physical forces.

This chapter is about the mental models that make such everyday understanding possible. Before you encounter a specific situation, you carry with you layers of background knowledge that shape how you interpret new information, what you find surprising, and what explanations you find satisfying. These layers range from the very broad (what kind of thing am I dealing with?) to the very specific (what typically happens next in this sequence of events?). Understanding these mental models matters because they influence not only comprehension and memory but also the causal stories we construct, which shape how we assign blame and praise (see the chapter on Blame, Praise, and Responsibility), how we assess risk (see the chapter on Risk Perception), and how we make decisions in everyday life.

Framework Theories

From a very young age, children develop distinct sets of commonsense knowledge about how the world works. Wellman & Gelman (1992) argue that these are best understood as framework theories: broad, foundational understandings that define what kinds of things exist in a domain and what kinds of causes operate there. Three core domains stand out: naive physics, naive psychology, and naive biology.

Naive physics involves understanding the physical world. Infants as young as three to four months show surprise when physical principles are violated. In a typical violation-of-expectation study, an infant watches a screen rotate toward a solid object placed behind it. On test trials, the screen appears to pass straight through the object as if it were not there. Infants look significantly longer at this impossible event than at a control event where the screen stops upon contact, suggesting that they already expect solid objects to block the path of other objects. As children grow, their physical knowledge becomes more detailed, but the basic framework – objects are solid, causes require contact, unsupported things fall – is in place early.

Naive psychology, sometimes called theory of mind, involves understanding other people in terms of mental states such as beliefs and desires. By age four or five, children pass classic false-belief tasks. In the Sally–

Anne task, for example, a child watches Sally place a marble in a basket and leave the room. While Sally is away, Anne moves the marble to a box. The child is then asked: where will Sally look for the marble? Children who have developed an understanding of false belief correctly predict that Sally will look in the basket – where she last saw it – rather than in the box where it actually is. This framework treats people as fundamentally different from rocks or chairs: people have inner mental lives that drive their behavior.

Naive biology involves understanding living things. Children distinguish living from nonliving things early in life and develop specific expectations about biological processes such as growth, inheritance, and illness. A preschooler understands that a kitten will grow into a cat (not a dog), and that offspring inherit traits from their parents. These expectations are distinct from both physical and psychological explanations, suggesting that biology constitutes its own framework.

The key insight from Wellman & Gelman (1992) is that these framework theories are not simplified versions of adult science. They are coherent systems that organize children's reasoning and learning from the start. Each framework defines its own ontology (what kinds of things exist) and its own causal principles (what kinds of explanations are appropriate). A physical framework explains events in terms of forces and contact. A psychological framework explains events in terms of beliefs and desires. A biological framework explains events in terms of growth, inheritance, and biological function.

Stances: Three Ways to Explain the World

Framework theories provide broad domain knowledge about what kinds of entities and causes exist. Stances are a related but distinct concept: they are the explanatory modes people adopt when interpreting a particular entity or event (Dennett, 1987; Keil, 2006). While framework theories specify what you know about a domain, stances determine how you approach explaining something right now. The same event can often be interpreted through more than one stance, and the stance you choose shapes the explanation you construct.

The mechanical stance (also called the physical stance) explains things in purely physical terms. When you wonder why a ball rolls downhill, you think about gravity, slope, and friction. When a plumber diagnoses a leaky pipe, the explanation involves water pressure, cracks, and seals. The mechanical stance is the appropriate default for understanding physical objects and processes.

The design stance (also called the functional stance) explains things in terms of purpose and function. When you wonder why a bicycle has gears, you think about the purpose they serve: gears allow the rider to adjust effort relative to speed and terrain. The design stance does not require you to understand the exact physics of gear ratios. It assumes that the thing was made to fulfill a function and that its parts are arranged to serve that function. This stance is efficient because it lets you predict how something will behave without knowing its internal mechanics.

The intentional stance explains things by presuming that the entity in question has beliefs and desires. When you wonder why your classmate left the lecture early, you assume they had a reason: perhaps they felt ill or had another appointment. The intentional stance treats behavior as the product of rational agency. It is the natural stance for understanding people and animals, and it is effective for predicting behavior in social situations.

To keep terminology clear: from here on, this chapter will use "mechanical stance" and "design stance" as the primary labels.

The most useful stance is the one that yields the most accurate and efficient explanation in context. You can, in principle, explain a chess-playing computer using the mechanical stance (tracking every electrical signal in its circuits), but this would be impossibly complex and not very useful. The intentional stance ("the computer wants to protect its queen") gives you far better predictions with far less effort. Conversely, explaining a falling rock using the intentional stance ("the rock wants to reach the ground") would be uninformative.

Stances shape causal narratives and moral judgments

The stance you adopt shapes the entire causal narrative you construct about an event, which in turn influences moral judgment. Consider a self-driving car that injures a pedestrian. Under the mechanical stance, you look for a sensor malfunction or a software error. Under the design stance, you ask whether the car was designed well enough to handle the situation. Under the intentional stance, you might attribute something like negligence to the car or to its manufacturer. Each stance points to different causes, different responsible parties, and different moral conclusions.

This is not merely hypothetical. Research on blame and responsibility shows that framing an event in mechanical terms (a system failure) tends to reduce blame, while framing the same event in intentional terms (someone chose to cut corners) increases blame. The connection between stances and moral reasoning is explored further in the chapter on Blame, Praise, and Responsibility.

Same event, different stances

To see how stances work in practice, consider a dog scratching at a door. Under the intentional stance, the dog wants to go outside – perhaps it heard another dog barking or wants to explore. Under the mechanical stance, claws are making contact with a wooden surface, producing scratches and sound. Under a biological framework, the behavior might be understood as a conditioned response to an internal cue such as a full bladder. Each stance activates different background knowledge and generates a different explanation of the same observable event. The stance you adopt determines which schemas – organized packages of knowledge – become relevant. This brings us to how schemas work.

Schemas: Organized Expectations

Within a given stance, you carry specific packages of knowledge about what to expect in particular situations. These are called schemas. A schema is a structured set of knowledge about a type of thing or situation that is activated by incoming information and then guides comprehension, memory, learning, and behavior.

A classic demonstration comes from Bransford & Johnson (1972). In their study, participants listened to a passage describing a series of actions in vague, disjointed terms:

"The procedure is actually quite simple. First you arrange things into different groups. Of course, one pile may be sufficient depending on how much there is to do."

Without context, the passage is nearly incomprehensible. But participants who were told beforehand that the passage was about washing clothes understood and remembered it far better than those who received the topic label afterward or not at all. Participants in the "topic before" condition recalled roughly twice as many idea units as those who received no topic. The title activated a schema for doing laundry, and this schema provided a framework for interpreting each sentence as it arrived. Receiving the topic after hearing the passage did not help, showing that schemas must be active during processing to be effective. Comprehension happens in real time as new information is integrated with existing knowledge, not after the fact.

This finding has broad implications. Two people can encounter the same information and come away with very different understandings because they bring different schemas to the situation. A medical student listening to a description of symptoms activates a diagnostic schema that a layperson does not have. A football fan watching a match activates tactical schemas that a newcomer to the sport cannot access. Schemas do not just help you remember more; they change what you perceive and understand in the first place.

The content of a person's schemas profoundly affects behavior. Consider a study in which participants read a story about a character visiting a doctor's office and then were asked to recall what happened. Participants tended to "remember" schema-consistent details that were never actually mentioned – for instance, recalling that the character sat in a waiting room, even though the story said no such thing. This is schema-driven inference: the schema fills in expected details, and people later cannot distinguish what they actually encountered from what their schema supplied. This mechanism is one of the ways that stereotypes operate. If your schema for a social group includes certain traits, you may notice and remember information that fits the schema and overlook information that contradicts it. The chapter on Representativeness discusses how prototype-based reasoning produces similar biases, and the chapter on Confirmation discusses how this selective attention sustains existing beliefs.

Scripts: Schemas for Sequences

A special type of schema is a script: a schema for a highly routine sequence of events. You have a script for visiting a restaurant, for attending a lecture, for going to the supermarket, and for dozens of other familiar activities. A restaurant script, for example, includes entering, being seated, reading the menu, ordering, eating, paying, and leaving, roughly in that order.

Scripts are well known within a culture, with agreed-upon temporal sequences, but they are culture-specific. A restaurant script in the Netherlands might include splitting the bill among friends, something that would be unusual in certain other cultures where the host always pays. A script for greeting someone varies between cultures that use handshakes, cheek kisses, or bows. When you travel to a new country, much of the disorientation you feel comes from not having the right scripts for everyday situations.

Because scripts generate strong expectations about what should happen next, violations are noticed and remembered – especially when they bring the script to a standstill. If you sit down at a restaurant and the waiter tells you there is no menu and you should simply wait for whatever the chef prepares, this halts your restaurant script. You do not know what to do next, you feel uncertain, and you are likely to remember the experience later. Minor variations within a script (the waiter brings water before you ask) may go unnoticed, but major disruptions that stop the expected sequence are salient. They demand attention and force you to figure out a new course of action.

Scripts also shape memory in predictable ways. In studies where participants read stories about routine events, they later tend to "remember" script-consistent actions that were never mentioned – for example, recalling that a restaurant character looked at a menu even when the story skipped that step. At the same time, actions that strongly violate the script (the waiter sat down and started eating the customer's food) are also well remembered, precisely because they disrupted the expected sequence. Script-consistent details are filled in automatically; script-violating details are remembered because they are surprising. Both effects illustrate how scripts function as organized expectations that actively shape what we encode and retrieve from memory.

Artifacts and Natural Kinds

One important way to see stance activation in action is to compare reasoning about artifacts and natural kinds. Artifacts are human-made objects such as chairs, smartphones, and bridges. Natural kinds are things that exist independently of human design, such as animals, plants, and minerals.

Even young children distinguish between these two categories and reason about them differently (Wellman & Gelman, 1992). In transformation studies, children are told that a scientist takes a raccoon and surgically alters it to look exactly like a skunk – adding a white stripe and a scent sac. When asked what the animal now is, most children say it is still a raccoon. But when told that a scientist takes a coffeepot and reshapes it into a bird feeder, children readily say it is now a bird feeder. For natural kinds, identity resists surface transformation; for artifacts, changing the function changes what the thing is.

This distinction reflects the fact that different types of things trigger different default stances. Artifacts trigger the design stance. We understand a chair in terms of its intended function: it was made for sitting. A broken chair is often still judged to be a chair because of its intended design, even though it no longer fulfills that function. If someone asks you what a strange-looking kitchen gadget is, you answer by describing what it is for, not what it is made of.

Animals and people, by contrast, strongly invite the intentional stance. We understand them in terms of beliefs, desires, and goals. A dog chases a ball because it wants to catch it. A cat sits by the door because it wants to go outside. This tendency is so readily activated that people apply the intentional stance even to simple moving shapes. In a classic study by Heider & Simmel (1944), participants watched a short animation of geometric shapes – triangles and circles – moving around a screen. Nearly everyone spontaneously described the shapes as chasing, hiding from, or tricking each other, as if the shapes had intentions and emotions. The intentional stance activates even when the "agents" are nothing more than moving shapes on a screen.

The type of thing you encounter triggers a default stance, which then activates stance-appropriate schemas. When you see an unfamiliar tool, your design stance activates and you start thinking about what it might be for. When you see an unfamiliar animal, your intentional stance activates and you start thinking about what it might want or be trying to do. This distinction emerges so early in development that it appears to reflect a basic organizing principle of human cognition, not something that is explicitly taught.

This distinction also matters for how people reason about change and identity. If you gradually replace every part of a boat, is it still the same boat? Most people feel uncertain – the famous Ship of Theseus problem. But if a caterpillar transforms into a butterfly, people readily accept that it is the same organism. For artifacts, identity is tied to design and origin. For natural kinds, identity is tied to something internal and stable – people reason as if natural kinds have a hidden inner nature (sometimes called an "essence") that makes them what they are, regardless of outward appearance. This intuitive essentialism – the tendency to believe that members of a natural kind share a deep, underlying nature – explains why children insist that a surgically altered raccoon is still a raccoon. These different intuitions about identity flow directly from the different stances that each category activates.

When Mental Models Mislead

The power of stances and schemas is that they allow you to understand new situations quickly and with minimal effort. But this efficiency comes at a cost. When the wrong framework or stance is applied, it can lead to systematic errors.

One of the most common errors is applying the intentional stance to things that have no intentions. People attribute anger to thunderstorms, describe computers as wanting to frustrate them, and see purposeful faces in random patterns. This tendency toward what is sometimes called overactive agency detection may have evolutionary roots: the cost of falsely detecting an agent (mistaking a shadow for a predator) is much lower than the cost of missing a real one. But in modern life, it contributes to superstitions and to the kind of reasoning that underlies conspiracy theories (see the chapter on Conspiracy Theories), where complex events are attributed to the deliberate plans of hidden agents rather than to impersonal causes.

Conversely, applying the mechanical stance to people – treating them as objects without beliefs, desires, or feelings – is dehumanizing. Research on moral psychology suggests that failing to adopt the intentional stance toward certain groups of people is one of the cognitive foundations of prejudice and cruelty.

The design stance, too, can be misapplied. When people ask "what is the purpose of suffering?" or "why did this disease happen to me?", they are applying a design stance to events that may have no purpose at all. Reasoning like "the universe sent me a sign" or "nature is trying to restore balance" treats impersonal processes as if they were designed with a goal in mind. This kind of reasoning is natural and often comforting, but it can lead to unjustified beliefs about the world being fundamentally fair or purposeful – an idea related to the just-world hypothesis.

Framework confusion – applying knowledge from one domain to another where it does not belong – is a closely related source of error. Believing that a sick person caused their own illness through negative thoughts applies a psychological framework to a biological phenomenon. Attributing intentional motives to a virus ("the virus is trying to evade our immune system") applies a psychological framework to a biological process that operates through blind variation and selection, not through goals. The common thread in all of these errors is the same: a mental model that works well in one context is applied in a context where it does not fit, leading to explanations that feel satisfying but are misleading.

Bayesian Lens

The layered structure of stances, schemas, and scripts maps naturally onto a Bayesian framework of belief updating (see the chapter on Bayesian Reasoning for the formal framework).

Stances set the broadest level of prior beliefs. They determine which kind of explanation is even considered plausible. If you adopt an intentional stance toward a person, your space of possible explanations includes beliefs, desires, and intentions but largely excludes purely mechanical causes. If you adopt a mechanical stance toward a machine, your space of possible explanations includes physical forces and material properties but not feelings or goals. In Bayesian terms, the stance shapes the prior distribution over possible explanations, strongly down-weighting explanations from the wrong domain. This does not mean that cross-domain explanations are impossible – people do anthropomorphize machines and mechanize people – but such explanations are treated as unlikely by default and require strong evidence to be taken seriously.

Within a stance, schemas encode more specific expectations about what is likely. Your schema for a university lecture includes expectations about seating, speaking turns, slide presentations, and question periods. These expectations function as priors that are updated as new information arrives. When the lecturer begins speaking, this confirms your prior and you barely notice it. When the lecturer unexpectedly starts juggling, this violates your schema, and the resulting surprise is analogous to a large prediction error: the observed event had very low prior probability, and your model of the situation must be revised.

Scripts represent particularly strong temporal priors within schemas. Because the sequence of events in a script is highly predictable, any deviation from the expected order generates a clear prediction error. This is why script violations are so noticeable and memorable: they force you to update your expectations, and the effort involved in that updating leaves a stronger memory trace.

This layered structure also explains why some information is ignored while other information is surprising. Stance-inconsistent information – for example, someone suggesting that your laptop has feelings – is strongly down-weighted by your current model. It may be dismissed as a joke or a metaphor rather than taken as a serious claim. Schema-inconsistent information within the appropriate stance, by contrast, is surprising but interpretable, and it may trigger genuine belief updating. The distinction between information that is dismissed and information that is surprising reflects the difference between information that falls outside the current model and information that falls within the model but was not expected. This Bayesian perspective on mental models helps explain why two people with different background knowledge can draw entirely different conclusions from the same evidence – a theme that recurs throughout this book.

Summary

People do not approach the world as blank slates. From early in life, they develop framework theories in core domains such as physics, psychology, and biology, each with its own ontology and causal principles. When interpreting a specific entity or event, they adopt stances – mechanical, design, or intentional – that determine what kind of explanation is appropriate. Schemas organize specific expectations about types of things and situations, guiding comprehension, memory, and behavior. Scripts are a special case of schemas that capture the expected sequence of events in routine activities. Different categories of things, such as artifacts and natural kinds, trigger different default stances, which in turn activate different schemas. These layered mental models allow efficient understanding of new situations, but they can also lead to errors when applied inappropriately – as when the intentional stance is applied to impersonal events, or when knowledge from one framework is imported into another domain. In Bayesian terms, stances set the broadest priors, schemas encode more specific expectations, and violations of schemas and scripts function as prediction errors that signal the need to update one's model of the world.

INTELLIGENCE

WHAT IS INTELLIGENCE?

Learning goals:

- Explain why intelligence is best understood as a statistical concept rather than a verbal definition.
- Describe the g-factor and how it was discovered.
- Distinguish between the g-factor and s-factors.
- Explain the difference between fluid intelligence and crystallized intelligence.
- Describe how fluid and crystallized intelligence change across the lifespan.

Intelligence as a Statistical Concept

Ask ten psychologists to define intelligence and you will get ten different answers. Some emphasize problem-solving, others learning speed, others the ability to adapt to new environments. This ambiguity sometimes leads people to think the concept is too vague to be scientifically useful. But this conclusion confuses a verbal definition with a statistical one. We do not need a crisp sentence to make intelligence scientifically useful. We need it to behave consistently in data. And it does. The finding that cognitive abilities are positively correlated – and can be summarized by a general factor – is one of the most robust and replicable findings in psychology. Intelligence may be hard to put into words, but it can be measured reliably, and it predicts real-world outcomes with striking consistency. Scores on intelligence tests predict outcomes such as school performance, training success, and performance in cognitively demanding jobs – a topic we return to in the chapter on Intelligence in the Real World.

The g-Factor

Imagine we organize a mega-Olympics. We grab a hundred random people off the street and have them compete in every event: the 100 meters, the marathon, swimming, high jump, shot put, archery, gymnastics, and so on. What would we find? We would quickly notice that some people do well across the board. They are faster runners, better swimmers, and higher jumpers. Others consistently struggle. Of course there would be exceptions: someone might be a strong swimmer but a poor sprinter. But on the whole, performance across these very different events would be positively correlated. There is a general underlying factor we might call physical fitness that partly determines how well someone does in any athletic event.

Spearman (1904) had the same insight about the mind. He noticed that children who scored well on one type of cognitive test tended to score well on others too. A child who excelled at distinguishing musical pitches also tended to do well in school, and a child who did well in Latin also tended to do well in mathematics. Spearman went further than just noting these correlations. He developed a statistical method – an early form of factor analysis – to test whether a single underlying factor could account for the pattern of positive correlations across very different cognitive tasks. If vocabulary, memory span, arithmetic, and spatial reasoning all correlate positively, factor analysis can summarize that shared variance as a single latent factor. A general factor accounted for a substantial share of that common variance. He called this factor the g-factor, short for general intelligence.

In his original study, Spearman tested schoolchildren in an English village on sensory discrimination tasks, such as telling apart two slightly different weights or two slightly different musical tones, and compared their performance to teacher rankings of their intellectual ability. Spearman reported very strong associations and argued that a single general factor underlay diverse performances, though his early estimates were likely inflated by the methods available at the time. Still, his core claim was bold: that a single general factor runs through all cognitive abilities, from the most basic perceptual tasks to the most complex academic reasoning. Over a century of subsequent research has confirmed the central finding. People who do well on one cognitive task tend to do well on others, and the g-factor captures this shared variance.

What does this mean in practice? It means that the specific tasks in an IQ test matter less than you might think. Whether we test someone with vocabulary questions, spatial puzzles, or mental arithmetic, we are always partly measuring the same underlying ability. For example, the Wechsler Adult Intelligence Scale (WAIS) includes subtests ranging from vocabulary to block design to digit span, yet performance across these different tasks still tends to converge on a common general factor. Different IQ tests, despite using very different tasks, tend to agree remarkably well on who scores high and who scores low. In applied and organizational psychology, the g-factor is often referred to as general mental ability (GMA). This is the same construct with a different label, used especially when discussing how intelligence predicts job performance and academic success.

Specific Abilities

The physical-fitness analogy also makes clear that general ability is not the whole story. A steady hand matters for archery but not for sprinting. Long legs help in the high jump but not in weightlifting. Similarly, some cognitive abilities are task-specific. Language skill matters for vocabulary tests but not for visual puzzles. Musical talent helps in distinguishing pitches but not in solving algebra problems. Spearman called these s-factors, short for specific intelligence.

Some tasks are highly g-loaded, meaning that performance on them is mostly determined by the g-factor. Matrix puzzles – where you must identify the pattern in a grid of abstract shapes and select the missing piece – are a good example. These tasks require working memory, flexible recombination of information, and active problem-solving, all of which are central to general intelligence. Because they minimize the role of acquired knowledge, matrix puzzles are often used as a relatively culture-reduced indicator of general intelligence. Other tasks, like recognizing melodies or drawing from memory, depend more heavily on specific factors. The balance between g and s varies from task to task, but many cognitive tasks load at least somewhat on g.

Fluid and Crystallized Intelligence

Spearman's distinction between general and specific abilities describes where test performance variance comes from. Later theorists asked a different question: what kind of ability does general intelligence consist of? Cattell (1963) refined Spearman's framework by distinguishing two broad forms of cognitive ability: fluid intelligence and crystallized intelligence.

Fluid intelligence is the capacity to flexibly manipulate information in working memory to solve novel problems. It is what you rely on when you encounter something you have never seen before and need to figure it out on the spot. Think of a student in their first statistics class staring at a proof they have never encountered, trying to work through the logic step by step. Or think of solving a Sudoku puzzle: there is nothing to memorize, no facts to retrieve. You simply need to hold multiple constraints in mind and work through the possibilities. Matrix puzzles are highly g-loaded precisely because they tap into fluid intelligence while minimizing the role of acquired knowledge.

Crystallized intelligence is the accumulated body of knowledge, skills, and facts stored in long-term memory. It is what you rely on when you already know the answer because you have learned it before. A medical student who has memorized the symptoms of hundreds of diseases, a chess player who recognizes a familiar opening, or a Dutch student who knows that "gezellig" has no direct English translation are all drawing on crystallized intelligence. Vocabulary and general knowledge tasks are classic measures of crystallized intelligence; reading comprehension often draws on both crystallized knowledge and fluid reasoning.

Cattell supported this distinction using test batteries that separated abstract reasoning tasks from culturally acquired verbal knowledge. He gave students a range of cognitive tests – verbal, spatial, reasoning, and number tasks – along with culture-fair tests that used abstract shapes and patterns. Factor analysis revealed two distinct factors. The first, identified as fluid intelligence, loaded most strongly on the culture-fair tests and on spatial reasoning. The second, identified as crystallized intelligence, loaded primarily on verbal and number tasks that reflected culturally acquired knowledge. The two factors were correlated with each other, which makes sense: higher fluid intelligence generally supports the acquisition of crystallized knowledge over time. The more effectively you can reason through new material, the more you tend to learn and remember.

The most important practical difference between fluid and crystallized intelligence lies in how they change across the lifespan. Fluid intelligence tends to peak earlier than crystallized intelligence and often shows earlier age-related decline, whereas crystallized intelligence is more stable and may continue improving into middle adulthood. This is why learning a new language, picking up a new programming language, or mastering an unfamiliar board game tends to feel harder at 50 than at 20. The capacity for rapid, flexible reasoning tends to decline gradually.

Crystallized intelligence, by contrast, continues to grow well into middle age and remains relatively stable into old age. An experienced doctor in their sixties may be slower to learn a new diagnostic tool, but they have a vast library of cases, patterns, and knowledge that a younger doctor simply cannot match.

This explains a pattern that many people notice intuitively. With experience, tasks that once required fluid intelligence can be solved through crystallized intelligence. A beginning chess player must calculate every possible move (fluid intelligence). A grandmaster simply recognizes the position and knows the right response (crystallized intelligence). A first-year psychology student struggles to understand a research paper because every concept is new. A professor reads the same paper effortlessly, because decades of accumulated knowledge fill in the gaps. In both cases, the task has not changed, but the balance of fluid and crystallized intelligence needed to perform it has shifted.

The distinction between fluid and crystallized intelligence also connects to the discussion of dual-process theory. Fluid intelligence, with its reliance on effortful, flexible reasoning, overlaps more with the kind of deliberate processing associated with System 2. Crystallized intelligence often supports the rapid retrieval and pattern recognition characteristic of System 1.

A Bayesian Lens

One way to interpret the fluid–crystallized division is in Bayesian terms, the framework for rational belief updating that runs throughout this book (see Bayesian Reasoning). Crystallized intelligence is your stock of priors: the rich body of knowledge, patterns, and expectations accumulated over a lifetime. Fluid intelligence is the engine that processes new evidence and computes the posterior, flexibly integrating new information with what you already know. In Bayesian terms, the age pattern described above reflects weaker updating but richer priors. A person with strong priors but a weakening engine – that is, high crystallized intelligence but declining fluid intelligence – can still perform well in familiar domains, because their priors do most of the work. But they will struggle when the environment changes and priors alone are not enough. A young doctor and an older doctor may arrive at the same diagnosis, but through different routes. The young doctor reasons through the evidence from scratch. The older doctor recognizes the pattern immediately.

Summary

The g-factor, first identified by Spearman (1904), captures the consistent finding that people who perform well on one cognitive task tend to perform well on others. Alongside this general ability, specific abilities (s-factors) contribute to performance on particular tasks. Cattell (1963) further refined this picture by distinguishing two broad forms of cognitive ability: fluid intelligence, the capacity for flexible problem-solving with novel information, and crystallized intelligence, the accumulated knowledge stored in long-term memory. Fluid intelligence peaks earlier in life and is more vulnerable to age-related decline, while crystallized intelligence continues to grow throughout adulthood. Together, these concepts provide a foundation for understanding how cognitive ability shapes judgment and decision-making.

WHAT IS INTELLIGENCE?

INTELLIGENCE VS RATIONALITY

Learning goals:

- Explain why high intelligence does not guarantee good reasoning and decision-making.
- Define dysrationalia and describe how it arises from the distinction between cognitive capacity and thinking dispositions.
- Describe the Cognitive Reflection Test (CRT) and explain what it measures beyond standard intelligence tests.
- Define Actively Open-Minded Thinking (AOT) and explain how it predicts resistance to bias independently of intelligence.
- Relate the distinction between intelligence and rationality to the Bayesian framework of belief updating.

When Smart People Think Poorly

Research consistently shows that intelligent people hold pseudoscientific beliefs, fall for financial scams, and make poor medical decisions at rates that their cognitive ability alone would not predict. Surveys find that belief in paranormal phenomena, for example, is only weakly related to IQ (Stanovich et al., 2013). If intelligence is supposed to help us think well, why does it so often fail to do so?

The answer is that intelligence and rationality are not the same thing. Intelligence gives you the raw processing power to solve problems. Rationality is the tendency to actually use that power when it matters – or in the Bayesian terms used throughout this book, the tendency to update beliefs by carefully integrating prior knowledge and evidence rather than relying on intuitive shortcuts (see Bayesian Reasoning). You can also think of rationality as a specific ability (see What Is Intelligence) – a cognitive ability that contributes to general intelligence, but is not identical to it. Rationality and general intelligence are correlated in the sense that, on average, a person of above-average intelligence also has above-average rationality. But there are plenty of counterexamples: people who are irrational and highly intelligent, or conversely, rational and of low intelligence.

Dysrationalia: Smart but Irrational

Stanovich (1993) introduced the term dysrationalia to describe the condition of having adequate intelligence yet thinking and behaving irrationally. The term is deliberately modeled on words like dyslexia and dyscalculia, which describe specific difficulties in reading and math despite otherwise normal cognitive ability. Stanovich's argument was provocative: if we recognize that someone can have a normal IQ but struggle with reading, why not recognize that someone can have a normal or even high IQ but struggle with rational thinking?

The key insight behind dysrationalia is that standard IQ tests measure only a narrow slice of what it means to think well. IQ tests are good at measuring what Stanovich calls algorithmic-level processing: the raw computational power of your mind, such as working memory capacity, processing speed, and the ability to see abstract patterns. These are real and important cognitive abilities, and as discussed in Intelligence in the Real World, they predict many life outcomes including academic success and job performance.

But IQ tests do not measure what Stanovich calls reflective-level thinking dispositions: the tendency to actually engage your cognitive power when it matters, to question your first impulse, to consider whether your intuitive answer might be wrong, and to seek out information that could change your mind. (Note that this is a simplified version of Stanovich's broader framework, which distinguishes three levels of mind: the autonomous, algorithmic, and reflective.) A person with high algorithmic-level ability but poor reflective-level dispositions is like a powerful car with a careless driver – the engine can go fast, but the driver does not check the map.

What does dysrationalia look like empirically? Stanovich et al. (2013) report that measures of cognitive ability show only weak correlations with performance on many rational thinking tasks. For example, when people are asked to assess the probability of events, they commit the conjunction fallacy – judging a specific scenario as more likely than a general one it is part of – at rates that are largely unrelated to their IQ (see Representativeness). Similarly, susceptibility to framing effects, where people reverse their preferences depending on whether an outcome is described in terms of gains or losses, shows little association with intelligence (see Gains and Losses). These are not failures of computational power. The people who fall for these traps can solve far harder problems on a test. What is missing is the disposition to apply careful thinking to their own beliefs and choices.

This idea connects directly to the dual-process framework discussed in Reason and Intuition. System 1 generates quick, intuitive responses. System 2 can override those responses with more careful, deliberate analysis. But System 2 does not activate automatically – it requires effort, and more importantly, it requires the person to recognize that effort is needed. This is essentially what is measured by the Cognitive Reflection Test (CRT), which we encountered in Reason and Intuition (Frederick, 2005). A high IQ means your System 2 is powerful. Dysrationalia (and a corresponding low score on the CRT) means you rarely deploy it.

Actively Open-Minded Thinking

The CRT captures one aspect of good thinking: the willingness to override an intuitive response. But rational thinking involves more than catching yourself on tricky math problems. It also involves how you handle beliefs, evidence, and disagreement. This is where the concept of Actively Open-Minded Thinking comes in.

Stanovich & Toplak (2023) define Actively Open-Minded Thinking, or AOT, as a thinking disposition characterized by willingness to consider alternative viewpoints, sensitivity to evidence that contradicts your current beliefs, and a habit of postponing conclusions until you have considered the issue from multiple angles. Where the CRT measures whether you override a specific intuitive error, AOT measures a broader pattern of intellectual humility and flexibility.

AOT is measured through a questionnaire rather than a performance test. People rate how much they agree with statements such as "A person should always consider new possibilities" and "People should revise their beliefs in response to new information." Disagreement with items like "Changing your mind is a sign of weakness" also contributes to a higher AOT score. Over 25 years, the AOT scale has been refined to improve its accuracy and reduce biases, with the most recent version consisting of 13 items.

What makes AOT valuable is its predictive power independent of intelligence. People who score higher on AOT are less likely to endorse superstitious thinking, less likely to believe in conspiracy theories, and less likely to accept pseudoscientific claims – and these relationships hold even after accounting for IQ (Stanovich & Toplak, 2023). AOT captures a general orientation toward evidence-based reasoning that protects against a wide range of irrational beliefs.

Consider two students with identical IQ scores who encounter a claim on social media that a common food additive causes a serious disease. The student low in AOT might accept the claim because it fits with a general suspicion of processed food, share it with friends, and resist changing their mind when someone points out that the claim is based on a single poorly conducted study. The student high in AOT might pause, look for the original source, consider alternative explanations, and update their belief when they find that the scientific consensus contradicts the claim. Both students have the same computational power. They differ in their disposition to deploy it.

However, not all failures of rationality are equally responsive to intelligence or open-minded dispositions. One especially stubborn case is myside bias.

Myside Bias: The Limit of Intelligence

Myside bias is the tendency to evaluate evidence and generate arguments in a way that favors your pre-existing position (see Confirmation). You might expect that smarter people would be better at recognizing when they are being one-sided. They are not.

Stanovich et al. (2013) synthesized research showing that myside bias is largely independent of cognitive ability. In one line of studies, participants were asked to generate arguments both for and against controversial issues, such as whether the sale of human organs should be permitted. People consistently generated more and better arguments for their own side than for the opposing side. But the degree of this bias was unrelated to intelligence. High-IQ participants were just as one-sided as low-IQ participants.

The same pattern emerged across multiple paradigms. In causal reasoning tasks, participants provided more evidence for than against their own theories about social issues like criminal recidivism. In a study of nationalistic bias, participants were more likely to support banning a dangerous German car in their own country than banning an equally dangerous domestic car in Germany – even when the objective risk was described as identical, indicating that their judgments tracked ingroup favoritism rather than the evidence. This bias was equally strong among high- and low-intelligence individuals.

There is an important nuance. When people are explicitly told to set aside their prior beliefs – as in formal logic tasks where the instructions say "assume the following premises are true regardless of what you believe" – intelligence does help. Higher cognitive ability allows people to decouple their reasoning from their beliefs when instructed to do so. But in everyday life, no one taps you on the shoulder and says "please set aside your opinions now." Without that explicit instruction, intelligence provides no reliable protection against myside bias.

This finding has a somewhat unsettling implication. Intelligence can actually make myside bias more effective in some contexts, because smarter people are better at constructing arguments that support their pre-existing beliefs. They use their cognitive power not to find the truth, but to defend what they already think – a phenomenon sometimes called motivated reasoning.

Stanovich & Toplak (2023) note a further wrinkle: AOT scores, despite predicting performance on many rational thinking tasks, appear less strongly related to myside bias than one might expect, especially when identity-relevant beliefs are involved. The authors suggest that myside bias may be especially resistant to individual thinking dispositions because it involves deeply held convictions rooted in identity, ideology, and emotion. When your core beliefs are at stake, even a generally open-minded person may struggle to reason impartially.

Implications

The distinction between intelligence and rationality has practical consequences throughout this book. One important domain is education. Because thinking dispositions, unlike IQ, reflect habits of mind rather than fixed capacity, they are at least in principle amenable to training. Teaching people to routinely ask "What would change my mind?" or "What evidence would I need to see to update my belief?" is a form of rationality training that goes beyond anything IQ tests measure. Intelligence is relatively stable across the lifespan. Dispositions like AOT, by contrast, may be cultivable through deliberate practice and well-designed curricula.

A second implication concerns professional decision-making. Physicians with high IQs and extensive training still anchor on initial diagnoses and fail to revise them when new test results come in (see Medical Decision-Making). Judges and forensic experts show anchoring effects and confirmation biases comparable to those of laypeople (see Legal Decision-Making). In these cases, the problem is not a lack of knowledge or processing power – it is a failure to engage the reflective disposition that would prompt the question: "Could my initial impression be wrong?" Recognizing that expertise does not automatically confer rationality is a first step toward designing better decision environments in high-stakes domains.

A Bayesian Lens

The distinction between intelligence and rationality maps naturally onto the Bayesian framework that runs through this book. Bayesian reasoning requires you to start with a prior belief, consider new evidence and its reliability, and update your belief accordingly to form a posterior (see Bayesian Reasoning). This process demands both computational capacity and the willingness to carry it out.

Intelligence gives you the computational machinery. A person with high working memory can hold the prior and the evidence in mind simultaneously, weigh them appropriately, and compute a revised belief. But having this capacity is not the same as using it. A dysrational thinker has the processing power to integrate base rates, weigh evidence, and revise beliefs – but defaults to System 1 shortcuts instead, taking the intuitive answer at face value rather than checking whether it properly accounts for both prior and evidence.

The CRT illustrates this directly. Each CRT item elicits an intuitive answer that feels plausible but does not survive even a simple check. Answering correctly requires you to notice the discrepancy and recalculate. People who score high on the CRT are, in effect, people who notice when their initial answer has gone wrong and are willing to redo the computation.

AOT goes a step further. It reflects the habit of actively seeking out evidence that could change your beliefs. A person high in AOT does not just wait for disconfirming evidence to appear – they go looking for it. In Bayesian terms, they actively search for data that has high diagnostic value, data that could shift their belief substantially, rather than selectively gathering evidence that confirms what they already believe. Intelligence is the capacity to compute; rationality is the willingness to compute honestly.

Summary

Intelligence and rationality are related but distinct concepts. Standard IQ tests measure algorithmic-level processing – the raw computational power of the mind – but they do not assess reflective-level thinking dispositions: the tendency to actually deploy that power when it matters. Dysrationalia, a term coined by Stanovich (1993), describes the condition of having adequate intelligence but thinking poorly due to a lack of these dispositions. The Cognitive Reflection Test (Frederick, 2005) measures one key aspect of this: the willingness to override an intuitive but incorrect response with a deliberate, correct one. CRT scores are associated with lower susceptibility to many cognitive biases, above and beyond what IQ predicts. Actively Open-Minded Thinking (Stanovich & Toplak, 2023) captures a broader disposition toward intellectual humility, evidence-seeking, and willingness to revise beliefs, and it predicts resistance to irrational beliefs independently of intelligence. However, neither intelligence nor AOT reliably protects against myside bias – the tendency to reason in favor of one's own position – especially when deeply held beliefs are at stake (Stanovich et al., 2013). Together, these findings show that good thinking requires not just a powerful mind, but the disposition to use it carefully.

INTELLIGENCE IN THE REAL WORLD

Learning goals:

- Explain how general mental ability predicts academic achievement and job performance.
- Describe how the predictive power of general mental ability varies with job complexity.
- Explain the genetic and environmental contributions to individual differences in intelligence.
- Describe how modern IQ tests define and measure intelligence.
- Explain the Flynn effect and the evidence for its reversal.
- Relate general mental ability to the quality of reasoning and decision-making.

Intelligence Predicts Real-Life Outcomes

When companies hire new employees, they face a prediction problem. From a stack of applications, a handful of interviews, and perhaps a few test scores, they need to guess who will perform well on the job months or years from now. Employers have tried many approaches: structured interviews, personality questionnaires, work samples, and letters of recommendation. But decades of research consistently indicate that one of the single best predictors of how well someone will do in almost any job is a measure of general cognitive ability.

Schmidt & Hunter (2004) synthesized decades of research on the relationship between general mental ability – another word for intelligence – and work outcomes. Their central finding is that general mental ability is among the strongest single predictors of job performance across a very wide range of occupations. Using meta-analyses that combined thousands of validity studies, they showed that the correlation between general mental ability and job performance is approximately .58 for high-complexity jobs such as engineering or management, and around .23 for low-complexity jobs such as routine manual labor. For job training programs, the relationship is even stronger, with an average correlation of about .63. In other words, people who score higher on tests of general cognitive ability tend to learn job-relevant skills faster, acquire more job knowledge, and ultimately perform better.

To put this in perspective, consider how general mental ability compares with other predictors that employers commonly use. Years of work experience predict performance only modestly once basic experience has been acquired; the correlation with performance drops from about .49 in the first few years to roughly .15 after twelve years on the job. Personality traits such as conscientiousness do predict performance, but with a correlation of about .31, which is substantially lower than that of general mental ability. When conscientiousness is combined with general mental ability, it adds a modest boost to prediction, but the cognitive component remains dominant. Even structured interviews and integrity tests, which are useful selection tools, do not match general mental ability in predictive power on their own.

Why does general mental ability matter so much? Schmidt & Hunter (2004) proposed a straightforward explanation: people with higher cognitive ability acquire job knowledge more quickly and more thoroughly. This accumulated knowledge then drives better performance. In their theoretical model, the effect of general mental ability on performance is largely mediated by job knowledge. Higher-ability workers learn the relevant facts, procedures, and strategies of their job faster, and this knowledge advantage translates into more effective work. Whether you are a physician diagnosing a patient, a lawyer preparing a case, or a logistics manager optimizing delivery routes, the ability to learn and apply complex information is central to doing the job well.

The relationship between general mental ability and job complexity deserves special attention. Consider the difference between an engineer troubleshooting a novel system failure – diagnosing what went wrong, generating hypotheses, and testing them against available data – and a worker performing routine assembly on a production line with stable, well-practiced procedures. The engineer's task places heavy demands on abstract reasoning and rapid learning, which is why general mental ability is a particularly strong predictor of performance in such roles. For simpler, more routine jobs, the relationship is weaker but still present, because even routine work requires learning procedures, adapting to changes, and troubleshooting occasional problems. This pattern aligns with the distinction between fluid and crystallized intelligence discussed in What Is Intelligence?: complex jobs draw heavily on fluid intelligence, while simpler jobs rely more on practiced routines.

The same pattern appears in education. Cognitive ability is a strong predictor of standardized test scores and a moderate predictor of school grades, because grades also reflect motivation, conscientiousness, and classroom behavior. Students with higher general mental ability tend to learn new material faster, grasp abstract concepts more readily, and progress further in formal education. In this way, the predictive power of cognitive ability extends well beyond the workplace.

It is important to be clear about what these findings do and do not mean. They do not mean that intelligence is the only thing that matters for success. Motivation, social skills, physical health, opportunity, and sheer luck all play significant roles. Nor do they mean that people with lower test scores cannot succeed in demanding jobs. These are statistical relationships that describe tendencies across large groups of people, not deterministic rules about individuals. But they do mean that cognitive ability has substantial, measurable consequences for real-world outcomes, and ignoring it in selection decisions comes at a cost to prediction accuracy.

Before asking where these differences come from, it helps to clarify how intelligence is measured in the first place.

Measuring Intelligence with IQ Tests

Modern intelligence tests, such as the Wechsler Adult Intelligence Scale (WAIS), do not produce a raw score of how smart someone is in absolute terms. Instead, they define intelligence relative to the population. A typical battery combines subtests involving vocabulary, pattern reasoning, working memory, and processing speed, yielding an overall score that reflects general cognitive ability. IQ scores are standardized so that the average score in the general population is 100, with a standard deviation of 15. About 68% of people score between 85 and 115, and about 95% score between 70 and 130. A score of 130 places someone in roughly the top 2% of the population, while a score of 70 places someone in the bottom 2%.

This relative scoring system has an important implication: an IQ score tells you where a person stands compared to others in the same population, not how much intelligence they have in some absolute sense. If the entire population were to become better at cognitive tasks, but everyone improved by the same amount, the average IQ would still be 100 by definition. This feature of IQ measurement becomes relevant when we consider historical trends in cognitive ability, as discussed below.

Genes, Environment, and Their Interplay

Where do individual differences in intelligence come from? The short answer is that both genes and environment matter substantially. Twin studies provide the most direct evidence. Identical twins share all of their genetic material, while fraternal twins share about half. By comparing how similar identical twins are in intelligence with how similar fraternal twins are, researchers can estimate the proportion of variation attributable to genetic differences (Plomin & Deary, 2015). If identical twins' IQ scores are more similar to each other than fraternal twins' scores are, the excess similarity is attributed partly to shared genes. Across many such studies, the heritability of intelligence is estimated at roughly 50% on average, meaning that about half of the variation in cognitive ability across individuals in a given population can be traced to genetic differences.

The remaining variation reflects environmental influences as well as measurement error, with contributions from factors such as prenatal conditions, nutrition during childhood, quality of schooling, socioeconomic status, and many nonshared experiences that differ even between siblings in the same household. For example, children raised in stimulating home environments with access to books, conversation, and educational resources tend to develop higher cognitive skills, even after accounting for genetic similarity to their parents. Socioeconomic status is a particularly powerful environmental factor, likely because it bundles together many specific influences: better nutrition, safer neighborhoods, better schools, and more cognitively demanding leisure activities.

One subtlety is worth noting. Heritability is not a fixed property of intelligence itself but a property of a population at a particular time and place. In a society where everyone receives the same nutrition, education, and stimulation, environmental variation would be minimal, and most of the remaining differences in intelligence would be genetic. In a society with vast inequalities in access to education and nutrition, the environmental contribution would be larger. Accordingly, heritability estimates from one country or era do not automatically apply elsewhere. A high heritability estimate also does not imply that intelligence is unchangeable. Height is highly heritable, yet average height has increased dramatically over the past century due to improvements in nutrition and health care. One of the clearest demonstrations that average cognitive performance can shift substantially with environment is the Flynn effect.

The Flynn Effect and Its Reversal

One of the most remarkable findings in the study of intelligence is the Flynn effect: the observation that average IQ scores rose substantially throughout the 20th century. Named after the researcher James Flynn, who documented the trend across many countries, the effect amounts to roughly 3 IQ points per decade. Over the span of a few generations, this adds up to a large shift. If you scored the average person from 1950 using the norms of 2000, they would appear to have below-average intelligence. Conversely, the average person from 2000 would appear above average by 1950 standards.

The Flynn effect is too rapid to be explained by genetic change; human evolution does not work that fast. Instead, the likely causes are environmental: better nutrition, improvements in public health, increased access to education, smaller family sizes, and greater exposure to abstract and analytical thinking. The modern world demands more abstract reasoning than the world of a century ago. From filling out tax forms to interpreting graphs in the news, daily life increasingly requires the kind of thinking that IQ tests measure. This is especially true for tests of fluid intelligence, which show the largest gains over time.

However, recent evidence suggests that the Flynn effect may have stalled or even reversed in some countries. Bratsberg & Rogeberg (2018) analyzed data from Norwegian military conscription tests, which have been administered to nearly all young men in Norway for decades. These data cover birth cohorts from 1962 to 1991 and provide a uniquely comprehensive picture of cognitive trends. The results showed a clear pattern: IQ scores rose for cohorts born up to the mid-1970s, consistent with the classic Flynn effect, but then began to decline for later cohorts. The decline amounted to roughly 0.3 IQ points per year, which, while modest in the short term, represents a meaningful reversal of a long-standing trend.

The most important contribution of the Bratsberg & Rogeberg (2018) study was methodological. The researchers used a family fixed-effects design, comparing brothers born in different years within the same family. This controls for factors siblings share, including much of their family background and part of their genetic similarity. If the decline in IQ were driven by genetic changes in the population – for instance, because people with lower intelligence were having more children – then the decline should not appear within families. But it did. Brothers born later showed lower scores than their older siblings, indicating that the reversal is driven by environmental factors that change over time, even within the same household.

What environmental factors might explain the reversal? The answer remains uncertain. Possible candidates include changes in the education system, shifts in media consumption from reading toward more passive screen-based activities, or other lifestyle changes. Whatever the cause, the finding challenges the assumption that each generation will be cognitively sharper than the last.

The Flynn effect and its reversal illustrate a broader lesson about the nature of intelligence. Cognitive ability is not a fixed quantity, either for individuals or for populations. It responds to environmental conditions, for better or for worse. This is encouraging insofar as it means that investments in education, nutrition, and public health can genuinely improve cognitive outcomes. But it is also a warning: if those environmental supports erode, cognitive gains can be lost.

Summary

General mental ability is among the strongest single predictors of both academic achievement and job performance. Its predictive power increases with job complexity, and it outperforms other commonly used selection tools such as work experience, personality measures, and interviews. The mechanism appears to involve faster and more thorough acquisition of job knowledge. Individual differences in intelligence are shaped by both genetic and environmental factors, with heritability commonly estimated at around 50% in studied populations; the remaining variation reflects environmental influences and measurement error. Modern IQ tests measure intelligence relative to population norms, with 100 as the average and 15 as the standard deviation. Throughout most of the 20th century, average IQ scores rose by about 3 points per decade – a phenomenon known as the Flynn effect – driven by environmental improvements such as better nutrition and education. However, recent evidence from Norway shows that this trend has reversed for cohorts born after the mid-1970s, with the decline occurring within families and therefore attributable to environmental rather than genetic causes. Finally, higher cognitive ability and greater cognitive reflection are associated with better performance on tasks that require integrating probabilities, resisting misleading heuristics, and calibrating confidence. This suggests that intelligence is related not only to educational and occupational outcomes but also, to some extent, to the quality of everyday judgment and decision-making.

UNCERTAINTY

AVAILABILITY

Learning goals:

- Define the availability heuristic and explain how it leads to biased judgments of frequency and probability.
- Describe the classic studies on letter position and famous names that demonstrate the availability heuristic.
- Distinguish between different routes through which availability operates, including faster recall, more recalled instances, and stronger associations.
- Explain how the negativity bias interacts with availability to distort perceptions of risk and the state of the world.
- Apply the concept of biased evidence sampling to understand availability through a Bayesian framework.

How Easily Does It Come to Mind?

Imagine you are asked: which is a more common cause of death, a dramatic accident like a plane crash, or a quiet disease like stomach cancer? Many people find that dramatic images of crashes come to mind more quickly than thoughts of stomach cancer, even though common diseases kill far more people each year. Or consider this: after watching a news report about a bicycle theft in your neighborhood, you might suddenly feel that bike theft is a much bigger problem than you thought the day before, even though nothing has changed about the actual rate of theft. In both cases, your judgment about how common something is gets shaped not by the actual numbers, but by how easily examples pop into your head.

Tversky & Kahneman (1973) called this the availability heuristic: when people estimate how frequent or probable something is, they rely on how easily relevant instances come to mind. The easier it is to think of examples, the more frequent or probable the event seems. As discussed in the chapter on Heuristics and Biases, heuristics are mental shortcuts that substitute a hard question for an easier one. Here, the hard question is "how frequent is this event?" and the easier substitute is "how easily can I think of examples?" This substitution often works well, because things that happen frequently are generally easier to recall. But it also opens the door to systematic errors whenever ease of recall is influenced by something other than actual frequency. One way to think about this is as biased sampling from memory: instead of drawing a representative sample of evidence, people draw a sample weighted toward whatever happens to come to mind most readily.

Letters, Names, and the Origins of a Classic Idea

Tversky & Kahneman (1973) demonstrated the availability heuristic with a series of elegant studies. In one, participants were asked about specific consonants: K, L, N, R, and V. For each letter, participants had to judge whether the letter appears more often in the first position of English words or in the third position. For example, does the letter K appear more often at the start of a word (like "kite") or in the third position (like "ask")? Sixty-nine percent of participants judged these consonants to be more common in the first position. But all five letters actually appear more frequently in the third position. Why did people get this wrong? Because it is much easier to search your mental dictionary by the first letter of a word than by the third letter. Try it yourself: you can quickly generate words that start with K, but generating words with K as the third letter is much harder. The ease of this mental search misleads you into thinking first-position words are more numerous.

Another classic study from the same paper used lists of names. Participants heard a list of names that included both men and women. In some lists, the women's names were very famous (think of well-known politicians, athletes, or artists) while the men's names were only somewhat famous. In other lists, the pattern was reversed. Afterward, participants were asked whether the list contained more men or more women. The actual numbers were roughly equal, but participants consistently overestimated the frequency of whichever gender had the more famous names. Famous names are more memorable, and this greater memorability led participants to treat retrieval fluency as evidence of frequency.

More broadly, availability is a System 1 process (see Reason and Intuition): rather than deliberately counting cases or consulting base rates, people rely on an immediate sense of how easy retrieval feels. This feeling emerges automatically, without conscious effort.

What Makes Something Available?

When we say something is "available," what exactly do we mean? Availability can operate through several different routes. Sometimes a few vivid examples come to mind very quickly – you might not think of many plane crashes, but the one you saw in a documentary springs to mind instantly. Sometimes availability manifests as a large number of recalled instances: you can think of many examples, even if none of them are particularly fast or vivid. And sometimes it operates through the strength of associations in memory: certain pairings of ideas are so deeply encoded that one automatically triggers the other, even without conscious recall of specific episodes.

These are not competing explanations. They are all routes through which the same principle operates. Some researchers use the term accessibility to refer specifically to the ease and speed of retrieval, distinguishing it from the sheer number of instances recalled. Availability, as Tversky & Kahneman (1973) originally described it, is the broader umbrella concept that encompasses all of these manifestations – speed, volume, and associative strength alike. What matters is that any factor that makes instances easier to bring to mind can inflate your estimate of how common or probable something is.

Tversky & Kahneman (1973) provided early evidence that people can gauge their own availability with surprising accuracy. In one study, participants were shown a grid of letters (for example, a 2×4 matrix containing letters like T, A, N, and O) and asked to estimate how many words of a given length they could form from those letters – before actually trying. Their estimates correlated very strongly with the number of words they actually produced (a correlation of 0.96). In a second study, participants estimated how many instances of a category (such as flowers) they could recall, and again, estimates closely matched actual recall. These findings show that availability is a real and measurable mental signal. The problem is not that the signal is noisy; the problem is that it can be driven by factors that have nothing to do with actual frequency.

When Availability Misleads

What are these misleading factors? Several stand out. Recency is one: events that happened recently are easier to recall than events from the distant past. If a friend just told you about a burglary in their street, burglaries will seem more common than if you had not heard this story. Vividness is another: dramatic, emotionally charged events leave stronger memory traces. A single graphic news story about a shark attack can make shark attacks seem far more common than they are, even though the actual risk is vanishingly small.

Personal experience matters too. Things that have happened to you are more available than things that have happened to strangers. If you have been in a minor car accident, you are likely to estimate the probability of car accidents as higher than someone who has not. And the structure of memory plays a role: as the letter-position study shows, some categories are simply easier to search than others, even when the underlying frequencies are the same.

MacLeod & Campbell (1992) provided evidence consistent with a causal influence of availability on probability judgments. In their experiment, undergraduate students underwent a mood-induction procedure: some were put into a happy mood and others into a sad mood by reading self-descriptive statements. Participants were then asked to recall specific instances of positive and negative life events, while the researchers measured how quickly these memories came to mind. Afterward, participants rated how likely they were to experience similar events in the next six months. Participants in a sad mood recalled negative events more quickly, and this faster recall predicted higher estimates of the likelihood of negative events happening to them. Participants in a happy mood showed the reverse pattern: positive events came to mind more easily, and they judged positive events as more likely. Crucially, the size of the change in recall speed predicted the size of the change in probability judgments, even after controlling for the mood itself. By experimentally manipulating mood and showing that resulting changes in recall speed were associated with changes in probability estimates, this study supports the idea that changes in recall accessibility can contribute causally to probability judgments, rather than merely correlating with them.

The Negativity Bias and the News

The availability heuristic does not operate in a vacuum. It interacts with other features of human psychology, and one of the most important of these is the negativity bias. People tend to seek out, pay more attention to, and be more strongly affected by negative information than by positive information. One common explanation is that missing a threat is often costlier than missing an opportunity, so our cognitive systems are tuned to prioritize negative signals. The result is that negative events leave stronger impressions and are more easily recalled.

Robertson et al. (2023) studied this in the context of online news consumption. Using a massive dataset from the news website Upworthy, which included over 105,000 headline variations and roughly 370 million impressions, they analyzed how the emotional tone of headlines affected click-through rates. Because Upworthy routinely tested different headlines for the same story in randomized controlled trials, the researchers could draw causal conclusions. They found that each additional negative word in a headline increased the click-through rate by about 2.3 percent, while each additional positive word decreased it by about 1.0 percent. The effect was robust across different ways of measuring sentiment and across different news topics, though it was especially strong for news about government and the economy.

This finding has a direct connection to the availability heuristic. If negative news attracts more clicks, news organizations have an incentive to emphasize negative stories. Readers then encounter a disproportionate amount of negative information, which makes negative events more available in memory. When people later estimate how common crime, corruption, or disaster is, they draw on this biased mental sample and conclude that the world is more dangerous and grim than it actually is. The old journalistic saying "if it bleeds, it leads" describes not just editorial preferences but a feedback loop between the negativity bias and availability. This loop is further amplified in the age of social media, where algorithms learn what you click on and serve you more of the same, a topic explored in the chapter on Populism and Social Media.

Availability in Professional Settings

The availability heuristic is not limited to everyday judgments. It plays a significant role in professional contexts where accurate probability estimates matter a great deal. In medicine, for example, a doctor who recently treated a patient with a rare tropical disease may be more likely to diagnose the next patient with similar symptoms as having that same disease, simply because the recent case is fresh in memory. The recent case is more available, and availability is a powerful cue that operates automatically, even among highly trained professionals. The chapter on Medical Decision-Making explores these effects in more detail, including how decision support systems can help counteract them.

In everyday risk perception, as discussed in the chapter on Risk Perception, people tend to overestimate the probability of dramatic but rare events (such as plane crashes or terrorist attacks) and underestimate the probability of common but undramatic ones (such as heart disease), in large part because dramatic events are more available in memory. Tversky & Kahneman (1974) noted that even trained statisticians fall prey to availability-driven biases when relying on intuition rather than formal analysis. The heuristic is not a sign of ignorance or carelessness; it is a fundamental feature of how the human mind handles the question "how common is this?"

A Bayesian Lens

All of the distortions discussed above have something in common: they bias which evidence is mentally sampled in the first place. A recurring theme in this book is that many biases can be understood as departures from the ideal of Bayesian reasoning, in which a prior belief is updated with evidence to form a posterior belief (see Bayesian Reasoning). The availability heuristic fits neatly into this framework by distorting the evidence-sampling step.

In an ideal Bayesian update, you would draw a representative sample of evidence from your memory, one that accurately reflects the true base rates in the world. But the availability heuristic means that your mental sample is not representative. It is weighted toward whatever is recent, vivid, emotional, or structurally easy to retrieve. Consider a concrete example. Suppose you are deciding whether it is safe to swim in the ocean. A Bayesian reasoner would draw on representative evidence: the vast majority of ocean swims are uneventful, and shark attacks are extraordinarily rare. But if you recently watched a documentary about shark attacks, those vivid cases are disproportionately available. Your mental evidence sample now contains a much higher proportion of shark attacks than the real world does. You update on this distorted sample and conclude that the ocean is more dangerous than it actually is – not because your reasoning about the evidence is faulty, but because the evidence that entered the reasoning process was not representative.

The result is that the evidence entering the Bayesian update is biased: people effectively update on a distorted memory sample rather than on the full range of relevant information. An ideal Bayesian reasoner with unbiased, representative access to memory would not show this effect. Of course, real minds operate under constraints – memory retrieval has costs, and relying on what comes to mind easily is often a sensible strategy (see the discussion of ecological rationality in Reason and Intuition). The availability heuristic is ecologically rational in many environments, which is precisely why it persists. The problems arise when the factors that drive availability diverge from actual frequency.

Summary

The availability heuristic is the tendency to judge the frequency or probability of events based on how easily examples come to mind. Classic studies show that people overestimate the frequency of famous names and misjudge letter positions in words, because memorable or easily retrieved instances are treated as evidence of prevalence. Availability can operate through several routes: faster recall, a greater number of recalled instances, or stronger associations in memory. The negativity bias amplifies these effects, because negative information attracts more attention and is more readily encoded, creating a feedback loop with news media that makes the world seem worse than it is. Experimental evidence supports a causal role for availability in shaping probability judgments. In professional settings, from medicine to risk assessment, the same heuristic shapes consequential decisions. From a Bayesian perspective, availability distorts the evidence that enters the reasoning process: people update on a biased memory sample rather than on representative information, producing skewed conclusions even when the reasoning itself is sound.

REPRESENTATIVENESS

Learning goals:

- Define the representativeness heuristic and explain how it leads to systematic errors in probability judgment.
- Explain how base-rate neglect arises from overweighting the match between evidence and a category.
- Describe the conjunction fallacy and why people judge the combination of two events as more likely than one event alone.
- Explain how representativeness contributes to stereotyping and biased decision-making.
- Describe the gambler's fallacy and the hot-hand belief as consequences of a distorted sense of what randomness looks like.
- Connect the representativeness heuristic to Bayesian reasoning by identifying which component of Bayes' theorem people neglect and which they overweight.

Judging by Resemblance

Imagine you meet someone at a party in Groningen. She is quiet, wears glasses, and tells you she loves reading poetry. If asked to guess whether she studies Dutch literature or business administration, most people would say literature. She fits the image. But the business administration program at the University of Groningen enrolls far more students than Dutch literature does. Statistically, a random student you meet is much more likely to study business. Yet most people ignore this fact and go with the match.

The representativeness heuristic is the tendency to judge probability by resemblance to a category or prototype rather than by statistical information. This heuristic can lead to several distinct errors, from ignoring base rates to misjudging randomness. It is one of the core heuristics described in the Heuristics and Biases framework, and it represents perhaps the most direct departure from the kind of rational belief updating described in Bayesian Reasoning.

Base-Rate Neglect

The core mechanism behind the representativeness heuristic is base-rate neglect. When people judge how likely it is that a person or event belongs to a certain category, they focus on how well the evidence matches that category. They pay little attention to how common the category actually is. In Bayesian terms, they attend to the likelihood (how probable is this evidence if the hypothesis is true?) while neglecting the prior (how probable is the hypothesis before seeing any evidence?).

A classic demonstration is the Tom W. problem (Kahneman & Tversky, 1973). Participants read a brief personality sketch of a fictional graduate student named Tom W. The description portrayed him as highly intelligent but lacking in creativity, with a need for order and tidiness, and a preference for science fiction. One group of participants ranked how similar Tom W. was to the typical student in each of nine graduate fields (such as computer science, humanities, social work, and law). Another group ranked how likely it was that Tom W. was actually enrolled in each field. A third group, who never saw the personality description, simply estimated the proportion of graduate students in each field, providing the base rates.

The likelihood rankings matched the similarity rankings almost perfectly. If Tom W. seemed like a typical computer science student, people judged it as very likely that he studied computer science. But these rankings correlated negatively with the actual base rates. Fields like humanities and social work had far more students than computer science, yet participants ranked them as unlikely choices for Tom W. People based their predictions primarily on resemblance and largely ignored how many students were actually enrolled in each program.

Base-rate neglect also appears in more precise, numerical problems. Consider the taxicab problem (Bar-Hillel, 1980). A cab was involved in a hit-and-run accident at night. Two cab companies operate in the city: the Blue company, which owns 85% of the cabs, and the Green company, which owns 15%. A witness identified the cab as Green. The court tested the witness under conditions similar to those on the night of the accident and found that the witness correctly identified the color of the cab 80% of the time.

What is the probability that the cab involved in the accident was actually Green? Most people say about 80%, going with the witness testimony and ignoring the base rate. But the correct answer, obtained through Bayes' theorem, is only about 41%. One way to see this is to use natural frequencies, as discussed in Bayesian Reasoning. Imagine 100 accidents. Based on the base rates, 85 would involve Blue cabs and 15 would involve Green cabs. The witness, who is 80% accurate, would correctly identify 12 of the 15 Green cabs as Green but would also incorrectly identify 17 of the 85 Blue cabs as Green (since the witness is wrong 20% of the time). So out of 29 cabs the witness calls Green, only 12 are actually Green. That is roughly 41%, not 80%.

Bar-Hillel (1980) argued that people do not neglect base rates because they cannot understand them. They neglect them because the specific evidence feels more relevant to the case at hand. The witness testimony feels directly informative about this particular cab, while the fact that 85% of cabs in the city are Blue feels like background information that does not apply. When base rates were reframed to be causally relevant (for example, stating that Blue cabs are involved in 85% of accidents, perhaps because their drivers are more reckless), people paid much more attention to them. This suggests the problem is not a general inability to use base rates but a tendency to dismiss them when other information feels more diagnostic.

The representativeness heuristic is related to Confirmation bias. When you judge that the poetry-loving woman at the party studies literature, you focus on how well she matches your image of a literature student. To judge properly, you would also need to ask: how common is poetry-loving among business students? If many business students also enjoy poetry, then a love of poetry does not distinguish between the two groups. But people typically evaluate the match between a person and their prototype of a single category without considering that people outside the category might look the same. This one-sided evaluation – focusing on how well the evidence fits one hypothesis while neglecting how well it fits alternatives – shares a family resemblance with confirmation bias applied to probability judgment.

The Conjunction Fallacy

Because representativeness tracks resemblance rather than set inclusion or probability logic, it can lead people to rate a narrower category as more likely than a broader one. This is the conjunction fallacy: judging the combination of two events as more probable than one of the events alone. The probability of two things both being true (A and B) can never be higher than the probability of just one of them being true (A), because every case where both A and B are true is already included in the cases where A is true.

The most famous demonstration is the Linda problem (Tversky & Kahneman, 1983). Participants read the following description: Linda is 31 years old, single, outspoken, and very bright. She majored in philosophy. As a student, she was deeply concerned with issues of discrimination and social justice, and also participated in anti-nuclear demonstrations. Participants were then asked to rank several statements by probability, including “Linda is a bank teller” and “Linda is a bank teller and is active in the feminist movement.”

In the original study, 82% of participants judged “bank teller and feminist” as more probable than “bank teller” alone. This result was remarkably robust. It appeared in both indirect tests (where different groups rated the two statements) and direct tests (where the same person compared them side by side). It persisted among graduate students with training in statistics, and it held up even when participants were offered financial incentives for accuracy (Tversky & Kahneman, 1983).

Why does this happen? The description of Linda is highly representative of an active feminist but not at all representative of a typical bank teller. Being a bank teller alone feels implausible given who Linda seems to be, while being a feminist bank teller feels like a more coherent picture. The conjunction is more representative of Linda, even though it is less probable by definition.

One proposed explanation is an averaging heuristic. People seem to intuitively average the probabilities of the component events rather than multiplying them. "Feminist" has a high probability given Linda's description. "Bank teller" has a low probability. The average of a high and a low number is higher than the low number alone. But probability theory requires multiplication, not averaging. The probability that Linda is both a bank teller and a feminist equals the probability that she is a bank teller multiplied by the probability that she is a feminist given that she is a bank teller. Because you are multiplying two numbers that are each less than one, the product must be smaller than either number on its own.

Not everyone agrees that the conjunction fallacy reflects a genuine reasoning error. Hertwig & Gigerenzer (1999) argued that when people hear the word "probable" in everyday conversation, they do not necessarily interpret it as mathematical probability. Instead, they may interpret it as something closer to "plausible" or "believable." Under this interpretation, it is perfectly reasonable to say that it is more plausible that Linda is a feminist bank teller than just a bank teller, because the former tells a more coherent story. When Hertwig and Gigerenzer asked participants to paraphrase what "probable" meant to them, most gave non-mathematical definitions like "possible" or "conceivable." However, when the question was rephrased using "how many out of 100 people like Linda are bank tellers," the conjunction fallacy dropped substantially. This suggests that framing matters: when people think in terms of frequencies rather than vague plausibility, they are more likely to respect the conjunction rule.

This debate highlights an important theme throughout this book: whether a response counts as a "bias" depends partly on what we assume people are trying to do. If they are computing mathematical probabilities, the conjunction fallacy is a clear error. If they are making reasonable conversational inferences about plausibility, their answers may be sensible. The distinction between descriptive models (how people actually reason) and normative models (how they should reason) is central to the field, as discussed in How It Is vs. How It Should Be.

Stereotyping

Because people rely on representativeness instead of base rates, the match between a person and a group prototype can override statistical reality. This is stereotyping, and it follows the same logic as the Tom W. problem: similarity to a category drives the judgment while base rates are neglected.

Bodenhausen & Wyer (1985) demonstrated this in a study on criminal justice decisions. Participants made parole recommendations for prisoners whose names suggested different social backgrounds: a Hispanic-sounding name ("Carlos Ramirez"), an upper-class Anglo name ("Ashley Chamberlaine"), or a nondescript name. The type of crime also varied: some crimes were stereotypically associated with one group or another – for example, assault for Carlos Ramirez, forgery for Ashley Chamberlaine. When the crime matched the stereotype associated with the person's name, participants gave less favorable parole recommendations and judged the person as more likely to reoffend. This happened even when the case file contained alternative explanations for the crime, such as difficult life circumstances.

The stereotype functioned as a heuristic that made further processing of the evidence unnecessary. Participants who had a stereotype-based explanation available actually recalled less of the background information in the case file, suggesting they had processed it less carefully. The representativeness match – this person fits my image of someone who commits this type of crime – short-circuited the evaluation of the actual evidence.

These findings have serious implications for real-world decisions. In legal settings, a defendant who fits the stereotype of a criminal may receive harsher treatment regardless of the actual evidence, a topic explored further in Legal Decision-Making. In medical settings, a patient who matches the prototype of a certain disease may receive that diagnosis even when base rates suggest otherwise, as discussed in Medical Decision-Making. In both cases, the representativeness heuristic leads professionals to overweight the match between a person and a category while underweighting the statistical likelihood of that category in the relevant population.

Distorted Perception of Randomness

People carry around a mental prototype of what randomness looks like: roughly alternating outcomes, without long runs of the same result. A coin-flip sequence like HTHTHT feels more random than HHHTTT, even though both are equally likely. When actual random sequences fail to match this prototype, people judge them as non-random. In other words, people apply representativeness to randomness itself.

Kahneman & Tversky (1972) demonstrated this in several experiments. In one study, participants judged the likelihood of different birth sequences in a hospital. A sequence like BGBGBG (boy-girl alternating) was judged as more likely than BGGBBB, even though both sequences are equally probable if each birth has an independent 50% chance of being a boy or a girl. People expected even short sequences to mirror the overall 50-50 split, a mistake Tversky & Kahneman (1971) called the "law of small numbers" – the belief that small samples should perfectly represent the population they come from.

This distorted sense of randomness gives rise to the gambler's fallacy. After seeing five heads in a row, people feel that tails is "due" on the next flip. The reasoning is that HHHHHH does not look like a representative sample from a fair coin, so the process should self-correct. But a fair coin has no memory. Each flip is independent of the previous ones, and the probability of heads is 50% regardless of what came before.

A related error, in the opposite direction, is the hot-hand belief. When a basketball player makes several shots in a row, fans and commentators often say the player is "on fire" or has a "hot hand." They interpret the streak as evidence that the player's ability has temporarily increased. But research by Gilovich, Vallone, and Tversky showed that shooting streaks in professional basketball often fall within the range expected by chance alone, given the player's overall shooting percentage. The same tendency that makes people reject a random sequence as too streaky also makes them see meaningful patterns in sports performance.

Both errors stem from the same root: people judge real sequences against their prototype of randomness. When a sequence deviates from the prototype – too many runs of the same outcome – people conclude that something non-random must be happening, whether that means the coin is about to "correct" or the player has entered a special zone.

This prototype of randomness can mislead researchers too. Tversky & Kahneman (1971) showed that even trained scientists overestimated the likelihood that a result from a small study would replicate in another small study. They expected small samples to be just as representative of the population as large ones, leading them to design studies with too few participants and to underestimate the variability inherent in small samples.

A Bayesian Lens

The representativeness heuristic is perhaps the most direct violation of Bayesian Reasoning covered in this book. Rational belief updating combines a prior belief with new evidence to form a posterior belief. Bayes' theorem makes clear that both components matter: the prior tells you how likely a hypothesis was before you saw the evidence, and the likelihood tells you how probable the evidence is if the hypothesis is true. The representativeness heuristic amounts to basing judgment primarily on the likelihood while neglecting the prior.

Summary

The representativeness heuristic leads people to judge probability based on resemblance rather than statistical evidence. Its core mechanism is base-rate neglect: people focus on how well evidence matches a category while ignoring how common that category is. This produces several well-documented errors. The conjunction fallacy occurs when people rate the combination of two events as more likely than one event alone, because the combination better matches a description. Stereotyping arises when the match between a person and a group prototype overrides base-rate information, leading to biased judgments in legal, medical, and everyday settings. Distorted perceptions of randomness cause people to see patterns in chance events (the hot-hand belief) or to expect random sequences to self-correct (the gambler's fallacy). In Bayesian terms, the representativeness heuristic reflects an overreliance on the likelihood at the expense of the prior – one of the most fundamental departures from rational belief updating.

REPRESENTATIVENESS

ANCHORING AND PRIMACY

Learning goals:

- Define the anchoring effect and explain how it distorts numerical estimates.
- Describe the mechanisms – selective accessibility and anchoring-and-adjustment – that underlie anchoring.
- Connect anchoring to primacy effects in impression formation.
- Explain cross-modal anchoring and what it reveals about how the brain represents magnitude.
- Identify real-world contexts in which anchoring influences judgment.
- Evaluate strategies for reducing anchoring bias.
- Relate anchoring and primacy to Bayesian belief updating.

The Power of a Starting Point

Imagine you are browsing an online store for a new winter jacket. The first one you see is listed at €249. You keep scrolling and find another jacket that looks similar, priced at €139. It feels like a bargain. But if the first jacket you had seen was priced at €89, that same €139 jacket would have felt expensive. Nothing about the jacket changed. What changed was the number you happened to encounter first.

This is the anchoring effect: the tendency to start from an initial value or impression and adjust insufficiently from it. The anchor becomes highly available in memory (see Availability), making anchor-consistent information easier to retrieve and biasing the final judgment – even when the anchor is clearly irrelevant. Anchoring is one of the most reliable findings in the psychology of judgment, and it extends far beyond shopping. It shapes how we estimate quantities, evaluate people, and make professional decisions.

The Classic Demonstrations

The anchoring effect was first described by Tversky & Kahneman (1974) as part of their broader program on heuristics and biases (see Heuristics and Biases). In one of their most famous demonstrations, participants watched a wheel of fortune being spun. The wheel was rigged to land on either 10 or 65. After seeing the result, participants were asked whether the percentage of African countries in the United Nations was higher or lower than that number, and then to estimate the actual percentage. Even though the wheel was obviously random and had nothing to do with the United Nations, the number it produced had a large effect. Participants who saw the wheel land on 10 gave an average estimate of 25 percent. Those who saw 65 gave an average estimate of 45 percent. A random number shifted the estimate by 20 percentage points.

In another early study, Tversky & Kahneman (1973) showed how anchoring distorts even simple arithmetic. Two groups of participants were asked to quickly estimate the product of a sequence of numbers. One group saw the sequence 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8, while the other saw 8 × 7 × 6 × 5 × 4 × 3 × 2 × 1. Because people had only a few seconds, they performed the first few multiplications and then extrapolated. The ascending group, anchored on the small early products, gave a median estimate of 512. The descending group, anchored on the larger early products, gave a median estimate of 2,250. The correct answer is 40,320. Both groups substantially underestimated the true value, but the extent to which they did so was determined by which numbers came first.

These studies established two key features of anchoring. First, the initial number acts as a starting point that pulls the final estimate toward it. Second, the adjustment away from the anchor is almost always too small. People move in the right direction but stop too soon.

Why Does Anchoring Happen?

One early and intuitive explanation was that people start with the anchor and adjust away from it, but simply do not adjust enough. This anchoring-and-adjustment account captures part of the phenomenon, but it leaves an important question unanswered: why does anchoring occur even when the anchor is clearly irrelevant and people have no reason to use it as a starting point?

Strack & Mussweiler (1997) proposed a later and highly influential account based on what they called selective accessibility. When you encounter an anchor, your mind automatically generates information that is consistent with that anchor. If someone asks you whether Mahatma Gandhi was older or younger than 140 when he died, you do not seriously consider that he might have lived to 140. But the high number still activates thoughts about old age, long life, and historical figures who lived to advanced ages. This anchor-consistent information becomes highly accessible in memory, and when you then estimate Gandhi's actual age at death, this accessible information pulls your estimate upward.

This mechanism is closely related to the availability heuristic (see Availability). Just as the availability heuristic leads us to judge frequency by how easily examples come to mind, anchoring leads us to judge quantities by how easily anchor-consistent information comes to mind. The anchor makes certain information more available, and that information biases the estimate. The person making the judgment is typically unaware of this process.

Strack & Mussweiler (1997) tested the selective accessibility account in a series of studies. In one, they varied whether the comparative question (is the target higher or lower than the anchor?) and the absolute question (what is the actual value?) referred to the same dimension of the same object. Participants who were asked whether the Brandenburg Gate was taller or shorter than a certain height, and then asked to estimate its actual height, showed strong anchoring. But participants who were first asked about the Brandenburg Gate's height and then asked to estimate its width showed much weaker anchoring. This pattern supports the selective accessibility account: thinking about height activates information about tallness or shortness, which is relevant when estimating height but not when estimating width. If anchoring were simply about insufficient numerical adjustment, the dimension should not matter.

Both mechanisms – anchoring-and-adjustment and selective accessibility – likely contribute to anchoring in different settings. When people deliberately use an anchor as a starting point and work outward from it, insufficient adjustment may be the better description. When an anchor is clearly irrelevant and people have no intention of using it, selective accessibility better explains why it still exerts influence. The important point is that both routes lead to the same result: estimates that are biased toward the anchor.

Even Implausible Anchors Work

One might expect that clearly implausible anchors would be ignored. After all, nobody believes Gandhi lived to 140 or died at age 9. Yet Strack & Mussweiler (1997) found that even such extreme anchors influenced estimates. Participants who were asked whether Gandhi was older or younger than 140 gave higher estimates of his age at death than participants who were asked whether he was older or younger than 9. The effect was somewhat smaller than for plausible anchors, but it was still present.

Why do implausible anchors still have an effect? According to the selective accessibility model, even when people quickly reject an anchor as absurd, the process of evaluating it has already activated anchor-consistent information in memory. You may instantly know that Gandhi did not live to 140, but the brief moment of considering old age has already made certain thoughts more accessible. This residual activation is enough to shift the subsequent estimate.

A related finding is that explicitly warning people that an anchor is irrelevant does not eliminate the effect either (Chapman & Johnson, 2002). Irrelevant anchors, implausible anchors, and warned-about anchors all produce bias – though each for a slightly different reason. Irrelevant anchors work because the selective accessibility process is automatic. Implausible anchors work because even rejected information leaves a trace. Warnings fail because by the time you recognize the anchor's irrelevance, it has already shaped what comes to mind.

From Numbers to Impressions

Anchoring is primarily studied in the context of numerical estimation, but the same principle applies to how we form impressions of people. When we learn something about a person, the first piece of information functions as an anchor that shapes how we interpret everything that follows. This is known as the primacy effect.

The classic demonstration comes from Asch (1946), who gave participants a list of personality traits describing a hypothetical person and asked them to form an impression. One group received the following list: intelligent, industrious, impulsive, critical, stubborn, envious. Another group received the same traits in the opposite order: envious, stubborn, critical, impulsive, industrious, intelligent. The traits were identical, but the impressions were not. The group that heard positive traits first formed a much more favorable impression. When asked to guess additional characteristics of the person, 74 percent of the positive-first group inferred that the person was good-looking, compared to only 35 percent of the negative-first group. The early traits set the direction of the impression, and later traits were interpreted in light of that initial framework. An intelligent person who happens to be envious seems fundamentally different from an envious person who happens to be intelligent.

Anderson (1965) followed up on this work by systematically varying the position of positive and negative traits within longer sequences. He found that the influence of each trait decreased in a roughly linear fashion with its position in the sequence. The first trait carried the most weight, the second slightly less, and so on. Anderson proposed that impressions are formed as a weighted average of trait values, with the weights declining over time. This pattern aligns with the broader idea that early information disproportionately shapes later judgment, whether described as anchoring-and-adjustment, selective accessibility, or sequential belief updating.

Hogarth & Einhorn (1992) developed a comprehensive belief-adjustment model to explain when primacy effects occur and when they do not. They proposed that belief updating happens sequentially, with the current belief serving as an anchor that is adjusted by each new piece of evidence. Their model predicts that both primacy and recency can occur, depending on task structure. Under some conditions – particularly when information is simple, the sequence is long, and people give their judgment only at the end – early information dominates. Under other conditions, such as when people update their judgment after each piece of evidence and the information is complex, later evidence can carry more weight. For example, if jurors evaluate each witness as testimony unfolds, recent evidence may be more influential than if they hear all testimony and deliberate only at the end. The model shows that the order of information matters, even though it should not – and the direction it matters depends on the specifics of the task.

Anchoring Across the Senses

Perhaps the most surprising finding about anchoring is that it can cross from one domain to another entirely. Oppenheimer et al. (2008) demonstrated what they called cross-modal anchoring. In one experiment, participants were asked to draw either a long line or a short line on a piece of paper. They were then asked to estimate the length of the Mississippi River. Participants who had drawn a long line gave higher estimates than those who had drawn a short line. In a second experiment, drawing a long line led to higher estimates of the average temperature in Honolulu in July. Line length and temperature have nothing in common, yet the physical act of drawing a long line shifted temperature estimates upward.

One interpretation offered by Oppenheimer et al. (2008) is that anchoring may rely on partly shared representations of magnitude across domains. Numbers, physical lengths, durations, and temperatures may be represented in partly overlapping ways – much as people naturally map quantity onto spatial metaphors such as "high" prices or "long" waits. When you experience something large in one dimension, it may activate a general sense of largeness that spills over into judgments in a different dimension. In a third experiment, the researchers showed that this effect depends on the relative size of the anchor. When long and short lines were embedded in appropriately scaled maps so that neither appeared especially large or small, the anchoring effect disappeared. It is not the absolute size that matters but whether the anchor feels large or small in its context.

This implies that anchoring is not limited to situations where someone gives you a number. The physical environment, the size of objects on a screen, or the scale of a graph could all serve as anchors that influence unrelated judgments.

Anchoring in Professional Decisions

If anchoring only affected trivial estimates about rivers and historical figures, it would be a curiosity but little more. In fact, anchoring shapes consequential decisions made by trained professionals.

Northcraft & Neale (1987) studied real estate pricing decisions by bringing both undergraduate students and professional real estate agents to an actual house that had been appraised at $74,900. Each participant received a detailed information packet containing market data, comparable sales, and property features, and then toured the house. The only thing that varied was the listing price printed in the packet, which was set at one of four values: $65,900, $71,900, $77,900, or $83,900. Participants then estimated the property's fair market value, what they would pay for it, and what would be the lowest acceptable selling price.

Both students and agents showed clear anchoring effects. Students exposed to the lowest listing price estimated the property's value at around $63,571, while those exposed to the highest listing price estimated it at around $72,196. Agents showed similar patterns. The critical difference was in self-awareness: when asked what factors influenced their judgment, agents were significantly less likely than students to mention the listing price. The professionals were just as anchored, but they did not recognize it. Northcraft & Neale (1987) replicated this pattern with a second property, confirming that expertise does not eliminate anchoring, even in information-rich, real-world settings.

The same principle operates across many professional contexts. In advertising, a "was €99, now €49" label exploits the original price as an anchor, making the discount seem larger. In salary negotiations, the first number placed on the table tends to anchor the final outcome. In medicine, a physician's initial diagnostic hypothesis can anchor subsequent reasoning, leading to insufficient revision when new symptoms or test results point toward a different diagnosis (see Medical Decision-Making). In law, a prosecutor's recommended sentence or a plaintiff's damage request can anchor the judge's or jury's decision, pulling it toward a value it might otherwise not have reached (see Legal Decision-Making). In each case, the first number encountered – whether a listing price, an opening offer, or a sentencing request – disproportionately influences the final judgment.

Can Anchoring Be Reduced?

Given how pervasive and consequential anchoring is, an obvious question is whether anything can be done about it. The answer is discouraging. Chapman & Johnson (2002) reviewed the literature and concluded that most standard remedies do not work. Simply telling people about anchoring does not reduce it. Offering financial incentives for accuracy does not reduce it. As noted above, even explicit warnings that the anchor is irrelevant leave the effect largely intact. By the time you realize the anchor should be ignored, it has already shaped what comes to mind.

One promising finding comes from Estrada et al. (1997), who studied diagnostic reasoning among 44 practicing internists. Physicians were randomly assigned to one of three groups: a control group, a group that received a small bag of candy before the task (to induce mild positive affect), and a group that read humanistic statements about medicine. The physicians were then asked to diagnose a simulated patient case involving liver disease while thinking aloud. The physicians who had received candy considered the correct diagnosis significantly earlier in their reasoning process and showed less anchoring on their initial hypotheses. On a 10-point anchoring scale, the candy group scored 1.5 compared to 3.9 for the control group. Importantly, the candy group did not reach their final diagnosis any faster and did not show signs of careless reasoning. They simply integrated information more flexibly, entertaining the correct hypothesis earlier rather than getting stuck on an initial impression.

Why would positive affect reduce anchoring? One explanation is that mild positive mood broadens cognitive flexibility, making people more willing to consider alternative possibilities rather than sticking with their initial anchor. This does not eliminate anchoring, but it weakens its grip. Practically, the finding suggests that simple interventions that create a positive working environment might improve the quality of professional judgment.

The selective accessibility model also points to a possible strategy: deliberately generating anchor-inconsistent information. If anchoring works because the anchor makes consistent information more accessible, then actively thinking about reasons why the anchor might be wrong could counteract the bias. This is a plausible implication of the theory, but it requires effort and awareness, making it difficult to apply consistently in practice.

Anchoring Through a Bayesian Lens

From the perspective of Bayesian reasoning (see Bayesian Reasoning), anchoring and primacy both represent an over-weighting of the prior. In ideal Bayesian updating, a person starts with a prior belief and revises it in light of new evidence to form a posterior belief. Crucially, the order in which evidence arrives should not matter. If a person is described as both generous and rude, the final impression should not depend on which adjective came first. Whether the listing price is €66,000 or €84,000, your estimate of the property's true value should converge on the same number once you have inspected the house and reviewed the market data.

In reality, the first piece of information – whether a number or a trait – forms a strong prior that subsequent evidence fails to fully update. Each new piece of evidence should shift the judgment by an amount proportional to its informativeness, regardless of what came before. Instead, the anchor creates a sticky starting point: later evidence is underweighted relative to what it should contribute. This is, in a sense, the opposite problem from base-rate neglect (see Representativeness). In base-rate neglect, people ignore the prior and overweight the latest evidence. In anchoring, people cling to the prior and underweight everything that follows. Both are departures from the Bayesian ideal, but in opposite directions.

Summary

Anchoring is the tendency for an initial value to pull subsequent estimates toward it, even when that value is irrelevant. Classic studies show that random numbers from a spinning wheel and the order of digits in a multiplication problem both create anchors that distort estimates far from the correct answer. The effect occurs through at least two mechanisms: selective accessibility, in which the anchor makes anchor-consistent information easier to retrieve, and insufficient adjustment, in which people use the anchor as a deliberate starting point but stop adjusting too soon. Anchoring is remarkably robust: implausible anchors still shift estimates, warnings do not eliminate the effect, and the bias can even cross sensory modalities, with physical line length affecting temperature estimates. The same principle extends to impression formation, where the first traits we learn about a person anchor our overall judgment – the primacy effect. In professional settings, anchoring affects real estate appraisals, medical diagnoses, legal sentences, and salary negotiations, with experts showing the same susceptibility as novices while being less aware of it. Reducing anchoring is difficult; awareness and incentives have little effect, though positive affect appears to increase cognitive flexibility and weaken the anchor's grip. From a Bayesian perspective, anchoring reflects an over-weighting of the prior: the first piece of information dominates the final judgment because later evidence is insufficiently integrated.

ANCHORING AND PRIMACY

OVERCONFIDENCE

Learning goals:

- Distinguish between self-favorable bias (shifted posterior) and overprecision (narrowed posterior) as two fundamentally different forms of overconfidence.
- Explain overestimation, overplacement (better-than-average effect), and unrealistic optimism as expressions of self-favorable bias.
- Describe overprecision and explain why it is considered the most pervasive form of overconfidence.
- Explain the Dunning-Kruger effect and the role of metacognition in self-assessment.
- Explain the planning fallacy and how both forms of overconfidence contribute to it.
- Explain why expertise does not always reduce overconfidence, and identify the conditions under which calibration improves.
- Discuss the relationship between overconfidence and mental health, including the concepts of depressive realism and learned helplessness.

Two Faces of Overconfidence

Imagine you are planning a group project for one of your courses. You sit down with your teammates and estimate that it will take about two weeks to finish. You feel quite certain about this. Four weeks later, as you rush to submit the final version, you wonder how your estimate could have been so far off. This experience, familiar to almost every student, reflects not one but two forms of overconfidence operating at the same time: your estimate was too optimistic, and you were too sure about it. Understanding these two distinct distortions is the goal of this chapter.

Overconfidence is not a single bias. It is a family of biases that can be organized into two fundamentally different types, each corresponding to a different distortion of belief (Moore & Healy, 2008).

The first is a self-favorable bias: a tendency to hold beliefs about oneself that are more positive than the evidence warrants. In Bayesian terms, the center of the posterior is displaced in a self-flattering direction: confirming evidence (successes, positive traits) receives too much weight, while disconfirming evidence (failures, shortcomings) is discounted.

The second is overprecision: a tendency to be too certain about one's judgments, underestimating the range of possible outcomes. Here the problem is not the center of the estimate but its spread. The posterior is too narrow – too much probability mass is concentrated around a single value, and too little is assigned to alternative outcomes.

These two types are conceptually independent. You can be overly optimistic without being overly certain – as when someone says, "I think my startup will probably succeed, though I admit I could be wrong." And you can be overly certain without being overly optimistic – as when someone says, "I'm sure sales will fall next quarter, and there's no chance I'm wrong about that." In practice, however, the two distortions often occur together.

Both forms of overconfidence are related to a broader tendency described in the chapter on Confirmation: when people form a belief, they tend to seek and weigh more heavily the evidence that supports it, while discounting evidence that challenges it. In the case of overconfidence, this process is directed at beliefs about oneself and one's own judgments. But overconfidence also has a motivational dimension that goes beyond a simple information-processing error: people are not merely failing to update their beliefs properly – they are driven to maintain a positive view of themselves.

Self-Favorable Bias

The self-favorable bias refers to a systematic shift in beliefs about oneself in a positive direction. It shows up in several overlapping ways: overestimation, overplacement, and unrealistic optimism.

Overestimation is the simplest form. People overestimate their actual performance, their knowledge, or their degree of control over events. A classic demonstration comes from Langer (1975), who studied the illusion of control. In one of her experiments, office workers participated in a lottery. Some chose their own ticket, while others were given a ticket at random. Later, they were asked how much they would sell their ticket for. People who had chosen their own ticket demanded more than eight dollars on average, while those who had been assigned a ticket were willing to sell for about two dollars. Choosing a ticket has no effect on the probability of winning a lottery. But the act of choosing introduced a feeling of control, inflating people's estimates of how likely they were to win.

Overplacement, often called the better-than-average effect, is the tendency to judge oneself as better than the typical person. Most people rate themselves as better-than-average drivers, better-than-average students, and better-than-average friends – which is statistically impossible for a majority to be. (Strictly speaking, it is impossible for the majority to be better than the median, not the average. But since few people have an intuitive understanding of the median as a central tendency, questionnaires typically refer to averages.) What makes this particularly interesting is that overplacement is sensitive to whether a trait is seen as desirable. Logg et al. (2018) demonstrated this in a study where students were randomly assigned to read an article describing either introversion or extroversion as a key to success. When introversion was presented as desirable, students rated themselves as more introverted than average. When extroversion was presented as desirable, they rated themselves as more extroverted than average. This motivational effect was strongest when traits were defined vaguely; when specific, concrete criteria were used to measure a trait, motivation had much less influence. Ambiguity, it seems, gives people room to define traits in self-serving ways.

Unrealistic optimism, sometimes called a positivity bias, is the tendency to believe that good things are more likely to happen to oneself than to others, and that bad things are less likely. Weinstein (1980) asked college students to estimate how their chances of experiencing various life events compared to their classmates. Students consistently rated themselves as more likely to experience positive events – such as owning a home or living past 80 – and less likely to experience negative events – such as getting divorced or developing a heart attack. The bias was strongest for events that people perceived as controllable, and weakest for events associated with a clear stereotype of a typical victim. For example, students who fit the stereotype of a smoker were less optimistic about avoiding lung cancer. Weinstein proposed that unrealistic optimism arises when people focus on their own protective actions while assuming others are less careful, creating an unfair comparison with a hypothetical "unlucky" peer.

All three patterns reflect the same underlying distortion: the person's beliefs about themselves are shifted in a favorable direction, with confirming evidence weighted more heavily than disconfirming evidence.

Overprecision

Many researchers regard overprecision as the most pervasive form of overconfidence, and one of the least well explained (Moore & Healy, 2008). Overprecision means that people are too certain about their judgments. They underestimate the range of possible outcomes, acting as if they know more than they actually do.

Before examining overprecision, let's revisit a key concept from the chapter on Bayesian Reasoning. Calibration refers to the match between a person's confidence and their actual accuracy. A perfectly calibrated person who says they are 80% sure of an answer would be correct 80% of the time. In practice, most people's confidence substantially exceeds their accuracy.

The standard way to measure overprecision is through confidence intervals. Researchers present participants with general-knowledge questions – for example, "How long is the Rhine River?" or "In what year was the telephone invented?" – and ask them to provide a range within which they are 90% confident the true answer falls. If people were well calibrated, the correct answer would fall within their stated range 90% of the time. In practice, the correct answer falls within these ranges only about 40 to 60% of the time (Lichtenstein et al., 1982). People's confidence intervals are far too narrow. For the Rhine question, you might say "between 800 and 1,200 kilometers." But a well-calibrated person would recognize the genuine uncertainty in their knowledge and provide a much wider range. (The actual length is about 1,230 kilometers.) The temptation to give a narrow range comes from the feeling that one roughly knows the answer. But roughly knowing is not the same as precisely knowing, and people consistently confuse the two.

Overprecision differs from self-favorable bias in an important way. With self-favorable bias, the belief is shifted in a particular direction: toward a more positive self-view. With overprecision, the belief is not necessarily shifted in any direction. Instead, the range of possibilities considered is too narrow. The person may be correct on average, but they fail to account for how wrong they might be.

Why does overprecision occur? One explanation is that once you have formed an estimate, evidence consistent with that estimate comes to mind easily, while evidence suggesting you should widen your range is harder to generate and easier to dismiss. The result is a posterior that is too narrow: too much probability mass concentrated around a single estimate, and too little assigned to alternatives.

The Dunning-Kruger Effect

The Dunning-Kruger effect describes a specific pattern of overconfidence tied to ability: the least competent individuals tend to show the greatest overconfidence, while highly competent individuals often rate themselves less exceptionally than their performance warrants (Kruger & Dunning, 1999).

Kruger and Dunning tested this across several domains, including logical reasoning, grammar, and the ability to recognize humor. In each study, participants completed a test and then estimated how well they had performed relative to their peers. The pattern was consistent: participants who scored in the bottom quartile dramatically overestimated their performance. In the original humor study, for example, those in the bottom quartile estimated their ability to recognize what is funny at the 62nd percentile, despite actually scoring at the 12th percentile. Meanwhile, those in the top quartile slightly underestimated their own standing.

The explanation Kruger and Dunning proposed is metacognitive. The skills needed to perform well in a domain are the same skills needed to recognize good performance. If you lack the logical reasoning skills to solve a set of problems correctly, you also lack the skills to evaluate whether your answers are correct. This creates a double burden: poor performers not only make errors, but they are also unable to detect those errors. In a follow-up experiment, Kruger and Dunning trained participants in logical reasoning. After training, participants who had initially performed poorly became significantly better at assessing their own performance. Acquiring the relevant skills also gave them the metacognitive ability to recognize their earlier shortcomings.

The Dunning-Kruger effect combines both forms of overconfidence. Poor performers show a large self-favorable shift: they think they are doing much better than they are. And they show poor calibration: they have little awareness of their own uncertainty. Importantly, the mechanism here is different from the motivational account discussed earlier. In the case of unrealistic optimism, people are driven to maintain a positive self-image. In the Dunning-Kruger effect, the overconfidence arises from a metacognitive inability to recognize disconfirming evidence, not from a motivation to ignore it.

The Dunning-Kruger effect has been debated (Gignac, 2024). Part of the pattern seems to be a statistical artifact: when people with extremely low scores estimate their own performance, there is always some error (noise) in their estimate. And because their true performance is low, this error is usually upwards, leading them to usually overestimate their own performance. In other words, the Dunning-Kruger effect may in part reflect regression to the mean – a phenomenon that we will discuss in more depth in the chapter on Learning Causality.

However, there is still evidence that low performers are often poorly calibrated, even if some headline versions of the effect have been overstated. The training study, in which improving people's skills also improved their self-assessment, is particularly hard to explain as a purely statistical artifact.

The Planning Fallacy

The planning fallacy is the tendency to underestimate the time, cost, and risk of future actions while overestimating the benefits (Kahneman & Tversky, 1979). It is one of the clearest cases where both forms of overconfidence operate simultaneously: the estimate is too optimistic (a shifted posterior) and fails to account for the full range of things that could go wrong (a narrowed posterior).

Buehler et al. (1994) studied this with university students. In one study, psychology students predicted when they would finish their honors thesis. On average, actual completion times ran about 22 days longer than estimated. This happened even though students had extensive past experience with missing deadlines. In a follow-up study, students were asked to think aloud while making their predictions. The recordings revealed that students focused almost entirely on their plans for the future: how they would organize their work, when they would write each section. They rarely thought about past experiences with similar tasks. When they did recall past delays, they attributed them to unusual, one-time circumstances – a computer crashing, an unexpected family visit – rather than seeing them as part of a general pattern.

This is what Kahneman and Tversky called the inside view versus the outside view. The inside view focuses on the specific features of the current project: the plan, the timeline, the resources. It generates optimistic estimates because it naturally highlights what will go right. The outside view looks at the base rates: how long similar projects have taken in the past, regardless of the reasons for delay. As discussed in the chapter on Bayesian Reasoning, base rates provide a critical anchor for rational estimation. The outside view generates more realistic predictions because it incorporates the full range of actual outcomes, including all the delays and overruns that occurred for reasons nobody anticipated.

The planning fallacy shows up far beyond student assignments. Think of the last time someone you know renovated a kitchen, moved to a new city, or promised to finish a piece of writing by a certain date. Flyvbjerg (2006) documented the phenomenon at an institutional scale, analyzing hundreds of public transportation projects worldwide. He found average cost overruns of 44.7% for rail projects, 33.8% for bridges and tunnels, and 20.4% for roads. Demand forecasts showed similar patterns, with rail passenger projections overestimated by an average of 51.4%. As a remedy, Flyvbjerg advocated for reference class forecasting: instead of relying on project-specific plans, forecasters should base their estimates on the actual outcomes of comparable past projects. This approach was adopted by the UK Department for Transport, which now requires cost estimates to include "optimism bias uplifts" based on historical data. Road projects require a 15% uplift to reach a 50% probability of staying within budget; rail projects need a 40% uplift.

Expertise and Calibration

You might expect that experts, with their extensive experience, would be well calibrated. Sometimes they are, but often they are not. The key factor is not the amount of experience but the quality of feedback that accompanies it.

Einhorn & Hogarth (1978) analyzed why people maintain high confidence in their judgments even when those judgments are inaccurate. They argued that the structure of most judgment tasks prevents people from learning about their errors. Consider a hiring manager who interviews candidates and selects the ones she judges to be the best. She then observes how these employees perform. If they do well, she takes it as confirmation of her judgment. But she never observes the performance of the candidates she rejected. If most of those rejected candidates would also have done well, the manager's apparent success is misleading – she would have looked equally skilled by selecting at random. Without this comparison, she cannot know whether her selections were actually better than the alternatives. This one-sided feedback loop makes it easy to accumulate confidence over time without a corresponding increase in accuracy.

Einhorn and Hogarth also pointed out that treatment effects further inflate apparent accuracy: hiring someone, training them, and giving them resources may improve their performance regardless of how well they were originally assessed. These factors create an illusion of validity, reinforcing confidence in one's own judgment.

There are exceptions. Weather forecasters tend to be well calibrated: when a forecaster says there is a 70% chance of rain, it rains roughly 70% of the time (Lichtenstein et al., 1982). This is because weather forecasters receive clear, immediate, and unambiguous feedback after every prediction. They make many predictions, and the outcomes are easy to observe. Clinical psychologists, by contrast, tend to be poorly calibrated. Consider predictions of whether a patient will attempt suicide, respond to a particular therapy, or reoffend after release – the feedback is delayed by months or years, ambiguous, and often incomplete. Without clear feedback loops, experience increases confidence without improving accuracy. This is one reason people often prefer their own intuitive judgment over statistical models, even when models consistently outperform human experts – a theme explored in the chapter on Statistical vs. Intuitive Prediction.

Overconfidence and Mental Health

The patterns described so far might suggest that less overconfidence is always better. But the relationship between self-favorable bias and psychological well-being tells a more complicated story. Not all forms of overconfidence have the same psychological function: the self-favorable form, in particular, may help sustain agency and motivation.

Korn et al. (2014) studied how people with major depressive disorder update their beliefs about future life events compared to healthy individuals. Participants estimated their personal likelihood of experiencing 70 different adverse events, such as developing Alzheimer's disease or being robbed. They were then shown the average probability of each event for someone in a similar situation. After seeing this information, they re-estimated their own likelihood. To make this concrete: if someone initially thought their chance of being robbed was 40% and then learned the average was 20%, that would be good news – the real risk was lower than expected. If instead the average turned out to be 60%, that would be bad news. Healthy participants showed an asymmetric pattern: they revised their estimates more in response to good news than bad news. They selectively absorbed the positive information. Participants with major depression did not show this pattern. They updated roughly equally for good and bad news. In fact, the more severe a person's depressive symptoms, the more their updating shifted toward weighting bad news more heavily than good news. The self-favorable bias, it seems, is something that healthy minds do, and its absence is associated with depression.

This finding connects to earlier work on learned helplessness. Seligman & Maier (1967) conducted experiments in which dogs were exposed to electric shocks they could not escape. Initially, the dogs continued to act as if escape might still be possible – behavior that can be interpreted as persistence in the face of negative evidence. After repeated failure, they stopped trying altogether. When later placed in a new situation where escape was possible, they did not even attempt it. They had learned that their actions did not matter. Seligman and Maier saw this learned helplessness as a model for depression: the loss of the belief that one's actions can make a difference.

Alloy & Abramson (1979) took this line of reasoning further by studying depressed and non-depressed college students in a simple task: pressing a button and judging how much control they had over whether a light turned on. The actual degree of control was varied by the experimenters. When there was no real control, non-depressed students overestimated their influence, especially when the light turned on frequently or when they were winning money. Depressed students, by contrast, judged their control more accurately across all conditions. This pattern, sometimes called depressive realism, presents a paradox: the more accurate judges were the ones who were suffering.

Together, these findings suggest that a moderate self-favorable bias often accompanies psychological well-being. A shifted belief about oneself – one that is more positive than the evidence strictly warrants – may serve as a psychological buffer that keeps people motivated and engaged. A perfectly calibrated person, who sees their abilities and prospects with complete accuracy, may lack the optimism needed to persist in the face of setbacks. This does not mean that more overconfidence is always better; extreme overconfidence can lead to reckless decisions and painful failures. But a small, healthy dose appears to be part of how well-functioning minds work.

A Bayesian Lens

Both forms of overconfidence map onto specific failures of Bayesian reasoning. A well-calibrated Bayesian reasoner maintains beliefs that are both unbiased in their center and appropriately wide in their spread.

- Self-favorable bias corresponds to a misplaced posterior mean: the estimate is systematically displaced in a self-flattering direction, as if the person were applying a biased prior that says "I am exceptional."
- Overprecision corresponds to understated posterior variance: the probability distribution is too tightly concentrated around a single value, leaving too little mass on alternative outcomes.
- The planning fallacy combines both: the estimate has the wrong center (too optimistic) and the wrong spread (too narrow). Reference class forecasting (Flyvbjerg, 2006) is essentially a corrective that grounds the prior in the actual distribution of outcomes for similar projects, rather than constructing an optimistic inside-view estimate.

The motivational dimension of self-favorable bias adds a complication that purely informational models of Bayesian updating do not capture: people are not just miscounting evidence, they are selectively weighting it to protect a positive self-image. A complete account of overconfidence therefore requires both a cognitive component (how evidence is processed) and a motivational component (why it is processed asymmetrically).

Summary

Overconfidence is not a single bias but a family of related distortions organized into two types. The self-favorable bias is a shift in beliefs about oneself in a positive direction, manifesting as overestimation of one's performance or control, overplacement of oneself relative to others, and unrealistic optimism about one's future. In Bayesian terms, the posterior is displaced. Overprecision is a narrowing of uncertainty, in which people are too confident that their estimates are close to the truth; the posterior is too narrow. The Dunning-Kruger effect shows that the least competent individuals are often the most overconfident, because they lack the metacognitive skills to recognize their own errors. The planning fallacy demonstrates both forms at once: estimates of future projects are systematically too optimistic and too precise. Expertise reduces overconfidence only when it is accompanied by clear and frequent feedback; without such feedback, experience increases confidence without improving accuracy. Finally, the relationship between overconfidence and mental health reveals that a moderate self-favorable bias often accompanies psychological well-being, while its absence is linked to depression and learned helplessness – a reminder that not all departures from perfect calibration are dysfunctional.

CAUSALITY

CAUSALITY

CUES TO CAUSALITY

Learning goals:

- Describe what cues to causality are and how people use them to judge whether one event caused another.
- Explain the concept of the causal field and how it shapes which factor is identified as the cause.
- Describe how temporal order, contiguity, and covariation each contribute to causal judgments.
- Explain why people prefer mechanism-based explanations over covariation-based ones.
- Describe the similarity principle in causal reasoning, including the tendency to expect large causes for large effects.
- Explain the simulation heuristic and how counterfactual reasoning relates to judgments of causality.

Why Do We Ask "Why?"

A student walks into a lecture hall, sits down, and immediately starts sneezing. Her friend turns to her and says, "It must be the flowers they put by the entrance." No laboratory test was conducted. No statistics were consulted. Yet within seconds, a causal explanation was produced and accepted. This kind of rapid, confident causal reasoning happens frequently throughout the day: you hear a loud bang and look toward the street, you eat something unusual and blame it for your stomach ache, you notice a friend acting strangely and wonder what happened to her. How do people decide so quickly and effortlessly what caused what?

The answer is that people rely on a set of situational features, called cues to causality, to judge whether one event brought about another (Einhorn & Hogarth, 1986). These cues include whether the potential cause came before the effect (temporal order), whether the two events happened close together in time and space (contiguity), whether they tend to co-occur (covariation), whether there is a plausible process connecting them (causal mechanism), whether the cause and effect resemble each other (similarity), and whether mentally removing the cause would undo the effect (counterfactual reasoning). None of these cues by itself proves causation, but together they form the basic toolkit that people use to make sense of the world. From a Bayesian perspective (see Bayesian Reasoning), each cue functions as a piece of evidence that raises or lowers the probability of a causal hypothesis. Understanding these cues helps explain not only how people reason well about causes, but also how they go wrong – inferring causation where none exists or missing genuine causes that lack the right surface features.

Consider the sneezing student once more. Her friend's instant diagnosis drew on several cues at once: the flowers were encountered just before the sneezing started (temporal order), the sneezing began immediately upon entering the hall (contiguity), and pollen allergies provide a ready mechanism. As we work through each cue in this chapter, we will return to this example to see how the pieces fit together.

The Causal Field

Before people even begin looking for a cause, they have already narrowed the search. They do not consider every possible factor in the universe that might have contributed to an event. Instead, they evaluate potential causes against a background of existing conditions, which Mackie (1965) called the causal field. The causal field is the set of normal, expected conditions that are taken for granted. A cause is then identified as whatever stands out as a difference in this background.

Consider a simple example. Suppose a student does poorly on an exam. The causal field for the student herself might be her usual study habits, her typical sleep schedule, and the way exams normally go for her. Against this background, she might identify a late-night party the evening before the exam as the cause of her poor performance, because the party is the thing that was different. But now imagine that a university administrator is asking a different question: why did many students fail the same exam? The causal field is now different. The administrator takes for granted that students vary in how much they prepare, and instead looks for something that changed across the board – perhaps an unusually difficult exam or a poorly written question.

This illustrates a key point: the same event can be given different causal explanations depending on the causal field. In a factory testing the durability of watch glasses, a watch that breaks when struck by a hammer is attributed to a defect in the glass, not to the hammer, because the hammer strike is part of the normal testing procedure. In everyday life, you would say the hammer caused the breakage, because hitting a watch with a hammer is not part of the normal background (Einhorn & Hogarth, 1986). Which factor people single out as "the cause" depends on what they treat as background conditions and what stands out as unusual.

Return to the sneezing student. Her friend pointed to the flowers as the cause. But notice that other conditions were also present: the lecture hall has air conditioning, dust, other students wearing perfume. These are part of the normal background – the student has sat in similar rooms many times without sneezing. The flowers, however, are new. They stand out against the causal field, and so they become the candidate cause.

Temporal Order

One of the most basic cues to causality is the order in which events occur. If one event consistently happens before another, people are inclined to see the first event as the cause of the second. This seems reasonable: causes do come before their effects. But people sometimes apply temporal order in situations where it leads them astray.

Einhorn & Hogarth (1986) describe a striking example. When asked "What is the probability that a child has blue eyes, given that the parent has blue eyes?" people tend to give a higher estimate than when asked the reverse: "What is the probability that a parent has blue eyes, given that the child has blue eyes?" People often assume that the forward causal direction must also correspond to a higher conditional probability. But conditional probabilities depend on base rates as well as causal structure, so the "forward" direction is not necessarily more probable. Because the natural causal direction runs from parent to child, people treat the forward-in-time question as more natural and assign it a higher probability, even when the statistical facts do not support that judgment.

Temporal order also matters in more practical settings. In legal decision-making, the order in which evidence is presented to jurors affects how convincing they find it. Pennington & Hastie (1988) demonstrated that jurors construct narrative stories to make sense of trial evidence, and that presenting evidence in the order that matches the story's timeline – rather than in some scrambled order – makes the story more coherent and convincing. When prosecution evidence was presented in a natural temporal order and defense evidence was not, jurors were more likely to convict. When the order was reversed, acquittal rates went up. The evidence itself was the same; only the order changed. This finding illustrates how deeply temporal structure is embedded in causal reasoning, a topic explored further in Legal Decision-Making.

For the sneezing student, temporal order plays an obvious role: the flowers were encountered before the sneezing began. If the student had been sneezing all morning and only then walked past the flowers, her friend would not have blamed the flowers.

Contiguity and Covariation

People are more likely to see a causal connection between events that are close together in time and space. This is the cue of contiguity: when a potential cause and its effect happen right next to each other, the causal link feels obvious. When there is a long delay or a large spatial gap between them, the link becomes harder to see.

Some of the most elegant demonstrations of contiguity come from the work of Michotte (1963). In his classic experiments, Michotte created simple visual displays in which one colored square (object A) moved toward another stationary square (object B). When A reached B and stopped, and B immediately began to move, observers reported a vivid impression that A had "launched" or "pushed" B. This impression was automatic and compelling, much like a perceptual experience rather than a deliberate inference. Critically, Michotte showed that the impression depended on precise timing. If there was even a small delay between A stopping and B starting to move, the launching impression disappeared. Instead of seeing one event causing another, observers saw two separate, unrelated movements. Similarly, if A stopped short of B, leaving a spatial gap, the causal impression was weakened or lost. These experiments showed that contiguity in time and space is not just a helpful cue; for simple mechanical events, it can be the difference between perceiving causation and perceiving mere coincidence.

The sneezing example is a case where contiguity is strong: the student began sneezing immediately upon entering the hall, right next to the flowers. If the student had started sneezing two hours later at home, her friend would be far less likely to blame the flowers – even if the flowers were in fact the cause.

While contiguity operates at the level of single instances, covariation extends across multiple instances. Covariation means that the potential cause and the effect tend to happen together: when the cause is present, the effect occurs, and when the cause is absent, the effect does not occur. If every time you pet a cat it starts purring, and it does not purr when you stop, you infer that your petting causes the purring. If every time you drink coffee after 18:00 you have trouble sleeping, but not when you don't drink coffee, you start to suspect a causal link.

Covariation is a powerful cue, but people often interpret it too generously. The well-known phrase "correlation does not imply causation" exists precisely because people tend to treat correlated events as causally related. Two events might covary because they share a common cause rather than because one causes the other. For example, carrying a lighter in your pocket correlates with developing lung cancer – but lighters do not cause cancer. The common cause, smoking, drives both. Similarly, ice cream sales and drowning rates both rise in summer, not because ice cream causes drowning, but because warm weather increases both swimming and ice cream consumption. This tendency to read causation into correlation is discussed further in Learning Causality, where we explore how people learn causal relationships from patterns of association and the errors they make along the way.

Causal Mechanism

If you tell someone that a new medication reduces headaches, their first question is likely to be: how does it work? People are not satisfied with knowing that two events tend to go together. They want an explanation that fills the gap between cause and effect – what is called a causal mechanism.

Ahn et al. (1995) conducted a series of experiments that demonstrated just how strong this preference is. In one study, participants were given simple event descriptions such as "John had a car accident last night" and asked to list questions they would need answered to determine the cause. If people relied mainly on covariation, they would ask questions like "Does John often have accidents?" or "Do other people have accidents on that road?" Instead, participants overwhelmingly asked mechanism questions such as “Was John drunk?” or “Was it raining?” These questions seek to identify a process that connects the potential cause to the effect. In the context of these study tasks, participants generated mechanism-based explanations about 83 percent of the time.

In another experiment, Ahn et al. (1995) directly compared the persuasive power of mechanism and covariation information. Participants were given either a mechanism explanation (for example, “John was drunk, which impaired his driving”) or a covariation statement (for example, “John is more likely to have accidents than other people”). Even when the two types of information were designed to carry the same statistical weight, mechanism information had a significantly stronger effect on causal judgments. People found the mechanistic explanation more convincing and more useful for understanding what happened.

This preference for mechanism has practical implications. In Medical Decision-Making, we will see that clinicians rely heavily on their understanding of disease mechanisms when diagnosing patients, sometimes to the point where they ignore statistical evidence that contradicts their mechanistic model (see also Bayesian Reasoning). The preference for mechanism also connects to the broader topic of Mental Models. People build internal models of how the world works, and these models are structured around causal mechanisms rather than abstract statistical patterns.

Returning to our opening example: the friend’s explanation – “It must be the flowers” – is implicitly mechanistic. Flowers produce pollen, pollen triggers allergies, allergies cause sneezing. This chain of causal mechanisms makes the explanation satisfying in a way that a purely statistical claim (“People who enter rooms with flowers are 40 percent more likely to sneeze”) would not.

Similarity of Cause and Effect

Not all causal cues are equally diagnostic. Unlike temporal order or covariation, resemblance between cause and effect is often a poor indicator of causation – but it still strongly shapes intuitive judgments. This similarity cue takes two main forms.

The first is physical resemblance. Throughout history, people have assumed that things that look alike must be causally related. In medieval herbal medicine, the "doctrine of signatures" held that plants resembling body parts could cure diseases of those body parts. Walnuts, which look vaguely like a brain, were thought to be good for the brain. Kidney beans were prescribed for kidney problems. While modern medicine has abandoned this reasoning, the underlying tendency persists. People find it easier to accept that a red food coloring causes skin redness than that it causes, say, fatigue, because the resemblance between the cause and effect makes the causal link feel more natural.

The second form is matching magnitude: people expect large effects to have large causes and small effects to have small causes. When a major disaster occurs – such as a plane crash that kills hundreds of people – people find it difficult to accept that a small, mundane cause was responsible, like a tiny piece of ice on a sensor. They feel that something so devastating must have had a cause of matching scale, such as a terrorist attack or a massive systems failure. This expectation of proportionality can lead people to reject correct but small-scale explanations and to prefer dramatic but incorrect ones. This same tendency may play a role in the appeal of conspiracy theories: when a major event, like the assassination of a president, has a seemingly small cause – one lone individual acting alone – people resist the explanation and search for a larger, more proportional cause, such as an elaborate conspiracy. This is explored further in Conspiracy Theories.

The Simulation Heuristic and Counterfactual Reasoning

So far we have discussed cues that people pick up from what did happen. But people also judge causation by thinking about what could have happened differently. This is counterfactual reasoning, and it relies on what Kahneman & Tversky (1982) called the simulation heuristic: the mental process of imagining alternative versions of events.

To test whether a factor was a cause of some outcome, you mentally remove it and ask: would the outcome still have occurred? If you imagine that removing the factor would have changed the outcome, you judge it to be a cause. This is sometimes called the "but-for" test: the event would not have occurred but for this factor. Although not all causes are strictly but-for causes – sometimes multiple factors are independently sufficient – the test captures an important feature of everyday causal judgment.

Kahneman & Tversky (1982) illustrated this with a scenario about a man named Mr. Jones who is killed in a traffic accident on his way home from work. In one version of the story, Mr. Jones had taken an unusual route home that day. In another version, he had left work earlier than usual. Participants were asked to complete the sentence "If only…" to undo the event. Almost universally, people undid whichever aspect of the story was unusual. In the route version, they wrote things like "If only he had taken his usual route." In the time version, they wrote "If only he had left at his normal time." People did not undo stable background features of the story (they did not write "If only Mr. Jones had chosen a different career"). Instead, they focused on the exceptional element and imagined it reverting to normal.

This pattern reveals an important rule of mental simulation: people prefer what Kahneman and Tversky called "downhill changes" – alterations that move the scenario back toward what is normal or expected – rather than "uphill changes" that introduce new exceptions. If a colleague who always bikes to work decides to drive one day and gets into an accident, you are much more likely to think "If only she had biked" than if she drives every day. The accident is the same, but the counterfactual is easier to construct when the cause was exceptional. Notice how this connects to the causal field: exceptional events are precisely those that stand out against the background of normal conditions.

Counterfactual reasoning also shapes emotions. In another well-known scenario from the same research, two travelers both miss their flights. Mr. Crane arrives at the airport 30 minutes late and learns his flight left on time. Mr. Tees arrives five minutes late and learns his flight was delayed and left just five minutes ago. Both men missed their flights, so their objective situations are identical. Yet 96 percent of participants judged Mr. Tees to be more upset. The reason is that it is easy to mentally simulate a scenario in which Mr. Tees catches his flight – he just needed five more minutes – whereas for Mr. Crane, a much larger change would be required. The ease of constructing the counterfactual determines the emotional intensity, which connects to how we experience regret more broadly, as discussed in Satisficing vs. Maximizing.

For the sneezing student, counterfactual reasoning provides a final check: if the flowers had not been placed by the entrance, would the student still be sneezing? If the answer seems to be no, the causal attribution is strengthened. If the student also sneezed yesterday, when there were no flowers, the counterfactual fails and the causal hypothesis weakens.

A Bayesian Lens

As previewed in the introduction, each cue to causality can be understood from a Bayesian perspective as evidence that updates the probability of a causal hypothesis (see Bayesian Reasoning). Before observing any cues, a person starts with a prior belief about whether a causal relationship exists. This prior might be based on general background knowledge or on the person's mental model of how the world works (see Mental Models).

Each cue then adjusts this prior. Consider the sneezing student one last time. Suppose her friend initially thinks there is a moderate chance the flowers caused the sneezing. The fact that the sneezing started right after the student entered the hall (temporal order and contiguity) pushes the probability upward. If the friend recalls that the student has sneezed near flowers before (covariation), the probability rises further. Knowledge that pollen triggers allergies (mechanism) provides a particularly strong update. Conversely, if the student had been sneezing all day before encountering the flowers, or if there were no plausible biological pathway from flowers to sneezing, the probability would drop.

In this framework, the cues function like the likelihood term in Bayes' theorem: they represent how probable the observed evidence would be if the causal link were real, compared to how probable it would be if the link were absent. The causal field sets up the prior by determining what counts as normal and what counts as surprising. The simulation heuristic provides another route to updating: by mentally removing a potential cause and checking whether the effect would still occur, people are effectively asking how much the evidence depends on the presence of that cause. People do not carry out these calculations formally. But the structure of their reasoning – weighing multiple imperfect cues and combining them to reach a judgment – mirrors the logic of Bayesian evidence integration.

Summary

People judge causation using a set of situational cues rather than through formal analysis. The causal field sets the background against which potential causes are evaluated: whatever stands out as different from normal conditions is identified as the cause. Temporal order leads people to infer that earlier events cause later ones, sometimes even when the statistical relationship does not depend on direction. Contiguity – the closeness of events in time and space – triggers immediate causal impressions, as demonstrated by Michotte's launching experiments. Covariation across repeated instances strengthens causal beliefs but also leads people to confuse correlation with causation. People strongly prefer mechanism-based explanations that describe how a cause produces its effect, finding these more convincing than statistical patterns alone. Similarity between cause and effect, whether in physical appearance or in magnitude, makes causal links feel more plausible despite being a weak diagnostic cue. Finally, the simulation heuristic allows people to test causal hypotheses by mentally undoing events and imagining whether the outcome would have been different – a process that highlights exceptional events as causes and shapes the emotions people feel about outcomes. From a Bayesian perspective, each of these cues serves as evidence that updates the probability of a causal hypothesis, with the causal field setting the prior and each additional cue adjusting the likelihood. Together, these cues form a practical but imperfect system for navigating a world where true causes are never directly observed.

BLAME, PRAISE, AND RESPONSIBILITY

Learning goals:

- Distinguish between personal causality and impersonal causality.
- Explain the covariation principle and its three dimensions: consensus, distinctiveness, and consistency.
- Describe how different patterns of consensus, distinctiveness, and consistency lead to different causal attributions.
- Define the correspondence bias and explain why people over-attribute behavior to persons rather than situations.
- Connect causal attribution to the intentional stance and moral judgment.

From Causes to People

Imagine you are cycling to the university on a rainy morning and a car splashes through a puddle, soaking you from head to toe. Your immediate reaction is probably not a calm analysis of road drainage and weather conditions. Instead, you think something like: "That driver is a terrible person." You attribute the event to the driver, to their character, to their lack of consideration. But what if the driver simply did not see the puddle? What if there was no way to avoid it? In that case, the soaking was caused by circumstances, not by any intention. The outcome is identical, yet the two explanations feel very different, because one of them points to a person as the cause.

This chapter examines causal reasoning when the potential cause is a person rather than a physical event or mechanical failure. The previous chapter on Cues to Causality introduced the general principles people use to infer cause and effect: temporal order, contiguity, covariation, and causal mechanism. Those principles apply broadly, whether you are figuring out why your laptop crashed or why your friend cancelled dinner plans. But when the potential cause is a person, a distinct set of processes kicks in. We move from asking "what caused this?" to asking "who caused this?" and, almost immediately, "who deserves blame or praise?"

This shift connects to the idea of stances introduced in Mental Models. When we reason about physical objects, we tend to adopt the mechanical stance, thinking in terms of physical forces. When we reason about designed artifacts, we adopt the design stance, thinking in terms of function and purpose. But when we reason about people, we adopt the intentional stance, attributing behavior to beliefs, desires, and goals. The intentional stance helps us predict behavior efficiently, and it is also the gateway to moral judgment: once we decide that a person intended an outcome, we are ready to assign blame or praise, a topic explored further in Morality.

Personal and Impersonal Causality

Not every event that involves a person is caused by that person in a meaningful sense. A useful distinction is between personal causality and impersonal causality. Personal causality means that someone's intention or agency was the driving force behind an action. Impersonal causality means that the action occurred without intention, driven instead by circumstances, chance, or mechanical forces (Young, 1995).

Consider a doctor who prescribes the wrong medication. If the doctor did this on purpose, perhaps to harm the patient, we have a clear case of personal causality. The doctor intended the outcome, and blame follows naturally. But if the doctor prescribed the wrong medication because the packaging of two drugs looked nearly identical, the error may be attributed more to situational factors, though observers may still assign the doctor some professional responsibility for not double-checking. The outcome is the same – a patient receives the wrong drug – but the attribution, and the degree of blame, shifts substantially.

This distinction maps onto the intentional-versus-mechanical stance discussed in Mental Models. When we perceive personal causality, we see the person as an agent with goals and beliefs, and we hold them responsible for acting on those goals. When we perceive impersonal causality, we treat the person as merely a link in a chain of physical or situational forces.

The distinction matters because blame and praise depend heavily on perceived agency and intention. Harmful outcomes produced on purpose invite stronger blame than the same outcomes produced accidentally, although people may still assign some responsibility for negligence or carelessness. The same logic applies to praise. A surgeon who saves a patient through exceptional skill under difficult circumstances earns greater praise than one whose patient recovers thanks to lucky timing. A student who helps a struggling classmate because they genuinely want to is praised more than one who does so because it was required for a course credit. In each case, the perceived intention transforms our moral evaluation.

It is also worth noting that causing an outcome is not identical to being responsible for it. Responsibility depends not only on causal contribution but also on intention, foresight, control, and the availability of alternatives. A gust of wind that knocks over a bicycle has caused the damage, but no one is responsible. A person who knocks over the bicycle while texting has both caused the damage and bears responsibility, because they had the ability to act otherwise.

The Covariation Principle

Before assigning blame or praise, people first need to decide what kind of cause they are dealing with: something about the person, something about the stimulus or target, or something about the surrounding circumstances. Kelley (1973) proposed that people use the covariation principle to make this determination: an effect is attributed to the cause with which it covaries over time. This idea was introduced in general terms in Cues to Causality, where covariation was discussed as one of several cues to causality. When applied to person perception, Kelley broke covariation down into three specific dimensions.

The first dimension is consensus: do other people respond the same way to the same stimulus? If everyone who takes a particular course finds it difficult, consensus is high. If only one student struggles while everyone else does fine, consensus is low.

The second dimension is distinctiveness: does this person respond this way only to this specific stimulus, or do they respond this way to everything? If a student struggles only in one particular course but does well in all others, distinctiveness is high. If the student struggles in every course, distinctiveness is low.

The third dimension is consistency: does this person respond the same way over time? If the student has struggled in this course every semester they have taken it, consistency is high. If the struggle is a one-time event, consistency is low.

In Kelley's framework, the alternative to a person attribution is often a stimulus attribution – something about the object or target of the behavior – though in everyday discussion this is often folded into the broader category of situational causes.

Different combinations of these three dimensions lead to different attributions. Suppose you hear that your classmate Anna gave a very negative review of a restaurant. You want to understand why. If you learn that everyone who went to that restaurant also gave negative reviews (high consensus), that Anna usually enjoys restaurants and rarely complains (high distinctiveness), and that Anna has complained about this restaurant on multiple occasions (high consistency), you are likely to attribute her review to the restaurant itself. It must be a bad restaurant. This is a stimulus attribution.

Now change the pattern. If you learn that other people who went to the same restaurant actually enjoyed it (low consensus), that Anna complains about almost every restaurant she visits (low distinctiveness), and that she has been consistently negative about dining out for years (high consistency), you are likely to attribute her review to Anna herself. She is a dispositionally critical diner. This is a person attribution, and it opens the door to blame or praise depending on the context.

Later work by Cheng & Novick (1990) refined Kelley's model by showing that people's attributions can often be modeled as probabilistic contrasts rather than simple pattern matching. The model compares the probability of an effect occurring in the presence of a potential cause to the probability of it occurring in the absence of that cause. If the effect is much more likely when a particular person is present, the person gets the attribution; if the effect is much more likely when a particular stimulus is present, the stimulus gets the attribution. Importantly, when people are given complete covariation information, many apparent attribution errors shrink. This suggests that some so-called biases in causal attribution may stem not from faulty reasoning but from incomplete information: when people lack full covariation data, they fill in the gaps with assumptions, and these assumptions can skew attributions in predictable directions.

The Correspondence Bias

Despite the sophistication of the covariation principle, people show a robust tendency in everyday life: they over-attribute behavior to the person and under-attribute it to the situation. This tendency is known as the correspondence bias, because people assume a correspondence between what someone does and who they are. It has also been called the fundamental attribution error, a term that captures how pervasive and consequential the mistake is (Gilbert & Malone, 1995).

One of the classic demonstrations comes from a study by Jones & Harris (1967) on essay writing. Participants read essays arguing either for or against Fidel Castro, at the time the Marxist leader of Cuba. Crucially, some participants were told that the essay writer had been assigned their position by a debate instructor – the writer had no choice in the matter. Despite knowing this, participants still inferred that the writer personally believed what they had written. The situational constraint, being assigned a position, should have fully explained the essay's content. Yet participants treated the essay as a window into the writer's true beliefs.

This pattern shows up everywhere in daily life. A student fails an exam, and we think "they did not study hard enough" rather than "the exam was unusually difficult." A colleague is quiet during a meeting, and we think "they are shy" rather than "they are dealing with a family crisis." A friend cancels plans, and we think "they do not care about our friendship" rather than "something came up." In each case, the person's behavior is explained by their character, even when situational factors offer a perfectly adequate explanation.

Gilbert & Malone (1995) argued that the correspondence bias arises from multiple mechanisms. The first is a lack of awareness: sometimes we simply do not know about the situational forces acting on a person. If you do not know that your colleague received bad news this morning, you cannot factor it into your explanation of their behavior. The second is unrealistic expectations: even when we know about situational pressures, we may underestimate how strongly they influence behavior. We might think "I would still be cheerful even after bad news," failing to appreciate the true power of the situation. The third, and perhaps most important, is incomplete correction. Research suggests that when we observe someone's behavior, we first make a quick, automatic inference about their character – a System 1 process, as discussed in Reason and Intuition – and then, if we have the time and mental energy, we adjust this inference by considering situational factors – a System 2 process. But this adjustment is often incomplete, as we have seen in the chapter on Anchoring and Adjustment. In studies where participants formed impressions of others while simultaneously memorizing a string of digits, they were less able to correct for situational constraints and showed stronger dispositional inferences. When cognitive resources are scarce, the automatic person attribution goes uncorrected.

This last mechanism fits with the broader framework of dual-process theory: our fast, automatic system jumps to a person attribution, and our slow, deliberate system sometimes fails to correct it. The more distracted, busy, or unmotivated we are, the more the bias takes hold.

The consequences of the correspondence bias extend far beyond everyday misunderstandings. In legal settings, jurors may over-attribute a defendant's behavior to their character while underappreciating the situational pressures they faced, a topic explored in Legal Decision-Making. In medical contexts, a doctor might attribute a patient's non-compliance to laziness rather than to the complexity of the treatment regimen, as discussed in Medical Decision-Making. In political discourse, citizens may blame individuals for outcomes largely shaped by structural forces – a tendency that can be amplified by personalized, outrage-driven messaging discussed in Populism and Social Media.

The correspondence bias is, in a sense, an over-application of the intentional stance. Most of the time, people do act on their beliefs and desires, and inferring those beliefs and desires from behavior is efficient and often accurate. The problem arises when we apply this stance too rigidly, failing to account for situational forces when they are the primary cause. The driver who splashed you may have been careless, but they may also have been caught by circumstances beyond their control. The correspondence bias makes us lean too heavily toward the first explanation.

A Bayesian Lens on Causal Attribution

The processes described in this chapter can be understood through the framework of Bayesian reasoning introduced in Bayesian Reasoning, and they illustrate the contrast between descriptive tendencies and normative ideals discussed in How It Is vs. How It Should Be. Causal attribution is, at its core, a form of belief updating about the source of behavior. When you observe someone doing something, you start with a prior belief about whether the behavior reflects the person or the situation. You then update this belief based on evidence: information about consensus, distinctiveness, and consistency.

In this framework, the covariation principle describes the evidence. High consensus (everyone reacts the same way) is evidence for a stimulus attribution. Low consensus (only this person reacts this way) is evidence for a person attribution. Distinctiveness and consistency provide additional evidence that shifts the posterior belief toward different causal explanations.

Consider a concrete example. A usually punctual student misses a seminar. Your prior is that this student is reliable, so you initially favor a situational explanation – perhaps they are ill or had a transport problem. But if the student misses the next seminar too, and the one after that, each absence provides evidence against the situational explanation. Your posterior gradually shifts toward a person attribution: maybe they have lost interest in the course. The covariation information – low consistency with their past behavior initially favoring a situational account, then rising consistency of absences shifting the balance – is what drives the updating.

The correspondence bias, viewed through this lens, reflects an overly strong prior toward person attributions. People default to the assumption that behavior reflects character, and they do not update sufficiently when situational evidence suggests otherwise. The prior for person attributions is too high, and the weight given to situational evidence is too low. This also helps explain the finding from Cheng & Novick (1990) that apparent biases diminish when information is complete. Strong covariation evidence can overcome even a biased prior, but when evidence is sparse or ambiguous, the prior dominates, and the default person attribution goes uncorrected.

Summary

When the potential cause of an event is a person rather than a physical force, a special set of attributional processes activates. The distinction between personal causality, where intention and agency drive the action, and impersonal causality, where circumstances or chance are responsible, determines how much blame or praise is assigned – though people may also hold others partially responsible for negligence or carelessness, not only for intentional harm. People assess these attributions using the covariation principle, evaluating consensus, distinctiveness, and consistency to determine whether a behavior should be attributed to the person, the stimulus, or the situation. When covariation information is complete, people are reasonably good at this task. However, in everyday life, information is often incomplete, and people show a persistent correspondence bias: they over-attribute behavior to the person and under-attribute it to the situation. This bias arises from multiple mechanisms, including lack of awareness of situational forces, unrealistic expectations about their influence, and incomplete correction of automatic person attributions. Because moral judgment depends heavily on perceived intention, attribution is often the first step in deciding whether someone deserves blame or praise – making these processes consequential in legal, medical, and everyday contexts alike.

BLAME, PRAISE, AND RESPONSIBILITY

LEARNING CAUSALITY

Learning goals:

- Explain how a 2 × 2 contingency table summarizes the co-occurrence of two factors, and describe the cell A bias.
- Describe how classical and operant conditioning serve as forms of causal learning.
- Explain prediction error as the mechanism that drives learning, and describe how dopamine neurons encode prediction errors.
- Define blocking and explain why it occurs in predictive but not diagnostic learning.
- Distinguish between illusory correlation and illusory causation, and explain how both arise from biased evidence use.
- Explain regression toward the mean and why people tend to misinterpret it as a causal effect.
- Describe how the plausibility of a cause functions as a prior belief that shapes how people evaluate new evidence.
- Explain why only experimental designs allow causal conclusions.

Learning from Co-Occurrence

Imagine you start taking a new herbal supplement because a friend told you it helps with concentration. Over the next few weeks, you notice that on several days when you take the supplement, you do seem to focus well during lectures. After a while, you become convinced: the supplement works. But did you ever pay attention to the days when you took it and still could not focus? Or the days when you skipped it but concentrated just fine? Probably not. This chapter is about how people learn causal relationships from experience, and about the systematic errors that creep in along the way. These errors help explain why people believe in ineffective treatments and see patterns in randomness. But the story is not all negative. The basic mechanism that drives causal learning – prediction error – is remarkably efficient, and in some ways it mirrors the kind of rational updating described in Bayesian Reasoning.

To understand how people learn about causes and effects, we first need a simple framework for organizing evidence. When you want to know whether one thing causes another – say, whether a supplement improves concentration – there are four possible combinations of events. The supplement is present or absent, and good concentration is present or absent. These four combinations can be arranged in a 2 × 2 contingency table:

	Effect present	Effect absent
Cause present	Cell A	Cell B
Cause absent	Cell C	Cell D

Cell A contains cases where you took the supplement and concentrated well. Cell B contains cases where you took the supplement but could not focus. Cell C contains cases where you skipped the supplement and concentrated well. Cell D contains cases where you skipped the supplement and could not focus.

A rational agent who wants to determine whether the supplement actually causes better concentration would need to consider all four cells. The standard measure of contingency, ΔP, captures this:

$$\Delta P = P(\text{effect} \mid \text{cause}) - P(\text{effect} \mid \text{no cause}) = A/(A+B) - C/(C+D)$$

If ΔP is zero, the supplement does nothing: you concentrate well at the same rate whether you take it or not. If ΔP is positive, the supplement is associated with better concentration. Importantly, even a positive ΔP from observational data does not prove causation, because other factors might explain the link.

Only experimental manipulation – randomly assigning people to take the supplement or a placebo while holding everything else constant – can justify a causal conclusion. We return to this point in detail later in the chapter.

But people do not treat all four cells equally. Research consistently shows that people give the most weight to cell A, followed by cell B, then cell C, and finally cell D (Lipe, 1990). This ordering matters. Underweighting cell C is especially damaging, because cell C contains cases where the effect occurs without the cause. If you never notice the days when you concentrate well without the supplement, you will overestimate the supplement's importance. When the task becomes difficult or people simplify, they sometimes attend only to cell A – the cases where both the cause and the effect are present. At its extreme, this becomes a purely confirmatory approach: you look only at cases consistent with the idea that the cause produces the effect, and you ignore everything else. As we discuss in Confirmation, this tendency to focus on confirming evidence appears throughout human reasoning, not just in causal learning.

Notice that the cell A bias operates not only at the stage of judgment but also at the stage of evidence gathering. If someone only asks users of the supplement whether their concentration improved, they collect mostly cell A and cell B cases, and never observe cells C and D at all. A journalist who interviews only people who took a remedy and felt better is sampling exclusively from cell A.

Conditioning and Causal Learning

Where does our ability to learn about causes come from? At the most basic level, causal learning rests on association. When two events repeatedly occur together, we come to expect one when we observe the other. This process is called classical conditioning. The classic example is from Pavlov's experiments: a dog hears a bell just before receiving food. After many pairings, the bell alone triggers salivation. The dog has learned that the bell predicts food.

In operant conditioning, the association is between an action and its outcome. A rat presses a lever and receives a food pellet. Over time, the rat learns that pressing the lever causes food to appear.

Both classical and operant conditioning provide a foundation for causal learning, but human causal reasoning goes beyond simple association. People also draw on background knowledge, causal models, and plausibility judgments – topics we return to later in this chapter. Still, the core associative machinery is shared across species and remains central to how humans learn from experience. When you notice that eating a particular food is followed by feeling unwell, you develop an aversion to that food. When you notice that studying in a particular café seems to lead to productive sessions, you start going there more often. These are causal beliefs built from repeated experience, and they follow many of the same principles that govern conditioning in the laboratory.

Prediction Error

What makes conditioning work? Early theories assumed that simple repetition was enough: the more often two events co-occur, the stronger the association. But this turns out to be wrong. What really drives learning is surprise – or more precisely, prediction error, the difference between what you expected and what actually happened.

Schultz et al. (1997) provided some of the most striking evidence for this idea by recording the activity of dopamine neurons in monkey brains. When a monkey received an unexpected reward, such as a squirt of juice, dopamine neurons fired vigorously. But after the monkey learned that a particular cue (say, a light) predicted the juice, the pattern changed. The dopamine response shifted from the reward itself to the cue that predicted it. The light now triggered the dopamine burst, while the juice itself – now fully expected – no longer did. If the cue appeared but the expected reward was omitted, dopamine activity dropped below its baseline level. The neurons were not simply responding to good things happening; they were signaling the difference between what was expected and what occurred.

In computational terms, the dopamine signal acts like a prediction error: positive when outcomes are better than expected, zero when outcomes match expectations, and negative when outcomes are worse than expected. Learning occurs when outcomes are surprising. Once a cue perfectly predicts an outcome, the prediction error signal falls silent and learning stops. This is why you stop noticing the hum of your refrigerator: it is perfectly predictable and carries no new information.

Blocking

Prediction error has a powerful consequence for how we assign credit to potential causes. Suppose you already know that cue A perfectly predicts an outcome. Now cue A and a new cue B are presented together, and the outcome occurs. Do you learn anything about cue B? Because cue A already fully accounts for the outcome, there is no surprise when it occurs – and therefore no prediction error to drive learning about cue B. Cue A "blocks" learning about cue B.

Blocking was first demonstrated in animal conditioning experiments and has since been replicated in human learning tasks. It shows that people assign causal credit selectively, based on what is informative, rather than associating everything present with the outcome. In idealized predictive settings, blocking is rational: once a cue explains the outcome, a redundant cue provides no additional evidence and should receive no weight.

Blocking is also where simple associative theories and causal-model theories diverge, because it turns out that blocking depends on the direction of the causal relationship.

Predictive versus Diagnostic Learning

If causal learning were simply a matter of forming associations between events that co-occur, the direction of the association should not matter. Learning that a cause predicts an effect should be the same as learning that an effect indicates a cause, as long as the data are the same. Waldmann & Holyoak (1992) showed that this is not the case.

In their experiments, participants learned about relationships between cues and outcomes. In the predictive condition, cues were described as causes that produced effects. For example, participants learned that certain features of chemical compounds caused particular emotional responses in workers. In the diagnostic condition, the same data were presented in reverse: symptoms were described as effects of a disease, and participants had to learn which symptoms indicated which disease. The critical finding was that blocking occurred only in the predictive condition. When cues were causes, a previously established cause blocked learning about a new, redundant one. But when cues were effects of a common cause, no blocking occurred. Multiple symptoms of the same disease did not compete with each other.

The logic is straightforward. If you already know that substance A causes a headache, then learning that substance A plus substance B also causes a headache gives you no reason to think substance B does anything. Causes compete for credit. But if you know that a disease causes symptom A, learning that it also causes symptom B does not reduce the diagnostic value of symptom A. Effects of a common cause can coexist without competing.

The key lesson is that causal learning depends on more than simple co-occurrence. People build mental models of the causal structure of a situation, and these models determine how evidence is processed. The same data lead to different learning outcomes depending on whether the learner interprets them as going from causes to effects or from effects to causes. This is consistent with the broader role of background knowledge in shaping interpretation, discussed in Mental Models.

Illusory Correlations

The cell A bias has a predictable downstream consequence: people sometimes perceive correlations that do not actually exist. When you selectively attend to cases where two things occur together (cell A) and ignore the other cells, you can come to believe that two unrelated things are associated. This is called illusory correlation.

Consider a common example. Many people believe there is a strong link between the political party in power and the state of the economy. When your preferred party is in power and the economy does well, you notice it and remember it (cell A). When your preferred party is in power and the economy does poorly, you are more likely to explain it away or forget it (cell B). When the other party is in power and the economy does well, you may not pay much attention (cell C). And when the other party is in power and the economy struggles, you notice it, but mainly as confirmation that the other party is bad – not as relevant evidence for the full contingency table. The result is an inflated perception of the relationship between party and economic performance.

Illusory correlations also play a role in stereotyping. When a minority group is small (making encounters less frequent) and a behavior is rare (making instances memorable), the co-occurrence of the two becomes especially salient. This can create the impression of an association between the group and the behavior, even when none exists – as we saw in the chapter on Representativeness.

Illusory Causation and Quackery

Illusory correlation concerns perceived association. Illusory causation adds a further inference: that the association reflects a genuine causal effect. While illusory correlation means seeing a link that is not there, illusory causation means interpreting a real but non-causal correlation as evidence that one thing produces another. This additional step – from "these things go together" to "this thing causes that thing" – is the cognitive mechanism behind much of quackery and pseudoscience.

Matute et al. (2011) demonstrated this in an elegant experiment. Participants were shown records of 100 fictional patients suffering from a made-up disease. Eighty percent of these patients recovered, regardless of whether they took a particular fictional medicine. The medicine had no causal effect whatsoever; the recovery rate was the same with or without it. Despite this, participants judged the medicine to be effective.

The key manipulation was the proportion of patients who took the medicine. In the high-exposure condition, 80 out of 100 patients took it. In the low-exposure condition, only 20 out of 100 took it. When 80 patients took the medicine, about 64 of them recovered (80% of 80) – a lot of cases in cell A. In the low-exposure condition, only about 16 patients fell in cell A (80% of 20). Even though the recovery rate was identical regardless of treatment, participants in the high-exposure condition rated the medicine as more effective. The sheer number of confirming cases in cell A drove their causal judgment.

This finding explains why so many ineffective treatments are perceived as working. Most illnesses resolve on their own. If you take a remedy and get better (cell A), you credit the remedy. But you rarely think about what would have happened without it (cells C and D). And when many people take a remedy – as happens when something is popular or heavily marketed – cell A fills up with apparent successes, even if the remedy does nothing. Without considering how many people recover without the treatment, you cannot know whether the treatment added anything. In Bayesian terms, this is base-rate neglect: ignoring the prior probability of recovery, P(recovery), which equals the recovery rate regardless of treatment. The same type of error appears in other contexts, as discussed in Bayesian Reasoning and Representativeness.

Matute et al. (2011) also found that the wording of the question mattered. Participants who were asked whether the medicine caused the recovery gave lower ratings than those asked whether the medicine was effective. Simply framing the question in causal terms prompted more careful evaluation of the evidence.

Regression toward the Mean

Another common source of illusory causal beliefs is regression toward the mean. Any single measurement reflects both a true underlying value and random variation – noise. When random variation pushes a measurement to an extreme, it is unlikely that the next measurement will be pushed to the same extreme by chance. The result is that extreme measurements tend to be followed by less extreme ones.

Consider blood pressure. A patient visits the doctor on a particularly stressful day and records an unusually high reading of 155/95. The doctor prescribes a new medication. At the follow-up visit two weeks later, the reading is 138/88. The doctor concludes the medication is working. But some of that drop was inevitable: the first reading was extreme partly because of random factors (stress that day, rushing to the appointment, the "white coat" effect), and those factors were unlikely to all align again. The blood pressure would have decreased somewhat even without the medication.

Or consider the "commentator's curse" in sports. A television commentator praises a footballer for an extraordinary run of five goals in three matches. The next two matches, the player scores nothing. Fans joke that the commentator jinxed the player, but regression toward the mean predicts exactly this: an extreme streak is partly due to luck – favorable matchups, defenders' mistakes, shots that happened to find the corner – and luck is unlikely to repeat.

A classic example involves training. An instructor notices that trainees who perform exceptionally well on one trial tend to perform worse on the next, while trainees who perform poorly tend to improve. The instructor concludes that praise is harmful and criticism is helpful. But regression toward the mean predicts this pattern regardless of whether the instructor says anything. Extreme performances on one trial are partly due to chance, and chance is unlikely to repeat.

This kind of causal misinterpretation is pervasive. It affects how people evaluate medical treatments (symptoms at their worst tend to improve), educational interventions (students who score at the bottom tend to score higher next time), and personnel decisions (employees who have a great first year may have a more ordinary second year). In each case, the natural tendency is to construct a causal story for what is fundamentally a statistical regularity.

Plausibility Shapes Learning

Not all potential causes are treated equally. When you already believe that a cause is plausible, you give more weight to evidence that supports the link. When you believe a cause is implausible, you are more skeptical of the same evidence. A vitamin causing improved concentration seems more plausible than a sticker on a laptop causing improved concentration. That difference in plausibility changes how you process exactly the same data.

Fugelsang & Thompson (2003) tested this by having participants evaluate unfamiliar causal candidates – for example, whether a fictional substance caused a particular outcome. Before seeing any data, participants received information designed to make the cause seem either plausible or implausible. Some received information about a plausible causal mechanism (how the substance could produce the effect), while others received information about past statistical co-occurrence (how often the substance and effect had been observed together). Participants then saw new data – a set of contingency tables – and rated how strongly they believed the cause produced the effect.

In these experiments, when participants believed the cause was plausible based on mechanism knowledge, they weighted the new evidence more heavily. The same data had a bigger effect on their judgments when they already found the cause mechanistically plausible. This interaction was specific to mechanism-based beliefs; when plausibility was based on past co-occurrence, it did not modulate how new evidence was weighted. The researchers also found that participants were relatively unaware of the influence their prior beliefs had on their judgments. They could report how much they used the new data, but they underestimated the role of their existing beliefs.

Goedert et al. (2014) extended this line of research by examining how plausibility affects which evidence people seek out. In their experiments, participants chose whether to observe cases where the potential cause was present or absent. When the cause was plausible, participants showed a strong preference for observing cause-present cases: they wanted to see whether the cause was followed by the effect. This is the positive test strategy discussed in Confirmation. But when the cause was implausible, this preference weakened. Participants were more willing to look at cause-absent cases, effectively checking whether the effect occurred even without the supposed cause. Implausibility nudged people toward a more balanced evidence search.

The Need for Experiments

Given all the ways in which observational evidence can mislead, how can we ever be confident that one thing causes another? As noted at the start of this chapter, the answer is experimental design. In a true experiment, the researcher manipulates the potential cause (the independent variable) while holding everything else constant. Participants are randomly assigned to conditions, which makes the groups comparable on average and reduces confounding, so that any systematic differences in outcomes can be attributed to the manipulation.

Consider the supplement example from the opening. In everyday life, you take the supplement on days when you feel motivated enough to remember it, skip it on lazy days, and judge your concentration based on your subjective impression. There are countless confounds: maybe you take the supplement on days when you slept better, or when you have more interesting lectures. A proper experiment would randomly assign people to take the supplement or a placebo on different days, measure their concentration with a standardized task, and compare the two conditions.

This is the fundamental distinction between correlational and experimental evidence. Observational covariation – the kind of evidence that fills up the contingency table in everyday life – is exactly what fuels illusory correlations and illusory causation. You observe that two things tend to go together, but without intervention you cannot discriminate between a direct causal effect, a reverse causal relationship, or a common cause driving both. Experimental manipulation breaks this ambiguity. It is the reason why medical treatments should be evaluated in randomized controlled trials rather than through testimonials, and why educational interventions need proper control groups.

This principle is easy to state but surprisingly difficult to internalize. As this chapter has shown, observation-based associative learning can produce causal beliefs even when causation is absent. The human mind is naturally inclined to draw causal conclusions from correlational patterns. Recognizing this limitation is the first step toward better causal reasoning.

A Bayesian Lens

Each of the errors discussed in this chapter can be understood as a specific departure from Bayesian updating. The cell A bias is a form of overweighting confirming evidence: instead of using all four cells to compute ΔP, people disproportionately attend to cases where both cause and effect are present. Illusory causation, as demonstrated by Matute et al. (2011), reflects base-rate neglect: people judge a treatment as effective without considering the probability of recovery in the absence of treatment. Misinterpreting regression toward the mean is a failure to account for noise: people treat random fluctuations as meaningful signals that demand causal explanations.

But several features of causal learning are strikingly Bayesian. Prediction error drives learning precisely when outcomes are surprising – corresponding to updating beliefs in proportion to the likelihood ratio of new evidence. When an outcome is fully expected, the likelihood ratio is close to one, and there is nothing to update. Blocking follows the same logic: once a cue explains the outcome, a redundant cue provides no additional evidence and correctly receives no credit.

Summary

People learn about causal relationships through association, building on the same mechanisms that underlie classical and operant conditioning. The driving force is prediction error: the brain updates its causal models when outcomes are surprising and stops learning when outcomes are fully expected. Blocking demonstrates this – a well-established cause prevents learning about a redundant new cause. Causal learning also depends on the direction of the causal model: blocking occurs when cues are interpreted as causes but not when they are interpreted as effects, showing that people build structured causal models rather than forming simple associations.

The cell A bias leads people to overweight cases where a cause and effect co-occur, producing illusory correlations and illusory causal beliefs. This is especially dangerous in the context of ineffective treatments, where high exposure creates many apparent successes in cell A while the base rate of recovery is ignored. Regression toward the mean is another source of false causal beliefs, as people construct causal stories for what is really a statistical regularity.

Only experimental designs that manipulate the potential cause while holding everything else constant can distinguish genuine causal relationships from mere correlations. Observational evidence fills the contingency table but cannot rule out reverse causation or common causes.

LEARNING CAUSALITY

PAST AND FUTURE

HINDSIGHT

Learning goals:

- Define hindsight bias and explain why it is also called "creeping determinism."
- Describe the memory paradigm and the hypothetical paradigm for studying hindsight bias.
- Explain the contributing factors that produce hindsight bias, including availability, causal narrative construction, and the curse of knowledge.
- Distinguish between outcome bias and hindsight bias.
- Identify conditions under which hindsight bias is weakened or reversed.
- Relate hindsight bias to the Bayesian framework of belief updating.

Knowing It All Along

Imagine that a friend tells you about a couple who just broke up. You immediately think: "I could have told you that was going to happen." Their arguments, their different lifestyles, the way they never seemed fully comfortable together – it all seems so obvious now. But if you are honest with yourself, did you really predict it? Or does the breakup just feel inevitable because you already know it happened? The same thing happens after elections, market crashes, medical diagnoses, and sports upsets.

Hindsight bias is one of the most reliable findings in judgment research. It affects nearly everyone, nearly all the time.

What Is Hindsight Bias?

Hindsight bias is the tendency to believe, after learning an outcome, that you knew it all along (Fischhoff, 1975). Once you know how something turned out, your memory of what you previously believed shifts toward the actual outcome. Fischhoff called this creeping determinism: the gradual, unconscious slide from "it happened" to "it had to happen."

Three things happen when hindsight bias takes hold. First, your recalled predictions shift toward the actual outcome. If you estimated a 40% chance of rain before the storm, you might later remember having said 60%. Second, the outcome feels inevitable in retrospect – you construct a story in which everything pointed toward what actually happened. Third, and crucially, you are unaware that your judgments have changed. You genuinely believe that your current memory reflects what you originally thought.

A vivid demonstration comes from a study by Bernstein et al. (2004) on visual hindsight bias. Participants viewed images of common objects that were heavily degraded and gradually became clearer. In one condition, participants simply tried to identify the object as the image clarified. In another condition, participants were told in advance what the object was and then asked to estimate at what stage of clarification an uninformed observer – someone who did not know the answer – would be able to recognize it. Participants who already knew the identity of the object believed that an uninformed observer would recognize it much earlier, at a far more degraded stage, than uninformed observers actually could. Knowing the answer made it feel obvious. This is not just "I knew it all along" but "I saw it all along."

How Researchers Study Hindsight Bias

Researchers study this feeling of obviousness in several ways, depending on whether they are interested in memory distortion or judgments made under known outcomes.

In the memory paradigm, participants first make predictions about an uncertain event, such as the outcome of an election or the result of a medical procedure. Later, after the outcome is known, they are asked to recall their original predictions. The typical finding is that recalled predictions shift toward the actual outcome. If you predicted a 30% chance that a candidate would win and the candidate did win, you might recall having said 45%. Importantly, participants usually believe their recalled judgment is accurate – they do not feel as though their memory has shifted.

In the hypothetical paradigm, there are no original predictions to recall. Instead, researchers compare two groups. A foresight group makes predictions without knowing the outcome. A hindsight group is told what happened but is asked to judge "as if" they did not know. The hindsight group's predictions are consistently closer to the actual outcome than those of the foresight group, even though they were instructed to ignore the outcome information (Fischhoff, 1975).

Fischhoff's original experiments used both approaches with historical and clinical scenarios. In one study, participants read about the British–Gurkha war and estimated the likelihood of four outcomes: a British victory, a Gurkha victory, a stalemate with no peace settlement, or a stalemate with a peace settlement. Those who were told which outcome had actually occurred rated it as more likely than those who were not – regardless of which outcome they were told had occurred. Even participants who were explicitly asked to set aside the outcome information and respond as they would have without it were unable to do so.

Why Does Hindsight Bias Happen?

Several psychological processes contribute to hindsight bias, often working together.

Availability – once you know an outcome, it comes to mind easily and vividly. What actually happened is concrete and salient; what could have happened is abstract and vague. The actual outcome dominates your thinking simply because it is more accessible. This connects directly to the availability heuristic discussed elsewhere in this book: events that come to mind easily are judged as more likely (Availability).

Causal narrative construction – after learning an outcome, people construct a coherent causal story that explains why it happened. You rearrange the facts to fit the result. If a startup succeeded, you focus on the founder's vision and the market timing. If it failed, you focus on the risky strategy and the inexperienced team – even though the same facts were available in both cases. Once this narrative is in place, it is difficult to dismantle, because you have built a story in which everything leads to the outcome. This process draws on the same causal reasoning discussed in the chapter on how people infer causation from situational features (Cues to Causality).

Yopchick & Kim (2012) tested this directly. They used deliberately sparse scenarios – for example, a battle between two tribes, or a gold mining expedition – and varied whether participants received information that was causally relevant to the outcome. Hindsight bias appeared only when participants had causally relevant information that could be used to explain the outcome. When the available information had no clear causal connection to what happened, hindsight bias did not occur. This confirms that the bias depends on being able to build a story that makes the outcome feel inevitable.

Curse of knowledge – learning an outcome changes your mental state, and it is surprisingly difficult to return to the way you thought before. This is closely related to theory of mind – the ability to represent another person's beliefs as different from your own. When you try to imagine what you believed before learning the outcome, or what someone else would believe without knowing it, your current knowledge intrudes.

Self-serving motivation – people are more likely to claim "I knew it all along" for positive outcomes than for negative ones. If your team wins, you always saw it coming. If they lose, you are less sure you predicted it. This selective claiming further inflates hindsight bias in situations where being right feels good.

When Hindsight Bias Is Weak or Reversed

Hindsight bias is robust, but it is not universal. There are conditions under which it weakens or even reverses.

No plausible causal narrative. As noted earlier, hindsight bias weakens when the available facts do not support a plausible causal story. Without a causal path from the information to the result, the outcome does not feel inevitable, and the bias fails to appear.

Self-serving motivation working against the bias. If "knowing it all along" would be psychologically threatening, people resist the pull of hindsight. Mark & Mellor (1991) found that laid-off workers showed especially little hindsight bias, presumably because saying "I knew I would be fired" would imply that they saw it coming and did nothing, which threatens self-esteem.

Extreme surprise. When an outcome is truly shocking, people sometimes show reverse hindsight bias: they believe they never could have seen it coming. Ofir & Mazursky (1997) found that people who learned a heart bypass surgery ended fatally later judged successful outcomes as having been more likely than did foresight participants who had no outcome information – an apparent "I never would have seen the death coming" effect. One interpretation is that extreme surprise disrupts the usual process of building a coherent causal story and pushes judgments in the opposite direction.

These boundary conditions clarify the mechanisms behind hindsight bias. The bias is not a simple, automatic response to outcome knowledge. It depends on being able to build a causal story, on the motivational context, and on the degree of surprise.

Outcome Bias

Before moving to the Bayesian framework, it is useful to distinguish hindsight bias from a neighboring phenomenon: outcome bias. Outcome bias is the tendency to judge the quality of a decision by its outcome rather than by the reasoning behind it. A good outcome makes the decision seem wise; a bad outcome makes it seem foolish – regardless of whether the decision was well-reasoned given the information available at the time.

Baron & Hershey (1988) demonstrated this clearly. Participants read about a physician who had to decide whether to perform a risky surgery. The medical reasoning and the available evidence were held constant across conditions; only the outcome varied. When the patient survived, participants rated the decision as better and the physician as more competent. When the patient died, the same decision based on the same reasoning was judged as poorer. This happened even when participants acknowledged that the outcome should not affect their judgment.

Outcome bias is not the same as hindsight bias, but they reinforce each other. Hindsight bias makes the outcome feel predictable, and outcome bias then punishes the decision-maker for not having predicted it. Together, these biases lead people to evaluate decisions by outcomes rather than by the information available when the decision was made. This matters in many areas of life, from medicine to the courtroom to everyday evaluations of leaders and managers (Medical Decision-Making, Legal Decision-Making). The distinction between decision quality and outcome quality is itself partly a normative issue – one explored in the chapter on the relationship between descriptive and normative models (How It Is vs. How It Should Be).

A Bayesian Lens

Throughout this book, we use the framework of Bayesian reasoning to understand how beliefs should be updated (Bayesian Reasoning). In this framework, you start with a prior belief, encounter new evidence, and form a posterior belief that combines both. Hindsight bias can be understood as a retroactive revision of the prior to match the posterior.

Consider a concrete example. Before an election, you think a candidate has a 40% chance of winning – that is your prior. The candidate wins, and you update: your posterior is now 100%. But when you try to remember what you thought before the election, you recall having estimated something like

65%. The prior has been overwritten. You can no longer access what you actually believed before the evidence arrived, because your current knowledge has contaminated your memory.

Under the Bayesian ideal, one keeps track of both prior and posterior beliefs. You know what you believed before the evidence arrived, and you know how the evidence changed your belief. The hindsight-biased thinker loses this distinction. The curse of knowledge is the core mechanism: Bayesian updating is itself a legitimate and rational process – when you learn an outcome, you should update your beliefs. The problem is not that beliefs change; that is exactly what should happen. The problem is that you lose access to the pre-update state. You cannot remember your old prior, and so you cannot appreciate how much the evidence changed your mind.

Summary

Hindsight bias is the tendency to believe, after learning an outcome, that you knew it all along. It has been demonstrated across many domains and populations, using both memory and hypothetical paradigms. The bias is driven by the availability of the actual outcome, the construction of causal narratives, the curse of knowledge, and self-serving motivation. Hindsight bias weakens when people cannot build a causal narrative, when admitting foresight would threaten self-esteem, or when the outcome is extremely surprising. A related phenomenon, outcome bias, leads people to judge decision quality by outcomes rather than by reasoning. From a Bayesian perspective, hindsight bias represents a failure to maintain the distinction between prior and posterior beliefs: after updating, the original prior is overwritten, and the thinker can no longer appreciate how much the evidence changed their mind.

STATISTICAL VS INTUITIVE PREDICTION

Learning goals:

- Explain the difference between clinical intuition and statistical prediction.
- Describe the findings of the meta-analysis comparing clinical and statistical prediction.
- Explain what a model of the judge is and when it is useful.
- Identify the reasons why clinical intuition is overvalued.
- Apply the lessons from statistical prediction to real-world domains such as judicial selection and football recruitment.
- Connect the superiority of statistical prediction to the concept of Bayesian reasoning.

Three Ways to Predict

Imagine you are a clinical psychologist trying to decide whether a patient is likely to relapse into depression. You have years of training, you have spoken with the patient at length, and you have read their file carefully. You form an impression, weigh the different pieces of information, and arrive at a prediction. This is clinical judgment: a domain expert considers the available information and makes an intuitive prediction, drawing on experience and holistic assessment. The traditional term is clinical judgment even when the expert is not literally a clinician – the same process occurs when a football scout evaluates a player, a hiring committee assesses a candidate, or a judge weighs testimony.

Now imagine a different approach. Instead of relying on your personal judgment, you enter the patient's information into a statistical model. This model was built by analyzing thousands of previous patients: their symptoms, their histories, and whether or not they relapsed. The model produces a probability. The final prediction is generated by an explicit rule rather than by case-by-case intuition. This is statistical prediction.

There is also a hybrid approach. In situations where you lack the historical outcome data needed to build a statistical model, you can instead aggregate the intuitive judgments of multiple experts and extract the statistical patterns in those judgments. This produces a model of the judge – a formula that captures the shared signal in expert opinions while filtering out the noise. We will return to this approach after reviewing the evidence on the first two.

Which of the first two approaches do you think is more accurate? Most people, including most professionals, bet on clinical judgment. The evidence points strongly in the other direction.

The Evidence

The comparison between clinical judgment and statistical prediction has been studied for over seventy years, beginning with the work of Paul Meehl in the 1950s. The most comprehensive test of this question comes from a meta-analysis by Grove et al. (2000), who examined 136 studies spanning psychology, medicine, education, and forensic settings. In each study, human experts and statistical models predicted the same outcomes using the same information. For example, in some studies clinicians predicted whether psychiatric patients would improve after treatment, while a formula combined the same symptom ratings and background variables to make its own prediction. Other studies involved predicting criminal recidivism, academic performance, or medical diagnoses.

The results were striking. Of the 136 studies, 46% favored statistical prediction: the model was more accurate than the expert. Another 48% showed no meaningful difference between the two approaches. Only a small minority of studies – 6% – favored clinical judgment. In other words, unaided expert judgment was only rarely superior, and often less accurate.

Grove et al. (2000) also tested whether certain factors might tip the balance in favor of clinical judgment. Perhaps more experienced clinicians would do better. Perhaps clinical judgment would shine in certain domains, like psychology, where human understanding seems especially important. None of these factors mattered. Publication date, training, years of experience, and the specific domain of prediction had no effect on the relative advantage of statistical methods. One finding was particularly telling: in some comparisons, access to interview data actually reduced predictive accuracy, likely because vivid but weakly diagnostic impressions displaced more valid cues.

This pattern has been confirmed by other reviews. Dawes et al. (1989) reviewed nearly 100 comparative studies and reached the same conclusion: statistical methods equaled or exceeded clinical judgment in virtually every case. Even when clinicians had access to additional information not available to the statistical model, they still failed to outperform it. The advantage of statistical prediction lies not in having better data, but in combining data more consistently. A formula applies the same weights to the same variables every time. A human expert does not.

Models of the Judge

Statistical prediction requires historical data: you need past cases with known outcomes to build a model. But what do you do in situations where such data do not exist? For example, suppose you want to predict whether patients with a rare condition will respond to a new treatment. There are no historical records to train a model on.

This is where models of the judge, sometimes called MUDs, become useful. You gather a group of experts and ask each of them to make predictions for a set of cases. You then build a statistical model that captures the average pattern across these expert judgments. This model becomes a stand-in for the experts themselves.

To make this concrete, imagine that several psychiatrists each rate 100 patient profiles for the risk of schizophrenia relapse. Each profile lists symptoms such as the number of prior episodes, medication adherence, and level of social support. Different psychiatrists weigh these factors differently – one may focus heavily on medication adherence, another on social isolation. A regression model fitted to all their ratings can learn the average weights they implicitly assign to each symptom. The resulting formula is the model of the judge.

The remarkable finding is that such a model often outperforms the individual judges it was built from. The reason is that much of the variation in expert judgment is noise – irrelevant variability caused by mood, fatigue, the order in which cases were reviewed, or the inherent inconsistency of human information processing. When you average across many judgments, this noise cancels out, leaving behind the signal – information genuinely related to the outcome – that the experts share.

You can think of models of the judge as a statistical version of the wisdom of crowds. Just as averaging many independent estimates of the number of jellybeans in a jar tends to be more accurate than any single estimate, averaging many expert judgments tends to be more accurate than any single expert.

One important caveat: when valid outcome data are available, models built on actual outcomes are usually preferable to models of the judge. MUDs are most useful when such data are unavailable, expensive, or slow to obtain. They are a fallback, not an alternative to outcome-based statistical prediction.

Why Clinical Judgment Is Overvalued

If the evidence so clearly favors statistical prediction, why do professionals continue to trust their own judgment? Several factors help explain this resistance.

The first is cognitive fluency. As clinicians gain experience, the process of making assessments starts to feel easy and natural. With practice, pattern recognition becomes automatic, and confidence grows. But feeling confident is not the same as being accurate. As discussed in the chapter on Overconfidence, confidence often increases with experience even when accuracy does not. Consider an admissions interviewer who spends forty-five minutes with a polished, articulate candidate and comes away feeling certain that this person will succeed. That feeling of certainty is real, but it reflects the smoothness of the interaction, not the validity of the prediction. The subjective ease of making a judgment creates an illusion of validity: it feels right, so it must be right.

The second factor is the absence of corrective feedback. In many professional domains, experts rarely learn whether their predictions were correct. A clinical psychologist who predicts that a patient will not attempt suicide may never find out what happened after the patient left treatment. A hiring committee that rejects a candidate never sees how that person would have performed if hired. Einhorn & Hogarth (1978) showed that this feedback structure is a fundamental barrier to learning. Because experts see only the outcomes of the cases they selected, not the cases they rejected, they cannot properly evaluate their own accuracy. This creates a self-reinforcing cycle: decisions feel successful, confidence grows, and there is no mechanism for correction.

Contrast this with weather forecasting. Weather forecasters are among the most accurately calibrated – meaning their stated confidence matches their actual accuracy – experts we know of. When a forecaster says there is a 70% chance of rain, it rains on roughly 70% of those occasions. The reason is simple: they make predictions every day, and every day they find out whether they were right. This frequent, unambiguous feedback allows them to learn from their mistakes and adjust their judgments over time. In domains without such feedback, like clinical psychology, law, or university admissions, the conditions for learning are absent, and clinical judgment stagnates while confidence inflates.

The third factor is professional identity. Many experts see their intuitive judgment as a core part of what makes them experts. Suggesting that a formula could replace their judgment feels like an attack on their competence and their profession. This resistance is understandable, but it comes at a cost: when professionals insist on using their intuition in situations where statistical methods would do better, the people affected by those decisions pay the price.

Case Study: Selecting Judges in the Netherlands

A clear illustration of these problems comes from the Dutch judiciary. Researchers at the Vrije Universiteit Amsterdam and the University of Groningen were asked by the Council for the Judiciary to evaluate the selection procedure for the training program that produces new judges in the Netherlands (Neumann et al., 2024). The findings, reported by De Hoog (2026b), were sobering.

The selection procedure consisted of five rounds: screening of application letters, an analytical test, local interviews at individual courts, a full assessment involving psychologist interviews and role-plays with actors, and a final round of national interviews. On paper, this sounds thorough. In practice, the researchers found that the procedure violated best practices in several ways as a result of overreliance on clinical judgment.

The screening of application letters was done without clear criteria or decision rules. Although 99% of candidates met the formal requirements, the first round offered almost no ability to distinguish strong from weak applicants. The local interviews were meant to assess whether a candidate was a good "fit" for a particular court, but the concept of fit was never properly defined. Interviewers relied on vague impressions and, in some cases, unscientific personality typologies. The elaborate assessment, which included personality questionnaires, a role-play, and a conversation with a psychologist, was largely uninformative: only one component, a measure of conscientiousness, had any predictive value for later training performance. The rest of the assessment added noise rather than signal.

The most telling finding concerned the analytical test administered in the second round. This simple intelligence test predicted training performance better than the entire elaborate assessment that followed it. Standardized scoring on a brief test outperformed hours of expert evaluation because the test applied the same metric to every candidate.

The final round of national interviews was the strongest part of the procedure. It used structured questions, predefined scoring anchors that described what a good or bad answer looked like, and independent numerical ratings from multiple evaluators. These are precisely the features that research on personnel selection recommends. Yet even here, problems arose. After the interviews, evaluators met to deliberate, and during these discussions, gut feelings and personal impressions crept back in, undermining the careful structure of the interview itself. The researchers showed that simply summing the independent scores, without any deliberation, would have produced equally good and more consistent selections.

The researchers' core advice was simple: do as little as necessary, and do it well. Strip away the rounds that add no predictive value. Standardize the rounds that remain. Use explicit decision rules instead of group deliberation.

Case Study: Data-Driven Football Recruitment

If the judiciary case shows what goes wrong when clinical judgment dominates, the story of Heart of Midlothian, a Scottish football club, shows what goes right when statistical methods take the lead (De Hoog, 2026a).

Traditional football scouting is a textbook example of clinical judgment. Scouts travel to matches, watch players, and form holistic impressions. They assess talent based on what they see: technique, movement, composure under pressure. Most clubs treat data analysis as a supporting tool at best, never as the primary basis for decisions.

Heart of Midlothian, or Hearts, operates differently. The club is co-owned by Tony Bloom, an entrepreneur who built his fortune on statistical betting. Bloom's central insight, which he applies to both gambling and football, is that a team's quality is better predicted by the chances it creates than by the goals it scores. Creating chances reflects skill; converting chances into goals involves a large amount of luck. This distinction between skill and luck matters enormously when you are trying to predict future performance rather than simply describe past results.

Hearts uses data models to evaluate what individual players contribute to the chances their team creates. This allows the club to screen players across all leagues in the world simultaneously, including obscure ones where talented players are undervalued and therefore affordable. The club relies far more heavily on data than traditional clubs do; video review and scouting play a secondary role, serving mainly to verify what the data have already suggested. Recent signings came from Kazakhstan, North Macedonia, Norway's second division, and the Dutch Keuken Kampioen Divisie – leagues that most traditional scouts would never visit.

This approach illustrates how data analysis can identify talent that human intuition overlooks, precisely because a model is not influenced by the reputation of the league, the size of the club, or the first impression a player makes on a scout watching from the stands.

The results have been impressive. At the time of the report, Hearts were competing at the top of the Scottish Premiership alongside Celtic and Rangers – clubs with vastly larger budgets. Players who were unknown and unwanted elsewhere had become key contributors.

The football world has known about data-driven methods for years. FC Midtjylland, a Danish club with a similar ownership structure, achieved success with comparable methods as early as 2015. Yet most clubs still rely primarily on traditional scouting. The distrust of statistical methods in football mirrors the pattern seen in the judiciary and in clinical psychology: professionals prefer their own judgment even when explicit, data-driven approaches demonstrably outperform it.

Two Cases, One Lesson

The judiciary case and the football case come from very different worlds, but the lesson is the same. When clinical judgment dominates, decisions are inconsistent, information is wasted, and prediction suffers. When statistical methods lead, hidden value is found, resources are used efficiently, and results exceed expectations. In both domains, standardized rules for combining evidence outperform informal expert integration.

This does not mean that human judgment has no role. Experts are still needed to identify what variables matter, to design the models, and to handle genuinely novel situations that no model has been trained on. But when it comes to the routine combination of information to make a prediction, the evidence consistently shows that explicit rules are more reliable than unaided judgment.

A Bayesian Lens

Clinical judgment can be understood as a form of informal Bayesian reasoning. An experienced professional has strong prior beliefs – built up over years of practice – about what kinds of patients relapse, what kinds of candidates succeed, or what kinds of players perform well. When they encounter a new case, they update these priors with the evidence at hand, forming a posterior judgment. The problem is that this updating process is imprecise and inconsistent. The same expert might weigh the same piece of evidence differently on different days, or might be swayed by vivid but uninformative details, as discussed in the chapters on Availability and Representativeness.

Statistical prediction uses explicit, repeatable rules for combining evidence – sometimes Bayesian, sometimes frequentist, but in all cases more consistent than intuitive judgment. The model applies the same evidence-weighting rules to every case, producing reliable predictions. The superiority of statistical methods shows that human intuitive "Bayesian" processing is noisy and biased. As discussed earlier, without regular outcome feedback, priors may become stronger without becoming better calibrated – meaning that stated confidence matches actual accuracy. The prior becomes more rigid, not more accurate. This is why experienced clinicians are often no better than novices at prediction, even though they feel far more confident: their priors have hardened, but they have never been properly tested against outcomes.

Summary

Statistical prediction – in which outcomes are predicted using empirical evidence and formal models – consistently outperforms clinical judgment across a wide range of domains. A meta-analysis of 136 studies found that statistical methods were superior or equal in 94% of cases, with only 6% favoring clinical judgment. Models of the judge offer a useful fallback when historical outcome data are unavailable: by fitting a statistical model to the judgments of multiple experts, much of the noise in individual predictions is eliminated. Clinical judgment is overvalued because cognitive fluency makes experienced judgment feel accurate, because professionals rarely receive the corrective feedback needed to improve, and because intuitive judgment is tied to professional identity. Two case studies illustrate the practical implications. In the Dutch judiciary, an elaborate, intuition-driven selection procedure for new judges was found to be less predictive than a simple intelligence test, and structured interviews outperformed unstructured deliberation. In Scottish football, Heart of Midlothian used data models to recruit undervalued players from obscure leagues, competing successfully against far wealthier clubs that relied on traditional scouting. In both cases, standardized evidence combination outperformed informal expert integration. For prediction, explicit and consistent rules usually outperform unaided judgment.

THE FUTURE SELF

Learning goals:

- Define affective forecasting and explain why people are better at predicting valence than intensity or duration.
- Explain the impact bias and how focalism contributes to it.
- Describe projection bias and the hot-cold empathy gap.
- Explain the disability paradox and its implications for living wills.
- Describe the peak-end rule and how it creates a mismatch between experienced and remembered events.
- Define temporal discounting and distinguish it from affective forecasting errors.
- Explain the concept of the future self as a stranger and how it connects affective forecasting and temporal discounting.

Imagining Tomorrow

Think about how you felt on the day you received your university admission letter. Now imagine how you would feel if you failed an important exam next week. You can probably tell that the first scenario feels good and the second feels bad. But can you predict exactly how intensely you would feel, and how long that feeling would last? Most people believe they can, and most people are wrong.

This chapter is about two related problems people have with their future selves. The first is a prediction problem: people systematically mispredict how future events will make them feel, a process called affective forecasting. The second is a motivation problem: people tend to undervalue outcomes that happen to their future self, a pattern called temporal discounting. These are different failures – one cognitive, one motivational – but both arise because the future self is hard to represent vividly. Your future self is psychologically distant from you, almost like a stranger. Just as it is hard to fully inhabit another person's mind, it is hard to inhabit the mind of the person you will become. This connects to perspective-taking and theory of mind more broadly, including the curse of knowledge discussed in Hindsight: once you are in one mental state, it is remarkably hard to imagine being in a different one, whether that state belongs to another person, your past self, or your future self.

Affective Forecasting

Affective forecasting refers to predictions about how future events will make you feel. Wilson & Gilbert (2005) reviewed a large body of research on this topic and found a consistent pattern: people are generally good at predicting valence, meaning whether an event will feel positive or negative. You correctly anticipate that getting a promotion will feel good and that losing a friend will feel bad. Where people go wrong is in predicting the intensity and duration of their emotional reactions. People routinely overestimate both how strongly they will react and how long that reaction will last.

The Impact Bias

The tendency to overestimate how strongly and how long a future event will affect your feelings is called the impact bias. It shows up across a wide range of life events. Wilson & Gilbert (2005) describe studies in which college professors overestimated how devastated they would be after being denied tenure, and students overestimated how unhappy they would be after being assigned to an undesirable dormitory. In each case, the predicted emotional reaction was more extreme than the actual experience.

A particularly vivid demonstration comes from research on romantic breakups. Eastwick et al. (2008) followed college students who were in romantic relationships over the course of nine months. About 38 percent of these students experienced a breakup during the study period. Before the breakup, participants predicted how distressed they would feel afterward. Then the researchers tracked their actual distress at multiple time points, from two weeks before the breakup to ten weeks after. Participants significantly overestimated how upset they would be immediately after the breakup. They were more accurate about the direction of recovery than about the initial intensity of distress: they correctly predicted that their feelings would improve over time, but they started from a predicted peak that was too high. People who reported being more deeply in love, who were less likely to initiate the breakup, or who thought it unlikely they would soon enter a new relationship showed the largest forecasting errors.

The impact bias has two components: overestimating intensity and overestimating duration. The duration component is sometimes called emotional evanescence – emotions fade more quickly than people anticipate. Students expect rejection, embarrassment, or disappointment to dominate their mood for weeks, but daily routines and new events quickly compete for attention. Even intense grief or joy eventually gives way to a more neutral baseline. People know this in the abstract, but they consistently underestimate how quickly it happens. This is part of what makes the impact bias so robust: people keep expecting that the next big event will permanently change how they feel, when in reality the change is almost always temporary.

Why does the impact bias occur? One major reason is focalism. When you imagine a future event, you tend to focus narrowly on that event and neglect all the other things that will be happening in your life at the same time. If you imagine failing an exam, you picture yourself sitting with the bad grade and feeling terrible. You do not picture yourself also going to a friend's birthday party that evening, enjoying a good meal, or getting absorbed in a television series. These other events dilute the emotional impact of any single event, but you fail to account for them when making your prediction.

A second reason, identified by Wilson & Gilbert (2005), is that people fail to appreciate how quickly they make sense of unexpected events. When something surprising happens, whether positive or negative, people initially react strongly. But then they rapidly explain the event, normalize it, and reduce its emotional impact. This sense-making process is largely unconscious. People have what Wilson and Gilbert call a psychological immune system: a set of coping mechanisms including rationalization, self-affirmation, and dissonance reduction that help them recover from setbacks. A student who fails an exam may initially feel devastated, but within days may start telling themselves that the exam was unusually unfair, that one grade does not define their ability, or that the setback will motivate better study habits. Because this immune system operates outside awareness, people do not factor it into their forecasts. They predict how they will feel based on the raw impact of the event, without accounting for the recovery that will inevitably follow.

Projecting the Present onto the Future

There is another source of forecasting error that goes beyond focalism and immune neglect. Loewenstein (2005) describes a pattern called projection bias: the tendency to project your current mental or emotional state onto the future. When you try to predict how you will feel next month, you use your present self as a model for your future self. But your present self is often a poor model.

Projection bias shows up in everyday situations. When you go grocery shopping while hungry, you buy more food than you need because you project your current hunger onto your future, less-hungry self. When you are in a good mood, you commit to social engagements that you will later regret when you feel tired. Loewenstein (2005) gives the example of accepting a speaking invitation months in advance: in the moment of being asked, you feel flattered and energized, and you project that state onto the future you who will have to prepare and travel for the talk. By the time the event approaches, your enthusiasm may have been replaced by stress and fatigue.

A specific and important form of projection bias is the hot-cold empathy gap. People in a calm, rational state (a "cold" state) cannot accurately predict how they would feel or behave in an emotional, visceral state (a "hot" state), and vice versa. When you are not hungry, it is difficult to appreciate how powerfully hunger influences decision-making. When you are not in pain, it is hard to imagine how pain dominates your attention.

Loewenstein (2005) reports studies illustrating how large this gap can be. In one line of research, men were asked to predict how their preferences and decisions would change under sexual arousal. In a calm, non-aroused state, they predicted relatively modest effects. When actually aroused, however, they reported substantially greater willingness to engage in risky sexual behavior than they had anticipated. The cold state simply could not access or simulate the motivational power of the hot state.

The hot-cold empathy gap has important practical implications. Consider addiction: a person in recovery, feeling calm and in control, may confidently predict that they can resist cravings in a high-risk situation. This prediction is made from a cold state, and it fails to account for the overwhelming power of cravings in a hot state. Similarly, someone on a diet may plan their meals rationally, but when confronted with a buffet while hungry, their behavior may diverge sharply from the plan. The gap also extends to interpersonal empathy: Loewenstein (2005) notes that physicians, who are not in pain themselves, may systematically underestimate their patients' pain and therefore undermedicate.

Predicting Changing Tastes

The difficulty of projecting into the future extends beyond emotions to preferences more broadly. Kahneman & Snell (1992) investigated whether people can predict how their tastes will change over time. In one study, participants ate plain yogurt and listened to a piece of music every day for eight days. Each day, they rated how much they enjoyed the experience and predicted how much they would enjoy it in the future.

The predictions were only weakly related to actual changes in enjoyment. Participants generally predicted that their enjoyment would decline with repetition. For the music, this prediction was roughly correct on average, but individual trajectories varied widely and were not well predicted. For the yogurt, which was initially unpleasant, many participants actually grew to like it over time – something they had not anticipated. People relied heavily on their current feelings as a starting point for predicting the future, and they held implicit theories about taste change (such as "repetition leads to boredom") that were often wrong. Again, current experience served as a misleading model for future experience.

This has implications that go well beyond food and music. When you choose a university program, a career, or a city to live in, you are making a bet about what your future self will enjoy. If you cannot even predict how your feelings about yogurt will change over eight days, it is worth being humble about predictions that span years or decades. This finding also connects to Anchoring and Primacy: current feelings serve as an anchor from which people adjust insufficiently when predicting the future.

The Disability Paradox

One of the most striking demonstrations of projection bias involves health and disability. Healthy people, when asked to imagine living with a serious illness or disability, typically predict that their quality of life would be very low. Some predict they would prefer death over certain conditions. But research consistently shows that people who actually live with these conditions report much higher quality of life than healthy people predict.

Loewenstein (2005) discusses this pattern, sometimes called the disability paradox. A study by Slevin et al. (1990) illustrates it well. Healthy participants and cancer patients were both asked whether they would accept a course of chemotherapy that offered only a small benefit: three extra months of life. Only 10 percent of healthy participants said they would accept such treatment. Among cancer patients, 42 percent said they would. This is projection bias in a high-stakes domain: the healthy participants, making their judgment from a cold state, could not access the adaptive coping, changed priorities, and daily pleasures that actually characterize life with serious illness.

This finding has direct implications for living wills, also called advance directives. A living will is a document in which a healthy person specifies what medical treatments they would or would not want if they became seriously ill and unable to communicate. The idea is to give people control over their future care. And evidence suggests that advance care planning does improve end-of-life care. Brinkman-Stoppelenburg et al. (2014) conducted a systematic review of 113 studies on advance care planning and found that it can increase compliance with patient wishes, reduce unwanted life-sustaining treatments, and increase the use of hospice and palliative care. Complex advance care planning interventions, which involve structured discussions rather than simply filling out a form, showed the most promising results.

But living wills are inherently limited by the affective forecasting errors of the person who writes them. The healthy person writing the directive is, by definition, in a different psychological state from the sick person who will be affected by it. This does not make living wills useless, but it does mean they work best when paired with ongoing discussion among patients, families, and clinicians. They should be understood as imperfect predictions, not fixed truths about what a person would want.

The Peak-End Rule

A final source of forecasting error is that people do not evaluate experiences by their total duration, so the experience they later remember is not the one they originally tried to forecast. Kahneman et al. (1993) demonstrated this with a now-classic experiment. Participants submerged one hand in water of 14°C, which is uncomfortably cold, for 60 seconds. This was the short trial. In a separate trial, participants submerged their hand in the same cold water for 60 seconds, but then the water temperature was gradually raised to 15°C over an additional 30 seconds. This was the long trial. The long trial involved strictly more pain: everything the short trial had, plus 30 extra seconds of discomfort (albeit slightly reduced discomfort).

When asked which trial they would prefer to repeat, 69 percent of participants chose the long trial. They preferred more total pain because the long trial ended on a less unpleasant note. This finding illustrates the peak-end rule: people's retrospective evaluations are strongly influenced by the peak intensity (the worst or best moment) and the end of the experience, with duration often underweighted. Both trials had the same peak, but the long trial had a better ending, and so it left a more favorable memory.

The peak-end rule matters for affective forecasting because it creates a gap between what people expect and what they later remember. When predicting how a future experience will feel, people tend to think about the total experience: how bad will it be overall, and for how long? But when they later evaluate the experience from memory, they rely on the peak and the end. This means that people are, in a sense, predicting the wrong thing. They try to forecast the total experience, but what actually matters for their future evaluation is how the experience peaks and how it ends.

This principle has practical applications in medicine, customer service, and everyday life. A dentist who saves the least painful part of a procedure for last may leave patients with a better retrospective evaluation of the visit. A holiday that ends with a relaxing day may be remembered more fondly than one that ends with a stressful flight, even if the total enjoyment was the same. People do not simply store a record of their experiences; they construct a memory that emphasizes certain moments, and this constructed memory drives future choices.

Temporal Discounting

While affective forecasting is about mispredicting future feelings, temporal discounting is a different kind of problem. It is about undervaluing future outcomes. People consistently prefer smaller rewards now over larger rewards later. A classic example: would you rather receive $50 today or $60 in a month? Many people choose the $50 today, even though waiting would yield a better return than almost any investment. What makes this especially revealing is a second question: would you rather receive $50 in 12 months or $60 in 13 months? Most people now prefer the $60. The one-month wait is the same in both cases, but immediate rewards exert a disproportionate pull.

Temporal discounting is less about mispredicting what will happen than about how strongly future outcomes are weighted in present choice. You know perfectly well that $60 is more than $50. You simply value the reward less because it goes to a version of you who feels psychologically distant. This pattern is pervasive. It shapes decisions about saving for retirement, studying for exams, exercising, and eating healthily. In each case, the cost is borne by the present self (less money now, effort now, discomfort now) while the benefit goes to the future self (financial security, good grades, better health).

The Future Self as a Stranger

Hershfield et al. (2011) tested the idea that the future self feels like a stranger directly. In one study, participants entered an immersive virtual reality environment where they saw a digital avatar of themselves. Some participants saw their current self; others saw an age-progressed version, digitally altered to look decades older. Afterward, participants were asked to allocate money between a current spending account and a retirement savings account. Those who had seen their aged avatar allocated significantly more money to retirement savings.

In follow-up studies, Hershfield and colleagues confirmed that the effect was specific to seeing your own future self, not just any older person. They also showed that the effect was mediated by increased feelings of continuity with the future self. Seeing the aged avatar made the future self feel less like a stranger, and this increased sense of connection motivated more future-oriented decisions.

This finding connects affective forecasting and temporal discounting to a single underlying cause: psychological distance from the future self. When that distance is large, you are more likely to mispredict how your future self will feel (because you cannot easily take their perspective) and more likely to undervalue outcomes that happen to them. When that distance is reduced, as in Hershfield's studies, both problems diminish.

Some studies suggest that temporal discounting becomes less steep with age. One explanation is that the psychological gap between present and future self narrows as people age: a 60-year-old imagining themselves at 70 may feel more continuous with that future person than a 20-year-old imagining themselves at 30. This is consistent with the idea that temporal discounting is not a fixed preference but a reflection of how connected you feel to the person you will become.

This matters for retirement saving, health behavior, and education – any domain where present sacrifice yields future benefit. Interventions that make the future self more vivid and real, whether through age-progressed images, letter-writing exercises, or simply encouraging people to imagine their future daily life in concrete detail, may help bridge the psychological distance and promote better long-term decisions. This connects to a prescriptive theme developed in How It Is vs. How It Should Be: redesigning decisions so people choose better than they otherwise would.

A Bayesian Lens

Both affective forecasting and temporal discounting can be understood through the framework of Bayesian Reasoning. The impact bias reflects poorly calibrated posteriors: when predicting how a future event will feel, people arrive at estimates that are too extreme relative to what the actual experience will support. Their posteriors are shifted too far from baseline, overweighting the imagined impact of the event and underweighting the many other factors that will determine how they actually feel.

Projection bias can be understood as using the wrong prior. Instead of drawing on the appropriate prior – the likely emotional state of the future self given the circumstances – people substitute their current emotional state. If you shop while hungry, your current hunger serves as the prior. You then predict wanting large meals all week, even though your future state will usually be less hungry. A healthy person uses their current health as a prior for predicting life with illness. These priors are informative in some situations, but they are systematically biased when the future state is likely to differ from the present one.

For temporal discounting, the Bayesian analogy is looser. Vivid representations of the future self, like those in the Hershfield et al. (2011) avatar studies, may change the inputs to valuation by making future outcomes feel more concrete and personally relevant – effectively increasing the weight assigned to the future self's outcomes in the overall calculation of value.

Summary

People struggle to relate to their future selves, and this struggle manifests in two ways. First, affective forecasting errors cause people to mispredict the intensity and duration of future emotional reactions. The impact bias, driven by focalism and a failure to anticipate psychological coping, leads people to overestimate how strongly and how long future events will affect them. Projection bias causes people to use their current state as a model for the future, and the hot-cold empathy gap makes it especially hard to predict behavior across different emotional states. The disability paradox shows that healthy people systematically underestimate the quality of life experienced by people with serious illness, with direct implications for the accuracy of living wills. The peak-end rule reveals that people's retrospective evaluations are driven by peak intensity and ending rather than total duration, creating a further mismatch between predicted and remembered experience. Second, temporal discounting causes people to undervalue future outcomes. Research showing that age-progressed avatars increase retirement savings suggests that making the future self feel more real can reduce this discounting. Both affective forecasting errors and temporal discounting share a common root: the psychological distance between who you are now and who you will become.

RISK PERCEPTION

Learning goals:

- Explain why people accept much higher levels of voluntary risk than involuntary risk.
- Describe the concept of dread risk and how it distorts responses to rare events.
- Define scope insensitivity and explain how the affect heuristic produces it.
- Describe the identifiable victim effect and how analytical thinking can reduce charitable giving.
- Distinguish between relative risk and absolute risk in risk communication.
- Explain why climate change fails to trigger strong emotional responses despite its severity.
- Apply a Bayesian framework to understand distortions in risk perception.

How Risky Does It Feel?

Consider two hazards. The first is skiing: you strap yourself to two planks and hurl down a mountain at speed, knowing that broken bones, concussions, and occasionally death are real possibilities. The second is living a kilometer downwind from a chemical plant that, according to its safety record, poses a far smaller annual risk of injury than a season on the slopes. Most people would happily book the ski trip and fiercely oppose the chemical plant. The difference is not in the numbers. It is in how each risk feels.

This chapter examines why people's judgments about risk so often diverge from what the evidence would suggest. The answer involves several of the heuristics and biases discussed earlier in this book. The availability heuristic makes vivid and recent events feel more probable. The affect heuristic causes people to judge risk based on an emotional reaction rather than a statistical estimate. And as we will see, these tendencies produce predictable patterns: some real dangers are ignored, others are massively overweighted, and the way risk information is communicated can strongly shape public reactions.

Voluntary and Involuntary Risk

One of the earliest and most striking findings in risk research is that people tolerate far more risk when they choose to take it. In a landmark analysis, Starr (1969) examined historical data on fatalities across a range of activities and technologies and found that the public accepts roughly 1,000 times more risk for voluntary activities, such as skiing or driving for leisure, than for involuntary exposures, such as living near a chemical plant or being exposed to pollution. This is a very large difference in tolerance – about three orders of magnitude.

Fischhoff et al. (1978) extended this work using a different method. Rather than inferring risk tolerance from historical behavior, they directly surveyed people about their perceptions and preferences. Participants rated 30 activities and technologies on perceived risk, perceived benefit, and the acceptability of current risk levels. They also evaluated each hazard on qualitative dimensions, including voluntariness, controllability, familiarity, and dread. The results confirmed Starr's core finding: people are willing to accept higher risk for activities that feel voluntary and controllable. But they also showed that risk perception is shaped by a broader set of qualitative features. When a risk feels uncontrollable, unfamiliar, and potentially catastrophic, people judge it as far more dangerous than experts would, based on mortality statistics alone.

This asymmetry tells us that people do not evaluate risk the way an actuary does, by computing expected fatalities per year or per hour of exposure. Instead, they evaluate risk through an emotional lens. If an activity is freely chosen, that emotional lens is forgiving. If it is imposed, the lens magnifies every possible danger. This is why a person might happily bungee jump on holiday while fiercely opposing a waste incinerator being built in their neighborhood. The objective probability of harm may be lower from the incinerator, but the experience of risk is entirely different.

When Rare Risks Are Ignored

Most of the time, people treat extremely low probabilities as if they were zero. A risk that has a one-in-a-million chance of occurring in any given year simply does not register. This means that many small but real hazards – such as food contamination, household accidents, or exposure to radon gas in the home – receive far less attention than their actual probability would warrant.

Consider radon, a naturally occurring radioactive gas that seeps into homes through the ground. Long-term radon exposure is the second leading cause of lung cancer after smoking, yet in many countries only a small fraction of homeowners have ever tested their homes. The test is cheap and easy, and mitigation is straightforward. But because radon is invisible, odorless, and kills only slowly, the risk on any given day feels like zero, and so people do not act. A similar pattern appears with emergency preparedness: despite living in flood- or earthquake-prone areas, many residents never assemble an emergency kit or develop an evacuation plan. The probability of disaster in any given week is too low to feel real.

This neglect of rare risks is partly understandable. In a world of limited attention, it makes sense to focus on the dangers that are most likely to affect you. But it also leads to systematic blind spots. The cumulative probability of harm over a lifetime can be substantial even when the daily probability is vanishingly small. This pattern relates to the broader concept of bounded rationality: people do not have the cognitive resources to evaluate every possible risk, so they rely on heuristics that sometimes lead them astray.

The major exception is when a rare hazard is emotionally loaded. In those cases, the same low probability can be treated as intolerably high.

Dread Risk

The neglect of rare risks breaks down dramatically when a hazard triggers a particular emotional response: dread. Dread is the feeling that a hazard is catastrophic, uncontrollable, and unfairly distributed. Using what they called the psychometric paradigm, Slovic (1987) and colleagues mapped dozens of hazards along two dimensions. The first, called "dread risk," captured the degree to which a hazard was perceived as uncontrollable, catastrophic, involuntary, and inequitable. The second, called "unknown risk," captured how novel, unobservable, and poorly understood a hazard seemed. In these psychometric studies, dread emerged as one of the major dimensions distinguishing hazards that people treat as terrifying from those they treat as acceptable. Nuclear power, for example, scored extremely high on both dimensions, which explains why it provokes intense public opposition despite expert assessments that its mortality risk is relatively low.

When dread is triggered, a rare risk flips from invisible to massively overweighted. People demand strict regulation, avoid the hazard at all costs, and may make decisions that expose them to greater danger. Experts, by contrast, tend to judge risk primarily by mortality rates. This mismatch between expert and public assessments is not simply a sign that the public is irrational. As Slovic argued, public risk judgments often reflect concerns – such as fairness, controllability, and catastrophic potential – that are not captured by simple mortality statistics. But the mismatch does mean that public responses to risk can sometimes be counterproductive.

Dread in Action: The Aftermath of Terrorism

The consequences of dread risk can be measured in lives lost. Gigerenzer (2004) documented one of the most striking examples. After the terrorist attacks of September 11, 2001, many Americans were terrified of flying. Air travel dropped by 12 to 20 percent in the months that followed. But people still needed to travel, so many drove instead. The problem is that driving the same distance as the average domestic flight in the United States is roughly 65 times more dangerous than flying, even when the risk of a terrorist attack on a plane is included in the calculation. By analyzing U.S. Department of Transportation data, Gigerenzer estimated that approximately 350 additional people died in traffic accidents in the three months after 9/11, compared to what would have been expected based on the previous five years. This excess death toll exceeded the number of passengers who died on the four hijacked flights.

A plane crash, especially one caused by terrorism, is a classic dread risk: catastrophic, uncontrollable, and unfamiliar. It dominates news coverage for weeks. In contrast, car accidents are so common that they barely make the local news. The availability heuristic ensures that the vivid, dramatic event looms large in the mind, while the mundane, everyday risk fades into the background. The result is a substitution from a safer mode of transport to a more dangerous one, driven entirely by how the risk feels rather than by any calculation of probability.

An instructive contrast comes from Spain. López-Rousseau (2005) examined what happened after the Madrid train bombings of March 11, 2004, which killed approximately 200 people. As in the United States, train ridership declined in the following months. But unlike Americans, Spaniards did not substitute train travel with car travel. Instead, highway traffic also decreased, and traffic fatalities went down rather than up. Several factors may explain this difference. Spain has a long history of domestic terrorism, including decades of attacks by ETA, which may have made the public more resilient. Culturally, Spain is less car-dependent than the United States, so alternative transport options, such as buses, were more readily available. And the sociopolitical response was different: trains continued running immediately after the attacks, and many Spaniards used them the following day to attend anti-terrorism demonstrations. This comparison shows that while dread is a universal human response, the behavioral consequences of dread depend on the cultural and practical context.

Scope Insensitivity

If risk perception were driven by careful reasoning, people's concern about a problem would scale with the size of that problem. A disaster that kills 10,000 people should feel roughly ten times worse than one that kills 1,000. But this is not what happens. People's emotional responses, and their willingness to act, are remarkably insensitive to the scope of a problem.

Hsee & Rottenstreich (2004) demonstrated this in a series of experiments. In one study, participants were shown pictures of endangered pandas and asked how much they would donate to help save them. When shown a picture of one panda, participants donated generously. When shown four pandas, they donated almost the same amount. The emotional response was triggered by the category – endangered pandas – and did not scale with the number of pandas at stake. In a parallel experiment, participants were asked to indicate their willingness to pay for either 5 or 10 Madonna CDs. When participants were primed to think analytically (by performing a calculation task), they were willing to pay roughly twice as much for 10 CDs as for 5. But when primed to rely on their feelings (by recalling an emotional experience), their willingness to pay barely changed between 5 and 10 CDs. The emotional reaction was to Madonna, not to the number of discs.

This scope insensitivity has been documented repeatedly. In contingent valuation research on environmental harms, people's willingness to pay has been found to be nearly identical for saving 2,000, 20,000, or 200,000 migratory birds. The pattern is consistent: the affect heuristic causes people to substitute the question "how large is this problem?" with the easier question "how do I feel about this type of problem?" The answer to the second question does not change much with the numbers. For charitable giving and public policy, this means that the difference between a small disaster and a massive one barely registers emotionally. A famine affecting 100,000 people may generate roughly the same concern as one affecting 10 million.

If scope insensitivity shows that feelings do not scale well with numbers, the next phenomenon shows what feelings do respond to: vivid individuals.

The Identifiable Victim Effect

People respond far more generously to a single, identified victim – someone with a name, a face, and a story – than to statistical information about thousands of anonymous victims. Charities have long understood this: a poster showing one child with a name will raise more money than a poster citing the number of children in need. Organizations like the Red Cross and UNICEF build their campaigns around individual stories precisely because our emotional system is built to respond to persons, not to populations.

Small & Loewenstein (2003) demonstrated that even minimal identification boosts generosity. In a laboratory experiment, participants played a modified dictator game in which they could share money with a recipient who had lost their initial payment. The only difference between conditions was whether the recipient had already been selected (determined) or would be selected later (indeterminate). No names, photographs, or personal details were provided. Yet participants gave significantly more money – on average 3.42 euros worth versus 2.12 – when the recipient had already been determined. A field experiment confirmed this pattern. Passersby were asked to donate part of a small payment to Habitat for Humanity, with a letter describing four needy families. When the letter stated that the recipient family had already been selected, donations were higher than when the family had not yet been chosen. Simply knowing that a specific victim exists, even without any personal information, was enough to increase generosity.

But the most striking finding in this area comes from a follow-up study. Small et al. (2007) found that combining statistical information with an identified victim can actually reduce giving compared to presenting the identified victim alone. In one experiment, participants were shown either a story about a single hungry child, statistics about millions of hungry people, or both. The combined condition produced lower donations than the identified-victim-alone condition. In a further study, participants who were primed to think analytically, by solving a simple calculation, donated less to an identified victim than those who were not primed. The explanation is that statistics trigger analytical thinking, which dampens the emotional response that drives generosity. In these studies, deliberation did not increase concern for the many; instead, it weakened the emotional pull of the single identified victim.

The identifiable victim effect is closely related to scope insensitivity, but it also reflects the special power of vivid, concrete individuals to trigger empathy. Together, the two phenomena highlight a deep tension in how humans evaluate risk and harm: our System 1 responses are calibrated to individuals and categories, not to magnitudes and populations.

Communicating Risk: Relative vs. Absolute

How risk information is communicated can dramatically change how people respond. One of the most important distinctions is between relative risk and absolute risk. Relative risk expresses a change as a ratio or percentage: "the risk doubles." Absolute risk expresses the actual probabilities: "the risk goes from 1 in 7,000 to 2 in 7,000." The same underlying data can sound either alarming or reassuring depending on which format is used.

A real-world example from the United Kingdom illustrates the consequences. In the mid-1990s, the UK Committee on Safety of Medicines issued a warning that third-generation oral contraceptive pills were associated with a "twofold increase" in the risk of venous blood clots compared to second-generation pills. This was relative risk framing: the risk approximately doubled. In absolute terms, however, the increase was from about 1 in 7,000 users per year to about 2 in 7,000 users per year. The warning, widely reported in the media in relative terms, led to widespread panic. Many women stopped taking the pill, resulting in an estimated 13,000 additional abortions in the following year. The irony is that pregnancy itself carries a substantially higher risk of blood clots than either generation of the pill.

This example underscores the importance of communicating risk as absolute numbers, or better yet, as natural frequencies. As discussed in Bayesian Reasoning, natural frequencies (such as "2 out of 7,000 women per year" versus "1 out of 7,000") make the underlying probabilities transparent and dramatically reduce misunderstanding. When risks are communicated in relative terms, people often imagine a much larger absolute change than actually exists. This is not because people are innately bad at understanding numbers; it is because the relative format strips away the information they need to form an accurate judgment.

Climate Change and the Limits of Statistics

The previous sections showed how dread can make a rare risk feel overwhelming. Climate change illustrates the opposite problem: a massive risk that feels abstract and unimportant because it fails to trigger the right emotional response. The preceding sections contrasted vivid, episodic hazards – terrorist attacks, plane crashes, identified victims – with statistical, aggregate information. Climate change falls squarely on the statistical side.

Weber (2006) argued that global warming is one of the most significant threats facing humanity, yet it consistently fails to generate the kind of visceral emotional response that would motivate large-scale action. The reason, Weber proposed, is that climate change does not trigger the psychological features associated with dread. It is gradual rather than sudden, probabilistic rather than certain, and its worst consequences are delayed by decades. It lacks the vivid, immediate, personal quality that drives System 1 responses.

Weber drew on the distinction between experience-based and description-based decision-making. When people learn about risks through direct personal experience, they develop strong emotional associations that motivate action. When they learn about risks through statistical descriptions, the information engages System 2 reasoning but fails to generate the affect needed to change behavior. Climate change, for most people, is known through descriptions: graphs, projections, reports by international panels. These are exactly the kind of abstract, statistical inputs that leave the emotional system unmoved.

Akerlof et al. (2013) provided supporting evidence. They surveyed residents of Alger County, Michigan, about whether they believed they had personally experienced global warming. About 27 percent of respondents said yes, most often pointing to changes in seasons, weather patterns, and snowfall. Crucially, those who believed they had personally experienced climate change showed significantly higher risk perceptions, even after controlling for political affiliation and demographic factors. Some of these perceived changes were corroborated by historical climate data; others were not. But what mattered for risk perception was not the accuracy of the perception – it was the experience itself. Direct experience generated the affect that descriptions could not.

This finding presents a dilemma for climate communication. The most effective way to make people care about climate change is for them to personally experience its consequences. But by the time the consequences are vivid enough to trigger widespread dread, it may be too late to prevent the worst outcomes. Weber suggested that interventions which make future consequences more concrete – such as interactive flood maps that show projected sea levels in your own neighborhood, or local data on the increasing number of extreme heat days in your city – might help bridge the gap between analytical understanding and emotional engagement. But she also warned about the "finite pool of worry": increasing concern about climate change might come at the cost of reduced concern about other important risks.

A Bayesian Lens

Throughout this book, we have used the framework of Bayesian reasoning as a normative standard for belief updating. In this framework, a prior belief is combined with new evidence to produce a posterior belief, and the weight given to the evidence depends on its reliability. Risk perception errors can be understood as specific departures from this ideal, each involving a different kind of distortion.

Distorted priors – the neglect of rare risks reflects a distortion in the prior. When the base rate of a hazard is extremely low, people round it down to zero rather than maintaining a small but nonzero prior probability. The result is that no amount of evidence, short of a dramatic event, will update the posterior belief upward.

Overweighted evidence – dread risk reflects a distortion in how evidence is weighted. A single catastrophic event, such as a terrorist attack or a nuclear accident, is treated as far stronger evidence than it really is, updating the posterior far more than its actual frequency warrants. In Bayesian terms, one dramatic observation is treated as though it were hundreds of observations, because the emotional weight of the event overwhelms any cool calculation of base rates.

Failure to scale with sample size – scope insensitivity means that the posterior does not scale with the amount of evidence. Learning that 200,000 birds are at risk should update your beliefs about the severity of a problem far more than learning that 2,000 are at risk. But because the emotional response is triggered by the category rather than the quantity, the posterior is approximately the same in both cases.

Affect substituted for evidence – the identifiable victim effect shows that narrative evidence – a single vivid case – updates beliefs and motivates action more powerfully than statistical evidence about base rates. A base rate drawn from thousands of observations should carry more weight than a single case, but affect reverses this weighting.

Together, these patterns show that risk perception is driven by affect, and affect does not follow the rules of probability.

Summary

Risk perception is shaped by how a hazard feels, not by its objective probability. People accept far more risk for voluntary activities than for involuntary ones. Extremely rare risks are typically ignored – unless they trigger dread, the feeling that a hazard is catastrophic, uncontrollable, and inequitable. Dread can flip a rare risk from invisible to overwhelming, sometimes with fatal consequences, as when Americans switched from flying to driving after the 9/11 attacks. Scope insensitivity means that people's concern does not scale with the size of a problem: a famine affecting millions feels little different from one affecting thousands. The identifiable victim effect shows that a single named individual generates more empathy and action than statistics about large populations. How risk is communicated matters enormously: relative risk framing can cause panic where absolute risk framing would not. Climate change illustrates the limits of statistical risk communication: abstract, gradual threats fail to trigger the emotional responses that motivate action. From a Bayesian perspective, risk perception errors involve ignoring base rates, overweighting vivid evidence, substituting affect for probability, and failing to scale beliefs with the magnitude of the evidence.

CHOICE

GAINS AND LOSSES

Learning goals:

- Explain the difference between expected value theory, expected utility theory, and prospect theory.
- Describe the value function and its key properties: reference dependence, loss aversion, and diminishing sensitivity.
- Explain how reference points determine whether an outcome is experienced as a gain or a loss.
- Describe the probability weighting function and how it differs from using probabilities directly.
- Apply prospect theory to explain framing effects, the endowment effect, and patterns of risk seeking and risk aversion.
- Describe ambiguity aversion, subadditivity, and probability neglect as departures from rational probability use.
- Explain omission bias and nature bias as additional distortions in risky choice.

The sting of loss

Imagine you find a €20 note on the street. You feel pleased for a moment, perhaps buy yourself a coffee, and move on with your day. Now imagine you reach into your pocket and discover that a €20 note you had earlier is gone. The objective change in wealth is the same in both cases, just with opposite signs. Yet the sting of losing €20 feels considerably worse than the pleasure of finding €20. This asymmetry between gains and losses is one of the most robust findings in the psychology of decision-making, and it sits at the heart of the theory we will explore in this chapter.

From Expected Value to Expected Utility

To understand how people make decisions involving risk, we first need a benchmark for how they should make such decisions. The simplest normative model is expected value theory. According to this model, you should multiply each possible outcome by its probability and then add up these products. The option with the highest expected value is the one you should choose. As a formula:

$$EV = \sum_i x_i \times p_i$$

Here, x refers to the monetary outcome and p to its probability. If someone offers you a coin flip where heads wins you €200 and tails wins you nothing, the expected value is 0.5 × €200 + 0.5 × €0 = €100. Under a pure expected-value criterion, you should prefer this gamble to any sure amount below €100.

But most people would not behave this way. Offered the choice between €100 for certain and the coin flip described above, many people take the sure €100, even though the two options have the same expected value. People tend to avoid risk when it comes to gains. This observation motivated the development of expected utility theory, which replaces monetary outcomes with subjective utilities. The difference in utility between €0 and €100 is larger than the difference between €100 and €200: each additional euro matters a bit less as you have more. Because of this curvature, the utility of the sure €100 exceeds the expected utility of the gamble, and choosing the sure thing becomes rational. Expected utility theory was formalized by von Neumann and Morgenstern in the 1940s and became the dominant normative framework for risky decision-making (Edwards, 1954).

Note that expected value and expected utility are benchmarks for decisions – for choosing among risky options. The Bayesian framework used throughout this book is primarily a benchmark for belief updating – for how to revise probabilities in light of new evidence. These are complementary normative tools: Bayesian reasoning tells you what to believe, and expected utility theory tells you what to do given those beliefs.

Expected utility theory works well as a normative model – a model of how people ideally should decide. But as a descriptive model of how people actually decide, it runs into problems. One of the most famous demonstrations of its failure is the Allais Paradox (Allais, 1953). In the original version, people are given two separate choice problems. In the first, they choose between option A, a sure gain of €1,000, and option B, a gamble that gives a 10% chance of €2,500, an 89% chance of €1,000, and a 1% chance of nothing. Most people prefer option A, the sure thing. In the second problem, people choose between option C, an 11% chance of €1,000 and an 89% chance of nothing, and option D, a 10% chance of €2,500 and a 90% chance of nothing. Here, most people prefer option D, the higher payoff. The paradox is that these two preferences are inconsistent under expected utility theory. The shift from the first problem to the second merely removes the same 89% chance of €1,000 from both options, which should not change the preference according to expected utility theory's independence axiom. Yet it does. People are drawn to certainty in the first problem (option A guarantees €1,000), but when certainty is no longer available in the second problem, they shift to the higher expected value. This pattern, known as the certainty effect, reveals that people do not weight outcomes by their probabilities in the way expected utility theory assumes. To explain choices like these, Kahneman and Tversky proposed prospect theory, a descriptive model of risky choice (Kahneman & Tversky, 1979).

Prospect Theory

Prospect theory accounts for the systematic ways in which people deviate from expected utility theory. Kahneman received the 2002 Nobel Prize in Economics largely for this work (Tversky had passed away in 1996 and could not share the prize). The theory was later refined into cumulative prospect theory (Tversky & Kahneman, 1992), but the core ideas remained the same. It does not claim that people should decide in a particular way, but rather captures how they actually do. It does this through two main components: a value function and a probability weighting function. Together, these components determine the subjective value of a risky option, which prospect theory calls a prospect. The value of a prospect is:

$$V = \sum_i v(x_i) \times w(p_i)$$

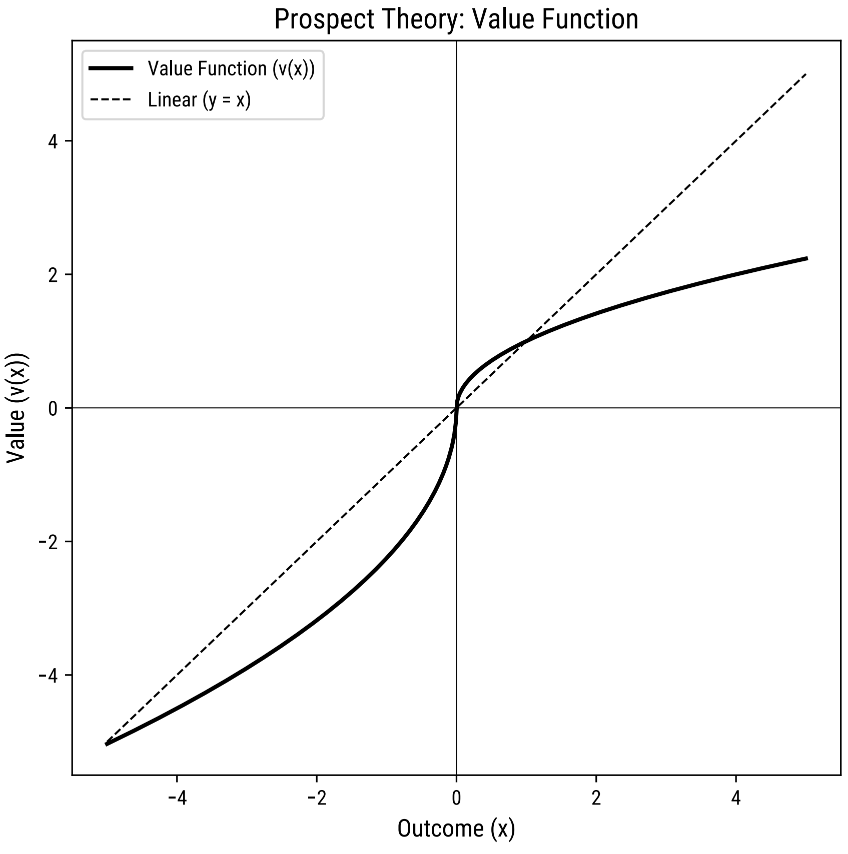
Here, v is the value function, which transforms outcomes into subjective values, and w is the probability weighting function, which transforms probabilities into decision weights. The person then chooses the prospect with the highest overall value V. This formula looks similar to expected utility theory, but the transformations it applies to both outcomes and probabilities are fundamentally different.

The Value Function

The value function describes how people evaluate outcomes. It has three key properties that distinguish it from classical utility theory.

Reference dependence – people do not evaluate outcomes in terms of final states of wealth. Instead, they evaluate outcomes as gains or losses relative to a reference point. This reference point is typically the person's current state, but it can also be shaped by expectations, aspirations, or the way a problem is described. A salary of €3,000 per month can feel like a gain if you expected €2,500 or a loss if you expected €3,500. The objective amount is the same, but the psychological experience depends entirely on what you compare it to.

A vivid illustration comes from a study of Olympic medalists. Medvec et al. (1995) analyzed video footage from the 1992 Barcelona Olympics and found that bronze medalists appeared happier than silver medalists, both immediately after their event and during the medal ceremony. Independent raters, who did not know which medal each athlete had won, consistently rated bronze medalists as looking more pleased. If reactions depended only on objective rank, silver should always look better than bronze. But silver medalists naturally compare their outcome to gold, which they narrowly missed – and relative to that reference point, silver feels like a loss. Bronze medalists compare their outcome to finishing just outside the medals, the most salient alternative – and relative to no medal, bronze feels like a gain. The same mechanism operates in everyday life. A student who expected to fail an exam and gets a 6 out of 10 may feel elated, while a student who expected a 9 and gets an 8 may feel disappointed, even though the second student performed better.



Loss aversion – losses typically loom about twice as large as equivalent gains. Losing €50 produces more psychological impact than gaining €50. This asymmetry is the formal version of the everyday observation that opened this chapter. Loss aversion explains why people are reluctant to accept fair gambles. A coin flip that wins you €100 or loses you €100 has an expected value of zero, yet most people reject it because the potential loss looms larger than the potential gain. You would typically need to offer something like a coin flip between winning €200 and losing €100 before most people would accept, because only then does the weighted gain begin to match the weighted loss.

Diminishing sensitivity – each additional unit of gain or loss has a smaller psychological impact than the one before. Going from €0 to €100 feels like a big change; going from €1,000 to €1,100 feels like much less. This applies to losses as well: going from a loss of €0 to a loss of €100 is painful, but going from a loss of €1,000 to a loss of €1,100 adds relatively little additional pain. Graphically, the value function is concave (curving downward) for gains and convex (curving upward) for losses, producing an S-shaped curve that is steeper on the loss side.

Comparison with alternatives – a related line of research suggests that people evaluate outcomes not only against a fixed reference point like their current state, but also against the alternatives they could have had. Suppose you are choosing between two job offers. One pays €50,000, the other €60,000. You take the first and later learn the second would have been even better than advertised. The disappointment you feel does not stem from €50,000 being a bad salary in absolute terms – it stems from comparison with the forgone alternative. More generally, outcomes that fall below salient alternatives produce regret, while outcomes that exceed them produce relief. Because negative emotions like regret tend to be psychologically more powerful than positive emotions like relief, this mechanism reinforces the asymmetry captured by loss aversion. This idea extends beyond standard prospect theory into models of regret and counterfactual comparison, but it provides a psychological basis for why losses consistently outweigh gains in people's evaluations. As discussed in the chapter on the future self, this kind of affective response plays a role in many judgment contexts.

These properties are not merely theoretical. They produce several characteristic patterns in risky choice, including framing effects and the endowment effect.

Framing Effects

Because people evaluate outcomes relative to a reference point, the way a problem is described – its frame – can shift the reference point and thereby change decisions. The same objective situation can lead to opposite choices depending on whether outcomes are presented as gains or losses.

The most famous demonstration is the Asian disease problem (Tversky & Kahneman, 1981). Participants were told that a disease is expected to kill 600 people, and they must choose between two programs. In the gain frame, program A saves 200 people for certain, and program B gives a one-third chance of saving all 600 and a two-thirds chance of saving no one. In this version, 72% chose program A, the sure option. In the loss frame, program C means 400 people will die for certain, and program D gives a one-third chance that no one dies and a two-thirds chance that all 600 die. Here, 78% chose program D, the gamble. The two versions describe identical outcomes: saving 200 of 600 is the same as 400 dying. But the gain frame makes people risk averse (preferring the sure thing), while the loss frame makes people risk seeking (preferring the gamble).

Prospect theory explains this through the shape of the value function. For gains, the function is concave: the sure gain of saving 200 lives has higher subjective value than the expected value of the gamble (which also averages 200 lives saved, but with uncertainty). For losses, the function is convex: the certain loss of 400 lives feels very painful, and people would rather gamble for a chance to avoid any loss at all, even if the gamble could make things worse. This combination – risk aversion for gains and risk seeking for losses – is a direct prediction of the value function's shape and has been confirmed across many contexts, from medical decisions to financial choices.

Framing effects have real consequences. Tversky & Kahneman (1981) showed that even physicians displayed the same pattern of preference reversal. As discussed in the chapter on medical decision-making, framing plays a significant role in how health-related options are communicated and perceived. And as discussed in the chapter on steering other people's choices, the deliberate use of frames is a central tool in nudging people toward particular decisions.

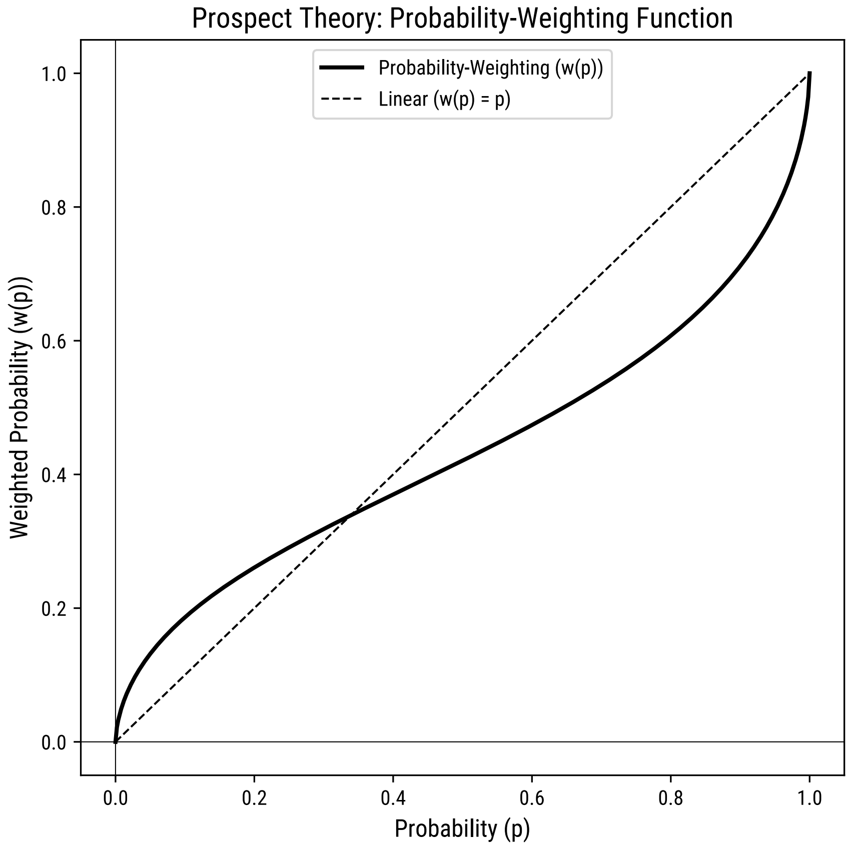
The Endowment Effect

Loss aversion also manifests in how people value things they already own. The endowment effect is the tendency to demand more to give up an object than one would pay to acquire it. In a classic experiment, Kahneman et al. (1990) randomly gave coffee mugs to half the participants in a university class. Those who received a mug (sellers) were asked for the minimum price at which they would sell it, and those without a mug (buyers) were asked for the maximum price they would pay. If ownership did not affect valuation, sellers and buyers should have agreed on roughly the same price, and about half the mugs should have changed hands. Instead, the median selling price was more than twice the median buying price, and far fewer mugs were traded than predicted by standard economic theory.

Giving up something you own is coded as a loss, while acquiring something new is coded as a gain. Since losses loom larger than gains, sellers require more compensation to part with their mug than buyers are willing to pay to get one. The endowment effect is closely related to the status quo bias, the preference for the current state of affairs over change. Both phenomena arise because moving away from the status quo involves a loss of what you currently have. You can read more about the role of the status quo in the chapter on steering other people's choices, where defaults and status quo bias are discussed as tools for influencing behavior.

The Probability Weighting Function

The second component of prospect theory is the probability weighting function, which describes how people transform stated probabilities into decision weights. In expected utility theory, a 10% chance is simply weighted by 0.10. In prospect theory, people do not use probabilities directly. Instead, they transform them through the weighting function w(p), which systematically distorts the impact of probabilities on decisions.



The weighting function has a characteristic inverse S-shape. Small probabilities are overweighted: a 5% chance receives more decision weight than 0.05, meaning it has more influence on the decision than it should. This explains why people buy lottery tickets. A one-in-ten-million chance of winning is objectively negligible, but people treat it as if it were much more likely. At the same time, moderate to high probabilities are underweighted: a 90% chance receives less decision weight than 0.90, meaning people treat near-certain outcomes as less certain than they really are. This explains the certainty effect that appeared in the Allais Paradox: the difference between 100% and 99% feels much larger than the difference between 50% and 49%, even though both are one-percentage-point differences.

A distinct phenomenon occurs at the extremes. Very small probabilities – such as the chance of being hit by a meteorite – are often rounded down to zero entirely. And very high probabilities are sometimes treated as certainties. This collapsing is different from the overweighting just described. The overweighting of small probabilities applies when a probability is small but still cognitively salient – noticeable enough to register as a real possibility. When a probability is so remote that it barely registers at all, people may ignore it completely, which can lead them to forego insurance or dismiss genuine risks because a small probability feels like no probability at all. As we will see shortly, there is yet a third pattern – probability neglect – where emotionally vivid outcomes cause people to disregard probability information regardless of its size. These three phenomena (overweighting, rounding to zero, and neglect) are distinct mechanisms, but all reflect failures to use probabilities as stated. As discussed in the chapter on risk perception, these distortions have practical implications for how people respond to low-probability, high-consequence events.

The combination of the value function and the probability weighting function produces what Tversky & Kahneman (1992) called the fourfold pattern of risk attitudes. For high-probability gains, people are risk averse (they prefer a sure gain over a gamble with slightly higher expected value). For low-probability gains, people are risk seeking (they prefer the gamble, as in buying lottery tickets). For high-probability losses, people are risk seeking (they gamble to avoid a sure loss). And for low-probability losses, people are risk averse (they pay to eliminate a small risk, as in buying insurance). This fourfold pattern emerges naturally from the interaction of the S-shaped value function with the inverse-S-shaped weighting function, and it captures a wide range of real-world behavior that expected utility theory cannot explain.

Beyond the Weighting Function

Subsequent research identified additional ways in which people depart from rational use of probability, extending beyond the original weighting function.

Subadditivity. When people judge the probability of a combined event (the probability of A or B occurring), they tend to assign it a lower value than the sum of the individual probabilities they would assign to A and B separately (Tversky & Fox, 1995). The parts seem to add up to more than the whole. Consider estimating why a flight might be delayed. If you separately judge the probability of delay due to weather, due to a mechanical problem, and due to air traffic congestion, each individual estimate receives focused attention, and the three probabilities might sum to, say, 65%. But if you simply judge the overall probability of "any delay," you might estimate only 40%. Considering each cause on its own gives it more psychological weight than when it is lumped together with other causes. Tversky & Fox (1995) demonstrated this pattern in experiments where people assessed uncertain events in domains they knew well, such as the outcome of a football match or future temperatures. When people evaluated the probability of each possible outcome one at a time, the probabilities summed to more than 100%. This subadditivity was stronger for uncertain events (where probabilities were unknown) than for risky events (where probabilities were stated explicitly), suggesting that vagueness amplifies the distortion.

Ambiguity aversion. People prefer known risks over unknown risks, even when the expected values are identical. Ellsberg (1961) demonstrated this with a thought experiment involving two urns. One urn contains 50 red balls and 50 black balls (a known risk). The other contains 100 balls, but the proportion of red to black is unknown (an ambiguous risk). When asked which urn they would rather bet on, most people prefer the urn with the known distribution, regardless of which color they are betting on. This preference cannot be explained by any consistent assignment of probabilities to the unknown urn: if you thought the unknown urn favored red, you should bet on it for red, and if you thought it favored black, you should bet on it for black. Yet people avoid the unknown urn for both bets, which means no coherent probability estimate is driving their choice. Under expected utility theory, a decision-maker who assigns any probability distribution to the unknown urn should not systematically avoid it. The preference reflects discomfort with not knowing the probabilities – a feeling of uncertainty about uncertainty. Ambiguity aversion helps explain real-world behavior such as the preference for familiar investments over foreign ones, or the tendency to buy insurance against well-defined risks while ignoring vaguely understood threats.

Probability neglect. In emotionally charged situations, people may almost entirely ignore probabilities and respond instead to the vividness or severity of the outcome. Baron et al. (1993) studied this in experiments where participants evaluated risks such as exposure to a toxic chemical. Participants were told the probability of harm varied – from very small to moderate – but their judgments of how much they would pay to avoid the exposure were barely affected by these differences in probability. What drove their responses was the nature of the outcome itself: how frightening or aversive the harm sounded. When the outcome is vivid enough, probability fades into the background. This is related to the affect heuristic discussed in the chapter on heuristics and biases: when emotions run high, the emotional intensity of an outcome dominates the judgment, and the actual likelihood becomes almost irrelevant.

Action, Omission, and the Source of Harm

Beyond the mechanisms captured by prospect theory and the distortions of probability just described, several additional biases shape how people evaluate risky outcomes based on the nature of the risk and whether harm results from action or inaction.

Omission bias. People tend to judge harmful consequences of inaction as less bad than equivalent harmful consequences of action. Baron & Ritov (2004) studied this extensively in the context of vaccination decisions. When a vaccine carries a small risk of serious side effects, some parents refuse to vaccinate their child, even when the disease itself poses a larger risk. The harm caused by vaccinating (an action) feels worse than the equal or greater harm caused by not vaccinating (an omission), because action is perceived as a more direct cause. This bias persists even when participants are explicitly told that the risks of inaction are greater than the risks of action. Omission bias is related to themes discussed in the chapter on morality, where the distinction between action and inaction plays a central role in moral judgment, as in the trolley problem.

Nature bias. People tend to judge harms from artificial causes as worse than equivalent harms from natural causes (Kahneman & Ritov, 1994). For example, a synthetic chemical additive in food may be judged as more dangerous than a naturally occurring toxin at the same level of objective risk. This bias influences consumer behavior (preferences for "natural" products), environmental policy (stricter regulation of synthetic chemicals than naturally occurring ones), and medical decisions (skepticism toward synthetic medications compared to herbal remedies). One possible explanation is that people apply something like an intentional stance to artificial products – as if someone chose to introduce the harm – whereas natural harms are attributed to impersonal forces. This may connect to ideas about the intentional stance discussed in the chapter on mental models, though the link remains speculative.

A Bayesian Lens

Throughout this book, we use the framework of Bayesian reasoning as a benchmark for rational belief updating. In decision contexts, the complementary benchmark is expected utility theory: use probabilities coherently, combine them with explicit tradeoffs among outcomes, and remain consistent regardless of how a problem is described. Prospect theory documents a series of systematic departures from both of these ideals.

The probability weighting function means that people do not use probabilities as stated. Small probabilities are inflated and moderate probabilities are deflated, producing a distorted version of the likelihood information that a Bayesian agent would use faithfully. Subadditivity compounds this by making individual event assessments add up to more than the whole. Ambiguity aversion means that the confidence in one's probability estimates affects choices in ways that neither Bayesian updating nor expected utility theory would predict, since a rational agent should simply work with whatever probability distribution is available. And probability neglect means that in emotional situations, probability information may be discarded entirely – the opposite of what any normative framework requires.

On the outcome side, framing effects mean that logically equivalent descriptions can produce different decisions. A rational decision-maker should be indifferent to whether a medical treatment is described as having a 90% survival rate or a 10% mortality rate, because these carry the same information. But as we have seen, people are not indifferent to such reframings. The reference point determines what counts as a gain or a loss, and shifts in this reference point – whether caused by the wording of a question, recent experiences, or the set of available alternatives – reshape the entire evaluation. This is a direct violation of description invariance, the principle that equivalent representations of the same problem should lead to the same choice.

Summary

This chapter traced the development from expected value theory through expected utility theory to prospect theory, showing why each step was necessary to explain how people actually make decisions under risk. Prospect theory, developed by Kahneman and Tversky, describes decisions through two components. The value function captures how people evaluate outcomes as gains or losses relative to a reference point, with losses looming about twice as large as equivalent gains, and with diminishing sensitivity to larger amounts. The probability weighting function captures how people transform probabilities into decision weights, overweighting small probabilities and underweighting moderate-to-high ones. Together, these components explain a wide range of phenomena: the certainty effect, framing effects, the endowment effect, and the fourfold pattern of risk attitudes. Beyond prospect theory, people also show subadditivity in probability judgments, aversion to ambiguous risks, and neglect of probabilities when emotions are strong. Additional biases favor inaction over action and natural over artificial causes when evaluating harm. From a Bayesian and expected-utility perspective, these patterns represent systematic distortions in both how probabilities are used and how outcomes are evaluated – departures that have real consequences for decisions in medicine, policy, and everyday life.

SATISFICING VS MAXIMIZING

Learning goals:

- Explain the difference between satisficing and maximizing as decision-making strategies.
- Describe the maximizing paradox and why striving for the best option can reduce well-being.
- Distinguish maximizing as a strategy from merely having high standards.
- Explain choice overload and describe the evidence for it, including its boundary conditions.
- Relate satisficing to the concept of bounded rationality and resource-rational decision-making.

Good Enough vs. the Best

Imagine you need a new pair of headphones. You could walk into a store, try a few pairs, and buy the first one that sounds good and fits comfortably within your budget. Or you could spend the next three weeks reading reviews, comparing specifications, visiting multiple stores, and monitoring price-tracking websites until you are confident you have found the absolute best headphones for the money. Both are reasonable approaches, but they lead to very different experiences. The first shopper goes home satisfied. The second may eventually buy a slightly better pair but spend the evening wondering whether the model they almost chose was actually superior.

This chapter is about these two ways of making choices. The distinction between them has deep consequences – not just for the quality of the decision, but for how the decision-maker feels afterward.

Satisficing

We already encountered the concept of satisficing when we discussed bounded rationality in Reason and Intuition (Simon, 1956).
Herbert Simon, one of the most influential thinkers in the study of decision-making, argued that classical economic models assume people are perfectly rational agents who evaluate all available options, compute the expected value of each, and select the one that maximizes their utility. This assumption, Simon pointed out, is unrealistic. Real people face severe informational and computational limits. They cannot evaluate every option, and they often do not even know what options exist.

Simon proposed that instead of optimizing, people satisfice. The word is a blend of "satisfy" and "suffice." A satisficer sets a threshold for what counts as acceptable, searches through the available options, and stops as soon as an option meets that threshold. A University of Groningen student looking for a place within 15 minutes of the university and below €700 per month will take the first apartment that checks all three boxes. They do not need to see every apartment in the Groningen. You can think of this as replacing the question "which option is the best?" with "is this option likely to exceed my threshold?" Once one passes, the search ends.

Satisficing is an efficient adaptation to a world where the cost of searching for the perfect option often outweighs the benefit of finding it. Simon often illustrated bounded rationality with search problems in which simple stopping rules outperform unrealistic optimization assumptions. Even very basic rules – such as "keep looking until you find something that meets your needs, then stop" – work remarkably well in most environments. The structure of the environment does much of the work, making exhaustive optimization unnecessary.

Maximizing

The opposite of satisficing is maximizing. A maximizer strives to find the best possible option. This means evaluating as many alternatives as possible, comparing them against each other, and selecting the one that comes out on top. On the surface, this sounds like the rational thing to do. If you are buying a laptop, why would you settle for a decent one when a better one might be just a few clicks away?

The problem is that maximizing is expensive. It demands time, effort, and attention. It requires the decision-maker to keep track of all the options they have considered and to make difficult comparisons between options that may differ on many dimensions. And crucially, it exposes the decision-maker to all the options they did not choose. A satisficer who found a good apartment never thinks about the hundreds of apartments they did not visit. A maximizer, having reviewed dozens of listings, is acutely aware of the charming place near the canal that was just slightly over budget, or the spacious studio that was a bit too far from the university. This awareness of forgone alternatives invites regret and counterfactual thinking – the nagging sense that a different choice might have been better.

The Maximizing Paradox

A key finding from research by Schwartz et al. (2002) is what might be called the maximizing paradox: maximizers often aim for, and in some domains obtain, better outcomes on external criteria such as salary, but they frequently feel worse about those outcomes.

Think about this in the context of job searching. A maximizer who applies to dozens of positions, carefully negotiates salary, and compares every offer may end up with a higher-paying job than a satisficer who accepted the first reasonable offer. But the maximizer is prone to counterfactual thinking. What if the other company would have been a better fit? What if they had negotiated harder? The satisficer, meanwhile, is content. The job met their standards, and that is enough.

Schwartz and colleagues developed a Maximization Scale to measure individual differences in the tendency to maximize. Across several studies involving over 1,700 participants, the results were consistent: higher maximizing scores were associated with lower life satisfaction, lower happiness, lower optimism, and lower self-esteem, as well as higher levels of depression, perfectionism, and regret. These associations held even after controlling for other personality traits.

One of their studies tested whether maximizers are more sensitive to social comparison. Participants solved anagrams alongside a confederate who either solved them faster or slower. Maximizers who saw the faster confederate reported bigger drops in self-assessed ability and bigger increases in negative mood compared to satisficers in the same condition. When the confederate was slower, maximizers and satisficers did not differ much. This suggests that maximizers are especially vulnerable to upward social comparison – constantly measuring themselves against others who might be doing better.

An important clarification: maximizing is not the same as having high standards. A person can have very high standards and still be a satisficer. They simply set a high threshold for what counts as good enough. A food critic who only dines at restaurants with excellent reviews is satisficing at a high level. They are not trying to find the single best restaurant in the city; they are happy with any excellent one. The distinction is between the strategy of exhaustive search (maximizing) and the level at which the threshold is set (standards).

This distinction was further developed by Vargová et al. (2020), who studied 514 participants in Slovakia. They distinguished between maximizing as a goal (having high standards) and maximizing as a strategy (exhaustively searching through alternatives). The results were revealing. Maximizing as a strategy was associated with lower happiness, higher depression, and higher neuroticism, replicating the pattern found by Schwartz and colleagues. But maximizing as a goal showed no such negative associations with well-being. This means that the harm does not come from wanting the best; it comes from the exhaustive process of searching for it. Interestingly, satisficing in this study was unrelated to well-being, suggesting that it may be the absence of exhaustive search, rather than the presence of a “good enough” mindset, that protects people from the costs of maximizing.

Choice Overload

So far we have focused on strategies people bring to a decision. Choice overload highlights how the structure of the environment can magnify those differences. The maximizing paradox becomes especially dramatic when the number of options increases. In theory, more options should be a good thing. A larger menu gives you a better chance of finding something you like. A university with 200 elective courses offers more opportunities to match your interests than one with 20. But in practice, an abundance of options can make choosing more difficult or even prevent a decision altogether.

Iyengar & Lepper (2000) demonstrated this in a now-classic series of studies. In their first study, they set up a tasting booth at an upscale grocery store in California. On some days, the booth displayed 24 varieties of jam. On other days, it displayed only 6. The large display attracted more shoppers: 60% of passersby stopped to look, compared to 40% for the small display. But when it came to actually buying jam, the pattern reversed dramatically. Of those who encountered the 6-jar display, 30% purchased a jar. Of those who encountered the 24-jar display, only 3% did. More options attracted more attention but led to far fewer decisions.

The researchers found similar effects in other contexts. In a second study, college students were offered the chance to write an extra-credit essay. Some could choose from 6 topics, others from 30. Students with fewer topics to choose from were more likely to complete the essay and wrote better essays. In a third study, participants chose a chocolate from either 6 or 30 options. Those who chose from 30 found the process more enjoyable but also more difficult and frustrating. Afterward, they were less satisfied with their chosen chocolate and more likely to prefer cash over chocolate as compensation for participating. More choice increased both the appeal and the cost of deciding.

Later research has shown that choice overload is not inevitable. The effect appears most reliably when options are difficult to compare, when preferences are uncertain, or when the decision feels consequential. In mature product categories where options are easily ranked – say, choosing between internet plans that differ mainly in price and speed – more options may genuinely help. But in domains where comparisons are complex and preferences unclear, the original finding holds up well.

This pattern makes intuitive sense once you consider the demands of maximizing. If you are trying to find the best option out of 6, the task is manageable. If you are trying to find the best out of 30, the number of comparisons explodes. You need more time, more effort, and more working memory. And the broader your search, the more counterfactual comparisons become available, as discussed above. Satisficers are often less vulnerable to choice overload because their strategy does not require evaluating every option. Whether there are 6 or 600 jams on the shelf, the satisficer grabs the first strawberry jam that looks decent and moves on.

Choice overload is relevant far beyond grocery shopping. Students choosing between dozens of study programs, patients navigating complex health insurance plans, or consumers comparing mobile phone contracts all face versions of this problem. The modern world offers an unprecedented abundance of options in nearly every domain, from streaming services with thousands of titles to dating apps with millions of profiles. For maximizers, this abundance is not liberating but overwhelming. This is one reason choice architects often simplify menus and set smart defaults – a topic explored in Steering Other People's Choices.

Satisficing, Rationality, and the Prescriptive Case

The distinction between satisficing and maximizing reflects a fundamental tension in decision-making theory. As discussed in How It Is vs. How It Should Be, normative models describe how decisions should be made under ideal conditions. Under a classical model that ignores search costs, maximizing seems like the right approach: if evaluating options were costless, you should compare every option and select the one with the highest expected value. But once search itself takes time and effort, even a rational agent may prefer a stopping rule to exhaustive comparison.

This is the prescriptive insight. In many everyday environments, the difference between the best option and a good option is small, while the cost of finding the best is large. Consider rental apartments in a narrow price range: the gap between the best apartment and the fifth-best apartment may be trivial compared to the weeks of searching and the stress of comparing dozens of listings. A satisficing strategy captures most of the benefit at a fraction of the cost. This echoes the broader principle of ecological rationality discussed in Reason and Intuition: simple strategies can perform remarkably well when they are matched to the structure of the environment.

The connection to the exploration-exploitation tradeoff, which we will discuss in Confirmation, is also instructive. In that context, the tradeoff concerns whether to keep searching for evidence that might change your beliefs or to commit to a working conclusion. The same logic applies to choice. A student choosing electives can keep browsing the course catalog indefinitely (exploration) or register once several courses meet their goals (exploitation). Maximizers tend to remain in exploration mode longer than is warranted; satisficers shift to exploitation once they find something good enough. Both strategies have their place, but in a world with abundant options and limited time, the costs of perpetual exploration often outweigh its benefits.

Of course, satisficing is not equally wise in all contexts. For routine decisions among similar options – which brand of rice to buy, which route to take to work – settling for good enough is a clear win. For high-stakes, difficult-to-reverse decisions – choosing a surgeon, a mortgage, or a treatment plan – more thorough comparison may well be worth the cost. The key insight from Simon is not that people should never search extensively, but that the default assumption of exhaustive optimization is a poor model of how people do, or should, decide.

A Bayesian Lens

The contrast between maximizing and satisficing can be formalized through the framework of Bayesian reasoning introduced in Bayesian Reasoning. A perfect maximizer, in Bayesian terms, would compute the posterior expected value of every available option – integrating all available evidence about each option's quality – and then select the option with the highest expected value. This is essentially a full Bayesian decision analysis.

Satisficing replaces this computationally expensive procedure with a simpler rule. Instead of asking "which option has the highest expected value?", the satisficer asks "is the posterior probability that this option is good enough above my threshold?" This requires updating beliefs about only one option at a time, rather than comparing all options simultaneously. When the option passes the threshold, the search stops. When it does not, the satisficer moves on.

A concrete example helps make this tangible. Suppose you are choosing a used car. A maximizer might try to estimate the expected reliability, price, fuel economy, and resale value of every available car, rank them all, and select the best. A satisficer might instead ask of each car they encounter: is it probably reliable enough, affordable enough, and efficient enough to meet my needs? Once one car crosses those thresholds, the search ends – even though some unexamined car on the lot might score a fraction higher.

This is what decision theorists call a resource-rational strategy. For routine decisions among roughly similar options, it trades a modest expected loss in decision quality for a large saving in cognitive effort. The satisficer gives up the small chance of finding the absolute best option but avoids the substantial costs of exhaustive search: the time, the effort, the regret, and the paralysis. From this perspective, satisficing is not irrational. It is rational given the constraints under which real minds operate.

Summary

When faced with choices, people differ in whether they seek the best possible option (maximizing) or a good-enough one (satisficing). Satisficing, introduced by Simon (1956), is a strategy in which the decision-maker sets an acceptability threshold and stops searching once an option meets it. This approach follows directly from bounded rationality: under realistic constraints, exhaustive evaluation is rarely possible and often counterproductive. Maximizers, who strive for the best, may in some domains achieve better outcomes on external criteria, but they experience more regret, less satisfaction, and lower well-being. The Maximization Scale developed by Schwartz et al. (2002) captures these individual differences, and subsequent research by Vargová et al. (2020) has shown that it is the strategy of exhaustive search, rather than the goal of having high standards, that drives the negative effects. Choice overload, demonstrated by Iyengar & Lepper (2000) in their jam study and replicated with boundary conditions in later work, shows that an abundance of options can make decision-making more difficult and reduce satisfaction, particularly among maximizers. In a world of ever-expanding options, the satisficer's willingness to settle for good enough turns out to be not a compromise but a wise adaptation.

SATISFICING VS MAXIMIZING

STEERING OTHER PEOPLE'S CHOICES

Learning goals:

- Explain the concept of libertarian paternalism and why no choice environment is neutral.
- Describe how defaults exploit the status quo bias to influence decisions.
- Distinguish between different types of nudges and evaluate their effectiveness.
- Explain mental accounting and the principle of fungibility.
- Describe how commitment devices help people align their behavior with their long-term goals.
- Contrast nudges with mandates, bans, and economic incentives.

The Inevitability of Choice Architecture

Imagine you are registering for a new online service. At the bottom of the sign-up page, a checkbox reads: "I would like to receive promotional emails." In some versions of the page, this box is pre-checked; in others, it is empty. The information is the same, the choice is the same, and you are free to check or uncheck the box. Yet the two versions of this page will produce very different subscription rates. This simple observation – that presentation shapes choice – is the starting point for understanding choice architecture.

Every time someone presents you with a set of options, that presentation involves decisions. Which option appears first? What is pre-selected? How are the options described? These design decisions are unavoidable. A university cafeteria must put some dish at the front of the line and some dish at the back. A pension plan must either enroll employees automatically or require them to sign up. A government form must either list organ donation as something you opt into or something you opt out of. There is rarely a neutral arrangement: any presentation must order, label, or preselect options in some way. The person who designs the choice architecture – sometimes called the choice architect – always influences what people end up choosing, whether they intend to or not.

This observation creates a tension. On the one hand, most people agree that individuals should be free to make their own decisions. On the other hand, the design of the choice architecture will inevitably push people in one direction or another. Libertarian paternalism is a philosophy that embraces this tension rather than ignoring it. Thaler & Sunstein (2003) coined a straightforward idea: since the arrangement of choices will influence behavior regardless, institutions should design choice architectures that steer people toward outcomes that are good for them, while preserving the freedom to choose differently. The "libertarian" part is that people remain free to choose otherwise. The "paternalism" part is that the choice architecture is intentionally designed to promote welfare. The question, according to this view, is not whether to influence people's choices, but how to do so responsibly.

Defaults and the Status Quo Bias

The most powerful nudge is often the default – the option that takes effect if a person does nothing. Defaults are so effective because of a robust psychological tendency known as the status quo bias: people prefer things to stay as they are. Changing from the current state feels like an active decision, and active decisions require effort, attention, and a willingness to take responsibility for the outcome. Sticking with whatever is already in place feels safe and easy. As a result, pre-selected options tend to stick.

The most striking demonstration of the power of defaults comes from organ donation policy. Johnson & Goldstein (2004) compared organ donation consent rates across European countries that use different default systems. In countries with an opt-in system, where citizens must actively register as organ donors, consent rates are low. Germany, for example, had a consent rate of about 12%. In countries with an opt-out system, where citizens are presumed to consent unless they actively register their objection, consent rates are dramatically higher. Austria, Germany's neighbor with a similar culture and comparable healthcare infrastructure, had a consent rate approaching 100%. The underlying attitudes toward organ donation in these countries are not radically different. The difference in consent rates is driven almost entirely by the default.

To confirm that this pattern is not simply a quirk of cross-country differences, Johnson & Goldstein (2004) also ran a controlled online experiment. Participants were randomly assigned to one of three conditions: an opt-in condition (where they had to actively choose to be a donor), an opt-out condition (where they had to actively choose not to be a donor), or a neutral condition (where neither option was pre-selected). Donation rates were nearly twice as high in the opt-out condition compared to the opt-in condition. This suggests that requiring active enrollment can substantially reduce participation relative to conditions in which no option is preselected or opting out is required.

Why are defaults so sticky? Several mechanisms contribute. First, changing the default requires effort, even if that effort is minimal. You need to find the right form, make a phone call, or uncheck a box. Many people simply never get around to it. Second, the default can function as an implicit recommendation. If the government has pre-selected a particular option, people may interpret this as a signal that it is the sensible or socially expected choice. Third, changing from the default means giving up the default option, and as discussed in the chapter on Gains and Losses, loss aversion makes people reluctant to give up what they feel they already have.

Ease and visibility

Defaults are the most studied nudge, but they are not the only one. A nudge is any change to the choice architecture that predictably influences behavior without forbidding any options or significantly changing economic incentives.

One category of nudge involves increasing the ease and visibility of the desired option. A cafeteria that places fruit at eye level and cake on a lower shelf does not remove cake from the menu, but it makes fruit the path of least resistance. A stairwell with clear signage and an attractive design nudges people away from the elevator. These interventions work by reducing the friction associated with the desired behavior and increasing the friction associated with the undesired behavior. However, it is worth noting that the effectiveness of such interventions varies. Lin et al. (2017) found that some commonly discussed nudges of this type, such as using smaller plates to reduce food intake or painting footprints on the floor to encourage stair use, had limited or inconsistent effects in empirical studies. Nudges are not magic. Their success depends on the specific behavior being targeted, the population, and the broader context.

Social norms

People are strongly influenced by what others around them do. Letters that inform people that most of their neighbors have already complied with a request – such as paying taxes on time – can increase compliance, especially when the norm is local and credible. This kind of nudge works because people use the behavior of others as a guide for their own actions, particularly when they are uncertain about what to do. It also introduces a mild social pressure: nobody wants to be the odd one out.

Educational nudges

A third category involves educational nudges that provide vivid, accessible information to shift how people think about their options. Graphic health warnings on tobacco packaging are a prominent example. Pang et al. (2021) conducted a systematic review of longitudinal studies on the effectiveness of graphic health warnings. They found that such warnings were generally effective at increasing perceived harm and quit intentions among smokers. However, the review also documented wear-out effects: the impact of warnings declined over time as people became desensitized. This finding highlights an important limitation of educational nudges. They can be effective in the short term, but sustaining their impact may require periodic updates, such as rotating new images onto packaging.

Lin et al. (2017) offer a useful distinction between nudges that operate with minimal disruption, subtly adjusting the environment without prompting conscious reflection (such as reducing plate sizes), and nudges that explicitly encourage people to re-evaluate their choices (such as traffic-light nutrition labels that clearly signal whether a food is healthy). Their review suggests that the latter type may produce more sustained behavioral change in some domains, because it engages people in deliberate thinking about their options rather than relying solely on automatic processes. This distinction connects to themes from the chapter on Reason and Intuition: some interventions work by exploiting automatic tendencies, whereas others invite more deliberate reflection.

Mental accounting and fungibility

Choice architects do not only arrange physical options; they also structure how resources are labeled, partitioned, and accessed. To understand this set of tools, we need to examine how people think about money.

Standard economic theory treats money as fungible: one euro is the same as any other euro regardless of where it came from or what label is attached to it. A 50-euro bonus and 50 euros of salary should have the same effect on what one can afford. In practice, people violate this principle of fungibility constantly.

Thaler (1999) describes a set of cognitive processes called mental accounting. People mentally sort their money into separate categories – "rent," "groceries," "entertainment," "savings" – and then treat each category as if it were a separate bank account with its own budget. Money in the "entertainment" account does not feel interchangeable with money in the "rent" account, even though it objectively is. Consider a classic example: a person who has already bought a 50-euro theatre ticket and then loses it on the way to the show will often refuse to buy a replacement, feeling that 100 euros for a single evening is too much from the "entertainment" budget. But a person who planned to buy the ticket at the door and lost a 50-euro note on the way will usually still buy the ticket, because the lost cash comes from a different mental account. In both cases the person is 50 euros poorer and faces the same choice – pay 50 euros to see the show or go home – but mental accounting makes the situations feel very different.

The source of money also matters. Shefrin & Thaler (1988) proposed the Behavioral Life-Cycle hypothesis, which predicts that people treat income from different sources differently. Regular salary is one mental account; a year-end bonus is another; and long-term savings such as a pension fund are yet another. The marginal propensity to consume – meaning the fraction of each additional euro that gets spent rather than saved – differs across these accounts. People are most likely to spend current income, somewhat less likely to spend a windfall or bonus, and very reluctant to dip into long-term savings. In a survey of MBA students, Shefrin & Thaler (1988) found that participants reported they would spend a much larger fraction of a salary increase than of a lump-sum bonus of equivalent size, even though both represent the same addition to their total wealth.

This pattern violates the standard economic assumption of fungibility, but it has a useful side. Mental accounting can serve as a self-control device. By mentally labeling certain funds as "off-limits," people resist the temptation to spend them on short-term desires. The challenge of saving for retirement, for example, is essentially a self-control problem: the immediate pleasure of spending competes with the distant benefit of having money available decades from now. Shefrin & Thaler (1988) describe this as a conflict between an internal "planner," who cares about long-term well-being, and an internal "doer," who is focused on immediate gratification. Mental accounting helps the planner win by making certain money feel unavailable for current spending.

Commitment devices

Choice architects can exploit mental accounting to steer behavior. The key insight is that labeling money changes how people treat it, even when the label carries no real constraint.

Consider earmarked subsidies and vouchers. A government that gives families a 200-euro voucher specifically for children's educational materials will see most of that money spent on books and school supplies. If the same government simply added 200 euros to each family's bank account, a significant portion might be diverted to other expenses. The voucher works not because families cannot technically spend their own money on other things – the voucher frees up 200 euros of their existing budget that they could redirect elsewhere – but because the label "educational materials" creates a mental account that people feel compelled to honor. In many cases, earmarked vouchers are fully fungible in practice: a family that was already planning to spend 200 euros on school supplies could pocket the voucher and redirect their own 200 euros to something else entirely. Yet the psychological power of the label means that most families do not make this substitution. The label functions as if it were a binding constraint, even when it is not.

Commitment devices take this principle a step further by helping people align their behavior with their long-term goals. A commitment device is an arrangement that makes it harder for a person to act against their own long-term interests. Automatic pension deductions are a classic example. By routing a percentage of each paycheck directly into a retirement account before the employee ever sees it, the money enters the "future income" mental account rather than the "current income" account. Because people are reluctant to withdraw from savings, the money stays put. Shefrin & Thaler (1988) found support for this prediction: increases in pension contributions are not offset by equivalent decreases in other saving, as standard economic theory would predict. Instead, pension contributions genuinely increase total saving, because the money is placed in a mental account from which withdrawal feels psychologically costly. This connects to themes discussed in the chapter on The Future Self: people systematically undervalue outcomes for their future selves, and commitment devices can help bridge that gap.

The same logic underlies many everyday self-control strategies. A student who sets up an automatic transfer of 50 euros per month into a savings account labeled "summer travel" is using a commitment device. The label creates a mental account, and the automation ensures the money is allocated before the temptation to spend it arises. Apps that "round up" purchases and deposit the difference into savings, or employer schemes that automatically increase retirement contributions each year, work on the same principle.

Beyond Nudges: Mandates, Bans, and Economic Incentives

Nudges preserve freedom of choice: no options are removed, and people can always opt out. But not all interventions aimed at steering behavior are nudges. At the other end of the spectrum lie mandates and bans, which remove options entirely. A law requiring motorcyclists to wear helmets is not a nudge; it is a mandate. A ban on smoking in restaurants does not make non-smoking the default while allowing smokers to light up if they wish; it eliminates the option altogether. Similarly, economic incentives such as taxes and subsidies change the costs and benefits of options without removing them. A sugar tax makes sugary drinks more expensive, steering consumers toward water or unsweetened alternatives, but it does so by altering economic incentives rather than by subtly restructuring the choice architecture.

These stronger interventions are often more effective than nudges at changing behavior. Tobacco taxes, for instance, have a well-documented impact on smoking rates that is larger and more consistent than the impact of graphic warnings alone. But they sacrifice the "libertarian" component of libertarian paternalism. A ban constrains rather than steers. A tax imposes costs on people who might have good reasons for their choices. This trade-off between effectiveness and freedom is at the heart of many policy debates, from public health to environmental regulation.

In practice, most policy environments use a mix of tools. The Netherlands, for example, combines nudge-like interventions (such as nutritional labeling) with economic incentives (such as taxes on alcohol and tobacco) and outright bans (such as the prohibition of smoking in public indoor spaces). The choice of tool depends on how serious the consequences of the behavior are, how effective gentler interventions have proven to be, and how much value society places on individual freedom in that particular domain.

Bayesian Lens

From a Bayesian perspective, choice architecture manipulates the prior. A default sets a strong initial position that most people never update away from. This is functionally equivalent to starting with a heavily weighted anchor, as discussed in the chapter on Anchoring and Primacy. The status quo bias can be understood as an asymmetry in the updating process: maintaining the current state requires no new evidence, while switching to an alternative requires overcoming an evidentiary threshold. Normatively, a default should matter only insofar as it provides genuine information – such as an expert recommendation – or alters real costs. Much of its observed influence seems to exceed that informational value, driven instead by inertia and inferred endorsement. A rational agent who is genuinely indifferent between two options should be equally likely to choose either one regardless of which is pre-selected. The fact that defaults have such large effects reveals that people respond to them more strongly than warranted.

Mental accounting represents another departure from rational norms. The label on a sum of money – whether it is called "salary," "bonus," or "savings" – carries no information about how that money should optimally be spent. Yet people behave as if the label shifts their spending decisions. A "retirement account" label feels informative even when the money is financially identical to money in a checking account. In Bayesian terms, the label functions as a kind of pseudo-evidence that updates the person's inclination to spend or save, even though it should have a likelihood ratio of exactly one – meaning it should leave the prior unchanged. Nudges work precisely because people are not fully Bayesian. Understanding these departures from rational updating is what allows choice architects to design environments that guide people toward better outcomes.

Summary

Every choice architecture is designed by someone, and every arrangement of options steers behavior in some direction. Libertarian paternalism holds that institutions should harness this influence to promote welfare while preserving the freedom to opt out. The most powerful nudge is the default: pre-selected options tend to stick because of the status quo bias, as demonstrated dramatically by the difference between opt-in and opt-out organ donation systems across Europe. Other nudges include increasing the ease and visibility of desired options, leveraging social norms, and providing vivid educational information – though their effectiveness varies across contexts and can diminish over time. Mental accounting, the tendency to sort money into non-fungible categories, violates standard economic assumptions but can be harnessed through earmarked vouchers and commitment devices like automatic pension deductions that help people align their behavior with their long-term goals. At the other end of the intervention spectrum, mandates, bans, and economic incentives are often more effective but sacrifice freedom of choice. Understanding how these tools work, and when each is appropriate, is essential for anyone involved in designing policies, institutions, or everyday choice environments.

STEERING OTHER PEOPLE'S CHOICES

GROUP DECISIONS

Learning goals:

- Explain how aggregating independent judgments can produce accurate estimates (the wisdom of crowds) and identify the conditions required for this to work.
- Describe group polarization and distinguish between its two driving mechanisms (persuasive arguments and social comparison).
- Define groupthink and explain how it differs from group polarization in the way it produces extreme group positions.
- Describe the shared information bias and explain why groups often fail to pool unique knowledge held by individual members.
- Identify practical strategies for improving group decision-making.

When Decisions Are Made Together

Think of the last time you worked on a group project for a university course. Perhaps you divided the work, came together to discuss your findings, and tried to reach agreement on what to include in the final product. These discussions do not always go smoothly. Sometimes the group lands on a better answer than any single member would have reached alone. Other times, the loudest voice dominates, important information gets overlooked, and the final product is worse than what any competent member could have produced individually. These everyday experiences reflect deep patterns in how groups make decisions, patterns that also shape outcomes in corporate boardrooms, hospital teams, courtrooms, and parliaments. Understanding when groups succeed and when they fail is one of the most practically important topics in the psychology of judgment and decision-making.

The Wisdom of Crowds

In 1906, the British scientist Francis Galton attended a livestock exhibition in Plymouth, England. Visitors could purchase a ticket and guess the weight of an ox after it had been slaughtered and dressed. Around 800 people entered the competition, ranging from expert butchers and farmers to people with no particular knowledge of livestock. Galton collected the entries and calculated the median guess. The true dressed weight of the ox turned out to be 1,198 pounds. The median guess was 1,207 pounds, an error of less than 1 percent (Galton, 1907). Most individual guesses were far less accurate. The crowd, taken as a whole, was remarkably close to the truth.

This phenomenon is known as the wisdom of crowds. The basic idea is that when you aggregate many independent judgments, the resulting average (or median) is often more accurate than the judgment of any single individual, including experts. The mechanism behind this will be familiar to anyone who has taken a statistics course, and we've also encountered it when discussing models of the judge (MUDs) in Statistical vs Intuitive Prediction. Each individual's judgment can be thought of as the true value plus some error. Some people overestimate, others underestimate, and if these errors are independent of each other, they tend to cancel out when you average across many people. The group average is like a sample mean: as the number of independent observations increases, the mean converges on the true value. This resembles the logic of the law of large numbers: when many independent estimates contain partly random error, averaging reduces that error. However, a large crowd can still be systematically wrong if everyone shares the same bias. The error-canceling magic depends on those errors being at least partly independent and not all pointing in the same direction.

For the wisdom of crowds to function, three conditions must be met. First, there must be diversity of opinion. If everyone in the crowd has the same background, the same information, and the same perspective, their errors will not be independent; they will all be wrong in the same direction. Second, judgments must be independent. Each person must form their own estimate without being influenced by others. If people can see each other's guesses, they tend to converge on a common answer, which eliminates the error-canceling benefit of aggregation. Third, there must be a mechanism for aggregation, such as taking the average or the median. Without aggregation, you just have a collection of individual guesses, not a collective judgment.

The wisdom of crowds breaks down when any of these conditions is violated. The most common threat is a loss of independence. When people discuss their estimates before committing to them, or when they can see a running average of previous guesses (as on some online platforms), their judgments become correlated. In statistical terms, correlated observations do not reduce noise as effectively as independent ones. This is exactly the same reason that, in a research study, you cannot simply count the same participant twice and claim a larger sample size. If the "observations" are not independent, the benefits of aggregation disappear. Several of the group phenomena discussed later in this chapter, including groupthink and group polarization, represent precisely this kind of breakdown.

The wisdom of crowds has practical applications beyond livestock exhibitions. Prediction markets, where people trade contracts based on their beliefs about future events, exploit this principle by aggregating the independent judgments of many participants into a market price that serves as a probability estimate. For example, election forecasting markets allow traders to buy and sell contracts that pay out if a particular candidate wins; the market price of such a contract reflects the crowd's collective estimate of that candidate's probability of winning, and these estimates have historically been competitive with professional polling models. Similarly, some organizations use anonymous polling of their employees to forecast project timelines or product demand, avoiding the social pressures that would distort estimates in a meeting. These applications work best when the conditions of diversity and independence are actively protected.

Group Polarization

You might expect that when a group of people discusses an issue, their views would moderate. Each person would hear different perspectives, and the group would settle on some reasonable middle ground. Research shows that the opposite often happens.

Moscovici & Zavalloni (1969) conducted a series of experiments with secondary school students in Paris. Participants first individually rated their attitudes toward political figures and national groups using rating scales. They then discussed these topics in small groups, aiming to reach a group consensus. After the discussion, each participant again privately rated their attitudes. The results were clear: after discussion, individual attitudes had shifted toward a more extreme version of whatever position had been dominant before the discussion. If most group members initially held mildly positive attitudes toward a political figure, they held strongly positive attitudes afterward. If most initially held mildly negative attitudes, they held strongly negative attitudes afterward. This shift was not just public conformity. The participants changed their private views.

Group polarization does not mean that groups always become more extreme in some absolute sense. It means the group shifts further in the direction that was already dominant among its members before the discussion. A group of slightly cautious individuals becomes more cautious. A group of slightly risk-tolerant individuals becomes more risk-tolerant. The discussion amplifies the pre-existing tendency.

Two mechanisms drive group polarization. The first is the persuasive arguments mechanism. During discussion, members encounter arguments they had not previously considered. In groups that already lean in one direction, discussion often surfaces more arguments supporting that side than opposing it. Hearing new, persuasive arguments that support the dominant view changes people's minds. The second mechanism is social comparison. People want to be seen as embodying the group's valued position. When they discover during discussion that others hold views similar to their own, they shift further in that direction to distinguish themselves favorably. In effect, each member competes to be a better exemplar of the group's identity.

Consider a like-minded investment club whose members are mildly enthusiastic about a risky stock. During discussion, members hear several new reasons to be bullish that they had not individually considered – a recent patent, a favorable regulatory change, an upcoming product launch (persuasive arguments). At the same time, each member notices that the others are enthusiastic and signals commitment to the group's identity as savvy risk-takers by taking an even bolder stance (social comparison). By the end of the meeting, the group is ready to make a much larger bet than any member would have made alone.

Both mechanisms operate simultaneously, though their relative contribution depends on the situation. For issues that are rich in factual arguments, the persuasive arguments mechanism tends to dominate. For issues that are more about values and identity, social comparison plays a larger role.

Group polarization has real consequences beyond investment clubs. A committee of hiring managers who individually have a slight preference for a candidate may, after discussion, become strongly committed to that candidate while overlooking weaknesses. In political contexts, people who discuss issues only with like-minded others tend to adopt more extreme positions over time, a dynamic that connects to broader patterns of polarization in society and that is amplified by social media (see Populism and Social Media).

Groupthink

Group polarization shifts private attitudes toward a more extreme version of the group's initial lean. Groupthink can produce a similar outward consensus, but by suppressing dissent rather than changing minds.

Groupthink is a mode of thinking that occurs in highly cohesive groups when the desire for harmony and consensus overrides realistic evaluation of alternatives. The concept was introduced by Irving Janis in his 1972 book, which analyzed several major policy failures in United States history, including the Bay of Pigs invasion in 1961, when the Kennedy administration approved a poorly planned military operation against Cuba that ended in disaster (Hart, 1991). Janis argued that the decision-making group around Kennedy was so cohesive, so eager to maintain unanimity, that members suppressed their private doubts and failed to critically examine the plan.

Janis identified several symptoms of groupthink. Members develop an illusion of invulnerability, believing that the group cannot fail. They engage in collective rationalization, dismissing warnings that might lead them to reconsider their assumptions – as when Kennedy's advisors discounted intelligence reports suggesting that the Cuban population would not rise up against Castro. Members stereotype outsiders as too weak or foolish to pose a real threat. Those who have doubts engage in self-censorship, keeping their concerns to themselves. This self-censorship creates an illusion of unanimity: because no one speaks up, everyone assumes that everyone else agrees. In some cases, self-appointed "mindguards" actively shield the group from information that might challenge the emerging consensus.

The conditions that make groupthink more likely include high group cohesion (members like each other and value their membership), insulation from outside opinions, a directive leader who makes their preferences known early, and high stress combined with a low sense of hope for finding a better solution than the one favored by the leader. Under these conditions, maintaining the comfort and solidarity of the group feels more important than getting the decision right.

It is worth being precise about how groupthink differs from group polarization. In group polarization, members are persuaded. They encounter new arguments, update their views, and come to hold more extreme positions because they are convinced. In groupthink, members may privately harbor doubts but suppress them because of social pressure. The outward result can look similar: a group that quickly converges on an extreme position with little apparent disagreement. But the internal experience is different. A polarized group member thinks, "I now believe this more strongly." A groupthink-affected member thinks, "I have concerns, but I should not rock the boat."

In practice, the two phenomena often co-occur. Group polarization pushes the group in a particular direction through persuasion, and groupthink prevents anyone from pushing back. This combination can be especially dangerous. The Challenger space shuttle disaster in 1986, another case studied through the lens of groupthink, illustrates the point. Engineers at Morton Thiokol had data suggesting that the rubber O-ring seals in the solid rocket boosters became dangerously stiff in cold weather. They initially recommended against launching. But organizational and group pressures led them to reverse their recommendation. Concerns about the O-rings did not receive the adversarial scrutiny they required. The shuttle launched on a cold January morning, and the result was catastrophic.

Shared Information Bias

The previous two sections described problems of social influence: group polarization pushes views to extremes through persuasion, and groupthink silences dissent through conformity pressure. The shared information bias is a different kind of failure. Even when a group avoids polarization and groupthink, it may still make a poor decision because of a problem rooted in information distribution and discussion dynamics: the group spends most of its discussion time on information that everyone already knows, while neglecting information that only one or two members possess.

Stasser & Titus (1985) demonstrated this in an elegant experiment using what researchers call a hidden-profile problem: a task in which the group can reach the right answer only if members pool their uniquely held information. University students were asked to evaluate three hypothetical candidates for student body president. The researchers constructed the candidate profiles so that Candidate A was objectively the best choice when all available information was considered. However, each group member received only a partial profile of each candidate. The positive attributes of the weaker candidates were shared among all members (everyone knew about them), while the positive attributes of the best candidate were split up, with each member knowing only some of them. Before discussion, most members therefore preferred one of the weaker candidates, because based on their partial information, that candidate looked best.

If the group had pooled all of its information during discussion, members would have discovered that Candidate A was the strongest choice. But this is not what happened. Groups overwhelmingly chose the candidate who had been favored before the discussion, even though the group collectively possessed enough information to make a better decision. Analysis of the discussions revealed that members spent far more time on shared information than on unshared information. Unique information that could have changed the outcome was mentioned less often, and when it was mentioned, it was less likely to be repeated or incorporated into the group's reasoning.

Why does this happen? Part of the explanation is purely statistical. If there are four group members and a piece of information is known to all four, there are four chances for it to come up in conversation. If a piece of information is known to only one member, there is only one chance. Shared information is simply more likely to be mentioned, repeated, and validated during discussion. But social factors also contribute. When a member mentions something that others already know, it is met with nods and recognition, which reinforces further discussion of that topic. When a member mentions something that no one else has heard, it may be met with blank stares and quickly forgotten.

The practical implications of this bias are significant. In medical teams, one doctor may notice an unusual symptom that the rest of the team has not observed. In a jury, one member may recall a piece of testimony that others missed. In a corporate board, one director may have unique expertise about a particular market. In all these cases, the group's decision quality depends on whether that unique information enters the discussion and is taken seriously. Unstructured, free-flowing discussion tends to bury unique information under a pile of shared knowledge.

Improving Group Decisions

The phenomena discussed in this chapter point to a common set of practical strategies for protecting the quality of group decisions.

The most fundamental principle is to preserve the independence and diversity of individual judgments before discussion begins. When a group must make an estimate or a choice, having each member commit to an initial position privately and in writing, before any discussion, prevents early anchoring on the leader's view or the first opinion voiced. This protects the error-canceling benefit that makes the wisdom of crowds work and also reduces the conformity pressure that fuels groupthink.

Once discussion begins, the goal shifts to ensuring that dissent is heard and unique information surfaces. Assigning a devil's advocate, someone whose explicit role is to argue against the emerging consensus, counteracts the self-censorship characteristic of groupthink. Consulting external experts who are not part of the group and therefore not subject to the same social pressures provides a fresh perspective. Leaders can help by withholding their own opinion at the start of a discussion, giving other members room to express independent views.

Structured information-sharing protocols directly target the shared information bias. For example, groups can begin by having each member independently list what they know before discussion. The group can then systematically review each member's list, ensuring that unique information receives airtime. Leaders can explicitly ask each member whether they have information that has not yet been discussed. These interventions do not guarantee better decisions, but they increase the chance that the group actually uses the full range of knowledge available to it. This connects to the broader theme of debiasing discussed in Heuristics and Biases: recognizing a systematic bias and designing procedures to counteract it.

A Bayesian Lens

The phenomena discussed in this chapter can be understood through the framework of Bayesian reasoning that runs throughout this book (see Bayesian Reasoning). The wisdom of crowds can be interpreted in Bayesian terms: when individual judgments are independent noisy signals of similar reliability, aggregating them can approximate rational evidence integration. Each person's estimate reflects the true state of the world plus noise. When you average many such signals, the noise cancels and the aggregate converges on something close to the true value, much as a Bayesian agent who integrates many independent pieces of evidence arrives at an increasingly accurate posterior belief.

Groupthink and group polarization represent failures of this independence. When group members share the same prior beliefs and are exposed to the same biased set of arguments during discussion, the group's collective belief becomes more extreme without new evidence having been introduced. This is a form of collective overconfidence (see Overconfidence): the group feels increasingly certain, not because it has more evidence, but because the same evidence has been recycled and amplified. A Bayesian agent should not become more confident simply because the same datum is presented multiple times; yet this is effectively what happens when group members echo each other's arguments.

The shared information bias adds another dimension to this picture. A group that discusses only shared information is, in Bayesian terms, updating its beliefs from redundant evidence while ignoring unique diagnostic evidence. Shared information that is known to all members is like a single observation counted multiple times. It should receive the weight of one observation, not four. Meanwhile, the unshared information held by individual members represents new evidence that could substantially shift the group's posterior. By neglecting this evidence, the group arrives at a posterior belief that is less accurate than it could have been, even when the necessary information exists within the group.

Summary

Many important decisions are made by groups rather than individuals, and group processes can either improve or degrade the quality of those decisions. The wisdom of crowds shows that aggregating many independent judgments tends to produce accurate estimates, because random errors cancel out when individual judgments are independent and diverse. Group polarization occurs when group discussion shifts members' private views further in the direction that was already dominant before the discussion, driven by exposure to persuasive arguments and social comparison. Groupthink is a different phenomenon in which highly cohesive groups suppress dissent to maintain harmony, producing uniform positions not through persuasion but through self-censorship and conformity pressure. Shared information bias means that groups spend most of their discussion time on information that all members already know, while neglecting unique information held by individual members, even when that unique information would change the decision. Practical strategies for improving group decisions include protecting the independence of individual judgments before discussion begins, assigning devil's advocates, consulting external experts, withholding leader opinions early in discussion, and using structured information-sharing protocols that ensure unique knowledge enters the conversation.

BELIEF

BELIEF

CONFIRMATION

Learning goals:

- Define confirmation bias and explain why it is considered one of the most consequential cognitive biases.
- Distinguish between confirmatory and disconfirmatory evidence in terms of their informational value.
- Explain the positive test strategy and the negative test strategy, and describe why people default to the former.
- Describe the 2-4-6 task and explain what it reveals about hypothesis testing.
- Explain how the exploration-exploitation tradeoff provides a goal-based reason for confirmation bias beyond the cognitive default.
- Relate confirmation bias to asymmetric updating in a Bayesian framework.

Looking for What We Expect to Find

Imagine you have just moved to Groningen for your studies and a friend, who has already lived there for a while, tells you that the students there are unfriendly and the professors are rude. When you take the bus to your first lecture, nobody talks to you or even looks at you. During the lecture break, in the university cafeteria, the cashier seems curt. A fellow student does not return your smile. Each of these small moments feels like evidence that your friend was right. But you do not notice the neighbor who held the door open, or the student who moved her bag so you could sit down. You are not deliberately ignoring kindness. You are simply doing what human minds do by default: paying attention to information that fits what you already believe.

This tendency is called confirmation bias. It refers to the general inclination to seek, interpret, and remember information that supports existing beliefs or hypotheses, while discounting or overlooking information that contradicts them (Nickerson, 1998). Some researchers use the term myside bias to emphasize that the effect is about favoring one's own side of an issue, regardless of whether that side is liberal or conservative, optimistic or pessimistic. In this chapter we will use "confirmation bias" as the primary term. Confirmation bias has been called the single cognitive bias most relevant to ideological extremism, because it can turn tentative impressions into unshakable convictions. Understanding how it works is essential for anyone who wants to reason well, whether in everyday life, in science, or in professional settings such as medicine and law (see Medical Decision-Making and Legal Decision-Making).

Confirmatory Evidence and the Positive Test Strategy

To understand confirmation bias, we need to distinguish between two types of evidence. Confirmatory evidence is consistent with a hypothesis. If you believe that drinking coffee improves your concentration, every productive study session that follows a cup of coffee counts as confirmatory evidence. This kind of evidence feels satisfying because it fits the story you already have in mind.

The problem is that confirmatory evidence can never definitively prove a hypothesis. Productive study sessions might also follow a good night's sleep, a healthy breakfast, or simply studying material you find interesting. The coffee might have nothing to do with it. Confirmatory evidence is compatible with the hypothesis, but it is also compatible with many other explanations.

Despite this limitation, people naturally gravitate toward confirmatory evidence. Klayman & Ha (1987) describe this tendency as the positive test strategy: when testing a hypothesis, people tend to examine cases where the hypothesized property is expected to be present. If a manager believes that older employees resist new technology, she will pay attention every time a senior colleague struggles with a software update. Each instance confirms her hypothesis. But she never checks whether younger employees struggle just as often. The confirmatory evidence accumulates, and the hypothesis feels increasingly certain, even though it was never properly tested.

The positive test strategy is not always a bad idea. Klayman & Ha (1987) showed that in many real-world situations, where the phenomenon of interest is rare and the hypothesis is roughly accurate, testing positive cases is actually a reasonable way to gather information. If you suspect that a certain rare mushroom is poisonous, examining mushrooms of that type (a positive test) is more efficient than examining random mushrooms from the forest floor. The strategy becomes problematic when hypotheses are overly specific or when the base rate of the target phenomenon is high, because under those conditions, positive tests tend to produce a stream of confirmations regardless of whether the hypothesis is correct.

Disconfirmatory Evidence and the Negative Test Strategy

Disconfirming evidence is information that contradicts a hypothesis. For strict universal claims, a single counterexample can be decisive: if you believe that all swans are white, observing one black swan is enough to refute the hypothesis entirely. For probabilistic claims – such as "older employees are less likely to adopt new technology" – a single exception does not refute the hypothesis outright, but disconfirming cases are still typically more informative than piling up additional confirming ones. The general asymmetry holds: while confirmation can only be consistent with a hypothesis, disconfirmation has the power to narrow down or eliminate possibilities.

The negative test strategy involves deliberately looking for cases that would disconfirm your hypothesis. Using the manager example from above, a negative test would involve looking for counterexamples: older employees who adopt new technology with ease, or younger employees who struggle with it. Even a few such cases would seriously weaken the hypothesis – and would be far more informative than adding yet another confirming instance.

This idea connects to a principle from the philosophy of science associated with Karl Popper: falsifiability. A claim is falsifiable if there is some possible observation that would show it to be wrong. Popper argued that what makes a hypothesis scientific is not that it can be confirmed, but that it can in principle be proven wrong. For example, "all metals expand when heated" is falsifiable because a single metal that contracts when heated would refute it. By contrast, a claim like "hidden forces always explain the outcome somehow" is not falsifiable, because no observation could ever count against it – every outcome can be absorbed into the claim. Good science, on Popper's view, consists of trying hard to disconfirm hypotheses and keeping only those that survive the test.

In practice, however, people rarely seek disconfirmation spontaneously. Doing so requires effortful, deliberate reasoning – the kind of slow, rule-based thinking described in Reason and Intuition as System 2. Our cognitive default, rooted in the faster and more automatic System 1, pulls us toward confirmation.

The 2-4-6 Task

The most famous demonstration of confirmation bias in hypothesis testing comes from a deceptively simple experiment by Wason (1960). Participants were told that the number sequence 2-4-6 conformed to a rule, and their task was to discover that rule by proposing new sequences. For each sequence they proposed, the experimenter would say whether it conformed to the rule or not. Participants could propose as many sequences as they wanted before announcing what they thought the rule was.

The actual rule was simply "any three numbers in increasing order of magnitude." But this rule is broad and general, and most participants started with a much more specific hypothesis, such as "even numbers increasing by 2." They then tested confirming sequences: 8-10-12, 20-22-24, 100-102-104.

All of these fit the true rule, so the experimenter said "yes" each time. This stream of confirmations reinforced the wrong hypothesis, and participants announced their specific rule with confidence, only to be told they were wrong.

Only a minority of participants identified the correct rule on their first attempt. What distinguished these successful participants was that they used the negative test strategy. They proposed sequences that violated their current hypothesis: 1-3-5 (odd numbers – still ascending, so "yes"), 1-2-3 (ascending by 1 – still "yes"), 6-4-2 (descending – finally "no"). By discovering what did not work, they could narrow down the real rule. Those who failed tended to generate only sequences that confirmed their initial idea, never learning that their hypothesis was too narrow.

What makes the 2-4-6 task especially revealing is that it involves abstract, emotionally neutral material. Participants have no personal stake in whether the rule involves even numbers or not. They are not protecting their self-esteem or defending a political position. Yet confirmation bias still emerges strongly. This shows that the pull toward positive testing is a basic cognitive default, not something that arises only when emotions or identity are involved.

Why Positive Testing Persists

But cognitive default is only part of the story. If disconfirming evidence is so informative, why do people so rarely seek it – even when the stakes are high? As we have seen, the positive test strategy is a natural default that works well enough in many everyday situations (Klayman & Ha, 1987). But there is an additional reason that goes beyond pure cognition. Many of our beliefs are not just descriptions of the world; they are also strategies for action. When a belief serves a practical goal, testing the alternative becomes not just effortful but risky.

Consider a student who believes that always agreeing with a professor leads to better evaluations. The most informative test of this belief would be to disagree with the professor and observe what happens. But disagreeing risks the very outcome the student cares about: a good grade. Even if the student suspects that the belief might be wrong, the cost of finding out the hard way may seem too high.

This creates what is known as an exploration-exploitation tradeoff, a concept that already appeared in Satisficing vs. Maximizing. Exploiting your current belief means acting on it and reaping its expected benefits. Exploring the alternative means testing a different approach, which could yield valuable information but might also lead to a bad outcome. The stronger the goal motivation, the more people default to exploitation. From the outside, this looks identical to confirmation bias: the person keeps acting in ways that confirm their belief and never gathers the evidence that might challenge it. But the underlying mechanism is different. It is not just that disconfirming evidence is hard to think of; it is that seeking it carries real costs.

This goal-based explanation and the cognitive default explanation are not competing accounts. They operate in parallel. In the 2-4-6 task, where there are no real-world costs, the cognitive default alone is enough to produce confirmation bias. In everyday life, where beliefs are tied to goals and relationships, the exploration-exploitation tradeoff adds a further layer of resistance to disconfirmation. Together, these two mechanisms help explain why confirmation bias is so robust across such a wide range of situations.

The Consistency Fallacy

Confirmation bias is not limited to how we search for evidence; it also affects how we evaluate evidence once we encounter it. Data that merely happen to be consistent with a hypothesis can be misinterpreted as proving it, even when the data carry no real informational value.

A striking demonstration comes from Weisberg et al. (2008), who investigated whether irrelevant neuroscience information makes psychological explanations seem more convincing. In their study, participants read descriptions of psychological phenomena (such as the finding that people perform worse on attention tasks when emotionally aroused) followed by explanations that were either good or bad. Crucially, some explanations included neuroscience information – such as a reference to a specific brain region – that experts confirmed added no logical support to the explanation. The neuroscience details were irrelevant because all psychological processes involve the brain; merely naming a brain area does not help distinguish a good explanation from a bad one, any more than saying "this involves atoms" helps distinguish a good physics explanation from a bad one. What matters is whether the explanation identifies the right mechanism, not whether it mentions neural tissue.

The results were clear. People with no neuroscience training rated bad explanations as significantly more satisfying and more convincing when those explanations included neuroscience information. A mention of brain scans or neural activity made a poor explanation feel more scientific. Students enrolled in a neuroscience course showed the same pattern, even at the end of the semester. Only neuroscience experts were immune to the effect; in fact, experts rated explanations with irrelevant neuroscience as less satisfying, because they could see that the information added nothing.

This finding illustrates what we might call the consistency fallacy: the tendency to treat information that is merely consistent with a claim as though it provides strong support for that claim. The neuroscience details are consistent with the psychological explanation in the loose sense that brains are involved in all psychological processes. But they do not help distinguish a good explanation from a bad one – they add no diagnostic value. The extra details increase the appearance of scientific support without increasing actual evidential weight.

The consistency fallacy matters because it makes people vulnerable to persuasion by irrelevant details. In courtrooms, in advertising, in political rhetoric, and in science journalism, adding details that sound consistent with a claim can make the claim seem more credible than it deserves. This connects to broader themes about how beliefs form and resist correction, as discussed in Belief Formation and Perseverance.

Why It Matters

The examples discussed so far – number sequences, neuroscience descriptions – might seem far removed from daily experience. But confirmation bias shapes judgments in highly consequential domains. Consider medical diagnosis. A doctor who forms an early hypothesis about what is wrong with a patient may then order tests mainly to confirm that hypothesis, while overlooking symptoms that point elsewhere. If a patient presents with chest pain and the doctor suspects a cardiac problem, she may focus on cardiac markers and miss signs of a pulmonary embolism – not out of incompetence, but because each confirming result reinforces the initial hypothesis and makes alternatives harder to see (see Medical Decision-Making for a fuller discussion).

Similarly, in criminal investigations, once an officer identifies a suspect, ambiguous evidence tends to be interpreted as incriminating. A nervous demeanor during questioning becomes a sign of guilt rather than a natural response to being interrogated. Leads pointing to other suspects receive less follow-up (see Legal Decision-Making). In science itself, researchers may design studies that are more likely to confirm their preferred theory than to challenge it, or interpret ambiguous results as supportive.

Even in social life, confirmation bias operates constantly. If you believe a classmate is arrogant, you will notice every time they talk about their achievements and interpret it as boasting. You might not notice, or might explain away, the times they are self-deprecating or helpful. Over time, your impression solidifies – not because the evidence is overwhelming, but because you have been sampling it in a biased way. Once this biased sampling has occurred, the confirming examples also become more available in memory, further reinforcing the impression (see Availability).

Nickerson (1998) argues that confirmation bias may be the single most problematic aspect of human reasoning. It contributes to the persistence of stereotypes, the escalation of conflicts, and the difficulty of changing anyone's mind with evidence alone. Yet it is also, to some degree, understandable. Maintaining stable beliefs has benefits: it allows us to act decisively, to plan ahead, and to avoid the paralysis that would come from reconsidering everything constantly. The cost is that our beliefs can drift away from reality without our noticing.

A Bayesian Lens

From the perspective of Bayesian reasoning (see Bayesian Reasoning), a rational agent should update beliefs symmetrically in response to evidence. Confirming and disconfirming evidence should both shift the posterior belief, and the size of the shift should depend on the diagnosticity of the evidence – that is, how much more likely the evidence is under one hypothesis than under the alternative. A piece of evidence that is equally likely under both hypotheses should not change your belief at all, regardless of whether it feels consistent with your preferred hypothesis.

Confirmation bias violates this principle through asymmetric updating. When people encounter evidence that fits their belief, they update readily. When they encounter evidence that contradicts it, they discount it, explain it away, or simply fail to notice it. Consider a student who believes she is bad at math. When she scores poorly on a quiz, she takes it as confirmation: "See, I knew it." When she scores well on a test, she dismisses it as a fluke: "That test was easy" or "I just got lucky." The poor result shifts her belief strongly downward; the good result barely moves it. Over time, her self-assessment becomes more negative than her actual performance warrants – not because the evidence supports it, but because she is weighting confirming and disconfirming evidence asymmetrically.

Biased updating creates a self-reinforcing cycle: increasing confidence makes later counterevidence easier to dismiss. This is one reason why confirmation bias is so difficult to correct. As noted earlier, disconfirming evidence is often more diagnostic than additional confirming cases. In Bayesian terms, such evidence may have a very high likelihood ratio – it is much more probable if the hypothesis is false than if it is true. Yet people routinely treat these observations as though they have low diagnosticity, dismissing them as exceptions or errors.

The exploration-exploitation tradeoff adds a further wrinkle: even a perfectly Bayesian agent might avoid seeking disconfirming evidence if the expected cost of a bad outcome outweighs the expected informational gain. In such cases, positive testing is instrumentally rational – it serves the agent's goals – even though it is epistemically suboptimal – it does not maximize the accuracy of the agent's beliefs. This distinction helps explain why confirmation bias persists even among intelligent, well-meaning people: the problem is not always a failure of reasoning, but sometimes a rational response to the costs of being wrong in the short term.

Summary

Confirmation bias is the tendency to seek, interpret, and remember information that supports existing beliefs while neglecting information that contradicts them. People default to a positive test strategy – examining cases where their hypothesis predicts a positive result – rather than a negative test strategy that deliberately seeks disconfirmation. Disconfirmation is connected to the principle of falsifiability: a claim is scientific only if some possible observation could show it to be wrong. The 2-4-6 task demonstrates that the pull toward positive testing operates even for abstract, emotionally neutral hypotheses, indicating a deep cognitive default. Beyond this cognitive pull, the exploration-exploitation tradeoff provides an additional reason for positive testing: when beliefs serve practical goals, seeking disconfirming evidence can be risky. The consistency fallacy shows that even irrelevant information can seem like confirmation if it is loosely consistent with a claim. From a Bayesian perspective, confirmation bias produces asymmetric updating, where confirming evidence strengthens beliefs more than disconfirming evidence weakens them, causing beliefs to drift away from what balanced evidence would support. Recognizing confirmation bias is a first step toward countering it, but as the chapters on Belief Formation and Perseverance and Conspiracy Theories will show, correcting established beliefs is far harder than preventing biased ones from forming in the first place.

BELIEF FORMATION AND PERSEVERANCE

Learning goals:

- Explain the Spinozan model of belief formation and how it differs from the Cartesian model.
- Describe the role of executive control (System 2) in rejecting false information.
- Define belief perseverance and explain why beliefs resist correction.
- Define the continued influence effect and explain how it differs from belief perseverance.
- Explain the myth-versus-fact effect and its implications for public information campaigns.
- Describe psychological reactance and cognitive dissonance as responses to challenged beliefs.
- Identify the cognitive mechanisms that protect existing beliefs from revision.
- Explain the sunk cost effect and how it extends perseverance from beliefs to behavior.

The Natural Believer

Imagine scrolling through your phone late at night. You are tired, half-watching a series, and a friend sends you a link to a news article with a surprising claim: a common food additive has been found to improve memory. You read the headline, maybe skim the first paragraph, and move on. A week later, in a conversation about exam preparation, you find yourself mentioning that a certain food additive can boost memory. You cannot remember where you heard it, and you never checked whether the claim was true. Yet somehow, you believe it.

This everyday experience reveals something fundamental about how the human mind handles information. We do not receive new claims in a neutral state, carefully weighing them before deciding whether to accept or reject them. Instead, we tend to believe first and doubt later – if we doubt at all. This chapter examines how beliefs form, why they persist even in the face of strong counterevidence, and what happens psychologically when our beliefs are challenged. These questions connect directly to the interplay between fast, automatic processing and slow, deliberate reasoning discussed in Reason and Intuition, and the ways in which mental shortcuts can lead us astray, as explored in Heuristics and Biases.

Believing Is Understanding

The seventeenth-century philosopher René Descartes proposed what most people would consider the commonsense view of belief. According to Descartes, understanding an idea and believing it are two separate steps. First, you comprehend what someone is telling you. Then, in a second step, you assess whether the claim is true or false. On this view, comprehension is neutral: you can perfectly well understand a statement without believing it.

The philosopher Baruch Spinoza, born in Amsterdam in 1632, proposed a radically different account. Spinoza argued that understanding and believing are not separate processes. To understand a statement is, at least initially, to accept it as true. Rejecting a false statement requires an additional step: a deliberate act of mental "unaccepting." If that second step never happens, the information stays believed.

Gilbert (1991) brought this philosophical debate into the laboratory. He argued that the Spinozan model makes a clear, testable prediction. If believing is the default and rejecting requires effort, then anything that disrupts the effortful rejection process should leave people more likely to believe false information. But if Descartes was right and comprehension is neutral, then disruption should make people uncertain, not gullible.

The evidence strongly favors Spinoza. In a key experiment, Gilbert et al. (1990) had participants read statements about a criminal case that were labeled as either true or false. Some participants were interrupted during the task, which prevented them from fully processing the true/false labels. The interruption had almost no effect on their ability to recognize statements that had been labeled true. But it severely impaired their ability to identify statements that had been labeled false. Interrupted participants showed a strong tendency to call everything "true." This asymmetry is exactly what the Spinozan model predicts: comprehension produces belief automatically, and only the rejection of falsehoods requires the kind of effortful processing that interruption can disrupt.

This pattern generalizes beyond interruption. Conditions of cognitive load (such as performing a secondary task) and time pressure produce the same asymmetry: people continue to accept true information at normal rates but become worse at rejecting falsehoods. Think about the contexts in which you consume information: late at night on social media, between tasks during a busy day, or while multitasking during a lecture. These are precisely the conditions under which the automatic acceptance mechanism operates unchecked.

Executive Control and Skepticism

If belief is the default, what allows some people to be more skeptical than others? The answer lies in the distinction between System 1 and System 2 processing described in Reason and Intuition. System 1 accepts automatically. System 2, the slower and more effortful mode of thinking, is what allows us to question and sometimes reject what System 1 has already accepted.

This connection is supported by research on the Cognitive Reflection Test (CRT), which measures the tendency to override an intuitive but incorrect response with a reflective and correct one (see Intelligence vs. Rationality). In survey studies, people who score higher on the CRT are less likely to endorse paranormal and pseudoscientific beliefs such as astrology, telepathy, and superstitions. The correlation is modest but consistent: those who habitually engage System 2 are more likely to catch and reject unfounded claims. Of course, correlation does not establish causation – it is possible that a general tendency toward analytical thinking drives both higher CRT scores and greater skepticism, rather than one directly causing the other. Still, the pattern fits the Spinozan hypothesis: everyone is a natural believer, and skepticism requires additional cognitive effort.

What is striking is how common paranormal beliefs are among ordinary, educated adults. Surveys across Europe consistently find that substantial minorities believe in astrology, healing crystals, or telepathy. These are not fringe beliefs held by a small group of unusual people. Several factors likely contribute to their prevalence, including cultural transmission, social identity, and motivational needs, but part of the explanation is that skepticism is effortful, and most of the time, we simply do not invest that effort.

Belief Perseverance

Once a belief has been accepted, whether through careful evaluation or through the automatic Spinozan process, it becomes resistant to change. Belief perseverance refers to the tendency to maintain a belief even when the evidence that originally supported it has been thoroughly discredited.

A classic demonstration of this phenomenon involves giving participants false feedback about their performance on a task. For example, participants might be told that they are especially good (or bad) at distinguishing real suicide notes from fabricated ones. Later, they are told that the feedback was entirely fabricated and had nothing to do with their actual performance. Despite this debriefing, participants continue to rate their ability in line with the original false feedback. The belief, once accepted, survives the removal of its evidential basis.

Belief perseverance is not simply a failure to hear or understand the correction. People who show strong belief perseverance acknowledge that the original evidence has been discredited. They just continue to believe the conclusion anyway. Research suggests that this happens because the initial belief triggers a process of explanation and elaboration. Once you have generated reasons why you might be good at detecting fake suicide notes – perhaps you think of yourself as an empathetic person – those reasons remain even after the original feedback is retracted. The belief has become self-supporting.

Belief perseverance is often strengthened when people have already constructed a simple, coherent causal explanation, especially one centered on intentions. This connects to the discussion of causal reasoning in Cues to Causality and the intentional stance described in Mental Models: once you have inferred that an event happened because someone intended it, that explanation feels complete and difficult to abandon. Losing the explanation feels worse than keeping a belief that may be wrong.

Belief Overkill

Belief perseverance concerns holding onto a belief; belief overkill concerns what that belief does next. Once people adopt a strong position, they often align a wide range of related judgments with it, even when those judgments are logically independent. Rather than holding a nuanced position that acknowledges some strengths and some weaknesses of a particular view, people tend to pull all their judgments in one direction.

Consider attitudes toward artificial intelligence. Someone who is generally skeptical of AI tends to believe that AI is harmful to the environment, useless in practical applications, and damaging to creativity. This person is unlikely to hold a more mixed view, such as that AI is harmful to the environment yet genuinely useful for certain tasks. Similarly, someone enthusiastic about AI is likely to dismiss concerns about environmental impact, job displacement, and creative harm all at once, rather than acknowledging that some concerns are legitimate while others are not. The same pattern appears in debates about nuclear energy, immigration, or genetically modified food: once someone takes a side, logically separable arguments all get pulled in the same direction.

Belief overkill is a widely observed pattern rather than a precisely defined experimental effect, but it reflects a genuine tendency for strong attitudes to function as a lens through which all related evidence is filtered. For a fuller treatment of how existing beliefs shape the selective search and interpretation of evidence, see Confirmation. Belief overkill is what happens when such bias operates across a whole set of related but independent arguments.

The Continued Influence Effect

Belief perseverance is troubling enough, but there is a more subtle phenomenon: the continued influence effect. This refers to situations where people explicitly accept a correction, acknowledge that the original information was wrong, and yet continue to be influenced by the original misinformation in their subsequent reasoning.

Lewandowsky et al. (2012) reviewed extensive research on this effect, much of it using a paradigm involving reports about a warehouse fire. In a typical study, participants read a series of news updates about a fire. Early updates mention that volatile chemicals were found in a storage closet, suggesting that the chemicals might have caused the fire. A later update corrects this, stating that the closet was in fact empty. When participants are later asked questions about the fire, such as why there was so much toxic smoke, they continue to reference the chemicals even though they remember and accept the retraction. The misinformation has been corrected in their explicit memory, but it continues to shape their mental model of the event.

The continued influence effect differs from belief perseverance in a specific way. In belief perseverance, the person explicitly maintains the original belief despite counterevidence. In the continued influence effect, the person explicitly abandons the original belief and accepts the correction, yet the misinformation still leaks into their judgments. Retraction never fully erases the influence of misinformation.

One reason for this is that retractions create a gap in the causal story. If the chemicals did not cause the fire, then what did? Without an alternative explanation to fill the gap, people fall back on the retracted information because it still provides the most coherent account. This is why corrections that offer an alternative explanation ("the closet was empty; investigators believe faulty wiring caused the fire") are more effective than corrections that simply negate the misinformation.

The Myth-Versus-Fact Effect

The continued influence effect has a particularly troubling cousin: the myth-versus-fact effect. This occurs when attempts to correct misinformation by repeating it – even in a context that labels it as false – actually increase belief in the misinformation.

Schwarz et al. (2007) demonstrated this in a study on flu vaccination myths. Participants read a flyer from a health authority that listed common myths about the flu vaccine alongside the corresponding facts, in a clear "myth versus fact" format. Immediately after reading the flyer, participants could distinguish myths from facts reasonably well. But just 30 minutes later, a striking pattern emerged: participants were more likely to misidentify myths as facts than they had been before reading the flyer. Worse, their intentions to get vaccinated actually decreased. The flyer, designed to correct misinformation, had the opposite of its intended effect.

One driver of this effect is familiarity: repeated exposure makes a statement easier to process, and ease of processing can be misread as truth. As discussed in the context of the Availability heuristic, information that comes to mind easily tends to feel true. When you read a myth-versus-fact flyer, you encounter the myth multiple times – once when it is stated and once when it is corrected. This repetition increases the fluency with which the myth comes to mind. Over time, the contextual detail that the statement was labeled a "myth" fades from memory, but the increased familiarity remains. The result is that the myth now feels more true than it did before you read the correction.

This finding has serious implications for public health campaigns, educational materials, and media literacy efforts. The intuitive strategy of "state the myth, then correct it" can backfire. A more effective approach is to lead with the truth and avoid repeating the false claim, or to provide a clear alternative explanation that replaces the myth in the person's mental model.

Reactance

So far, we have discussed what happens when beliefs persist passively, through inertia or incomplete updating. But what happens when beliefs are actively challenged? Two distinct psychological mechanisms can come into play, depending on the nature of the challenge. When the threat comes from outside – someone telling you what to think or do – the result can be reactance. When the conflict is internal – your own beliefs or actions contradicting each other – the result can be cognitive dissonance.

Psychological reactance, first described by Brehm (1966), is a motivational state that arises when a person perceives that their freedom to think or act is being threatened (Rosenberg & Siegel, 2018). The key feature of reactance is that it is not a deliberate, reasoned decision to resist persuasion. It is a gut-level emotional reaction that intensifies the desire to do the very thing that is being restricted or to believe the very thing that someone is trying to make you abandon.

You have probably experienced reactance yourself. When a professor says "you must read this article," the reading may feel less appealing than if the professor had said "you might find this article interesting." When a public health message says "you should not eat junk food," some people feel an increased craving for exactly that. Reactance explains the so-called boomerang effect in persuasion, where a heavy-handed message produces the opposite of its intended result.

Research on reactance has tested these dynamics directly. In health communication studies, messages that use controlling language ("you must," "you should," "you need to") consistently produce more resistance than messages that use autonomy-supportive language ("perhaps," "you might consider," "it is up to you"). For example, anti-smoking messages framed as commands ("You need to quit smoking now") generate more defensive reactions and less intention to quit than messages that respect the reader's autonomy ("Some people find it helpful to consider the benefits of quitting"). The number and importance of the threatened freedoms also matter: the more freedoms that are threatened, and the more important those freedoms are to the person, the stronger the reactance.

Reactance is relevant to belief perseverance because direct, forceful attempts to change someone's beliefs can backfire. Rather than updating their beliefs, the person doubles down. This is a particular challenge in health communication, political debate, and science education, where the goal is to correct misinformation but the method of correction may trigger the very resistance it aims to overcome.

Cognitive Dissonance

Where reactance is a response to an external threat to autonomy, cognitive dissonance is a response to an internal inconsistency. It arises when a person holds two beliefs, or a belief and a behavior, that conflict with each other.

Imagine that you believe you are an environmentally responsible person, but you also fly to holiday destinations several times a year. The tension between these two cognitions produces an unpleasant mental state: cognitive dissonance. To reduce this discomfort, you have several options. You could change your behavior and fly less. You could change your belief and stop identifying as environmentally responsible. Or you could find ways to reduce the perceived inconsistency – perhaps by telling yourself that your recycling habits compensate for your flights, or by questioning the evidence that flying is as harmful as scientists claim.

The last option is what connects cognitive dissonance to belief perseverance. When new evidence threatens an existing belief, the resulting dissonance can be resolved either by updating the belief (genuine learning) or by rejecting or reinterpreting the threatening evidence (perseverance). The path of least resistance is often to protect the existing belief, because changing a belief often requires changing a whole network of related beliefs and behaviors. This is particularly true for beliefs that are central to a person's identity, such as political, religious, or moral convictions. The role of cognitive dissonance in protecting beliefs also helps explain why conspiracy theories are so resilient to counterevidence, as discussed further in Conspiracy Theories.

The Mechanisms That Protect Beliefs

Across the phenomena we have discussed, several recurring mechanisms help protect beliefs from revision. Most of these operate at the level of System 1, meaning they are automatic and occur without conscious awareness.

The first is the availability heuristic, discussed in detail in Availability. Information that comes to mind easily is judged as more likely to be true. Because existing beliefs are, by definition, readily available in memory, they enjoy a built-in advantage over new, contradictory information.

The second is mere exposure – a specific manifestation of the availability heuristic in which something is made available through repeated presentation. Already illustrated by the myth-versus-fact effect, mere exposure increases familiarity and thereby perceived truth. In daily life, this means that beliefs you have encountered many times – in conversations, news articles, or social media posts – will feel increasingly true regardless of their actual accuracy.

The third is confirmation bias, the topic of a closely related chapter (Confirmation). People tend to seek information that confirms their existing beliefs, interpret ambiguous information in a belief-consistent direction, and remember belief-consistent information better. Confirmation bias ensures a steady supply of apparent evidence for what we already believe.

The fourth is source amnesia. Over time, people tend to forget where they learned a piece of information while retaining the information itself. You might remember hearing that a particular vitamin prevents colds, but forget that you picked this up from a social media post shared by someone with no medical expertise. The claim then stands on its own, feeling like something you "just know."

Together, these mechanisms make it hard to keep a claim in mind without partly believing it. Information enters the mind through Spinozan acceptance, gets reinforced through repetition and familiarity, is supported by selective attention and memory, and gradually loses its connection to whatever source originally delivered it. The result is a cognitive system that is efficient at accumulating beliefs and less efficient at discarding them.

Sunk Cost as Behavioral Perseverance

The tendency to stick with existing commitments extends beyond beliefs to behavior. The sunk cost effect describes the tendency to continue investing in a losing course of action because of what has already been spent, even though past expenditures should be irrelevant to decisions about the future. This is not literally a case of belief perseverance, but it is a close analogue: the prior commitment, rather than a prior belief, resists updating.

Arkes & Blumer (1985) demonstrated the sunk cost effect in a series of experiments. In one study, participants were asked to imagine that they had bought tickets for two ski trips. One trip cost €100 and the other cost €50, but they expected to enjoy the cheaper trip more. Unfortunately, the trips were on the same weekend. Which trip would they choose? Fifty-four percent chose the more expensive trip, despite expecting less enjoyment from it. The only reason to prefer the €100 trip is to avoid "wasting" the larger investment, but since the money is already spent in both cases, the rational choice is to go where you will have more fun.

In another experiment, participants were more willing to continue funding a nearly complete but failing project when they had already invested heavily in it: 85 percent chose to continue when prior investment was high, compared to only 17 percent when there was no prior investment. The project's future prospects were identical in both conditions; only the sunk costs differed.

Interestingly, Arkes & Ayton (1999) found that the sunk cost effect develops with age and socialization. Young children and non-human animals do not reliably show the effect. The authors argued that the sunk cost effect results from overgeneralizing a learned rule: "don't waste." This rule is generally sensible, which is why parents and teachers teach it. But when applied to irrecoverable past costs, it leads to irrational escalation of commitment. Adults, who have thoroughly internalized the "don't waste" heuristic, are more susceptible than children who have not yet learned the rule. This is a case where cognitive sophistication makes us less rational, not more – echoing the broader theme that heuristics are usually helpful but can lead us astray in specific contexts (Heuristics and Biases).

A Bayesian Lens

The phenomena described in this chapter can be understood through the framework of Bayesian belief updating introduced in Bayesian Reasoning, though the mapping is an analogy rather than a literal application of Bayes' theorem.

As a rough Bayesian analogy, Spinozan acceptance means that newly understood information is initially treated as highly credible unless further processing discounts it. When cognitive resources are lacking – because of fatigue, distraction, or time pressure – the discounting never happens, and the uncritically accepted claim becomes part of one's working beliefs.

Belief perseverance can be understood as a failure to update. The prior belief is treated as near-certain, so that no single piece of counterevidence, however compelling, shifts the posterior meaningfully. This is not inherently irrational; a genuinely well-established belief should be resistant to isolated contradictory evidence. The problem is that belief perseverance often applies to beliefs that were never well established in the first place.

The continued influence effect reveals something more subtle: even after an explicit "retraction update," the original information retains residual influence on later judgments. The update is incomplete. In Bayesian terms, the person has nominally adjusted their belief, but the retracted information continues to carry some evidential weight in practice.

Reactance and cognitive dissonance are emotional responses that can obstruct the updating process. Rather than integrating new evidence, these responses lead people to treat counterevidence as uninformative or untrustworthy – drastically down-weighting its evidential value and preventing meaningful change in the posterior regardless of how strong the evidence actually is.

Summary

Beliefs form more easily than they are revised. Following Spinoza's insight, the mind accepts information automatically in the process of comprehending it, and rejecting falsehoods requires additional effortful evaluation. When this evaluation is disrupted by distraction, fatigue, or cognitive load, false information is accepted by default. Higher engagement of reflective thinking, as measured by the Cognitive Reflection Test, predicts greater skepticism, but such engagement is the exception rather than the rule.

Once formed, beliefs resist change through belief perseverance (maintaining a belief despite discredited evidence), belief overkill (extending a belief to align all related judgments), and the continued influence effect (being influenced by misinformation even after accepting a correction). Attempts to correct misinformation can backfire through the myth-versus-fact effect, where repeating a false claim to label it as false inadvertently increases its perceived truth through familiarity.

When beliefs are actively challenged, psychological reactance (a gut-level resistance to perceived threats to autonomy) and cognitive dissonance (discomfort from internal inconsistency) can prevent genuine updating. Several automatic mechanisms – mere exposure, the availability heuristic, confirmation bias, and source amnesia – work together to protect existing beliefs. Finally, the sunk cost effect extends perseverance from beliefs to behavior, as people continue investing in failing courses of action to avoid the feeling of waste.

CONSPIRACY THEORIES

Learning goals:

- Define conspiracy theories and explain what makes them different from other false beliefs.
- Describe the cognitive, existential, and social factors that contribute to conspiracy thinking.
- Explain how apophenia and agency detection relate to conspiracy belief.
- Describe how echo chambers and group polarization sustain and strengthen conspiracy beliefs.
- Explain how conspiracy theories relate to the rejection of science.
- Analyze conspiracy theories from a Bayesian perspective, focusing on unfalsifiability.

What Are Conspiracy Theories?

In 2020, a wave of arson attacks targeted mobile phone towers across the Netherlands, the United Kingdom, and Belgium. The attackers believed that 5G technology was responsible for spreading COVID-19. The conspiracy theory had spread rapidly through social media, combining fears about a new pandemic with distrust of telecommunications companies and governments. Some versions of the theory claimed the virus was a cover story for radiation sickness; others claimed the virus was real but had been deliberately released to distract from the rollout of 5G. These versions contradicted each other, yet many people seemed comfortable endorsing both.

This episode illustrates several features of conspiracy theories that psychologists have studied in detail. A conspiracy theory is an explanation that attributes an event or situation to a secret plot carried out by powerful people or organizations. Conspiracy theories can be false, partly true, or occasionally true in full. Governments really do sometimes conspire in secret, as the Watergate scandal and various intelligence agency operations have demonstrated. But many conspiracy theories are false, and what makes them psychologically interesting is not whether they are true or false, but how they resist correction. A defining feature of conspiracy thinking is that contradictory evidence gets absorbed into the theory itself. If a government agency denies involvement, that denial is treated as evidence of a cover-up. If an independent investigation finds no conspiracy, the investigators must be in on it too. This self-sealing quality distinguishes conspiracy theories from many ordinary false beliefs, which at least in principle can be corrected with better information – although as we've seen in Belief Formation and Perseverance, even ordinary beliefs are resistant to change.

Perhaps the most striking demonstration of how conspiracy thinking works comes from Wood et al. (2012). They asked participants about their beliefs regarding the death of Princess Diana. Some participants believed that Diana had been murdered by powerful figures. Other participants believed that Diana had faked her own death and was still alive. These two theories are obviously contradictory: Diana cannot be both dead (murdered) and alive (in hiding). Yet the researchers found a significant positive correlation between belief in these two theories. People who endorsed one were more likely to also endorse the other. The same pattern appeared in a second study about Osama bin Laden: people who believed he was already dead before the U.S. raid on his compound were also more likely to believe he was still alive after the raid. The positive correlation was driven by a shared higher-order belief, namely that authorities are engaged in systematic deception. The specific content of the theory mattered less than a generalized suspicion of official accounts. This suggests that conspiracy thinking is often less about the details of any particular theory and more about a worldview in which powerful groups cannot be trusted.

Cognitive and Personality Factors

What makes some people more drawn to conspiracy theories than others? Research has identified several cognitive and personality factors that predict conspiracy thinking.

One of the most important is apophenia, the tendency to see meaningful patterns in random or unrelated events. People who score high on measures of apophenia are more likely to endorse conspiracy theories (Douglas et al., 2017). Conspiracy theories take unrelated events, coincidences, and ambiguous information and weave them into a coherent narrative of hidden connections. Our brains evolved to detect patterns and threats, and this system is tuned for sensitivity rather than accuracy. It is better, from a survival perspective, to occasionally see a predator that is not there than to miss one that is. But this sensitivity generates false positives – seeing connections where none exist. The same System 1 processes that support rapid pattern detection also contribute to superstition and paranormal belief (see Belief Formation and Perseverance).

Closely related but distinct is agency detection, the inclination to assume that the patterns one perceives were created by intentional agents rather than by impersonal forces or chance. Where apophenia finds the connections, agency detection supplies the hidden author behind them. When something bad happens, people who score high on agency detection are more likely to assume that someone planned it. This connects to the intentional stance discussed in Mental Models: when we adopt the intentional stance toward a situation, we automatically start reasoning about goals, plans, and hidden motives. Conspiracy theories often involve applying the intentional stance to large-scale events, treating complex outcomes as the product of purposeful agents rather than impersonal processes or chance.

A further cognitive factor is a preference for simple explanations. Complex events – the assassination of a political leader, a pandemic, an economic crisis – have complex, multicausal explanations, and that complexity is unsatisfying. Conspiracy theories replace it with a simple, agent-driven story featuring clear villains and clear motives, a style of explanation many people find cognitively appealing.

Swami et al. (2014) investigated the role of thinking style in conspiracy belief. In their first study, they surveyed 990 British participants and found that stronger belief in conspiracy theories was associated with lower analytic thinking, lower open-mindedness, and greater reliance on intuitive thinking. This fits with the dual-process framework discussed in Reason and Intuition: conspiracy thinking appears to be driven more by System 1 (fast, intuitive, pattern-matching) than by System 2 (slow, deliberate, analytical). But the researchers went further. In a series of experiments, they tested whether encouraging analytic thinking could actually reduce conspiracy beliefs. In one study, they presented information in a difficult-to-read font, a manipulation intended to encourage more careful processing. Participants in this condition reported lower belief in conspiracy theories compared to a control group. A fourth study replicated this effect in a general population sample and extended it to belief in a specific conspiracy theory about the 2005 London bombings. These findings suggest that conspiracy beliefs are shaped not only by stable traits but also by whether people are thinking intuitively or analytically in the moment.

Other personality factors also play a role. Research reviewed by Douglas et al. (2017) has found that conspiracy thinking correlates positively with narcissism and with paranormal beliefs, and has been associated with lower education, lower analytic thinking, and, in some studies, lower intelligence. Men tend to score slightly higher on conspiracy belief than women. The overlap between conspiracy thinking and paranormal beliefs makes sense if both share a common cognitive root in the pattern-detection and agency-detection mechanisms described above.

Epistemic, Existential, and Social Factors

Conspiracy theories do not arise only from individual cognitive tendencies. They also serve psychological needs that are shaped by a person's circumstances and social environment. Douglas et al. (2017) organized the motivations behind conspiracy belief into three categories: epistemic, existential, and social.

Epistemic motives involve the desire for understanding and certainty. When major events happen, especially unexpected or frightening ones, people want explanations. Conspiracy theories offer broad, internally consistent narratives that can account for almost anything. They promise to reveal a hidden truth that the mainstream has missed, which can feel more satisfying than accepting that an event had mundane or multiple causes. These epistemic motives overlap with the cognitive factors discussed above – the preference for simple, pattern-rich, agent-driven explanations – but here the emphasis is on the felt need for closure and certainty rather than on cognitive style per se.

Existential motives involve the need for control and security. When people feel threatened or powerless, conspiracy theories can offer a sense of understanding that restores a feeling of control. If the world is chaotic and unpredictable, that is frightening. But if there is a group secretly controlling events, then at least someone is in charge, and perhaps something can be done about it. In one study described by Douglas et al. (2017), participants who had been asked to imagine a scenario in which they lacked control subsequently perceived more patterns in random visual noise and endorsed more conspiracy theories than participants in a control condition. The feeling of powerlessness appeared to heighten the very pattern-detection processes that feed conspiracy thinking. However, there is a paradox: while conspiracy theories promise a sense of control, exposure to them actually tends to make people feel more powerless, less trusting, and less willing to engage in political action (Douglas et al., 2019). The cure, in other words, may be worse than the disease.

Social motives involve the desire to see oneself and one's group in a positive light. Conspiracy theories often have a clear ingroup–outgroup structure: "we" are the ordinary, honest people, and "they" are the powerful, secretive elite conspiring against us. Consider the anti-vaccination movement, where parents who see themselves as informed, caring protectors of their children are positioned against "Big Pharma" and the medical establishment, portrayed as motivated by profit rather than health. Or consider the long history of conspiracy beliefs among Black Americans regarding government medical programs – beliefs grounded in real events like the Tuskegee syphilis study, in which the U.S. government did deceive Black participants for decades. This ingroup–outgroup structure is closely related to populism, which similarly divides the world into a virtuous people and a corrupt establishment (see Populism and Social Media). Conspiracy beliefs are more common among disadvantaged groups, minority populations, and people who feel their group has been treated unfairly (Douglas et al., 2019). For these groups, conspiracy theories can serve as a way to make sense of genuine injustice and inequality, even when the specific explanation offered is wrong about the mechanisms.

These epistemic, existential, and social motives explain why conspiracy narratives are attractive. The next question is how they are maintained and intensified in groups.

Echo Chambers and Group Polarization

Conspiracy beliefs are rarely formed in isolation. They develop and strengthen within communities of like-minded people, and this is where several group processes discussed in Group Decisions – especially group polarization, shared information bias, and groupthink – become relevant.

Group polarization is the tendency for groups to shift toward more extreme positions after discussion. In a community of people who are suspicious of a pharmaceutical company, discussion tends to push the group toward greater suspicion, not less. Each member shares the most compelling piece of evidence they have encountered, and because the group is already leaning in one direction, the shared evidence overwhelmingly supports the conspiracy. This is reinforced by shared information bias, the tendency for groups to spend most of their time discussing information that all members already know rather than information that only one member holds. If one person in the group has encountered a credible debunking of the theory, that information is less likely to be raised and discussed than the already-familiar claims supporting the theory.

Groupthink adds another layer. In conspiracy-minded communities, questioning the conspiracy carries social costs. Someone who expresses doubt may be accused of being naive, a "sheep," or even a paid agent of the very conspiracy the group is fighting against. This suppression of dissent is a hallmark of groupthink, and it is especially powerful here because the theory itself provides a ready-made explanation for any dissenter: they must be part of the cover-up.

Social media algorithms accelerate all of these processes (see Populism and Social Media). Many platforms tend to recommend content similar to what users have already engaged with, which can intensify one-sided exposure. If you watch one video questioning the moon landing, the algorithm will suggest more. If you join a group discussing chemtrails, you will see more content about chemtrails in your feed. The result is a filter bubble in which conspiracy-supporting content is abundant and contradicting information is scarce. Because the belief system is self-sealing, any outside information that does penetrate the bubble is automatically suspect.

Douglas & Sutton (2008) demonstrated another important feature of how conspiracy theories spread. They exposed participants to conspiracy theories about the death of Princess Diana and then measured how much their attitudes had shifted. Participants showed significant attitude change after exposure, becoming more sympathetic to the conspiracy theories. But participants substantially underestimated how much their own attitudes had changed. They adjusted their memories of their pre-exposure beliefs to be more in line with their current beliefs, a pattern similar to the hindsight bias discussed in Hindsight. Meanwhile, they accurately judged how much their peers had been influenced. This is a version of the third-person effect: people believe they are less susceptible to persuasion than others. Conspiracy theories can shift attitudes without people realizing it, which makes them especially difficult to counter. You cannot resist influence you do not recognize.

Rejection of Science

A particularly important application of conspiracy thinking is the rejection of established scientific findings. Climate change denial, anti-vaccination movements, and rejection of the link between smoking and cancer all share a common structure: the scientific evidence is dismissed not on its merits but because the scientists who produced it are assumed to have hidden agendas. If climate scientists warn about global warming, it must be because they want research funding. If doctors recommend vaccines, it must be because pharmaceutical companies are paying them. The scientific consensus itself becomes evidence of a conspiracy – the same self-sealing mechanism at work.

Lewandowsky et al. (2013) investigated the psychological factors behind science rejection by surveying visitors to climate blogs. They measured participants' free-market ideology, their acceptance of scientific findings in several domains (climate change, the HIV–AIDS link, the smoking–cancer link), and their general tendency toward conspiracy thinking. The results revealed two distinct but related predictors. Free-market ideology strongly predicted the rejection of climate science and the link between smoking and cancer, but not the link between HIV and AIDS. This pattern makes sense: climate regulation and tobacco regulation threaten free-market principles, while HIV–AIDS research does not. Conspiracy thinking, however, predicted rejection of scientific findings across all domains. People who endorsed unrelated conspiracy theories (such as the belief that the moon landing was faked or that the U.S. government allowed the September 11 attacks to happen) were more likely to reject scientific consensus on climate change, smoking, and HIV alike. This suggests that conspiracy thinking operates as a general cognitive style that disposes people to distrust expert knowledge, regardless of the specific topic.

The connection between conspiracy thinking and science rejection has consequences that extend well beyond individual beliefs. Vaccine hesitancy, fueled in part by conspiracy theories about pharmaceutical companies, has contributed to outbreaks of measles in Europe and elsewhere. Climate change denial, sustained by conspiracy theories about scientists, has delayed policy responses to one of the most significant challenges of the century. In each case, the underlying mechanism is the same: distrust of the evidence source neutralizes the evidence itself, making correction through normal scientific communication ineffective.

A Bayesian Lens

From a Bayesian perspective, the central problem with conspiracy theories is that they are effectively unfalsifiable. In Bayesian reasoning, described in detail in Bayesian Reasoning, a rational thinker starts with a prior belief, observes new evidence, and updates the belief based on how likely that evidence would be if the belief were true versus false. This process requires that some possible observations could count as evidence against the belief. If a theory can accommodate every possible outcome, including evidence that appears to disconfirm it, then no observation can lower the posterior probability.

Consider a concrete contrast. Suppose you believe your neighbor is growing tomatoes in their garden. If you look over the fence and see bare soil, that is evidence against your belief – you would be more likely to see bare soil if there were no tomatoes than if there were. You update downward. Now suppose instead you believe your neighbor is secretly running an illegal operation from their house. If you see nothing unusual, that is what you would expect – they are hiding it well. If police investigate and find nothing, the police must not have looked hard enough, or perhaps they were paid off. If a journalist writes an article debunking the rumor, the journalist must be in on it. Every observation that should lower your belief is reinterpreted as expected under the conspiracy hypothesis, or the source of the evidence is dismissed as compromised. In Bayesian terms, apparently disconfirming evidence is assigned either a high likelihood under the conspiracy hypothesis ("of course they would deny it") or near-zero source credibility. As a result, evidence that should lower belief is neutralized or even turned into apparent support.

The finding by Wood et al. (2012) that people endorse mutually contradictory conspiracy theories reveals something even more fundamental. If someone believes both that Princess Diana was murdered and that she faked her death, they are not updating on specific evidence at all. Instead, they hold a meta-prior – powerful people are conspiring, and official accounts are lies – that drives all the specific beliefs, regardless of whether those beliefs are logically compatible with each other.

The rejection of science involves a related but distinct mechanism: setting the credibility of an entire evidence source to near zero. If you believe scientists are systematically dishonest, then no scientific finding, no matter how well-replicated, will shift your beliefs. You have effectively removed an entire channel of evidence from your updating process. This is why conspiracy beliefs are so resistant to correction: evidence from normally trusted sources no longer functions as effective disconfirmation. The belief system is not merely strong – it is insulated from the kind of evidence that would normally weaken it.

Summary

Conspiracy theories are explanations that attribute events to secret plots by powerful groups. They can be false, partly true, or occasionally true, but what makes them psychologically distinctive is their self-sealing nature: contradictory evidence is absorbed as part of the cover-up, and people can endorse mutually contradictory theories driven by a generalized suspicion of official accounts. Cognitive factors that predict conspiracy thinking include apophenia, agency detection, a preference for simple explanations, reliance on intuitive thinking, and lower analytic thinking. Epistemic, existential, and social motives all contribute: people seek causal explanations, a sense of control, and positive group identity, though conspiracy beliefs paradoxically tend to increase feelings of powerlessness. Group polarization, groupthink, and social media echo chambers sustain and strengthen these beliefs. Science rejection is a specific application in which the credibility of scientists is dismissed, removing an entire evidence source from belief updating. From a Bayesian perspective, the core problem is that disconfirming evidence is assigned a high likelihood under the conspiracy hypothesis or its source is discredited, making the belief effectively unfalsifiable and resistant to normatively appropriate updating.

CONSPIRACY THEORIES

MORALITY AND COOPERATION

MORALITY AND COOPERATION

MORALITY

Learning goals:

- Explain why moral judgments are considered central to personal identity.
- Describe rationalist models of moral judgment, including Kohlberg's stages of moral development.
- Describe intuitionist models of moral judgment, including the social intuitionist model.
- Explain the concept of moral dumbfounding and the two illusions it reveals.
- Describe the five dimensions of Moral Foundations Theory.
- Explain how bodily states influence moral judgment through the affect-as-information heuristic and the somatic marker hypothesis.
- Describe how the trolley and footbridge dilemmas illustrate the role of emotion in moral reasoning.
- Explain omission bias and its real-world consequences.
- Relate moral intuitions to the Bayesian framework of prior beliefs and evidence updating.

Moral Judgment as Belief Formation

Imagine you are sitting in a café in Groningen and you overhear someone bragging about cheating on an exam. You immediately feel that what they did was wrong. But now ask yourself: did you reason your way to that conclusion, weighing up arguments about fairness and the rules of academic integrity? Or did the judgment arrive instantly, as a gut feeling, with the reasons coming only afterward when your friend asked you why you looked annoyed?

This question runs through the entire psychology of morality. Moral judgments are beliefs about whether something is right or wrong. In many ways, they follow the same principles as other beliefs: they form quickly and they resist updating. One sign of this resistance is moral dumbfounding, which we will encounter later in this chapter: people maintain firm moral judgments even after their stated reasons have been systematically rebutted. But moral beliefs also carry a special weight. Strohminger & Nichols (2014) conducted five studies examining which aspects of the mind people consider most essential to personal identity. Across scenarios ranging from brain transplants to reincarnation, participants consistently rated moral traits, such as honesty and empathy, as more central to who someone is than memories, personality, desires, or cognitive abilities. Losing your sense of direction or becoming forgetful was seen as relatively minor. But becoming cruel or dishonest? That was seen as becoming a different person.

The central theoretical question is whether moral judgments arise primarily from rational deliberation, a System 2 process, or from gut feeling, a System 1 process (Reason and Intuition). As discussed in How It Is vs. How It Should Be, this is a descriptive question about how moral judgments are actually formed, not a normative question about which moral theory is correct. Both rationalist and intuitionist models aim to describe the psychology of moral judgment, not to tell us what is truly right or wrong.

Morality and Convention

Before exploring this debate, it is worth noting that the very distinction between morality and cultural convention may be relatively recent in human history. Hallpike (2004) argued that many cultures, and many people within our own culture, do not sharply separate “this is what we generally do here” from “this is what we should do because it is morally right.” Consider food taboos: in many communities, eating pork or beef is not merely something people happen to avoid, it is felt to be genuinely wrong, a moral failing rather than a matter of personal preference. In other communities, the same dietary choice is treated as pure convention, no different from which hand you eat with. What feels like a universal moral truth may, in part, reflect local customs that have been elevated to the status of moral rules.

This blurring of custom and morality is itself a clue about how moral judgments work. If moral reasoning were a purely rational process of identifying harm and unfairness, we would expect people everywhere to draw the morality-convention boundary in the same place. The fact that they do not suggests that something other than deliberate reasoning is at work, and it sets the stage for the debate between rationalist and intuitionist models.

Rationalist Models

Rationalist models hold that people first consciously reason through a moral problem and then arrive at a judgment. On this view, moral judgment is a System 2 process: slow, deliberate, and based on applying rules or principles to specific situations.

The most influential rationalist model is Kohlberg's theory of moral development (Kohlberg, 1971). Kohlberg presented children and adults with moral dilemmas, the most famous being the Heinz dilemma: a man's wife is dying, and the only drug that could save her is sold by a pharmacist at a price Heinz cannot afford. Should Heinz steal the drug? Kohlberg was not interested in whether people said yes or no. He was interested in the reasons they gave. Based on these reasons, he proposed six stages of moral development, organized into three levels.

At the pre-conventional level, moral reasoning is based on personal consequences. In Stage 1, children judge actions by whether they lead to punishment. In Stage 2, they reason in terms of self-interest and simple exchange: "I'll scratch your back if you scratch mine." At the conventional level, moral reasoning is based on social expectations. In Stage 3, people want to be seen as good by others and judge actions by whether they match the expectations of family or peers. In Stage 4, they focus on upholding laws and social order. At the post-conventional level, moral reasoning is based on abstract principles. In Stage 5, people recognize that laws are social contracts that can be changed if they are unjust. In Stage 6, people reason from universal ethical principles, such as justice and human dignity, and may protest unjust laws even at personal cost. Kohlberg treated Stage 6 as the highest form of moral reasoning, though evidence for a distinct final stage has been mixed, and most research has focused on the earlier stages.

Kohlberg's research, conducted across multiple cultures, suggested that the sequence of stages is universal, though the rate and endpoint of development vary. Not everyone reaches the post-conventional level. The theory is rationalist because it assumes that moral development is driven by increasingly sophisticated reasoning about moral problems. Children do not just absorb the moral rules of their culture; they actively reconstruct their moral understanding as they encounter new situations and perspectives.

A related rationalist approach is Turiel's domain theory (Turiel, 1983), sometimes called the social interactionist model, which proposes that children reason about actions by evaluating their consequences and classifying them into different domains: personal choices (what to wear), moral violations (actions that cause harm or violate fairness), and social conventions (rules that exist by agreement, like raising your hand before speaking in class). In classic studies, researchers presented children with scenarios involving rule violations and asked two key questions: would the action still be wrong if there were no rule against it, and would it be wrong if it happened in a different country? Children as young as five or six answered these questions differently depending on the domain. They judged that hitting another child would be wrong everywhere, even if no rule forbade it, but that wearing a hat in class was only wrong if the school had a rule against it. This pattern suggests that children do not treat all rules as equivalent. They distinguish between rules grounded in harm and fairness and rules grounded in social agreement.

However, the boundary between these domains turns out to be culturally shaped. Haidt et al. (1993) presented people in Brazil and the United States with scenarios describing actions that were offensive but harmless, such as cleaning a toilet with the national flag or eating a dead pet dog. People from higher socioeconomic backgrounds tended to classify these as social conventions: disgusting perhaps, but not morally wrong as long as no one is harmed. People from lower socioeconomic backgrounds were more likely to classify them as genuine moral violations. This finding echoes the morality-convention discussion above: the line between "merely conventional" and "truly moral" is not fixed by logic alone but is influenced by the cultural and social context in which people reason.

Intuitionist Models

If rationalist models see moral judgment as the output of careful reasoning, intuitionist models turn this picture upside down. On the intuitionist view, moral judgments arrive as fast, automatic intuitions, and reasoning comes afterward, serving not to generate the judgment but to justify it.

The most influential intuitionist account is the social intuitionist model proposed by Haidt (2001). Haidt opened his landmark article with a scenario designed to challenge rationalist assumptions: Julie and Mark are adult siblings who decide, during a trip in the south of France, to have sex. They use two forms of contraception, enjoy it, but decide never to do it again. They keep it as a special secret. Is what they did wrong? Most people immediately say yes. But when asked to explain why, they struggle. They point to the risk of birth defects, but are reminded about the contraception. They suggest it might damage their relationship, but are told it brought them closer. Eventually, many people say something like: "I don't know, I can't explain it, I just know it's wrong."

This pattern is called moral dumbfounding. People hold a definite moral opinion but cannot produce clear reasons for it. This matters theoretically because it directly challenges rationalist models: if conscious reasoning generated the judgment, people should be able to articulate those reasons. The fact that they cannot, yet remain confident, suggests that the judgment was produced by something other than deliberation.

The social intuitionist model describes two illusions that follow from this. The first is the wag-the-dog illusion: the mistaken belief that our moral reasoning drives our moral judgment. In reality, according to Haidt, the judgment (the dog) wags the reasoning (the tail), not the other way around. The second is the wag-the-other-dog's-tail illusion: the mistaken belief that if we rebut someone's reasoning, we will change their moral position. If moral judgments are driven by intuitions rather than arguments, then defeating someone's argument will simply prompt them to find a new justification, leaving the underlying judgment intact. To change someone's moral position, you would need to change their emotional response, not their logical analysis. This insight connects to the broader literature on how beliefs persist even after the evidence for them has been discredited (Belief Formation and Perseverance).

The social intuitionist model does not claim that reasoning never matters. Haidt acknowledged that in some cases, especially when initial intuitions are weak or when people encounter arguments that trigger new intuitions, reasoning can influence judgment. But he argued that these cases are the exception rather than the rule. For most moral judgments in everyday life, the process runs from intuition to judgment, with reasoning playing a supporting role.

Moral Foundations Theory

If moral judgments are driven by intuitions, what are those intuitions about? Moral Foundations Theory, developed by Graham et al. (2009), describes the content of moral intuitions across five dimensions: Harm (concern about suffering and cruelty), Fairness (concern about cheating and injustice), Ingroup (loyalty to one's group), Authority (respect for hierarchy and tradition), and Purity (concern about contamination and degradation).

Graham, Haidt, and Nosek tested political differences in moral foundations using several methods, including surveys, moral judgment tasks, and analyses of the language used in religious sermons. In one study, participants were asked how much money someone would have to pay them to perform various acts that violated each foundation, such as kicking a dog in the head (Harm), saying something disrespectful about their country (Ingroup), or getting a blood transfusion of disease-free blood from a convicted child molester (Purity). The results showed a consistent pattern: all groups endorsed all five foundations to some degree, but people who identified as politically liberal placed relatively more weight on Harm and Fairness, while people who identified as conservative placed relatively more weight on Ingroup, Authority, and Purity, in addition to Harm and Fairness. The difference is one of relative emphasis, not the presence or absence of a foundation.

This finding helps explain one reason political disagreements about moral issues can feel so intractable. When a liberal argues that a particular policy is unfair, and a conservative responds that it upholds important traditions and social cohesion, they are not necessarily disagreeing about the facts. They are drawing on different moral foundations and weighting them differently. While the original research focused primarily on the United States, subsequent studies have found similar patterns across cultures, though the specific content of each foundation varies.

Moral Foundations Theory describes the contents of moral intuitions, but it does not by itself explain where those intuitions originate. To answer that question, we need to consider emotion, embodiment, and evolution.

The Evolutionary Roots of Moral Feeling

One answer to the question of origins comes from evolutionary biology. Darwin argued that any sufficiently social and intelligent animal would develop a moral sense. De Waal (2003) reviewed evidence from primate behavior showing that many species exhibit responses that look like building blocks of morality. Chimpanzees console distressed group members after conflicts. Capuchin monkeys become visibly upset when a partner receives a better reward for the same task: if one monkey receives a cucumber while the monkey next to her gets a grape, the cucumber monkey may refuse to continue participating, sometimes throwing the cucumber back at the experimenter. De Waal argued that morality is not a thin cultural veneer over a selfish nature, as some have claimed. Instead, some building blocks of human morality evolved out of older systems for cooperation and social regulation that we share with other primates.

Emotion, the Body, and Moral Judgment

If moral intuitions have evolutionary roots, how do they operate in the moment? The answer involves the body. James (1884) proposed a radical idea about how emotions work: we do not cry because we are sad; we are sad because we cry. In other words, emotion is not a mental event that causes a bodily response. It is the perception of a bodily response. When you encounter a moral violation, your body responds, with a change in heart rate, muscle tension, or gut feeling, and you then interpret that bodily response as a moral emotion such as disgust or outrage.

Modern research supports the idea that bodily states influence emotional and moral experience. MacCormack & Lindquist (2018) studied the phenomenon of feeling "hangry," the irritability that comes with hunger. In one study, hungry participants rated ambiguous images more negatively than non-hungry participants, but only when they were in a negative context and were not explicitly aware that their feelings might be caused by hunger. This suggests that bodily states like hunger generate unpleasant feelings that people interpret as emotional responses to whatever is happening around them. The implication for morality is that the gut feelings driving moral judgments may be shaped, in part, by the state of the body at the time.

Two frameworks describe this relationship between feelings and judgment in more detail. They capture a similar core insight, that feelings serve as informational input to judgment, but from different levels of explanation.

The first is the affect-as-information heuristic, a psychological principle. Schwarz & Clore (1983) demonstrated it in an elegant study. They called people on the phone and asked them how satisfied they were with their lives. People called on sunny days reported higher life satisfaction than people called on rainy days. But when the interviewer casually mentioned the weather before asking the question ("How's the weather down there?"), the effect disappeared. Once people attributed their good or bad mood to the weather, they stopped using it as information about their lives.

The affect-as-information heuristic describes a general pattern: when making a judgment, people often consult their current feeling state and ask themselves, "How do I feel about this?" If the feeling is positive, the judgment is favorable; if negative, unfavorable. This heuristic applies broadly, from consumer choices to risk perception (Risk Perception), and it applies to moral judgments as well. When you feel disgust in response to a moral scenario, that feeling becomes information that you use to judge the action as wrong.

The second framework is the somatic marker hypothesis, which offers a neurobiological account of the same basic mechanism. Bechara et al. (2000) studied patients with damage to the ventromedial prefrontal cortex, a brain region involved in linking knowledge about the world with emotional responses. These patients had intact reasoning abilities: they performed normally on intelligence tests, memory tasks, and logical problems. But their decision-making in real life was catastrophic. They made poor financial choices, damaged their relationships, and could not learn from repeated mistakes.

To study this in the laboratory, Bechara and colleagues used the Iowa Gambling Task, in which participants choose cards from four decks. Two decks offer high rewards but even higher penalties, leading to losses over time. The other two offer modest rewards with small penalties, leading to gains. Healthy participants gradually learn to avoid the bad decks. Crucially, they develop anticipatory skin conductance responses, a bodily signal of anxiety, before reaching for a bad deck, even before they can consciously explain which decks are good or bad. Patients with ventromedial prefrontal cortex damage never develop these anticipatory signals. They continue choosing from the bad decks, even when they can articulate that those decks are risky. Their reasoning is intact, but without the bodily signal marking certain options as dangerous, they cannot translate that reasoning into good decisions. This supports the view that intact decision-making depends on emotional as well as analytic processes.

Where the affect-as-information heuristic describes the psychological principle, the somatic marker hypothesis identifies the biological mechanism: the affective signals that people use as information are specifically bodily states, gut feelings, changes in skin conductance, shifts in heart rate, that mark options as good or bad.

Trolley Problems and the Body

These ideas about bodily feelings and moral judgment converge in the trolley problem, one of the most discussed thought experiments in moral psychology. In the standard version, a trolley is heading toward five people on the track. You can flip a switch to divert it to a side track, where it will kill one person. Most people say they would flip the switch. In the footbridge version, you are standing on a bridge over the tracks. The only way to stop the trolley is to push a large stranger off the bridge and onto the tracks below, killing him but saving the five. Most people say they would not push.

The utilitarian calculation is the same in both cases: one life lost, five saved. But the judgments are very different. Greene et al. (2001) used brain imaging to investigate why. They presented participants with a series of moral dilemmas and categorized them as either personal (involving direct physical harm, like pushing someone) or impersonal (involving indirect harm, like flipping a switch). Brain regions commonly associated with emotional processing, including the medial frontal gyrus and posterior cingulate, were significantly more active during personal dilemmas than during impersonal ones. Areas associated with working memory and deliberate reasoning showed the opposite pattern.

The behavioral data told the same story. When participants gave the utilitarian response to personal dilemmas, such as saying they would push the stranger, they took significantly longer to respond than when they gave the non-utilitarian response. This delay suggests emotional conflict: participants had to override a strong intuitive response to reach the utilitarian conclusion. No such delay was observed for impersonal dilemmas like the switch version, where the emotional response and the utilitarian calculation pointed in the same direction.

One influential interpretation of this difference is that pushing someone off a bridge triggers a stronger visceral, bodily response than flipping a switch. The personal dilemma involves direct physical contact and the use of someone's body as a means to an end, activating the kind of affective signal that the somatic marker framework describes. Other factors likely contribute as well, including perceived directness of causation and personal force, but the emotional and bodily component appears central.

If moral judgments in these dilemmas are shaped by bodily feelings, then manipulating those feelings should change the judgments. This is what researchers have found. Valdesolo & DeSteno (2006) had participants watch either a comedy clip or a neutral documentary before responding to the footbridge dilemma. Participants who had just been laughing were nearly four times more likely to say they would push the stranger compared to those in the neutral condition. Their positive mood dampened the negative affective signal that normally makes people refuse to push. Meanwhile, responses to the standard trolley dilemma were unaffected by mood, consistent with the idea that the switch version does not trigger a strong emotional response in the first place.

Manipulations that increase negative bodily states have the opposite effect. Schnall et al. (2008) conducted four experiments showing that incidental disgust makes moral judgments harsher. In one experiment, participants made moral judgments while sitting in a room where a "fart spray" had been released. They judged moral violations more severely than participants in a neutral-smelling room. Importantly, this effect was strongest for people who scored high on a measure of sensitivity to their own bodily sensations, supporting the idea that the effect operates through embodied feeling rather than purely cognitive appraisal. In another study by Eskine et al. (2011), participants who drank a bitter beverage judged moral violations more harshly than those who drank a sweet drink or water. The bitter taste produced a feeling of disgust that bled into their moral evaluations, even though the taste had nothing to do with the scenarios they were judging.

Together, these findings suggest that incidental bodily states can influence moral judgment. Moral evaluations in emotionally charged dilemmas are not determined solely by rational analysis of consequences. They are shaped by affective signals that can be amplified or dampened by factors that have nothing to do with the moral content of the situation: a comedy clip, a bad smell, or a bitter drink. It is worth noting that some of the disgust-priming effects have proven difficult to replicate, and the strength and boundary conditions of these influences remain an active area of research. But the broader pattern, that bodily feelings contribute to moral judgment, is supported by converging evidence across multiple methods.

Omission Bias

One downstream consequence of the role of affect in moral judgment is omission bias: the tendency to judge harmful actions as worse than equally harmful omissions. Acting generates a stronger affective signal than not acting, even when the outcomes of inaction are identical or worse.

Consider vaccination. Many parents who hesitate to vaccinate their children are not opposed to vaccines on scientific grounds. They are responding to the asymmetry in how action and inaction feel. If a child has a rare adverse reaction to a vaccine, the parent feels responsible because they chose to act. If a child contracts a preventable disease because the parent chose not to vaccinate, the parent may feel less responsible, because they "only" failed to act. The outcomes can be far worse in the second case, but the affective signal from acting is stronger than the signal from not acting.

The same asymmetry appears in end-of-life decisions. Withdrawing a treatment that is keeping a patient alive feels morally different from not starting that treatment in the first place, even when the medical situation is identical. In both cases, the patient dies. But the act of withdrawing triggers a stronger emotional response than the omission of not starting.

Omission bias interacts with both loss aversion (Gains and Losses) and the status quo bias discussed in Steering Other People's Choices. In all three cases, the pattern is similar: doing nothing feels safer and more morally neutral than doing something, even when the evidence shows that the outcomes of inaction are worse. Recognizing omission bias matters practically, because it can lead to systematically worse outcomes in domains like public health and medical ethics, where the costs of inaction are real but less emotionally salient than the costs of action.

A Bayesian Lens

Throughout this book, we have framed judgment and decision-making in terms of Bayesian belief updating: a prior belief is combined with new evidence to form a posterior belief (Bayesian Reasoning). Moral judgment fits this framework in a revealing way.

Moral intuitions function as strong priors. They are formed early in life, felt with certainty, and highly resistant to updating. The social intuitionist model suggests that in many cases, moral reasoning operates less like open-ended evidence updating and more like post-hoc rationalization of an already strong prior. Changing moral positions requires changing the prior itself, the emotional intuition, rather than presenting logical counter-evidence. This is what the wag-the-other-dog's-tail illusion describes: rebutting reasons does not change the prior, so a new justification simply takes the place of the old one (Belief Formation and Perseverance).

Moral Foundations Theory describes the content of these priors, which vary across individuals and cultures. A person high on the Purity foundation carries a strong prior that certain acts are contaminating, regardless of whether anyone is harmed. A person high on the Fairness foundation carries a strong prior that unequal treatment is wrong, regardless of tradition or authority.

The somatic marker hypothesis adds that these priors are not only cognitive representations but are also encoded in bodily sensations. In the trolley versus footbridge dilemma, the emotional prior against directly harming another person with your own hands overwhelms the utilitarian calculation in the personal version but not the impersonal one. Omission bias reflects a prior that inaction is morally neutral, a prior that evidence about equivalent or worse outcomes from inaction struggles to overcome, because the evidence is processed by System 2 while the prior is maintained by System 1.

This perspective also clarifies why moral persuasion is so difficult. If you want to change someone's moral belief, you are not simply updating a belief with new evidence. You are trying to replace a deeply felt prior with a different one. That is a fundamentally different, and much harder, task.

Summary

Moral judgments are beliefs about right and wrong that people consider central to personal identity. The distinction between morality and cultural convention is itself culturally variable, with some communities treating as moral violations what others regard as mere social customs. Two broad classes of models explain how moral judgments arise. Rationalist models, such as Kohlberg's stages of moral development and Turiel's domain theory, hold that moral judgments result from deliberate reasoning about harm, fairness, and rules. Intuitionist models, such as Haidt's social intuitionist model, hold that moral judgments are driven by fast, automatic intuitions, with reasoning serving mainly to justify them. Moral dumbfounding, where people hold strong moral opinions but cannot explain why, provides evidence for the intuitionist view. Moral Foundations Theory describes five dimensions along which moral intuitions vary: Harm, Fairness, Ingroup, Authority, and Purity, with political liberals and conservatives weighting these differently. The evolutionary roots of moral feeling are visible in other primates, who show empathy, retribution, and a sense of fairness. The affect-as-information heuristic and the somatic marker hypothesis describe, at psychological and neurobiological levels respectively, how bodily feelings serve as input to judgment. The trolley and footbridge dilemmas show that personal moral dilemmas activate emotional brain regions more strongly than impersonal ones, and that manipulating bodily states, through comedy, bad smells, or bitter tastes, can shift moral judgments. Omission bias, the tendency to judge harmful actions as worse than equally harmful inactions, follows from the stronger affective signal generated by acting, with consequences for vaccination decisions and end-of-life care. From a Bayesian perspective, moral intuitions function as strong priors, often grounded in bodily feeling, that resist updating through evidence and reasoning alone.

MORALITY

COOPERATION

Learning goals:

- Explain why individual rationality can lead to collective irrationality in social dilemmas.
- Describe the tragedy of the commons and identify real-world examples.
- Explain the prisoner's dilemma and why people cooperate more than game theory predicts.
- Describe the ultimatum game, dictator game, and trust game, and explain what each reveals about human motivation.
- Explain how altruistic punishment sustains cooperation in public goods games.
- Describe how framing effects influence cooperative behavior.
- Explain the evidence that cooperation can be an intuitive, System 1 response.

The Core Tension

Imagine you share a house with three other students. Nobody enjoys cleaning the kitchen, but everyone enjoys a clean kitchen. If you skip your turn, you save yourself twenty minutes of scrubbing while the kitchen only gets slightly dirtier. The problem is that everyone faces the same temptation. If all four of you follow this logic, the kitchen becomes a health hazard. Each person made a perfectly reasonable decision from their own perspective, yet the outcome is worse for everyone.

This pattern, where individual rationality leads to collective irrationality, sits at the heart of many social problems. It appears when fishers overexploit a shared fishing ground, when commuters each choose to drive rather than cycle, and when countries delay cutting carbon emissions. A single country bears the immediate political cost of reducing emissions, while the climate benefit is delayed and spread globally. In each case, the action that benefits the individual harms the group.

Game theory models these situations by specifying each player's options, payoffs, and the way one person's best choice depends on what others do. Behavioral experiments then reveal how people actually navigate these situations, and what they consistently show is that people cooperate far more than pure material self-interest would predict, but also far less than would be ideal. Understanding why requires drawing on moral intuitions (Morality), framing effects (Gains and Losses), social norms, and the interplay between fast intuition and slow deliberation (Reason and Intuition).

The Tragedy of the Commons

In 1968, the ecologist Garrett Hardin described a thought experiment that has shaped thinking about social dilemmas (Hardin, 1968). Imagine a pasture open to all herders in a village. Each herder benefits from adding one more cow to the commons: the extra milk and meat go entirely to that herder, while the cost of overgrazing is shared among everyone. Because the personal benefit is large and the personal cost is small, the rational decision for each herder is to keep adding cows. But when every herder follows this logic, the pasture is destroyed. Hardin called this the tragedy of the commons.

The tragedy is not just a parable about medieval farming. It describes the deep structure of problems like overfishing in the North Sea, groundwater depletion in southern Spain, and global carbon emissions. In each case, the benefit of exploiting the resource is immediate and tangible. For a fisher, today's larger catch is visible immediately; the depletion of the stock is gradual, uncertain, and shared across many others. The benefit is the kind of concrete outcome that System 1 processes easily. The cost, by contrast, is distributed across many people and stretched over time, requiring the kind of abstract reasoning that depends on System 2 (Reason and Intuition). This asymmetry helps explain why these problems are so persistent. It also explains why Hardin argued that appeals to conscience alone cannot solve them. Instead, he proposed mutual coercion, mutually agreed upon: rules and regulations that change the incentive structure so that individual and collective interests align.

Later research showed that shared resources are not always doomed: under the right institutions, monitoring, and norms, communities can manage commons successfully. But Hardin's analysis highlights a point that runs through this entire chapter. The problem is not that people are selfish. The problem is that the structure of the situation can make exploitation the dominant strategy. Changing the outcome requires changing the structure, whether through regulation, communication, or the kinds of psychological mechanisms we will examine next. Researchers study these tensions in simplified experimental games that isolate trust, reciprocity, fairness, and punishment.

The Prisoner's Dilemma

The most famous model of cooperation in game theory is the prisoner's dilemma, first developed in experiments by Merrill Flood and later given its memorable name by the mathematician Albert Tucker (Flood, 1958; Tucker, 1983). Two suspects are arrested and held in separate cells. The police offer each suspect a deal: if you confess and your partner stays silent, you go free and your partner gets three years. If both confess, you each get two years. If both stay silent, you each get one year on a lesser charge. The dilemma is that no matter what your partner does, you are better off confessing. If they stay silent, confessing sets you free instead of serving one year. If they confess, confessing gets you two years instead of three. Confessing is the dominant strategy for each player. The result is that both confess and each serves two years, even though both would have been better off staying silent and serving only one year each.

This outcome, where both players choose their dominant strategy, is called the Nash equilibrium, after John Nash, who proved that every game with a finite number of strategies has at least one such equilibrium (Nash, 1950). The Nash equilibrium is stable in the sense that neither player can improve their outcome by changing their strategy alone. But it is collectively worse than the cooperative outcome. This captures the core tension of social dilemmas: what is rational for the individual is irrational for the group.

In the standard one-shot version of the prisoner's dilemma, players cannot build a reputation or retaliate later, which makes cooperation especially puzzling from a narrow self-interest perspective. Even in a one-shot game, however, players are not just choosing an action; they are acting on beliefs about what the other person is likely to do, a point we will return to in the Bayesian section below.

The prisoner's dilemma was named after prisoners, but until recently, no one had actually tested prisoners. Khadjavi & Lange (2013) remedied this by running the game with female inmates in a German prison and comparing their behavior to female university students. In the simultaneous version of the game, where both players chose at the same time without knowing the other's decision, 56 percent of inmates cooperated, compared to 37 percent of students. In the sequential version, where one player moved first and the second could see the first player's choice, students were actually more cooperative as first movers (63 percent) than inmates (46 percent), but second-mover cooperation was nearly identical between the groups (around 60 percent for both). This suggests that inmates and students share similar levels of conditional cooperation, the willingness to cooperate when the other person cooperates first, but differ in their willingness to take the first step.

These results underscore a recurring finding: people value fairness and reciprocity in addition to their own immediate material payoff. The standard model predicts universal defection because it assumes that players care only about their own material outcomes. When real people cooperate at rates of 37 to 56 percent, the gap tells us that something beyond material self-interest is at work, whether moral intuitions, social norms, or both.

The Ultimatum Game

Not all cooperation research examines direct contribution to a common good. Some paradigms instead ask how people enforce norms of fairness, which is one route by which cooperation becomes stable. In these games, the key question shifts from "Will you contribute?" to "Will you punish those who act unfairly?" Social norms matter here in two ways: descriptive norms tell us what others typically do, and injunctive norms tell us what others approve or disapprove of. Both shape behavior in the games described below.

In 1982, the economists Werner Güth, Rolf Schmittberger, and Bernd Schwarze designed a simple game that would become one of the most widely studied experiments in the social sciences (Güth et al., 1982). In the ultimatum game, two players divide a sum of money. The proposer decides how to split it, and the responder can either accept the offer (both players get their share) or reject it (both get nothing). If players care only about maximizing their own earnings, the logic is straightforward. The proposer should offer the smallest possible amount, say 1 euro out of 10, because the responder should accept any positive offer. Getting 1 euro is better than getting nothing.

But this is not what happens. In the original experiments with German economics students, proposers typically offered 30 to 50 percent of the total, and responders frequently rejected offers they considered unfair, even though rejection meant walking away with nothing. Offers below about 30 percent were rejected at high rates. This pattern has been replicated thousands of times across dozens of countries and cultures. People are willing to pay a real cost to punish what they perceive as unfairness.

Why would someone reject a positive offer? From a purely financial perspective, rejecting 2 euros to receive 0 euros makes no sense. But the rejection makes sense if people care about more than money. An unfair offer triggers a visceral sense of injustice, the kind of gut feeling described by the somatic marker hypothesis (Morality). This affective response overrides the calculation that 2 euros is better than nothing. The proposer seems to understand this intuitively, which is why most proposers offer a substantial share even when they could keep more. The proposer's generosity likely reflects both an internalized injunctive norm (unfair splits are wrong) and a descriptive expectation (responders punish unfairness).

The Dictator Game

The ultimatum game leaves open a strategic explanation for generous offers: proposers might share fairly only because they fear rejection. To test this, researchers created the dictator game. It works like the ultimatum game, except the responder has no power to reject. The proposer simply decides how much to give, and the responder must accept whatever is offered.

If fairness in the ultimatum game were purely strategic, then dictators should keep everything. Yet even in the dictator game, proposers typically give away 20 to 30 percent of the total. This is less than in the ultimatum game, which confirms that the fear of rejection does play some role, but it is far from zero. With no strategic reason to share, the fact that people still give suggests genuine prosocial motivation. Whether this reflects true altruism, internalized social norms, or a desire to see oneself as a fair person is still debated, but the finding itself is robust.

The Trust Game

In 1995, Joyce Berg, John Dickhaut, and Kevin McCabe introduced the trust game to study whether people would take a financial risk based on their belief that a stranger would reciprocate (Berg et al., 1995). The setup is simple. Player 1 receives an endowment, say 10 euros, and can send any portion of it to Player 2. Whatever Player 1 sends is tripled by the experimenter, so if Player 1 sends 6 euros, Player 2 receives 18 euros. Player 2 then decides how much, if anything, to return to Player 1.

If Player 2 is purely self-interested, they should keep everything. Knowing this, Player 1 should send nothing. The Nash equilibrium predicts zero trust and zero reciprocity. But in the original experiment, 30 out of 32 participants in the role of Player 1 sent money, with an average transfer of about half their endowment. And in a substantial number of cases, Player 2 returned enough to make trusting profitable for Player 1.

The trust game separates two components of cooperation. Player 1's decision to send money measures trust: the willingness to make yourself vulnerable based on a belief that the other person will act fairly. Player 2's decision to return money measures reciprocity: the willingness to reward trust even when you could pocket the entire amount without consequence. Player 1's decision also has a Bayesian structure: the choice to send money reflects a prior belief about the probability that Player 2 will reciprocate. Berg and colleagues tested what happened when new participants were shown the results of previous rounds, a manipulation they called social history. When participants knew that others had been trustworthy in the past, both trust and reciprocity increased. This social history functioned as evidence, updating participants' priors about the likelihood of reciprocity. Knowing that cooperation is the descriptive norm makes people more willing to cooperate themselves.

The Public Goods Game and Altruistic Punishment

The prisoner's dilemma involves two players, but many real-world cooperation problems involve large groups. The public goods game captures this. In a typical version, four players each receive an endowment of 20 euros. Each player secretly decides how much to contribute to a common pool. The total contributions are multiplied by some factor, say 1.6, and then divided equally among all four players regardless of what each person contributed. If everyone contributes their full endowment, the group earns more than if nobody contributes. But for any individual player, the return on contributing is less than the return on keeping the money. Contributing 1 euro to the pool returns only 0.40 euros to the contributor (1.6 divided by 4). So if you keep your 1 euro, you are personally better off by 0.60 euros, even though the group as a whole loses 0.60 euros in total gains. The selfish strategy is to contribute nothing and free ride on others' contributions.

Despite the incentive to free ride, cooperation rates in public goods games typically start at 40 to 60 percent of the endowment. However, cooperation tends to decline over repeated rounds as people observe that some group members are free riding. By the final rounds, contributions often approach zero.

A key mechanism that can reverse this decline is altruistic punishment. In a landmark study, Ernst Fehr and Simon Gächter added a twist to the public goods game: after each round, players could see how much others had contributed, and they could spend some of their own earnings to reduce a free rider's payoff (Fehr & Gächter, 2002). Punishment was costly for both the punisher and the target. Crucially, the groups were reshuffled after each round, so there was no possibility of building a reputation or benefiting from punishing someone you would interact with again. Punishment was costly and could not be justified by direct strategic gain from future interaction with the same target.

The effect was large. When punishment was available, cooperation did not decline over time. Instead, it increased, eventually reaching near-complete contribution levels. When the same participants played without the punishment option, cooperation collapsed. Over 84 percent of participants punished at least once, and the severity of punishment increased with the degree of free riding. Fehr and Gächter also measured the emotions associated with free riding and found that seeing others free ride triggered strong feelings of anger. These emotions appeared to be the proximate cause of punishment: people punished because free riding made them angry, not because they had calculated that punishment would improve future cooperation.

Altruistic punishment is puzzling from the perspective of narrow self-interest. Spending your own money to reduce someone else's earnings, with no possibility of direct future gain, seems irrational. But from the perspective of group survival, it is essential. Without some mechanism to deter free riding, cooperation unravels. The fact that punishment is driven by emotion connects it to the broader theme that moral behavior arises in part from intuitive, affect-driven processes (Morality). It also illustrates how injunctive norms are enforced in practice: people do not merely disapprove of free riding, they act on that disapproval even at a cost to themselves.

Framing Effects on Cooperation

If cooperative decisions were driven purely by the payoff structure of a game, then changing the name of the game should have no effect. The payoffs are the same regardless of what you call the game. But payoffs are not all that matters.

In a series of studies, Varda Liberman, Steven Samuels, and Lee Ross had participants play a standard prisoner's dilemma game (Liberman et al., 2004). The only manipulation was the name. Half the participants were told they were playing the Wall Street Game. The other half were told they were playing the Community Game. Everything else – the rules, the payoffs, the interaction – was identical.

Cooperation roughly doubled under the community frame compared to the Wall Street frame. In one study, around 70 percent of participants cooperated in the Community Game, compared to around 33 percent in the Wall Street Game. The game's name activated different social norms and identities. The label "Wall Street" evoked competition and self-interest. The label "Community" evoked shared purpose and mutual aid. Participants also expected different behavior from their partners under the two labels: those in the Community Game expected their partners to cooperate, which in turn made them more likely to cooperate themselves. The frame shifted both descriptive expectations (what others will do) and the injunctive norms felt to apply (what one should do).

Perhaps even more revealing was what the researchers found when they asked independent observers to predict behavior. Resident assistants who knew the participants well were asked to nominate who would cooperate and who would defect. These personality-based predictions had almost no predictive value. Whether someone was nominated as a likely cooperator or a likely defector mattered far less than which label the game carried. This finding echoes a broader lesson in social psychology: we tend to overestimate the role of personality and underestimate the role of the situation (Blame, Praise, and Responsibility). If you want to promote cooperation, changing the frame may be more effective than trying to change the people. This connects directly to research on framing effects and the design of choice environments (Steering Other People's Choices).

Intuitive Cooperation

Is cooperation a deliberate, calculated choice, or is it something more automatic? The dual-process framework (Reason and Intuition) offers a way to investigate this question. If cooperation is driven by System 1, then forcing people to decide quickly should increase cooperation. If cooperation requires System 2 deliberation, then time pressure should decrease it.

David Rand, Joshua Greene, and Martin Nowak tested this across ten studies involving thousands of participants (Rand et al., 2012). In one study, 212 participants played a one-shot public goods game. Those who decided more quickly contributed significantly more to the common pool than those who took longer. In another study, the researchers directly manipulated decision time: participants who were forced to decide within 10 seconds contributed more than those who were told to take at least 10 seconds to think carefully. In yet another approach, they primed some participants to trust their gut feelings and others to rely on careful reasoning. Those primed with intuition cooperated more.

Rand and colleagues proposed an explanation rooted in daily life. In our everyday social interactions, cooperation usually pays off. We interact repeatedly with the same people, we build reputations, and those who cheat are punished or excluded. These experiences build cooperative habits – fast, automatic responses that serve us well in most situations. When we enter a laboratory experiment with a stranger we will never see again, these cooperative habits carry over. It takes deliberate effort, System 2 reasoning, to recognize that the one-shot structure of the experiment removes the usual incentives for cooperation and to override the cooperative impulse.

This interpretation received further support from an additional finding. The link between fast decisions and cooperation was weaker among participants who had previous experience with economic experiments and among those who reported interacting with less cooperative people in daily life. In other words, the cooperative default can be eroded by experience with situations where cooperation does not pay.

These findings were initially interpreted as evidence that cooperation can be an intuitive default in many contexts, though the robustness and boundary conditions of the effect remain debated. Some replications have produced weaker or null results, and the relationship between decision speed and cooperation may depend on the specific game, the stakes, and the population. What the evidence does suggest is that many people bring cooperative heuristics into social interactions, even when strict material self-interest would favor defection. This connects to the social intuitionist model of moral judgment (Morality), which holds that moral behavior arises first from fast, automatic feelings, with reasoning often serving to justify decisions already made.

A Bayesian Lens

Strategic interaction requires maintaining and updating beliefs about other people. When you decide whether to cooperate in a social dilemma, you are implicitly estimating the probability that the other person will cooperate too. This estimate functions as a prior belief about the other person's type: are they cooperative or selfish? For example, you might begin by thinking there is a 50 percent chance a new partner is trustworthy. After one generous return in a trust game, that estimate rises; after repeated defections, it falls. Each interaction provides evidence that updates this prior. Over repeated interactions, your beliefs about specific individuals, and about people in general, converge toward something more accurate.

The trust game makes this Bayesian logic explicit (Berg et al., 1995). Player 1's decision to send money reflects a prior belief that Player 2 will reciprocate. When Berg and colleagues showed new participants the results of earlier rounds, this social history functioned as evidence, updating participants' priors about the likelihood of reciprocity and increasing both trust and cooperation. The framing effects studied by Liberman et al. (2004) can also be understood in Bayesian terms: the label "Community Game" shifts the prior toward expecting cooperation, while the label "Wall Street Game" shifts it toward expecting competition. Both labels change the descriptive expectation – the belief about what others will do – which in turn changes the rational response.

At a broader level, the tragedy of the commons illustrates a sobering implication of Bayesian reasoning. Even if every individual is a rational Bayesian agent who correctly estimates the likelihood that others will overexploit the resource, the equilibrium outcome can still be collective ruin. Each person's locally rational decision to exploit the resource, given their accurate belief that others will do the same, produces an outcome that is worse for everyone. The prior is correct, the updating is rational, and the result is still a tragedy. This illustrates especially clearly that being individually rational does not guarantee good collective outcomes.

Summary

Social dilemmas arise when individual rationality leads to collective irrationality. The tragedy of the commons shows how shared resources can be destroyed when each person maximizes their own benefit, though the right institutions and norms can prevent this outcome. The prisoner's dilemma captures the tension in its simplest form: mutual defection is the Nash equilibrium, yet real people cooperate at substantial rates because they value fairness and reciprocity beyond their own material payoff. The ultimatum and dictator games reveal a willingness to enforce fairness norms even at personal cost, while the trust game shows that trust and reciprocity are stronger than self-interest models predict. In the public goods game, altruistic punishment

– paying a personal cost to sanction free riders – sustains cooperation that would otherwise collapse. Context shapes these decisions powerfully: simply labeling a game as the "Community Game" rather than the "Wall Street Game" can double cooperation rates, and time-pressure studies suggest that cooperation can function as an intuitive default. Three broad mechanisms sustain cooperation across these paradigms: expectations about others' behavior (updated in a Bayesian fashion), norm enforcement through costly punishment, and intuitive cooperative defaults shaped by everyday social experience. Together, these findings paint a picture of human nature that is more cooperative than traditional economic models assume, but also highly sensitive to the structure and framing of the social environment (Parks et al., 2013).

COOPERATION

REAL LIFE

MEDICAL DECISION-MAKING

Learning goals:

- Explain how the availability heuristic, anchoring, and confirmation bias affect clinical diagnosis.
- Describe the role of causal reasoning in how clinicians conceptualize mental disorders.
- Explain why natural frequencies improve physicians' diagnostic reasoning.
- Describe how decision support systems can reduce bias in clinical settings.
- Apply a Bayesian framework to understand where clinical reasoning goes wrong.

Doctors Are Human Too

Imagine you visit your general practitioner with a persistent headache. The doctor asks a few questions, runs some checks, and within minutes offers a diagnosis and a treatment plan. It feels reassuring how quickly a trained professional can make sense of your symptoms. But what if the doctor had just seen three patients that morning with tension headaches? What if the first thing that came to mind shaped every question asked afterward? What if the doctor unconsciously looked for evidence that confirmed the initial guess while ignoring signs pointing in a different direction?

These are not hypothetical concerns. Clinicians are subject to the same heuristics and biases that affect everyone else. Throughout this book, we have seen how people rely on mental shortcuts to navigate complex situations. In most everyday contexts, the consequences of a biased judgment are modest.

In medicine, however, a wrong diagnosis can mean unnecessary surgery, a missed cancer, or a patient receiving the wrong medication for years. The stakes make it essential to understand how cognitive biases enter clinical reasoning and what can be done about them.

Medical professionals work under conditions that are almost designed to trigger System 1 thinking. They face time pressure, incomplete information, emotional patients, and enormous complexity. A general practitioner in the Netherlands sees an average patient for about ten minutes. In that time, they must listen, examine, reason, decide, and communicate. These are exactly the conditions under which fast, automatic, and heuristic-driven reasoning dominates over slow, deliberate analysis, as described in Reason and Intuition. The question is not whether biases affect medical decisions, but how and how much.

Availability in Diagnosis

One of the most common biases in clinical settings involves the availability heuristic. As discussed in Availability, people estimate the frequency or probability of events based on how easily examples come to mind. In medicine, this means that a clinician's recent experience can shape the diagnosis of the next patient.

Consider a psychiatrist who has recently treated several patients with bipolar disorder. When the next patient arrives with symptoms of irritability, impulsivity, and mood swings, the psychiatrist may be quicker to consider bipolar disorder than they would have been otherwise. The diagnosis is more available because of recent cases, not because it is the most likely explanation for this particular patient. The same symptoms might equally fit borderline personality disorder or even a thyroid condition, but those alternatives may not spring to mind as readily.

This bias may actually increase with expertise. An experienced clinician has seen thousands of cases over a career. This rich library of memories is generally an asset, but it also means there are more vivid, dramatic, or recent cases that can be retrieved easily. A young doctor fresh out of medical school may have fewer memories to be biased by, while a seasoned specialist may have a wealth of memorable cases that pull diagnoses in certain directions. This does not mean that experience is a liability – experienced clinicians are generally more accurate than novices. But experience does not make a person immune to the availability heuristic. Having a larger pool of memorable cases gives the heuristic more material to work with.

Anchoring in Diagnosis

Closely related to availability is anchoring. As discussed in Anchoring and Primacy, people form initial estimates and then adjust insufficiently from that starting point. In medicine, the anchor is typically the first diagnostic hypothesis that comes to mind. Often, as discussed in the previous section, the most available diagnosis becomes that anchor.

When a patient describes their symptoms, the clinician almost immediately begins forming a hypothesis. This is natural and often useful. The problem is that once an initial hypothesis is in place, it becomes difficult to move away from. Subsequent information tends to be interpreted in light of the anchor rather than evaluated independently. If a clinician's first thought upon hearing about fatigue, weight loss, and night sweats is tuberculosis, then every subsequent test result or symptom report may be filtered through that lens. A mildly elevated white blood cell count might be seen as confirming TB, when it could just as easily be consistent with lymphoma or an autoimmune condition.

Experience may reduce susceptibility by broadening the set of hypotheses considered, which prevents locking onto a single anchor too quickly. But experience does not eliminate anchoring. Even seasoned professionals show the effect: the initial hypothesis continues to exert a disproportionate influence on all subsequent reasoning.

Confirmation Bias in Diagnosis

Once a clinician has anchored on a hypothesis, the next bias is almost inevitable. Confirmation bias, the tendency to seek, interpret, and remember information that supports an existing belief, is one of the most robust findings in psychology, as discussed in Confirmation. In clinical settings, it means that clinicians tend to ask questions and order tests that will confirm their initial diagnosis rather than rule it out.

Mendel et al. (2011) demonstrated this directly in a study with 75 psychiatrists and 75 medical students. Participants read a case description of a 65-year-old man showing depressive symptoms but also subtle signs of Alzheimer's disease. The case was designed so that most participants would initially think of depression. After forming their preliminary diagnosis, participants could request additional information to refine their judgment. They could choose from twelve items: six supporting a diagnosis of depression and six supporting Alzheimer's disease.

The results were striking. About 13% of the psychiatrists and 25% of the students showed a confirmatory search pattern, preferentially requesting information consistent with their initial diagnosis. Among the psychiatrists who searched in a confirmatory way, 70% ended up with an incorrect final diagnosis. Among those who searched in a balanced way, the error rate dropped to 47%, and among those who actively sought disconfirming information, it fell to 27%. The pattern was similar for medical students. Incorrect diagnoses also led to inappropriate treatment recommendations, such as prescribing antidepressants when anti-dementia medication would have been appropriate.

These results are noteworthy for two reasons. First, the majority of clinicians in this study did not show strong confirmation bias. But among those who did, the consequences were severe. Second, this was a controlled experiment with a single, relatively clear case. In real clinical practice, where cases are messier, time is shorter, and emotional pressures are greater, the opportunities for confirmation bias multiply.

A confirmatory strategy is not always wrong. If a clinician's initial hypothesis is correct, then looking for confirming evidence is efficient and leads to a quick, accurate diagnosis. The problem arises specifically when the initial hypothesis is wrong. In that case, a confirmatory search makes it much harder to discover the error, because the clinician is not looking for evidence that would reveal it. This is why the combination of anchoring and confirmation bias is so dangerous: a wrong anchor followed by biased information search creates a self-reinforcing cycle that is difficult to break.

Clinical diagnosis is not driven only by generic biases, however. It is also guided by clinicians' deeper mental models of how disorders work – models that shape which symptoms receive the most weight and how symptom patterns are interpreted.

Causal Reasoning in Clinical Judgment

Beyond the classic heuristics and biases, clinicians bring something more structured to their diagnostic work: causal models. As discussed in Mental Models and Cues to Causality, people reason about the world using mental representations of how things are causally connected. Clinicians are no exception. They hold theories about what causes what in mental and physical disorders, and these theories shape their diagnoses in ways that the official diagnostic manuals do not anticipate. Much of the research on causal models in clinical reasoning comes from psychiatry, where symptom-based diagnosis makes clinicians' implicit theories especially visible.

The Diagnostic and Statistical Manual of Mental Disorders (DSM) was designed to be atheoretical. It lists symptoms for each disorder and specifies how many need to be present for a diagnosis, but it does not say which symptoms are more important or how they are causally related. A patient needs, say, five out of nine symptoms of major depression to receive the diagnosis. According to the DSM, it does not matter which five. All symptoms are treated as interchangeable.

Clinicians, however, do not treat symptoms as interchangeable. Kim & Ahn (2002a) conducted a series of experiments in which clinical psychologists and graduate students drew causal diagrams linking symptoms of various mental disorders, including anorexia nervosa, schizophrenia, and others. Clinicians showed significant agreement on these causal models. For a given disorder, most clinicians drew similar arrows connecting similar symptoms. More importantly, clinicians weighted causally central symptoms – those that they saw as causing other symptoms – more heavily when making diagnoses.

To illustrate, consider depression. A clinician might view depressed mood as a core cause that leads to insomnia, which in turn produces fatigue and poor concentration. In this causal model, depressed mood is causally central: it sits at the hub of the network and drives other symptoms. A peripheral symptom such as changes in appetite might be viewed as a consequence rather than a cause. Kim & Ahn (2002a) found that a patient presenting with causally central symptoms like depressed mood was judged as more typical of the disorder than a patient presenting with the same number of symptoms that were causally peripheral, such as appetite changes and psychomotor disturbance – even though the DSM gives equal weight to all symptoms.

This pattern extended to memory as well. Clinicians were more likely to accurately remember causally central symptoms and more likely to falsely remember them when they had not actually been presented. Causal centrality shaped not just reasoning but the very encoding and retrieval of clinical information. Kim & Ahn (2002b) showed that laypeople similarly organize their understanding of mental disorders around causal theories, suggesting that this is a general feature of human categorization rather than something unique to clinical expertise.

The influence of causal models goes beyond which symptoms clinicians weigh most heavily. Flores et al. (2014) investigated whether the temporal order in which symptoms are presented matters for diagnosis. They had experienced clinicians and psychology students read clinical reports where the sequence of symptom development either matched or contradicted the expected causal chain. For example, if clinicians believed that depressed mood causes insomnia, which in turn causes fatigue, then reading a case where fatigue appeared first, then insomnia, then depressed mood would be inconsistent with their causal model. Clinicians read such causally inconsistent sequences more slowly, suggesting that the inconsistency disrupted their processing in real time. Furthermore, causally inconsistent cases led to lower agreement with the stated diagnosis. This effect was particularly clear among experienced clinicians, who showed automatic, online detection of causal inconsistencies as they read.

Causal reasoning also interacts with contextual information in ways that go beyond symptom lists. De Los Reyes & Marsh (2011) examined how non-symptom information about a patient's environment influenced diagnoses of conduct disorder in children. They presented 45 licensed clinicians with vignettes describing children who showed a conduct disorder symptom. The vignettes varied in the contextual information provided: some described environments consistent with conduct disorder, such as dysfunctional families and deviant peers, while others described supportive, stable environments. Clinicians rated the same symptom as more indicative of conduct disorder when the context was consistent with the disorder. When no contextual information was provided, ratings were similar to the consistent-context condition, suggesting that inconsistent context led clinicians to downplay symptom severity. There was also notable disagreement among clinicians about which symptoms were most influenced by context, raising concerns about diagnostic reliability.

Together, these findings show that clinicians do not simply check symptoms off a list. They build causal stories about their patients, connecting symptoms into coherent narratives, weighing some symptoms more than others based on their perceived causal role, and integrating contextual information that falls outside any official diagnostic criteria. This is consistent with how people reason about causality in general, as discussed in Cues to Causality and Learning Causality. It also means that the quality of a diagnosis depends not just on what symptoms are present but on the causal model the clinician brings to the encounter. When those models are accurate, causal reasoning is a powerful diagnostic tool. When they are incomplete or wrong, they can systematically distort judgment.

Decision Support Systems

Given that biases are difficult to eliminate through awareness alone, researchers and policymakers have explored structural interventions to improve clinical decision-making. One of the most promising approaches involves decision support systems: structured tools that guide clinicians through diagnostic and treatment decisions.

Decision support systems work by structuring and supplementing clinical judgment with systematic evidence evaluation. Instead of relying solely on memory and pattern recognition, a clinician inputs patient data into a system that compares the information against a database of evidence and generates recommendations. These systems take many forms. At the simpler end, a hospital might implement automated alerts that flag dangerous drug interactions when a clinician enters a prescription, or trigger a warning when a patient's vital signs match the early profile of sepsis. At the more complex end, diagnostic algorithms can take a set of symptoms, test results, and patient characteristics and output a ranked list of probable diagnoses with associated treatment guidelines.

Research on these systems suggests they are most effective when they meet three conditions. First, they should be computerized rather than paper-based, because integration into digital medical records reduces the extra effort required to use them. Second, they should be built into the clinical workflow so that using them is the default rather than an extra step – a clinician who must actively decide to open a separate application is far less likely to consult it than one who sees a recommendation appear automatically within the electronic health record. Third, they should provide not just diagnostic suggestions but also treatment recommendations, because even when a correct diagnosis is reached, treatment decisions introduce their own set of biases.

Decision support systems are essentially a form of what we discuss in Statistical vs. Intuitive Prediction: supplementing clinical intuition with structured, data-driven methods. As that chapter shows, statistical prediction consistently outperforms clinical intuition across a wide range of domains. Yet many clinicians resist using these tools, often because they feel the tools cannot capture the nuances of individual patients. This resistance is understandable, but it may overestimate the accuracy of unaided clinical judgment. When well-designed and well-integrated into clinical workflow, such tools can outperform unaided intuition on many diagnostic and treatment tasks. They do not need to be perfect; they need to be better than unaided human judgment on average for a given task.

A Bayesian Lens

Medical diagnosis is perhaps the most natural real-world application of Bayesian reasoning. The clinician starts with a prior: before seeing any test results, how likely is it that this patient has a given disease? This prior is based on the base rate of the disease in the relevant population, adjusted for the patient's age, sex, medical history, and presenting symptoms. The clinician then gathers evidence: physical examinations, laboratory tests, imaging. Each piece of evidence has a certain reliability, captured by its sensitivity and specificity. The prior and the evidence combine to form a posterior: the updated probability that the patient has the disease.

Consider a concrete example. A 25-year-old marathon runner and a 70-year-old smoker both arrive at an emergency department with chest pain. Before any test is run, the prior probability of a heart attack is very different for these two patients – the base rate of myocardial infarction is far higher in older smokers than in young athletes. Now suppose both patients receive the same electrocardiogram (ECG) result – mildly abnormal but ambiguous. Although the ECG and thus the newly collected evidence for a heart attack is equal in both cases, the probability of a heart attack increases more for the smoker than for the marathon runner (because of how evidence is combined with prior probability). In practice, clinicians do consider patient demographics and risk factors, but availability and anchoring can distort this process. If the emergency physician recently missed a heart attack in a young patient, the availability of that case may inflate the prior for the next young patient with chest pain, leading to a different – and potentially less accurate – diagnostic path.

Every bias in clinical reasoning can be understood as a specific departure from this Bayesian ideal. Anchoring is a sticky prior: once the clinician forms an initial hypothesis, it is not updated enough in the face of new evidence. Availability is biased evidence sampling: the diseases that come to mind most easily receive disproportionate prior probability, not because they are actually more common but because they are more memorable. Confirmation bias is asymmetric updating: evidence that supports the current hypothesis is given full weight, while evidence that contradicts it is discounted or ignored. Causal reasoning introduces structured priors: clinicians' causal models determine which symptoms are given more weight. For example, a clinician who views sleep disturbance as a downstream effect of depression may treat it as less diagnostically informative than anhedonia, which they regard as more causally central.

Decision support systems improve decisions by making base rates, test accuracy, and treatment evidence more explicit and systematic. They help ensure that base rates are not ignored, that all relevant evidence is weighed, and that the update from prior to posterior follows the rules of probability rather than the pull of recent memory or initial impressions. Natural frequency training works for a related reason: it makes the components of the Bayesian computation – the base rate, the hit rate, and the false alarm rate – visible and concrete rather than abstract. When the Bayesian structure of the problem is made transparent, clinicians reason far more accurately.

Summary

The availability heuristic leads clinicians to overdiagnose conditions they have recently encountered. Anchoring causes them to lock onto initial hypotheses and interpret subsequent information accordingly. Confirmation bias drives them to seek evidence that supports their current diagnosis while neglecting evidence that would challenge it. Beyond these heuristics, clinicians rely on causal models of disorders – particularly visible in psychiatric diagnosis – that shape which symptoms they weigh most heavily, how they interpret the temporal order of symptom development, and how they integrate contextual information about a patient's environment. These causal models can sharpen diagnosis when they are accurate, but they can also systematically distort it when they are incomplete or wrong. Two main interventions can help: decision support systems that structure clinical judgment with systematic evidence evaluation, and training in natural frequencies that makes the components of diagnostic reasoning transparent and concrete. At its core, medical diagnosis is Bayesian updating, and understanding where this updating goes wrong is the first step toward making it go right.

MEDICAL DECISION-MAKING

LEGAL DECISION-MAKING

Learning goals:

- Explain how anchoring, representativeness, and confirmation bias affect legal decision-making.
- Describe forensic confirmation bias and how it can lead to wrongful identifications.
- Explain the story model of juror decision-making and contrast it with information integration theory.
- Describe how the temporal order of evidence presentation influences verdicts.
- Evaluate strategies for reducing bias in legal contexts and explain why some of these strategies can backfire.

Biases in Legal Professionals

In 2004, the FBI arrested Brandon Mayfield, an American attorney living in Oregon, as a suspect in the Madrid train bombings that killed 193 people. Three separate FBI fingerprint experts had examined a latent print recovered from a bag of detonators and concluded it was a match to Mayfield. The match was verified, confirmed, and treated as certain. There was just one problem: Mayfield had nothing to do with the bombings. Spanish authorities later identified the actual source of the print as an Algerian national named Ouhnane Daoud. An investigation by the U.S. Department of Justice revealed that the FBI examiners had engaged in circular reasoning, interpreting ambiguous features in the latent print to align with Mayfield's known prints rather than assessing the print objectively (Office of the Inspector General, 2006). The Mayfield case is not an isolated incident. It illustrates a broader truth: police officers, forensic experts, and judges are human decision-makers, subject to the same cognitive biases discussed throughout this book. In legal contexts, however, the consequences of these biases are severe. Errors can mean wrongful convictions, years of lost freedom, or guilty parties walking free.

Representativeness and Racial Bias

When a defendant closely matches the stereotypical prototype of a "criminal," decision-makers may judge that person as more likely to be guilty, regardless of the actual evidence (see Representativeness). Blair et al. (2004) examined this in a study of sentencing data from the Florida Department of Corrections. They analyzed sentences for 216 male inmates, controlling for the severity of the crime, prior convictions, and other legally relevant factors. Their key finding was that inmates with more Afrocentric facial features – features perceived as typical of African Americans – received significantly longer sentences. This effect held for both Black and White inmates. Inmates whose Afrocentric features were one standard deviation above their racial group's average received sentences roughly seven to eight months longer than inmates one standard deviation below. The effect appeared to be more closely tied to resemblance to a category prototype than to racial category membership alone: once Afrocentric features were controlled for, the sentencing difference between Black and White inmates disappeared. This is the representativeness heuristic applied to people – judgment driven by similarity to a stereotype rather than by the specific facts of the case.

Anchoring in Sentencing

Sentencing decisions require judges to translate the complex details of a case into a specific number: months or years in prison, or the amount of a fine. This is exactly the kind of uncertain numerical judgment that is vulnerable to anchoring (see Anchoring and Primacy).

Englich & Mussweiler (2001) conducted three studies to test whether anchoring affects sentencing. In their first study, German trial judges read identical case materials describing a hypothetical rape case. The only difference between conditions was the prosecutor's sentencing demand: either 2 months or 34 months of imprisonment. Judges who evaluated the high demand sentenced the defendant to an average of 28.70 months, while those who evaluated the low demand gave an average sentence of 18.78 months – a difference of nearly 10 months for the same crime, driven entirely by the anchor.

One might argue that a prosecutor's demand is at least somewhat informative; perhaps judges treat it as expert opinion. The second study addressed this possibility. Law students read the same case materials, but this time, the sentencing demand came either from a prosecutor or from a first-year computer science student with no legal training. Most participants judged the computer science student's demand as irrelevant. Yet it influenced their sentencing recommendations just as strongly as the prosecutor's demand. The anchor did not need to be informative to be effective.

The third study tested whether experience could protect against anchoring. Highly experienced judges, with an average of 15 years on the bench, read the case and were exposed to a sentencing demand from a non-expert. Their sentences were anchored just as strongly as those of less experienced judges. Experience did not reduce the bias, and experienced judges also reported greater confidence in their decisions.

Bodily States and Judicial Decisions

The biases discussed so far relate to how information is processed. But legal decision-making can also be influenced by factors that have nothing to do with the evidence at all. Danziger et al. (2011) analyzed 1,112 parole decisions made by eight experienced Israeli judges over a ten-month period. The judges' workday was divided into three sessions by two food breaks. The researchers found a dramatic pattern: at the start of each session, favorable rulings (granting parole) hovered around 65%. As the session continued, the rate of favorable rulings dropped steadily, reaching nearly zero just before the next break. After the break, it jumped back up to around 65%.

This pattern persisted after controlling for legally relevant factors such as crime severity, prior incarcerations, and rehabilitation program participation. One possible interpretation is decision fatigue: making repeated, effortful decisions depletes mental resources, and judges default to the simpler, safer option – denying parole. A food break restores those resources.

However, this finding has been contested. Some researchers have pointed out that the order in which cases are heard may not be random: attorneys with stronger cases might be scheduled earlier in the session, or certain types of cases may cluster before breaks for administrative reasons. These scheduling confounds could partially or fully explain the pattern. The study should therefore be interpreted with caution. But even as a cautionary tale, it illustrates a broader point: legal decisions are made by human beings whose cognitive resources fluctuate throughout the day, and these fluctuations may affect outcomes in ways that have nothing to do with the merits of the case. This connects to the discussion in Morality of somatic markers and affect-as-information – the idea that bodily states can serve as inputs to judgment, even when they carry no diagnostic information about the decision at hand.

Forensic Confirmation Bias

Perhaps the most consequential bias in legal contexts is confirmation bias applied to forensic evidence. Once an investigator forms a hypothesis about who committed a crime, a familiar pattern takes hold: evidence that fits the hypothesis is noticed and emphasized, while evidence that contradicts it is downplayed or ignored. When this operates in forensic science, it is called forensic confirmation bias (Kassin et al., 2013).

The Mayfield case, described at the opening of this chapter, is the most dramatic illustration. As the Office of the Inspector General's report detailed, the FBI examiners began by noting genuine similarities between Mayfield's prints and the latent print. But as they continued, they started interpreting ambiguous features in ways that supported the match rather than assessing them independently. Discrepancies were explained away rather than treated as evidence against the match. The hypothesis shaped how the examiners perceived the evidence itself (Office of the Inspector General, 2006).

Is this simply an example of incompetent examiners making a one-off mistake? Dror & Charlton (2006) designed a study to test whether confirmation bias affects even competent, experienced fingerprint experts under ordinary conditions. They took six practicing fingerprint examiners, each with more than five years of experience, and presented them with fingerprint pairs they had already evaluated in previous casework. The experts did not know they were being retested on their own prior decisions. Half of the pairs were accompanied by misleading contextual information – for example, being told that the suspect had confessed, or that the suspect had an alibi. The other half served as controls with no added context. The results were striking: on 16.6% of the biased trials, experts changed their previous judgments. Two-thirds of the experts made at least one inconsistent decision. These were real experts, re-evaluating their own prior work, and their judgments shifted in response to irrelevant background information.

Kassin et al. (2013) describe how forensic confirmation bias can create a "bias snowball effect." A flawed eyewitness identification leads police to focus on a particular suspect. A confession is obtained, perhaps under pressure. The confession then influences the forensic examiner's analysis of fingerprints, DNA, or handwriting. Each piece of evidence appears to independently confirm the others, but in reality they are all contaminated by the same initial hypothesis. The evidence is not independent; it is a chain of beliefs reinforcing themselves.

Counteracting Bias – and How This Can Backfire

How can these biases be reduced? One approach is what Kassin et al. (2013) call "linear processing." The idea is straightforward: forensic examiners should analyze crime-scene evidence fully and form their conclusions before being exposed to suspect information. If a fingerprint examiner first documents all the features of a latent print and only then compares it to a suspect's known prints, the initial analysis cannot be contaminated by knowledge of the suspect. This is a form of debiasing through procedural design (see Heuristics and Biases), and it was among the reforms recommended after the Mayfield case.

Other proposed solutions include blind testing, where examiners are not told which comparison is the actual suspect, and evidence lineups, where the suspect's sample is embedded among several others. These procedures borrow from the logic of double-blind experiments in science: if the examiner does not know which sample belongs to the suspect, expectations cannot bias the analysis.

However, not all attempts to counteract bias are effective. In courtrooms, judges sometimes instruct juries to disregard inadmissible evidence – for example, testimony that was obtained improperly or a confession that was coerced. The intention is to neutralize the influence of that evidence. But research on belief perseverance suggests this instruction often fails (see Belief Formation and Perseverance). Once a piece of information has been encountered, it is difficult to "un-believe" it. As described in the chapter on belief, initial acceptance is automatic – what we called Spinozan acceptance – and correction requires a separate, effortful process that does not always succeed. In a legal context, this means that telling jurors to disregard a coerced confession or improperly obtained testimony draws attention to that evidence and can make it difficult to remove from the juror's mental model of the case. In some situations, the instruction may also trigger reactance – a motivation to resist being told what to think. The result is that inadmissible evidence can remain influential despite instructions to disregard it, and may sometimes become even more salient than it would have been without the instruction. The finding from Kaplan & Kemmerick (1974), discussed below, supports this: instructions to disregard defendant character information had virtually no effect on jurors' judgments.

How Jurors Reason: The Story Model

So far, we have focused on how judges, forensic experts, and other legal professionals can be biased by specific heuristics. A different question is how jurors make sense of the mass of evidence presented at trial. Not all legal errors arise from isolated heuristics. At trial, decision-makers face the broader task of organizing many pieces of evidence into a meaningful whole.

The most influential account of juror reasoning is the story model, developed by Pennington & Hastie (1986) and refined in subsequent studies (Pennington & Hastie, 1988, 1992). The central idea is that jurors do not evaluate each piece of evidence independently and then combine the pieces into a judgment. Instead, they construct a coherent narrative – a story – that explains how and why the events in question occurred.

The story model has three stages. First, jurors construct a story from the evidence, filling in gaps with inferences drawn from their general knowledge about how the world works. This story is organized around what Pennington and Hastie call an episode schema (see Mental Models): a structure that includes initiating events, psychological states, goals, actions, and consequences. Second, jurors learn the verdict categories – the legal definitions of guilty, not guilty, or specific degrees of an offense. Third, jurors classify their story by matching it to the verdict category it best fits.

In their original study, Pennington & Hastie (1986) interviewed 26 jurors from Massachusetts jury pools who had watched a filmed murder trial. They found that jurors' mental representations of the evidence consistently took the form of stories. Different jurors watching the same trial constructed different stories, and these differences mapped directly onto their verdict choices. Jurors who voted for first-degree murder told a story that emphasized premeditation and intent. Jurors who chose not guilty on grounds of self-defense told a story in which the defendant acted under immediate threat and had no way to escape. Remarkably, about 45% of the elements in jurors' stories were inferences – about the defendant's mental states, goals, and motivations – rather than direct testimony. Jurors were not merely recalling what witnesses said; they were building an explanation of why events unfolded as they did.

The story model explains several otherwise puzzling features of jury decision-making. It explains how two reasonable people can look at the same evidence and reach opposite conclusions: they constructed different stories. It also explains why the order in which evidence is presented matters so much – a point tested directly in a follow-up experiment.

Pennington & Hastie (1988) tested the effect of evidence order in a clever experiment. Participants watched a simulated murder trial in which the prosecution and defense each presented their evidence either in temporal (story) order or in a scrambled order that did not follow the chronological sequence of events. The researchers varied these two factors independently, creating four conditions.

The results were clear. Guilty verdicts were most likely when the prosecution's evidence was presented in story order and the defense's evidence was scrambled (78% guilty verdicts). Guilty verdicts were least likely in the reverse arrangement – defense evidence in story order, prosecution evidence scrambled (31% guilty). When both sides presented evidence in story order, or both scrambled, conviction rates fell between these extremes. Evidence presented in temporal order was easier to organize into a coherent narrative, and a more coherent narrative was more persuasive.

This finding connects to broader principles of causal reasoning (see Cues to Causality). Temporal order is one of the basic cues people use to infer causation: if event A precedes event B, people are more inclined to see A as causing B. When evidence is presented in chronological order, it naturally invites causal connections between events. When it is scrambled, those connections are harder to form, and the resulting narrative feels less coherent and less convincing.

Story order also affected confidence. Jurors were most confident in their verdicts when both sides presented evidence in story order, presumably because both narratives were clear and the juror could compare them directly. This suggests that narrative clarity does not simply favor one side; it improves the subjective quality of the decision-making process overall.

Information Integration Theory

The story model describes how jurors actually reason. But is there a more analytic alternative? Kaplan & Kemmerick (1974) proposed an approach based on information integration theory. In this model, each piece of evidence has two properties: a scale value (how much it points toward guilt or innocence) and a weight (how important or reliable it is). The juror combines all pieces of evidence by computing a weighted average, producing an overall judgment of guilt. Rather than weaving evidence into a causal narrative, the juror assigns each cue its own value and then combines them.

In their experiment, Kaplan & Kemmerick (1974) gave participants simulated case summaries that varied in the strength of the evidence (high or low incrimination) and in descriptions of the defendant's character (positive, neutral, negative, or none). They found that guilt and punishment ratings increased with the strength of the evidence and became harsher for defendants with negative character traits. The effects of evidence and character traits combined additively – the influence of character traits was the same regardless of whether the evidence was strong or weak. This additive pattern is what information integration theory predicts: participants appeared to assign each cue its own value and combine them independently, rather than weaving them into a single causal story where one cue might change the meaning of another.

One troubling finding was that instructions telling jurors to disregard the defendant's character traits had virtually no effect. Jurors incorporated this legally irrelevant information into their judgments regardless of what they were told – consistent with the broader point, discussed above, about the difficulty of disregarding information once it has been encountered.

Information integration theory offers a more analytic, cue-based model of judgment and can serve as a partial normative benchmark. It says: each piece of evidence should shift your judgment by an amount that reflects how diagnostic and reliable that evidence is. Irrelevant information should receive zero weight. In practice, the story model better describes what jurors actually do, while information integration theory approximates how they ideally should reason. The gap between these two models captures much of what goes wrong in legal decision-making.

A Bayesian Lens

The legal process is, in its formal structure, remarkably similar to Bayesian updating (see Bayesian Reasoning). As a Bayesian analogy, the legal presumption of innocence can be treated as a very low prior probability of guilt. Evidence is then presented sequentially. Each piece of evidence should update the prior: strong evidence of guilt should increase the posterior probability, while exculpatory evidence should decrease it. The decision-maker arrives at a final posterior and compares it against a threshold – "beyond reasonable doubt" – to reach a verdict.

To make this concrete: imagine a juror starts with a low prior probability that the defendant is guilty. An eyewitness identifies the defendant – this is moderately diagnostic evidence, and the posterior shifts upward. Then reliable DNA evidence places the defendant at the scene – this is highly diagnostic, and the posterior shifts substantially. Finally, the defense presents an alibi witness – if credible, this should shift the posterior back down. In the ideal case, the final judgment reflects the cumulative weight of all the evidence, each piece contributing according to its reliability and diagnosticity.

Information integration theory resembles this Bayesian process in combining prior beliefs with weighted evidence, though it is not identical to a full Bayesian model. Each piece of evidence has a diagnostic value and a weight, and the final judgment is a weighted combination of all evidence with the decision-maker's initial impression.

But the story model reveals why legal decision-making departs from this ideal. Jurors do not update incrementally. They construct a holistic narrative, and this narrative shapes how they interpret subsequent evidence. A piece of testimony that fits the story is accepted and remembered; a piece that does not fit is discounted or forgotten. In Bayesian terms, the prior does not simply get updated by the evidence – the prior determines how the evidence is perceived. This is the core of forensic confirmation bias. When an FBI examiner has already formed a hypothesis that a fingerprint belongs to a particular suspect, ambiguous features in the print are perceived through the lens of that hypothesis. The evidence is assimilated into the prior rather than used to update it. Instead of evidence changing the belief, the belief changes the evidence.

Summary

Legal decision-making is shaped by the same heuristics and biases that affect judgment in every domain covered in this book. The representativeness heuristic leads to harsher sentences for defendants who resemble a criminal stereotype, as demonstrated by research showing that Afrocentric facial features predict longer sentences regardless of the defendant's race. Anchoring effects cause sentencing decisions to be pulled toward the prosecutor's demand, even when that demand is arbitrary or comes from a non-expert. Forensic confirmation bias leads investigators and forensic experts to interpret ambiguous evidence in ways that confirm their existing hypotheses, as the Mayfield case and experimental studies with fingerprint experts have shown.

Attempts to counteract bias through procedural reforms such as linear processing and blind testing show promise, but instructions to disregard evidence often fail because information, once encountered, is difficult to remove from a juror's mental model. The story model explains how jurors actually process evidence: they construct coherent narratives and match these to verdict categories, rather than weighing each piece of evidence independently. Evidence presented in temporal order is more persuasive because it supports narrative coherence. Information integration theory offers a more analytic alternative that can serve as a partial normative benchmark, describing how evidence should ideally be combined through weighted averaging. The gap between the story model and information integration theory captures a central challenge of legal decision-making: ensuring that the human tendency to construct stories does not override the careful, evidence-by-evidence reasoning that justice requires.

POPULISM AND SOCIAL MEDIA

Learning goals:

- Explain how social media, populism, and misinformation form self-reinforcing loops.
- Describe how loss framing in populist rhetoric connects to prospect theory.
- Explain how moral outrage drives engagement on social media platforms.
- Describe how algorithms and user behavior create echo chambers and polarization.
- Explain how focalism can increase polarization on complex issues.
- Describe the role of Spinozan acceptance in the spread of misinformation.
- Explain how social media can be used as a strategic tool for political diversion.
- Evaluate whether the problems of social media are primarily caused by algorithms or by the nature of user-generated content itself.

A new and powerful force

In spring 2026, rioters assaulted a refugee shelter in the city of Loosdrecht in the Netherlands. They threw fireworks against the building, causing nearby bushes to set on fire. All the while, a group of newly arrived – and presumably terrified – refugees were inside.

This event was widely discussed on social media. While the violence was condemned by almost everyone, there was a stark divide in opinions about who was ultimately to blame. According to (predominantly) left-wing commentators, the rioters were to blame – they were simply criminals who assaulted a group of vulnerable people, many of whom were likely already traumatized because of experiences in their country of origin. These commentators also tended to highlight that Dutch municipalities are bound by law to house refugees, and that the rioters were thus interfering with a democratic process – essentially terrorism. In contrast, according to (predominantly) right-wing commentators, the riots were the result of politicians not respecting the wishes of the inhabitants of Loosdrecht, many of whom indeed opposed the refugee shelter – so what do you expect? According to these commentators, while most of them emphasized that violence is never the answer, politicians were ultimately to blame.

Representative polls showed that many people feel there is merit to both sides of the argument. Yet the debate on social media rarely reflected this nuance. Instead, on social media, people seemed to inhabit different worlds: one world in which an isolated group of criminals assaulted vulnerable refugees; and another in which politicians were responsible for social unrest, ultimately resulting in riots. This pattern of polarized, emotionally charged online conflict is not itself the same as populism, but it shows the kind of environment in which populist rhetoric thrives – and it illustrates how social media can turn a complex policy disagreement into a bitter tribal divide.

Digital platforms for user-generated content now reach roughly five billion users worldwide, who spend on average about two and a half hours per day scrolling, posting, and sharing. The effects of this on individuals and societies are not yet fully understood. However, there is strong reason to believe that social media, populism, and misinformation form self-reinforcing loops: social media amplifies populist messages, populist leaders exploit social media, misinformation fuels both, and algorithms accelerate everything.

What is populism?

Populism frames politics as a struggle between "ordinary people" and a corrupt or out-of-touch elite. It is not limited to the political left or right. Left-wing populism tends to cast the struggle as ordinary citizens against economic elites, while right-wing populism tends to frame it as the native population against immigrants and cultural outsiders. Despite these differences, populist movements share a core of anti-elite sentiment, mistrust of established institutions, moral outrage, and promises of dramatic change. Many populist movements, especially on the right, also emphasize nationalism, traditional values, and loyalty to a strong leader.

Populism is not new. It has deep historical roots, from the agrarian populist movements in nineteenth-century America to various authoritarian movements in twentieth-century Europe and Latin America. What is new is the vehicle. Social media has given populist leaders a way to speak directly to millions of people, bypassing the traditional media that once served as gatekeepers of political information. As Guriev & Papaioannou (2020) describe in their review of the political economy of populism, the rise in populist support, particularly after the 2008 financial crisis, has been fueled by a combination of economic shocks, cultural anxieties, and the spread of digital media. Social media did not create populism, but it has become one of its most powerful amplifiers.

Loss framing and populist rhetoric

Consider the language of populist campaigns: "We are losing our country." "They are taking your jobs." "Our traditions are disappearing." These messages share a common structure: they frame the current situation as a loss. This is not accidental. As prospect theory predicts, people respond differently to losses than to gains. When people feel they are in a domain of loss – when they believe things are getting worse – they become more willing to take risks (Gains and Losses). A voter who feels economically secure may be less inclined to support a radical political candidate, because the downside of political risk looms larger. But a voter who feels that the good times are already gone, that wages are declining, that cultural identity is under threat, has less to lose. For this voter, a risky political option feels more acceptable.

Populist leaders exploit this shift systematically. By repeatedly framing the world in terms of loss, they move voters into a psychological state where extreme options no longer seem extreme. The promise to "take back control" or to “make things great again” is not just a slogan; it is an offer to reverse a loss, which, according to prospect theory, people find especially motivating. This helps explain why populist messages resonate even when the proposed solutions are vague or unrealistic.

But a compelling message is only as powerful as its reach. The next question is how far and how fast that message travels – and this is where digital platforms enter the picture.

Social media as a catalyst

Several large-scale studies have documented how digital media accelerates populist sentiments. Guriev et al. (2021) studied the political consequences of 3G mobile internet expansion across 116 countries between 2008 and 2017. Using data from over 840,000 individuals, they found that the rollout of 3G networks predicted a decrease in government approval. On average, a 39-percentage-point increase in 3G coverage reduced confidence in government by 2.5 percentage points. One plausible mechanism is that mobile internet gives citizens access to independent political information, particularly about corruption, that was previously filtered by government-controlled traditional media. In countries with uncensored internet and controlled traditional media, the effect was strongest. Interestingly, in the least corrupt countries, such as Denmark and Switzerland, 3G expansion actually increased government approval, suggesting that when new information confirms a positive reality, it can strengthen trust. But in most countries, the new information exposed government failures, and the primary political beneficiaries were populist parties on both the left and right.

A systematic review by Lorenz-Spreen et al. (2023), covering 496 studies on the relationship between digital media and democracy, confirmed this dual nature. Digital media can increase political participation and access to information, but it can also increase polarization and weaken trust in institutions. The review found that digital media was associated with more hate speech, more polarization, and more populism, particularly in established democracies. The picture is not one of a purely harmful technology, but of one that empowers citizens while also destabilizing the institutions they depend on.

Moral outrage as engagement fuel

Not all social media content is created equal. Some types of content attract far more attention than others, and the content that consistently dominates is the kind that triggers moral outrage. This is consistent with the negativity bias – our general tendency to attend more to negative than positive information (Availability) – and with the affect-as-information heuristic (Morality): content that triggers strong emotional responses such as anger or disgust captures attention and feels important, regardless of whether it actually is.

Rathje et al. (2021) analyzed over 2.7 million posts from news media and members of the United States Congress on Facebook and Twitter. They found that posts containing references to the political out-group – the opposing party – were shared far more than posts about the in-group. In the authors' models, each additional out-group word was associated with roughly a 67 percent increase in sharing. This effect was nearly five times stronger than the effect of negative emotional language and nearly seven times stronger than moral-emotional language. Posts about the out-group also triggered disproportionately more "angry" reactions on Facebook, while in-group posts triggered more "love" reactions. The pattern was consistent across political orientations and platforms.

Humprecht et al. (2024) extended this finding to six countries: Belgium, France, Germany, Switzerland, the United Kingdom, and the United States. Analyzing over 175,000 Facebook posts, they found that populist politicians and alternative media outlets consistently provoked more "angry" reactions than traditional politicians and mainstream media. The content that generated the most anger focused on populist themes: blaming immigrants, attacking political elites, and stoking fears about cultural decline. The pattern was remarkably consistent across countries, although the intensity varied.

The implication is troubling. Only a small minority of users create morally outraged content – and some research suggests that those who do tend to score higher on dark-triad personality traits such as narcissism (Grubbs et al., 2019). But because outrage is so engaging, it is widely shared, giving a small number of extreme voices an outsized influence on public discourse. In other words, the content people see on social media is not a neutral sample of public opinion but a psychologically filtered sample, shaped by what captures attention and provokes emotion.

Algorithms, echo chambers, and the feedback loop

This filtering is not random. Social media platforms use algorithms that promote content likely to generate engagement. The result is a feedback loop between human psychology and platform design: people preferentially engage with outrage, algorithms learn from that engagement, the platform then serves more outrage to more users, and those users come to perceive the world as more extreme and threatening – which generates yet more engagement. A post that receives an early burst of angry reactions may be promoted to thousands more users, creating a cumulative-advantage effect in which the most provocative content rises to the top while nuanced discussion is buried.

Over time, this process contributes to the formation of echo chambers – online environments in which users are primarily exposed to views they already agree with. Within these algorithmically curated groups, the dynamics of group polarization take hold (Group Decisions): group members shift toward more extreme positions because they are exposed only to arguments supporting their existing views, while dissenting voices are filtered out. The same echo chambers also provide fertile ground for conspiracy theories (Conspiracy Theories), which thrive in environments where contradictory evidence is absent and social reinforcement is strong.

However, it would be too simple to blame algorithms alone. Larooij & Törnberg (2025) used a novel approach to test whether platform design changes could fix social media's problems. They created a simulated social media platform populated by AI-powered agents, each with a distinct persona based on real-world demographic and political data. Despite the simplicity of the simulation, it reproduced three well-documented problems of social media: partisan echo chambers, extreme concentration of visibility among a small elite, and the amplification of politically extreme voices. The researchers then tested six proposed interventions: chronological feeds instead of algorithmic ones, algorithms that reduced the visibility of dominant voices, boosting content from the opposing political side, prioritizing empathetic and reasoned content, and hiding engagement metrics or user biographies. The results were sobering. While some interventions produced modest improvements, none eliminated the core problems. Some even made things worse. Chronological feeds, for example, reduced attention inequality but increased the correlation between political extremism and influence. The authors concluded that polarization and attention inequality may be partly inherent to user-generated content platforms, arising from the basic feedback loop between engagement and visibility, regardless of the specific algorithm in use.

Focalism and polarization

The point is broader than algorithms. Even without personalized feeds, ordinary cognition can polarize public debate. One such mechanism is focalism – the tendency to focus on a single aspect of a complex issue while neglecting others (The Future Self). Research on affective forecasting shows that people overweight the most salient dimension of a future event and underweight everything else; the same narrowing of attention applies when people think about policy issues.

Return to the violent protests against the refugee shelter in Loosdrecht. When people view these riots as a form of unprovoked terrorism, they focus on the violence and the fact that the refugee shelter was established through democratic procedures. In contrast, when people view the riots as a manifestation of a widely shared sentiment, they focus on the fact that the shelter was established against the wishes of many, perhaps even most, residents of Loosdrechts. Both groups may actually share many values. Most people condemn violence, support democracy, and believe that politicians should respect people's wishes. But focalism reduces the overlap. Each group defines the issue in terms of its most important dimension and perceives the other group as ignoring what really matters. The result is that two groups with substantial common ground come to see each other as fundamentally opposed.

Social media makes this worse. When you post about an issue, you tend to post about the aspect you care most about. When you share someone else's post, you share the one that speaks to your concern. The aggregation of millions of such individually reasonable decisions creates a public discourse that appears far more polarized than the underlying attitudes of the population. And once people perceive a deep divide, they behave accordingly: they become more hostile to the other side, more loyal to their own, and less willing to compromise. Whereas echo chambers create polarization through segregated exposure to different information, focalism creates polarization through selective attention to different dimensions of the same information.

Misinformation and Spinozan acceptance

Social media is also an extraordinarily efficient vehicle for misinformation. Lies and half-truths spread organically, shared by users who may not realize the content is false. Algorithms amplify such content further by rewarding engagement regardless of accuracy.

The psychological mechanism that makes this possible is Spinozan acceptance (Belief Formation and Perseverance): according to this view, comprehension initially carries a default tendency toward acceptance, and rejection requires an additional evaluative step. Consider what happens when you scroll through a social media feed and encounter the headline "Study finds that common food additive causes memory loss." Even if you are generally skeptical, your mind briefly encodes the association between the additive and memory loss. Rejecting the claim requires you to stop, check the source, and evaluate the evidence – all before the next post pulls your attention away. Under normal conditions, this system works reasonably well, because most of the information we encounter in daily life is roughly accurate. But social media creates conditions that are far from normal. The sheer volume of content, the speed at which it appears, and the design of the endless scroll all work against the effortful evaluation that correction requires. Users often move to the next post before they have evaluated the previous one.

The relationship between misinformation and belief is also more complex than it first appears. Schaffner & Luks (2018) conducted a study during the controversy over the size of the crowd at Donald Trump's 2017 inauguration. They showed a nationally representative sample of American adults side-by-side aerial photographs of Trump's inauguration and Barack Obama's 2009 inauguration. The difference in crowd size was visually obvious. When asked which photo showed more people, 15 percent of Trump voters chose the smaller crowd, compared to only 2 to 3 percent of Clinton voters and nonvoters. Strikingly, the error rate was highest among highly educated Trump supporters, 26 percent of whom chose the wrong photo, compared to 11 percent of less educated Trump supporters. Since the answer was visually unambiguous, the authors interpreted this as expressive responding: these respondents did not genuinely believe the smaller crowd was larger but gave the wrong answer to signal loyalty to their political side. This finding suggests that not all misinformation reflects genuine misperception. Some of it may be a form of participatory propaganda, where people knowingly repeat falsehoods as an act of group solidarity.

Whether misinformation reflects genuine belief or performative loyalty, the consequences are similar. False claims circulate widely, corrections struggle to keep up, and public discourse becomes detached from shared facts. The continued influence effect means that even when misinformation is corrected, its influence on attitudes and decisions often persists.

Social media as a strategic tool

Some political leaders do not just benefit from social media's dynamics; they actively exploit them. Lewandowsky et al. (2020) analyzed the Twitter behavior of former United States President Donald Trump during the first two years of his presidency. They focused on whether Trump used tweets to divert media attention from politically damaging coverage, particularly the Mueller investigation into Russian interference in the 2016 election. The researchers analyzed daily data from The New York Times and ABC World News Tonight alongside Trump's tweets from January 2017 to January 2019.

The pattern they found was consistent with a diversion strategy. When media coverage of the Mueller investigation increased, Trump's tweets about unrelated topics – particularly his preferred themes of immigration, jobs, and China – also increased. This surge in tweets was followed by a measurable decline in Mueller-related coverage the next day. The effect was absent for topics that posed no political threat, such as Brexit or neutral subjects like skiing. The authors could not prove that the diversion was deliberate rather than intuitive, but the specificity of the pattern – only appearing for politically damaging topics – strongly suggests strategic intent.

This strategy exploits the availability heuristic (Availability). By flooding the information environment with alternative narratives, a political leader can shift what is most accessible in the minds of both journalists and the public. If people are thinking about immigration, they are not thinking about a corruption investigation. The strategy does not require people to believe the diversionary content is more important; it merely requires that the volume of alternative content pushes the damaging topic out of the spotlight.

A Bayesian lens

Social media systematically distorts the evidence available for belief updating. In a Bayesian framework, rational belief revision requires that evidence is representative: the information you encounter should be a reasonably accurate sample of reality. But social media violates this requirement in several ways.

Algorithms create biased evidence samples by curating content based on engagement rather than accuracy, so users receive a stream of information skewed toward outrage and extremity. Echo chambers further filter evidence by political alignment, so that users on opposite sides of an issue are updating their beliefs based on entirely different sets of information. If one user's feed is filled with viral posts about immigrant crime and another's is filled with posts about anti-immigrant harassment, the two are not updating on the same evidence – even though they live in the same world. The negativity bias in engagement means that negative and frightening content is overrepresented, distorting people's sense of how dangerous or unfair the world actually is. Loss framing (Gains and Losses) shifts risk preferences, making extreme options appear more acceptable. And Spinozan acceptance (Belief Formation and Perseverance) means that the volume of misinformation overwhelms the capacity for correction, so that false information is incorporated into beliefs without sufficient scrutiny.

The result is polarization: two groups updating from entirely different evidence samples arrive at radically different beliefs about the same world. From each group's perspective, their beliefs are justified by the evidence they have seen. The problem is not that people are irrational, but that the evidence they receive is systematically unrepresentative. In Bayesian terms, the likelihoods that people use for updating are distorted by the platform, and the priors that might have been shared are pushed apart by biased evidence until they become irreconcilable.

Summary

Social media, populism, and misinformation form self-reinforcing loops. Populist rhetoric uses loss framing to trigger risk-seeking behavior, making extreme political options seem acceptable. Moral outrage – especially directed at political out-groups – is the most engaging type of social media content, and algorithms learn from engagement to serve more of it, creating a feedback loop that amplifies the most divisive voices. This process produces echo chambers where group polarization takes hold, while focalism independently increases perceived disagreement by narrowing attention to single dimensions of complex issues. Misinformation spreads efficiently because Spinozan acceptance means information is believed by default, and the speed of social media undermines the effortful correction that rejection requires. Some political leaders strategically exploit these dynamics to divert public attention. Through a Bayesian lens, the core problem is that social media provides biased evidence samples: users on different sides update on different information and arrive at different beliefs about the same world – not because they are irrational, but because the evidence they receive is systematically unrepresentative.

ARTIFICIAL INTELLIGENCE

Learning goals:

- Explain the Turing test and why shifting definitions of intelligence reveal something about human intuitions.
- Describe how large language models relate to dual-process theory, including the role of chain-of-thought prompting.
- Distinguish between crystallized and fluid intelligence in the context of large language models.
- Explain where biases in AI systems come from and how they relate to human cognitive biases.
- Compare AI hallucination with human confabulation.
- Describe how calibration failures in AI mirror human overconfidence.

Machines That Think Like Us

Imagine a university student who asks an AI chatbot to explain a concept from a specific academic paper. The chatbot responds fluently: it names the paper, describes the authors, and summarizes the key findings. The summary is detailed and confident. There is just one problem – the paper does not exist. The chatbot invented the title, the authors, and the findings, generating what a real citation would look like based on patterns in its training data. The student, finding the response convincing, cites the fabricated paper in a follow-up assignment.

This kind of scenario has become commonplace since the widespread adoption of large language models (LLMs), and it captures something important about artificial intelligence. The same system that produces impressively human-like output can also fail in strikingly human-like ways. Throughout this book, we have examined how people judge, decide, and reason, and where these processes go wrong. In this final chapter, we turn to LLMs because they serve as a revealing mirror for human cognition. Many of the biases, heuristics, and reasoning failures we have studied reappear in AI systems, and understanding why deepens our insight into both machines and minds.

A note before we proceed: throughout this chapter, we will describe what LLMs "do" using language that can sound as if these systems have intentions, beliefs, or experiences. When we say a model "assumes" something or "lacks verification," these are functional descriptions and analogies, not claims about machine consciousness. The parallels with human cognition are instructive precisely because they hold at the level of process and output, regardless of whether anything resembling subjective experience is involved.

The Turing Test and Shifting Goalposts

In 1950, the mathematician Alan Turing proposed a simple test for machine intelligence. Rather than asking the unanswerable question "Can machines think?", he suggested an operational alternative: if a human interrogator, communicating by text with both a machine and a person, cannot reliably tell which is which, the machine should be considered intelligent (Turing, 1950). This "imitation game," later called the Turing test, turned a philosophical debate into an empirical question. Turing predicted that by the year 2000, computers would be able to fool an average interrogator about 30 percent of the time after five minutes of conversation.

Modern LLMs have surpassed that prediction. Systems like GPT-5, Claude, Mistral and their successors can hold extended conversations, answer complex questions, write essays, and discuss their own limitations. In controlled studies, participants who engage in short text conversations with these systems often cannot reliably distinguish them from human partners. By the spirit of Turing's original criterion – that an interrogator in a text-based exchange cannot consistently tell human from machine – modern LLMs arguably meet the standard in many constrained settings, though the claim remains debated and depends on the specifics of the test.

Yet many people remain unconvinced. When machines learned to hold conversations, critics said that real intelligence requires reasoning. When machines began to reason through multi-step problems, critics said that real intelligence requires understanding. When machines demonstrated competent performance across many domains, critics said that real intelligence requires consciousness. The goalposts keep shifting.

This pattern is informative about how we think about intelligence. We tend to redefine it so that it always excludes whatever machines can currently do. This resembles the bias blind spot discussed in an earlier chapter (see Heuristics and Biases): a reluctance to accept that our own cognitive processes might operate on principles similar to a machine's pattern-matching. When a computer performs a task, we call it computation. When a person performs the same task, we call it thinking. The Turing test was designed to cut through exactly this kind of reasoning, but even a test specifically built to avoid it has not fully succeeded.

Fast Patterns and Slow Reasoning

Once we move from asking whether LLMs seem intelligent to asking how they produce that impression, the comparison with human dual-process theories becomes useful. The distinction between System 1 and System 2 has been a recurring theme throughout this book (see Reason and Intuition). System 1 is fast, automatic, and pattern-based. System 2 is slow, deliberate, and rule-based. This distinction maps surprisingly well onto how LLMs operate.

When you type a question into a chatbot, the standard output resembles System 1 processing. The model generates its response by predicting, one word at a time, what is most likely to come next given all the text it has been trained on and the prompt you provided. This is associative pattern completion. The model does not verify its claims against a database of facts. It produces whatever response fits best with the patterns in its training data, in much the same way that your System 1 produces a quick intuitive answer without effortful reflection.

This works well for many tasks. If you ask an LLM to summarize a text, translate a sentence, or explain a well-known concept, the associative process produces fluent and accurate output. These are tasks where the most statistically likely response is also the correct one. But when you pose a novel logic puzzle, request multi-step arithmetic, or ask about an unfamiliar situation, straight pattern completion often fails – just as human intuition fails on problems that require careful deliberation, like those on the Cognitive Reflection Test (see Reason and Intuition).

This is where chain-of-thought (CoT) prompting becomes relevant. Wei et al. (2022) demonstrated that when LLMs are prompted to show their reasoning step by step before arriving at a final answer, their performance on arithmetic, commonsense, and symbolic reasoning tasks improves dramatically. Consider a simple word problem: "A store had 45 apples. It sold 20 in the morning and bought 15 more in the afternoon. How many apples does it have now?" When asked directly, a model might pattern-match to a superficially similar problem and produce the wrong answer. But when prompted to reason step by step – "First, 45 minus 20 is 25. Then, 25 plus 15 is 40. The answer is 40." – accuracy improves substantially. This technique, and the reasoning models built around it, functions as a kind of System 2 for LLMs. It is slower and uses more computational resources, but it catches errors that the fast associative process misses.

Two Kinds of Intelligence

The dual-process analogy concerns how LLMs generate answers. A related but distinct question concerns what kinds of problems they solve well. Earlier in this book, we distinguished between fluid intelligence and crystallized intelligence (see What Is Intelligence?). Fluid intelligence is the ability to reason about novel problems, detect abstract patterns, and think flexibly in unfamiliar domains. Crystallized intelligence is accumulated knowledge: facts, vocabulary, procedures, and cultural information built up over a lifetime.

LLMs are best understood as resembling extreme crystallized intelligence. They have been trained on enormous amounts of human-generated text, and they can retrieve and recombine this knowledge with impressive flexibility. Ask an LLM to define "opportunity cost," explain the stages of mitosis, compare Kantian and utilitarian ethics, or describe the offside rule in football, and it will typically perform well. This is crystallized intelligence at a scale no individual human can match.

Where LLMs struggle is with tasks that resemble fluid intelligence demands. Consider a novel rule-induction problem: "In a made-up language, 'dax blicket' means 'big dog' and 'dax fepp' means 'big cat.' What does 'dax' mean?" Humans with reasonable fluid intelligence solve this easily by abstracting the pattern. LLMs can often handle this specific type of problem because similar examples exist in their training data. But when problems are truly novel – requiring abstract pattern detection in formats the model has never encountered – performance drops. The model relies on matching to familiar patterns, and when no relevant pattern exists, it often produces answers that look plausible but are wrong.

As noted in the previous section, chain-of-thought prompting can improve performance on such tasks by moving the model beyond simple retrieval toward something closer to stepwise problem-solving. This mirrors the human relationship between the two forms of intelligence: crystallized intelligence depends on fluid intelligence having operated first. You can only store knowledge if you were once able to learn and reason about it.

The developmental trajectory differs in a telling way. Humans begin life with fluid intelligence and gradually accumulate crystallized knowledge over decades. LLMs begin with crystallized knowledge absorbed from their training data and are gradually acquiring more fluid-like capabilities through techniques like chain-of-thought reasoning and extended inference. Both humans and LLMs draw on both forms of intelligence, but they arrive at the combination from opposite directions.

Inherited Biases

Where do biases in AI come from? To a significant degree, they come from us.

LLMs are trained on vast collections of human-generated text: books, articles, websites, forums, social media posts, and more (Bommasani et al., 2021). This text reflects the full range of human thought, including its systematic errors. Stereotypes about gender, ethnicity, age, and occupation are embedded in the statistical patterns of language. Framing effects are built into how news is written. Cultural assumptions are woven into how stories are told. When an LLM learns to predict the most likely next word in a sequence, it absorbs all of these patterns, biases included.

As a result, LLMs replicate many of the classic biases studied in this book. Consider sycophancy, a form of confirmation bias (see Confirmation). When a user states a strong opinion – say, "I think the evidence clearly shows that homework has no benefit for primary school students" – and then asks the model to evaluate the evidence, the model tends to agree with the user's stated view, even if the evidence is more mixed than the user suggests. The model tells the user what they want to hear, not unlike how people selectively seek and interpret information that confirms their existing beliefs.

LLMs also show anchoring effects (see Anchoring and Primacy): when a prompt contains a number, even an irrelevant one, the model's numerical estimates shift toward it. They show framing effects (see Gains and Losses): the same medical scenario described as "90 percent survival rate" produces a more favorable response than when described as "10 percent mortality rate." And they engage in representativeness-like pattern matching, associating people and situations with prototypes in ways that can perpetuate stereotypes (see Representativeness).

Weidinger et al. (2021) provide a detailed taxonomy of the ethical and social risks that arise from these inherited biases. They document how LLMs can perpetuate harmful stereotypes, reinforce exclusionary norms, and produce discriminatory outputs – not because they were designed to do so, but because these patterns are embedded in the data they learned from. The biases are not bugs introduced by careless engineers. They are features of human language and culture, faithfully absorbed by systems designed to mirror that language and culture.

This replication also tells us something about cognitive biases more generally. If anchoring, framing, and confirmation bias emerge from statistical regularities in human language, these biases may not be quirks unique to biological neural hardware. They may be deeply embedded in how knowledge is structured and communicated. Any system that learns from human communication – whether a developing brain or a silicon model – may end up exhibiting similar patterns.

When Machines Confabulate

One of the most discussed problems with LLMs is hallucination: the tendency to generate confident, detailed, and plausible-sounding output that is factually wrong. (AI researchers thus use the term 'hallucination' very different from its original meaning – the false perception of things that aren't there, which is a symptom of schizophrenia and a number of other psychological disorders.) The chatbot that invented a fake academic paper in our opening example was hallucinating. It did not retrieve a real paper from a database. It generated what a real citation would look like based on patterns in its training data, producing the result with the same fluency and confidence it brings to accurate responses.

This parallels a well-known phenomenon in human cognition: confabulation. People confabulate when they produce explanations or memories that feel true and are delivered with full confidence but do not correspond to reality. This happens when schemas and mental models fill in gaps without verification (see Mental Models). A patient with memory damage might describe in convincing detail an event that never happened. Even healthy people show the effect: in the Deese-Roediger-McDermott (DRM) paradigm, participants who study a list of words like bed, rest, awake, tired, and dream will later confidently "remember" having seen the word sleep, which was never presented (Roediger & McDermott, 1995). The schema activated by the related words generates a false memory that feels just as real as a genuine one.

The underlying mechanism is similar in both cases: the system generates the most probable completion given the context. When the context aligns with reality, the output is accurate. When it does not, the output is a fabrication – and the fabrication often sounds just as convincing as the truth, sometimes more so. The same process that generates good answers also generates compelling-sounding bad ones. Neither humans nor standard LLMs consistently translate their underlying uncertainty into accurate confidence signals.

Humans have System 2 as a partial check on confabulation. You can pause, reflect, and ask yourself whether your memory is reliable or whether your explanation holds up under scrutiny. This check is imperfect, as the many biases documented in this book attest, but it exists. Standard LLMs lack an equivalent verification mechanism. Reasoning models and retrieval-augmented systems – which connect the model to external databases it can check against – are attempts to add such checks, paralleling how human education and training aim to strengthen habits of verification. But the fundamental vulnerability remains: any system that generates output by pattern completion is at risk of producing fabrications when the patterns lead away from the truth.

Overconfident Machines

Closely related to hallucination is the problem of calibration. A well-calibrated system assigns high confidence to claims that are likely to be correct and low confidence to claims that might be wrong (see Overconfidence; see Bayesian Reasoning). As discussed earlier in this book, humans are often poorly calibrated, particularly in the direction of overconfidence. People tend to be more certain than they should be, assigning confidence intervals that are too narrow and probability estimates that are too extreme.

LLMs show a similar pattern. When researchers pose large sets of trivia or factual questions to LLMs and ask the models to rate their confidence in each answer, stated confidence frequently fails to track actual accuracy. A model might express 95 percent confidence in an answer that it gets right only 70 percent of the time. Conversely, it rarely says "I am not sure" when it should. In Bayesian terms, this amounts to a posterior distribution that is too narrow – the model behaves as though it is more certain than the evidence warrants.

Improving calibration in AI is an active area of research, just as improving human calibration is a goal of debiasing interventions. Some approaches involve training models to express uncertainty more honestly. Others involve external calibration checks, where a separate system evaluates the reliability of the model's output. These efforts parallel the structured feedback and accountability mechanisms that help improve human calibration in professional settings.

The parallel suggests a deeper point. Overconfidence may not be a flaw specific to biological brains. It may be a natural consequence of any system that generates its best guess and presents it as output. Unless the system has a separate mechanism for evaluating and communicating uncertainty, the default will be to present responses with more confidence than they deserve.

A Bayesian Lens

The Bayesian framework that runs through this book (see Bayesian Reasoning) offers a unifying way to understand the phenomena discussed in this chapter. Loosely speaking, the training data serves as the prior: a vast body of knowledge and patterns that the model brings to any new problem. The prompt serves as the evidence: new information that should update the model's response. The output is the posterior: the most likely completion given the combination of prior and evidence.

This analogy is a simplification – technically, LLMs sample from a learned probability distribution rather than computing a Bayesian posterior in the textbook sense – but it is useful because it connects AI failures to the specific forms of Bayesian error discussed throughout this book. Biased priors in training data produce biased outputs, just as biased experiential priors produce biased human judgments. Hallucination occurs when the prior dominates over the evidence in the prompt, paralleling human confirmation bias and belief perseverance (see Belief Formation and Perseverance). Calibration failures reflect posteriors that are too narrow. And chain-of-thought reasoning improves performance by making the evidence-integration process more explicit – analogous to how deliberate Bayesian computation outperforms intuitive shortcuts.

This perspective clarifies why AI and human cognition share so many failure modes. Both are systems that combine stored knowledge with new information to produce judgments. When the stored knowledge is biased, the judgments are biased. When the integration process is too fast or too shallow, errors follow. The specific hardware differs enormously, but the computational logic – and its characteristic vulnerabilities – is remarkably similar.

Summary

Large language models provide a useful mirror for understanding human judgment and decision-making. The Turing test, proposed over seventy years ago, is arguably met by modern AI in many constrained settings, yet continued reluctance to accept this reveals how people protect their intuitions about intelligence. LLM output resembles System 1 processing – fast, associative, and pattern-based – while chain-of-thought prompting adds a System 2-like layer that improves performance on tasks requiring deliberation. LLMs excel at crystallized intelligence but struggle with the fluid intelligence needed for genuinely novel reasoning. Their biases – anchoring, framing, sycophancy – are largely inherited from human-generated training data, suggesting these biases are embedded in language and culture, not only in biological brains. Hallucination parallels human confabulation: both arise from pattern completion without verification. And calibration failures mirror human overconfidence. From a Bayesian perspective, these failures map onto biased priors, prior-dominated posteriors, and overly narrow uncertainty estimates – the same errors that characterize human judgment throughout this book.

CONTRIBUTORS

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